

Related U.S. Application Data

continuation of application No. 17/521,380, filed on Nov. 8, 2021, now Pat. No. 11,495,160, which is a continuation of application No. 17/345,666, filed on Jun. 11, 2021, now Pat. No. 11,189,212, which is a continuation of application No. 17/180,441, filed on Feb. 19, 2021, now Pat. No. 11,037,481, which is a continuation-in-part of application No. 17/009,408, filed on Sep. 1, 2020, now Pat. No. 11,043,157, which is a continuation-in-part of application No. 16/887,807, filed on May 29, 2020, now Pat. No. 10,950,162, which is a continuation-in-part of application No. 16/860,769, filed on Apr. 28, 2020, now Pat. No. 10,950,161, which is a continuation-in-part of application No. 16/853,203, filed on Apr. 20, 2020, now Pat. No. 10,997,896, which is a continuation-in-part of application No. 16/831,157, filed on Mar. 26, 2020, now Pat. No. 10,950,160, which is a continuation of application No. 16/659,307, filed on Oct. 21, 2019, now Pat. No. 10,607,527.

- (60) Provisional application No. 62/876,878, filed on Jul. 22, 2019, provisional application No. 62/847,630, filed on May 14, 2019, provisional application No. 62/805,705, filed on Feb. 14, 2019, provisional application No. 62/750,673, filed on Oct. 25, 2018.

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CPC **G09G 3/3413** (2013.01); **G09G 5/02** (2013.01); **G09G 2300/0452** (2013.01); **G09G 2320/0242** (2013.01); **G09G 2340/06** (2013.01)

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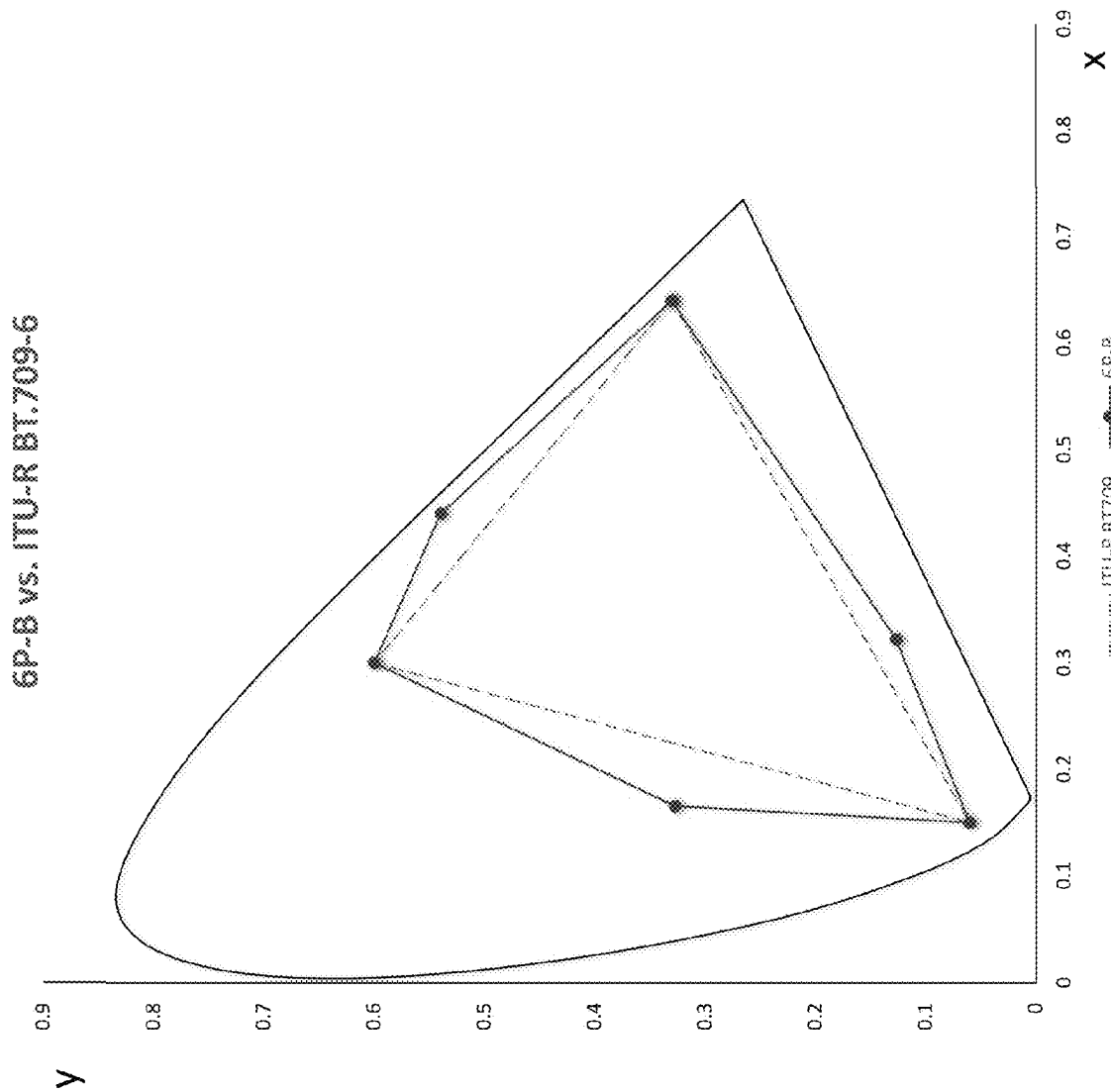


FIG. 1

6P-C Ref D60 vs. SMPTE RP431-2

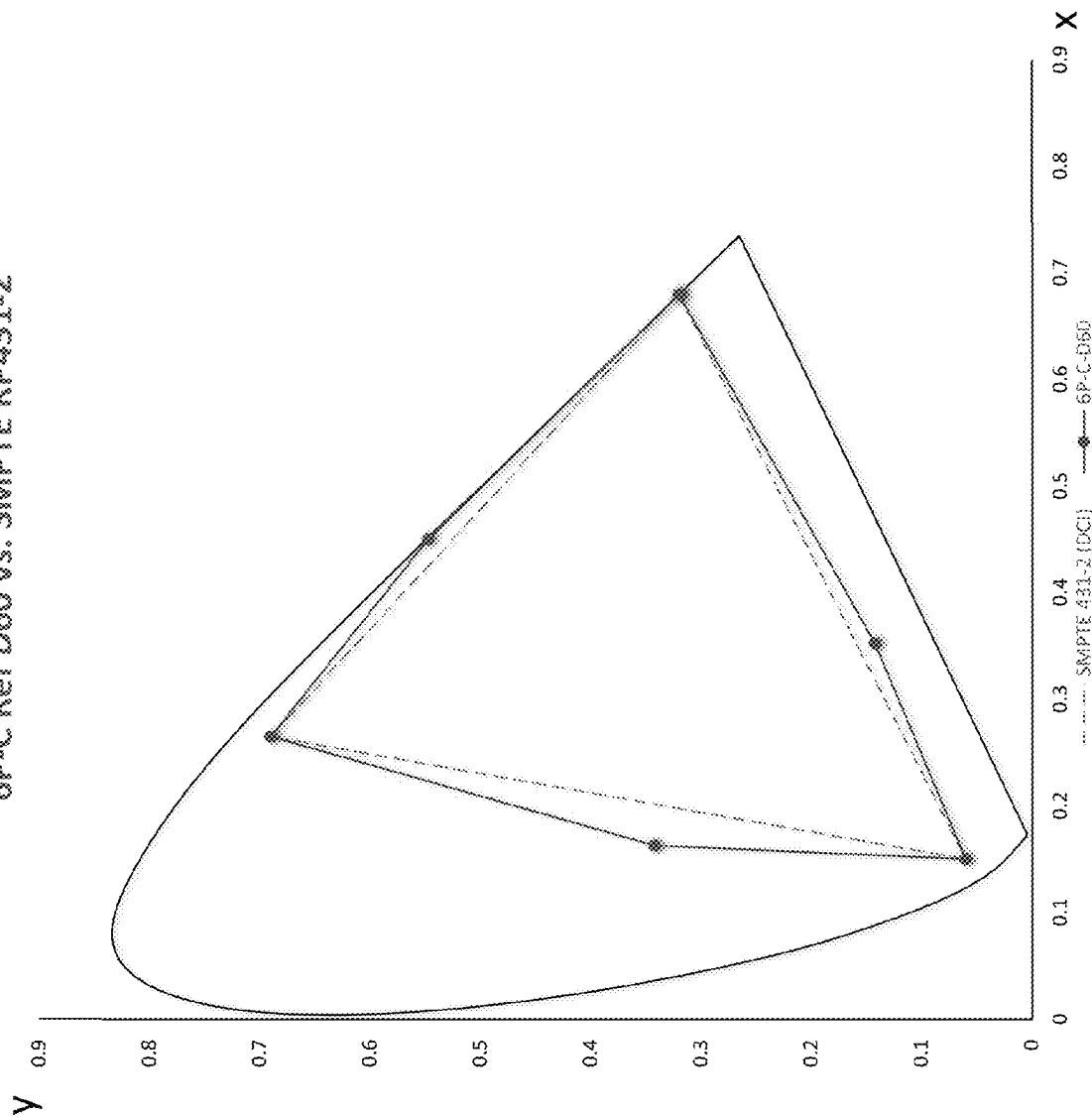


FIG. 2

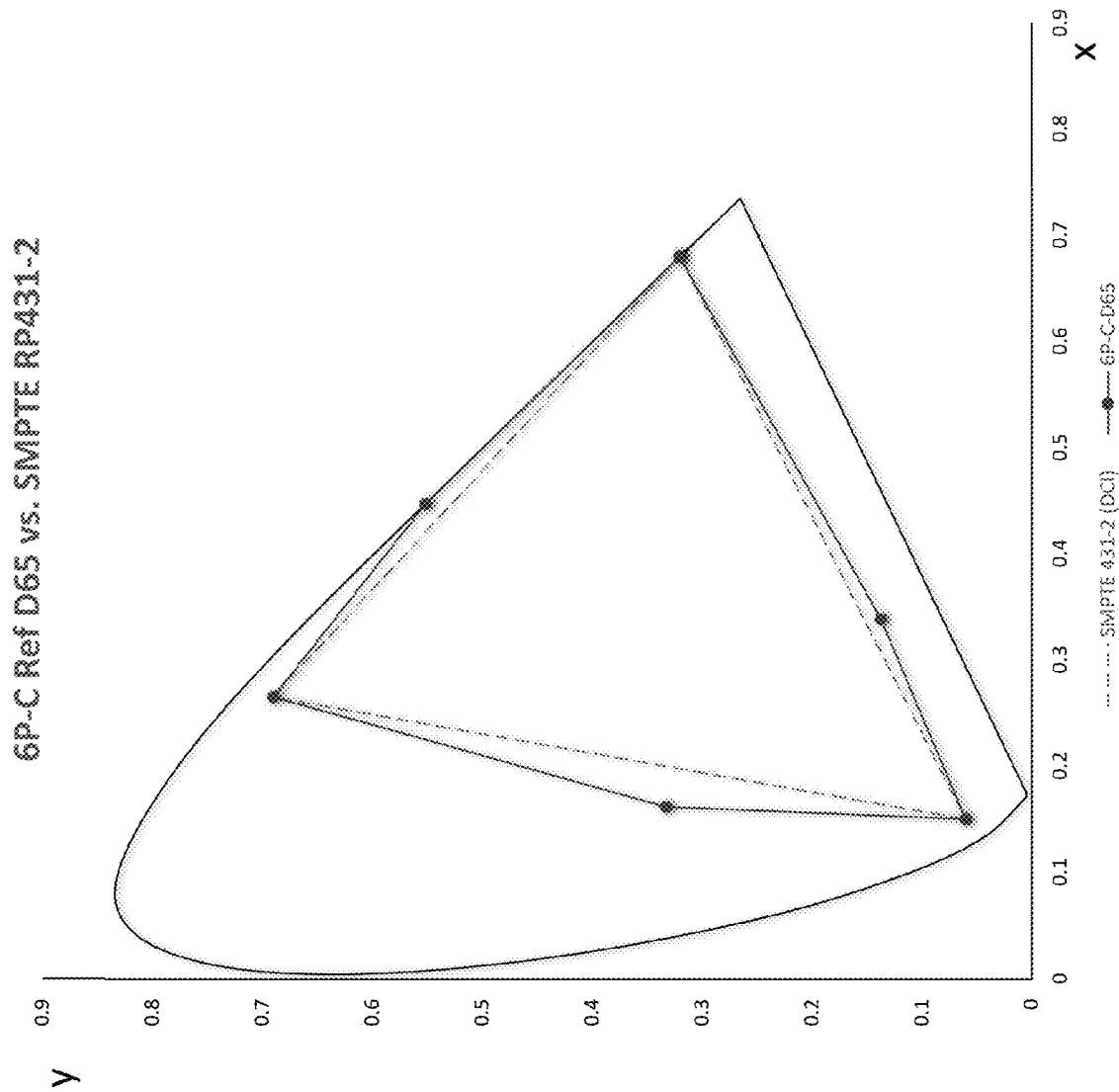


FIG. 3

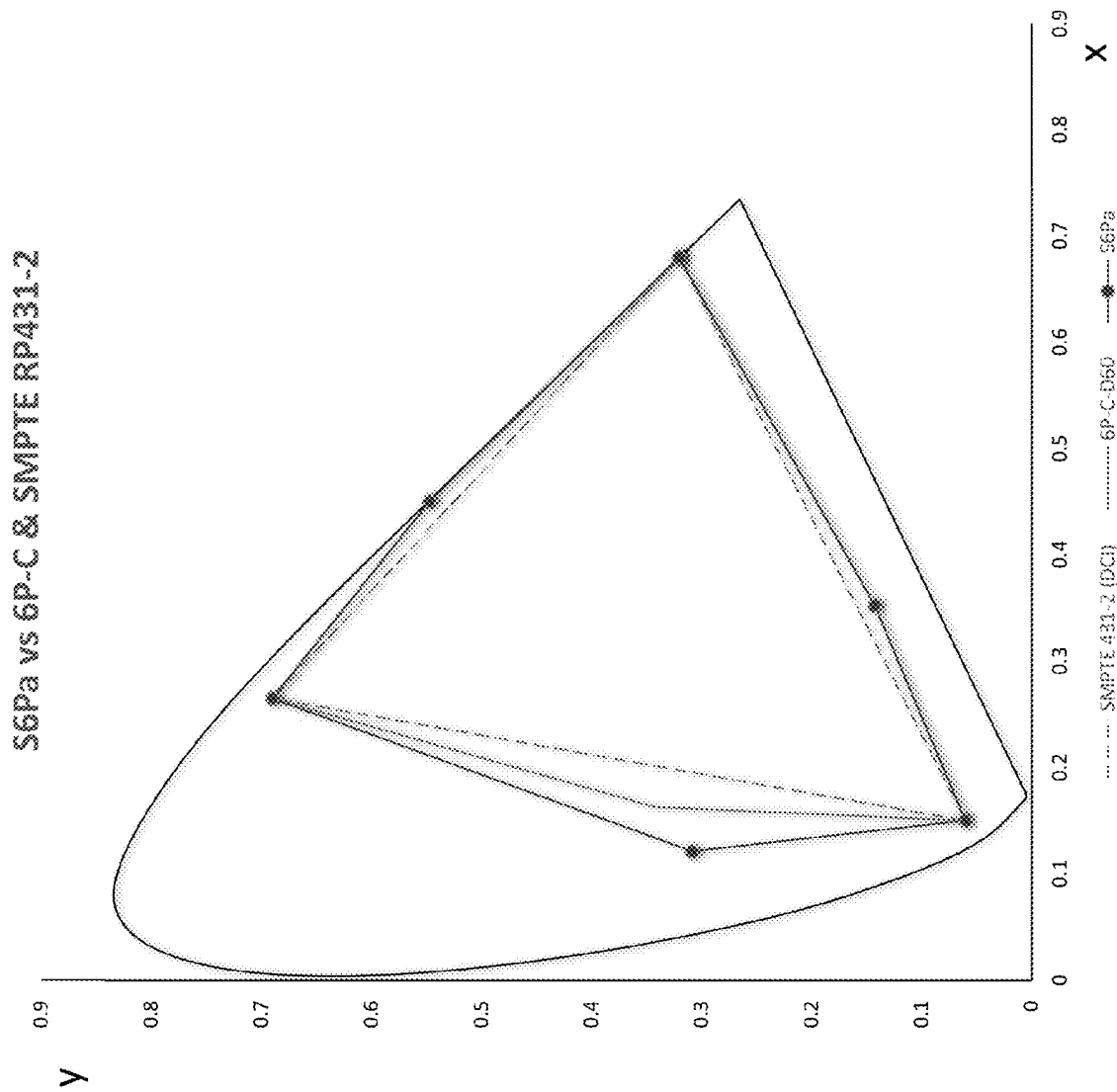


FIG. 4

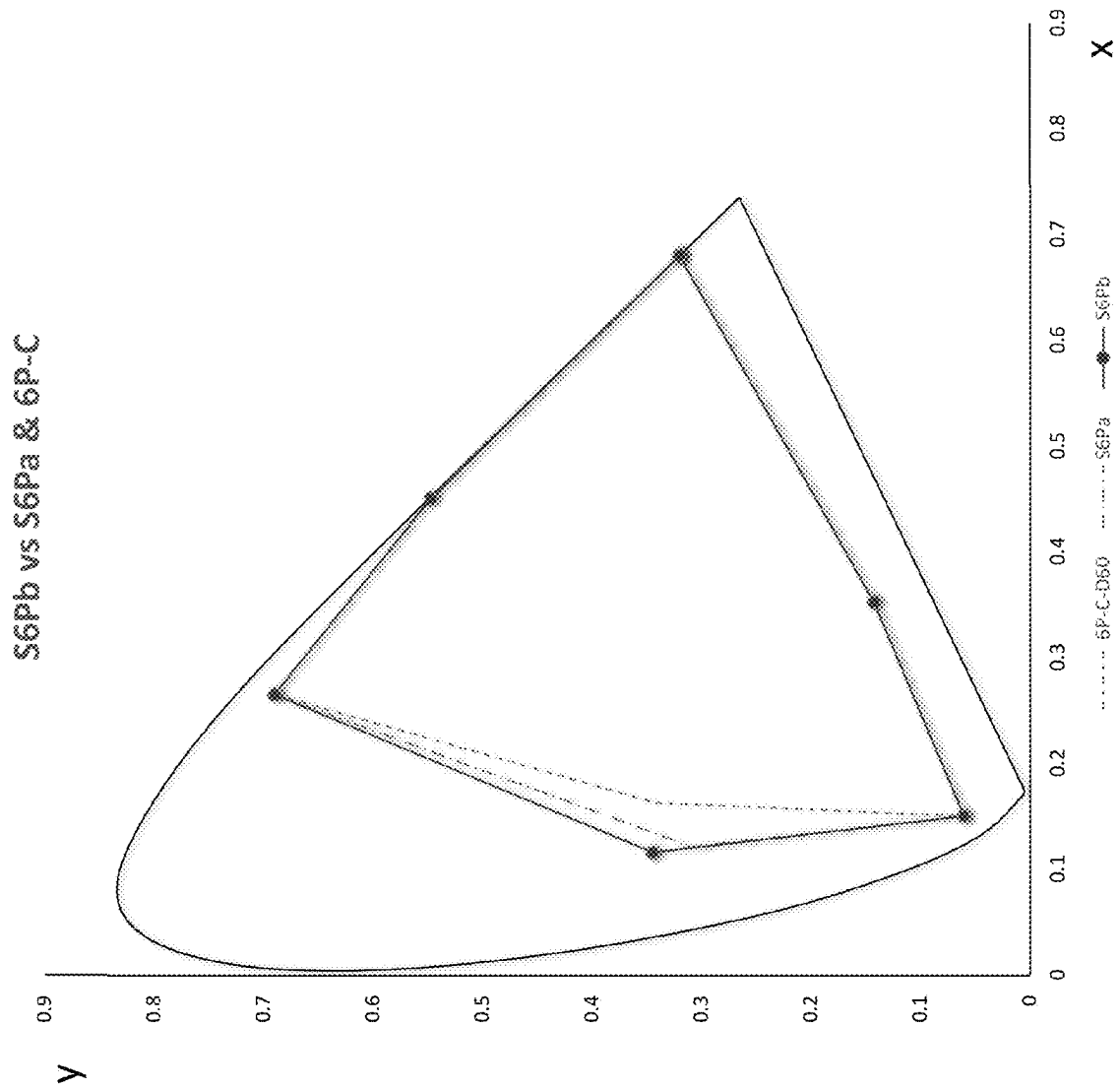
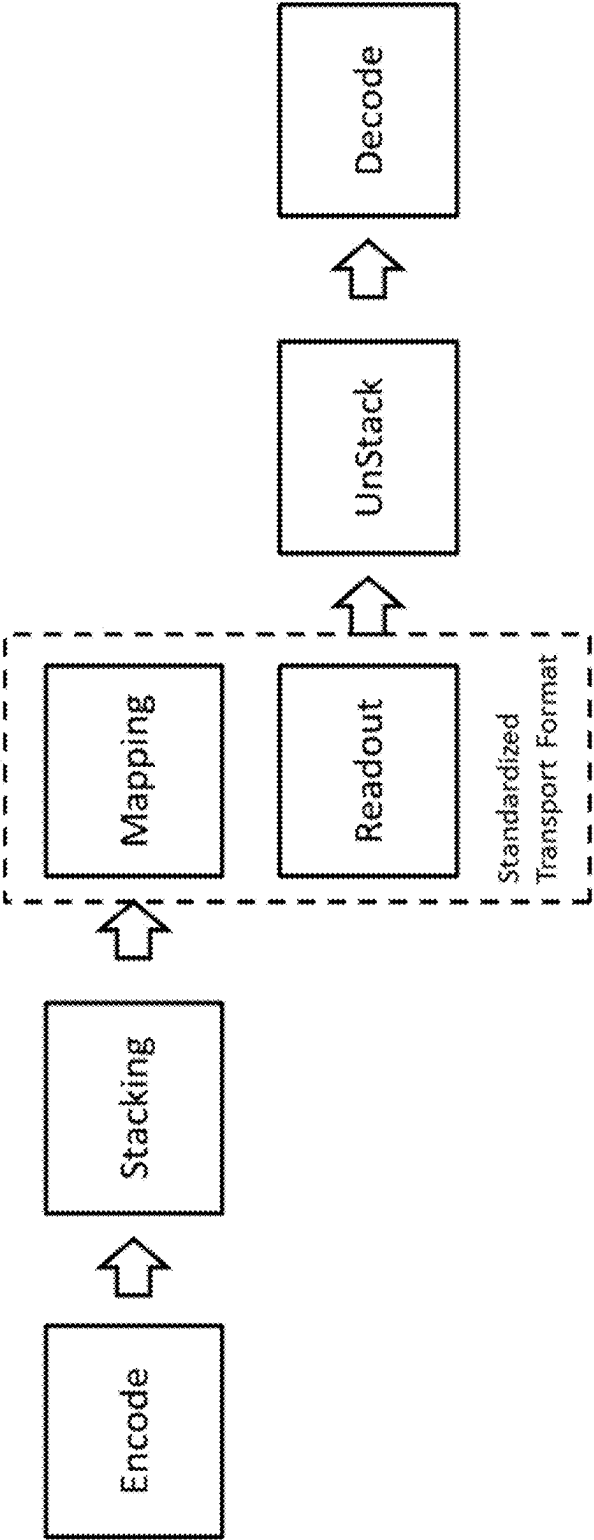
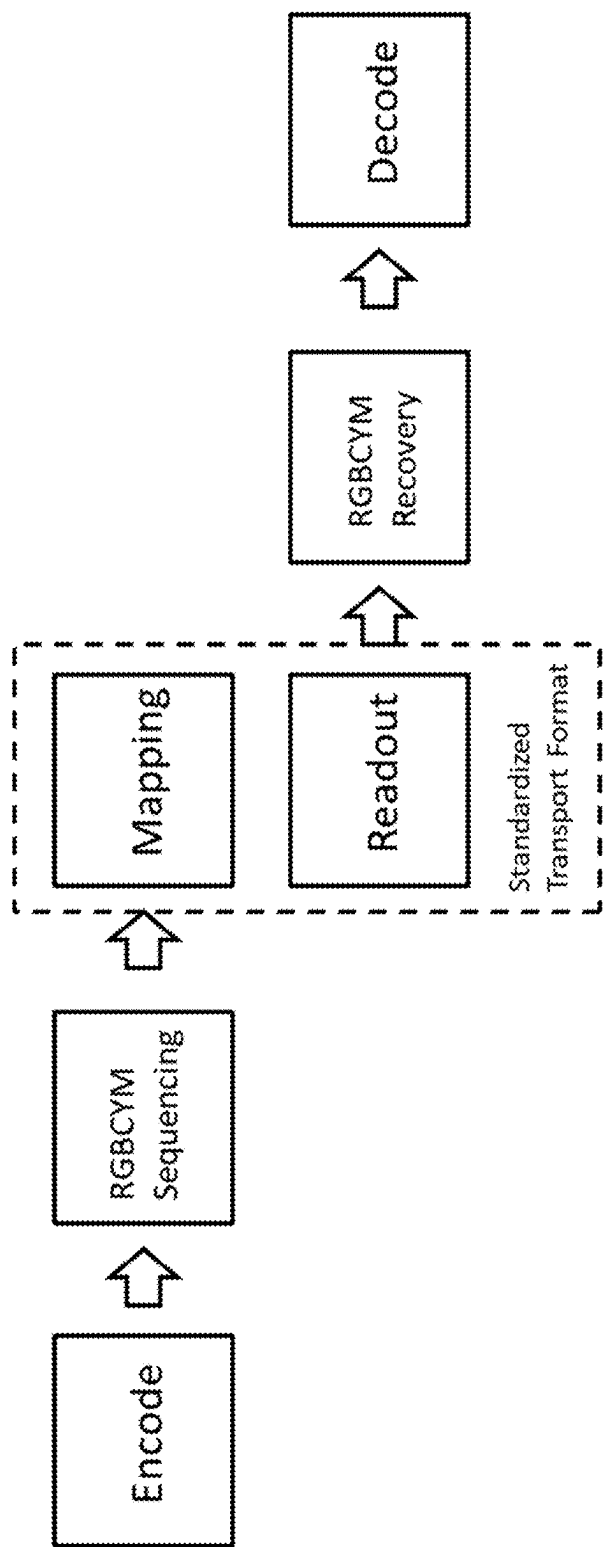


FIG. 5



System 1

FIG. 6



System 2

FIG. 7

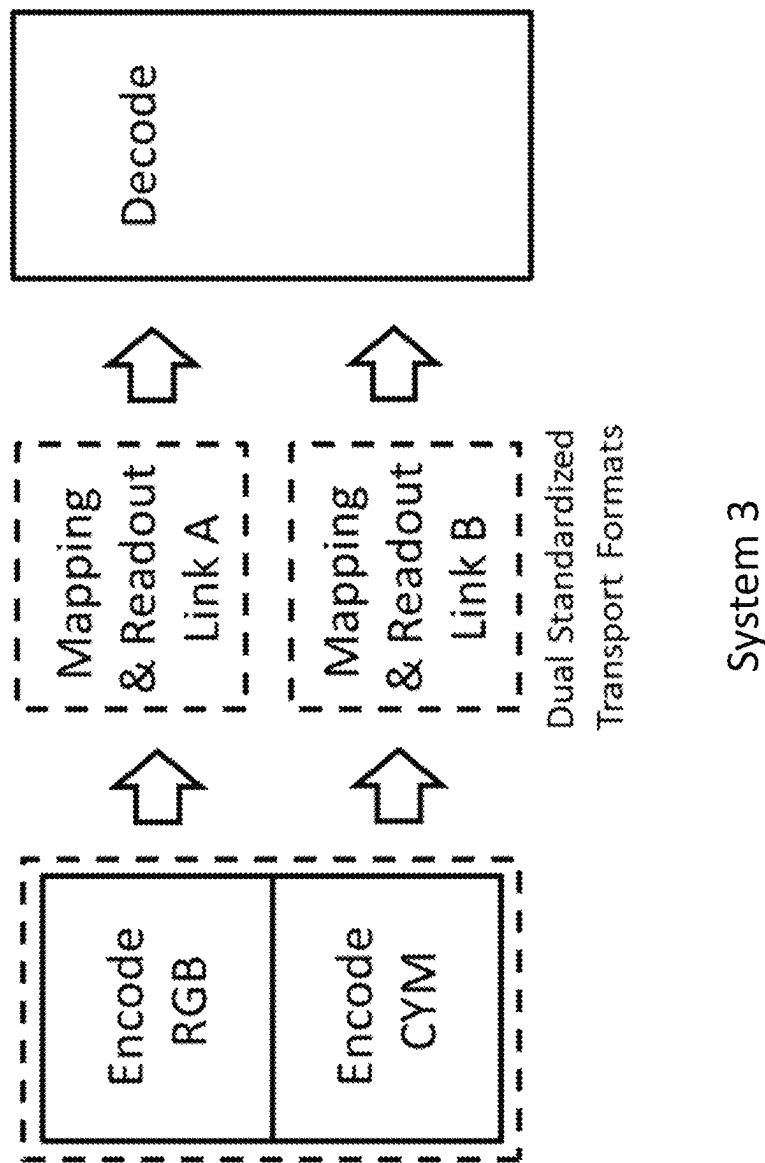


FIG. 8

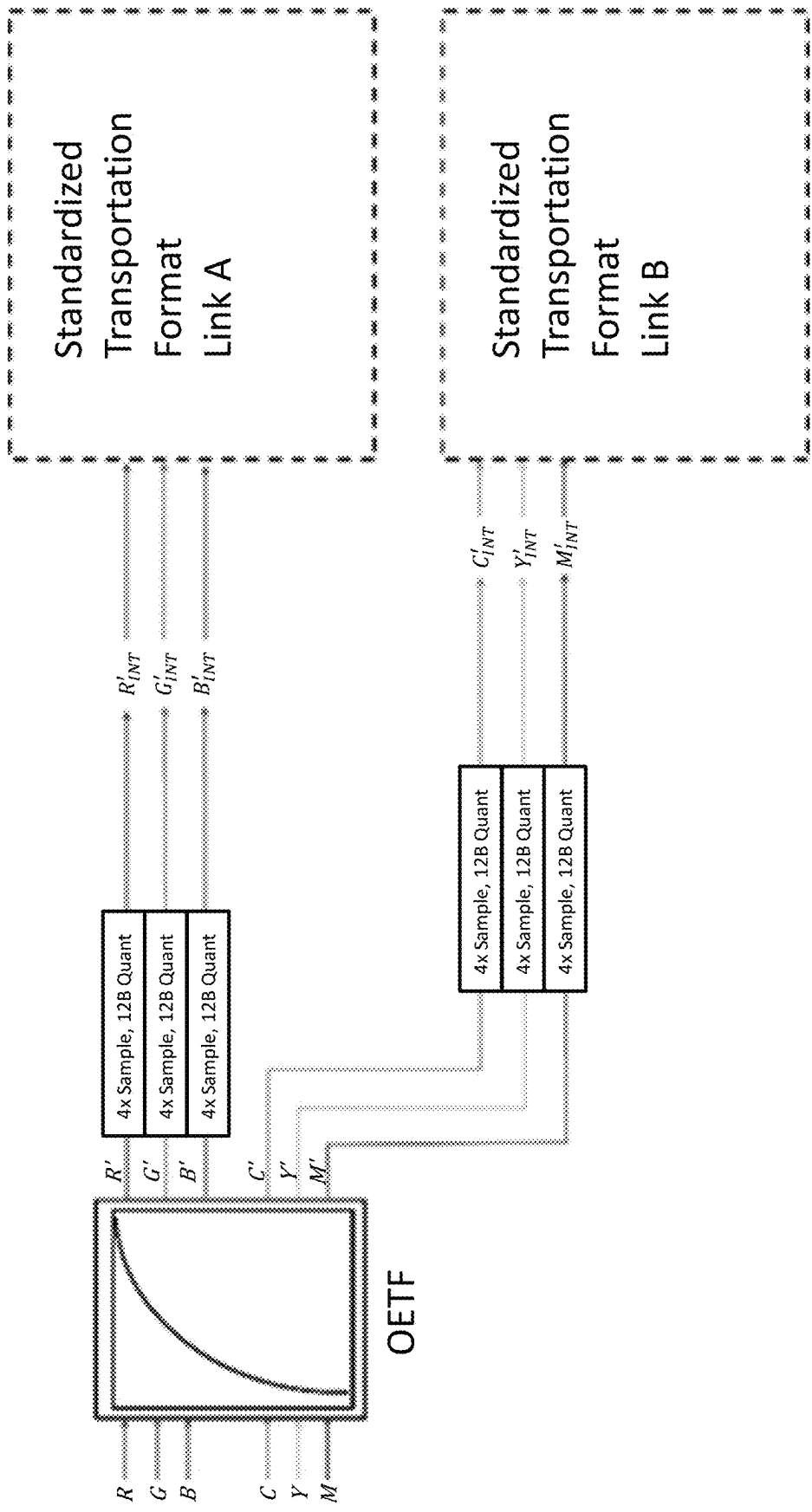


FIG. 9

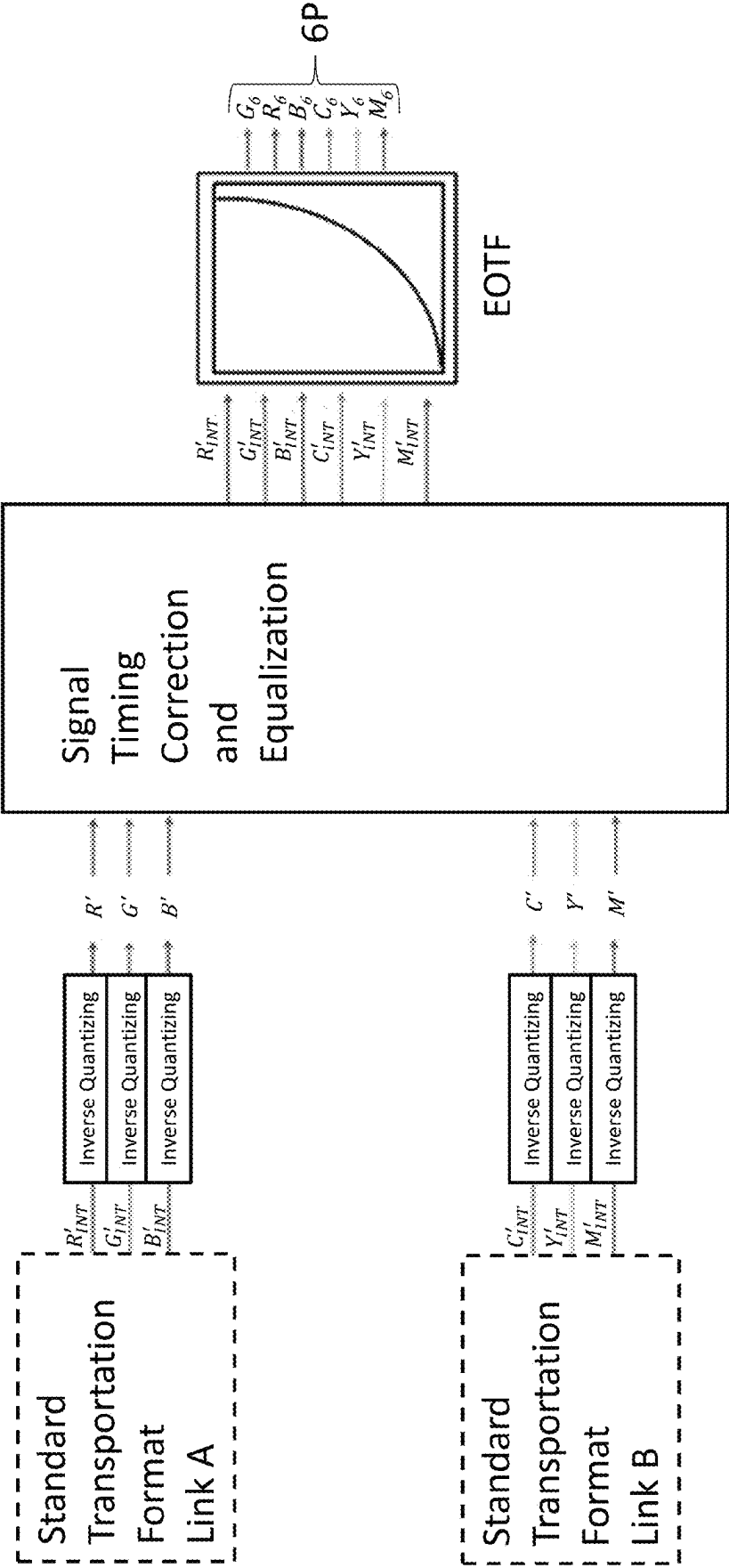


FIG. 10

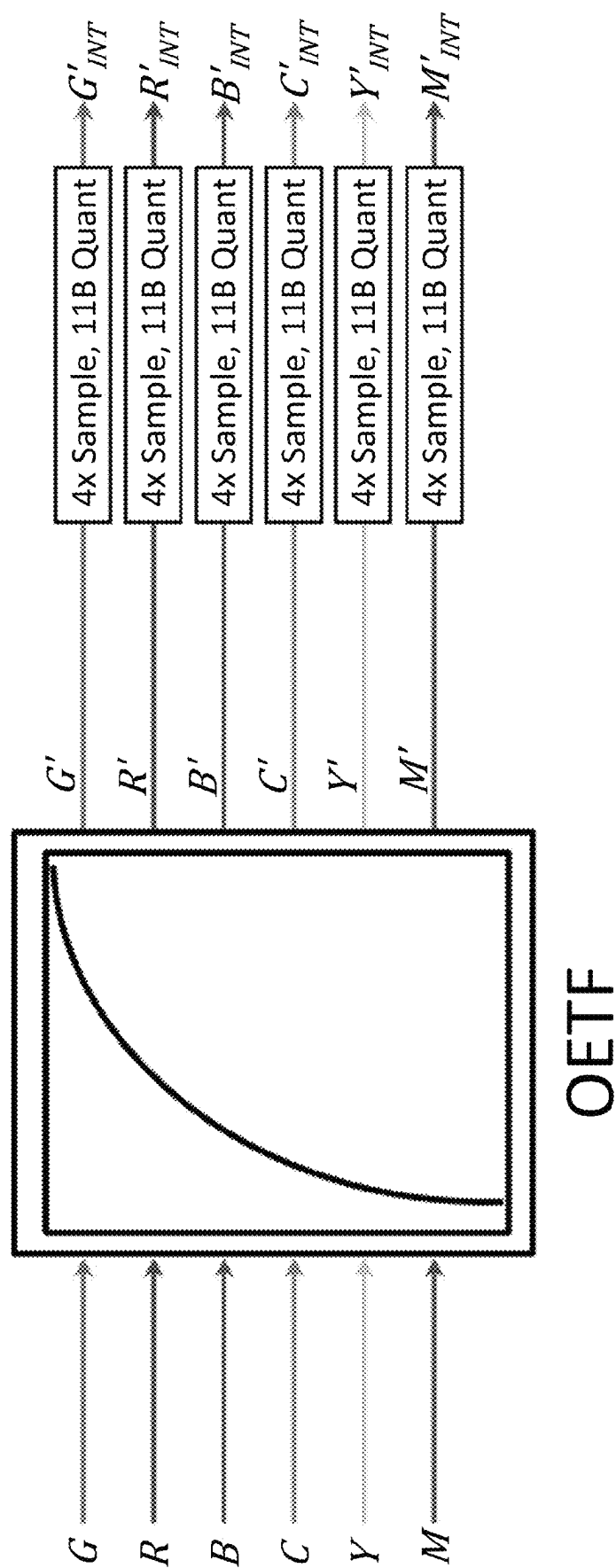


FIG. 11

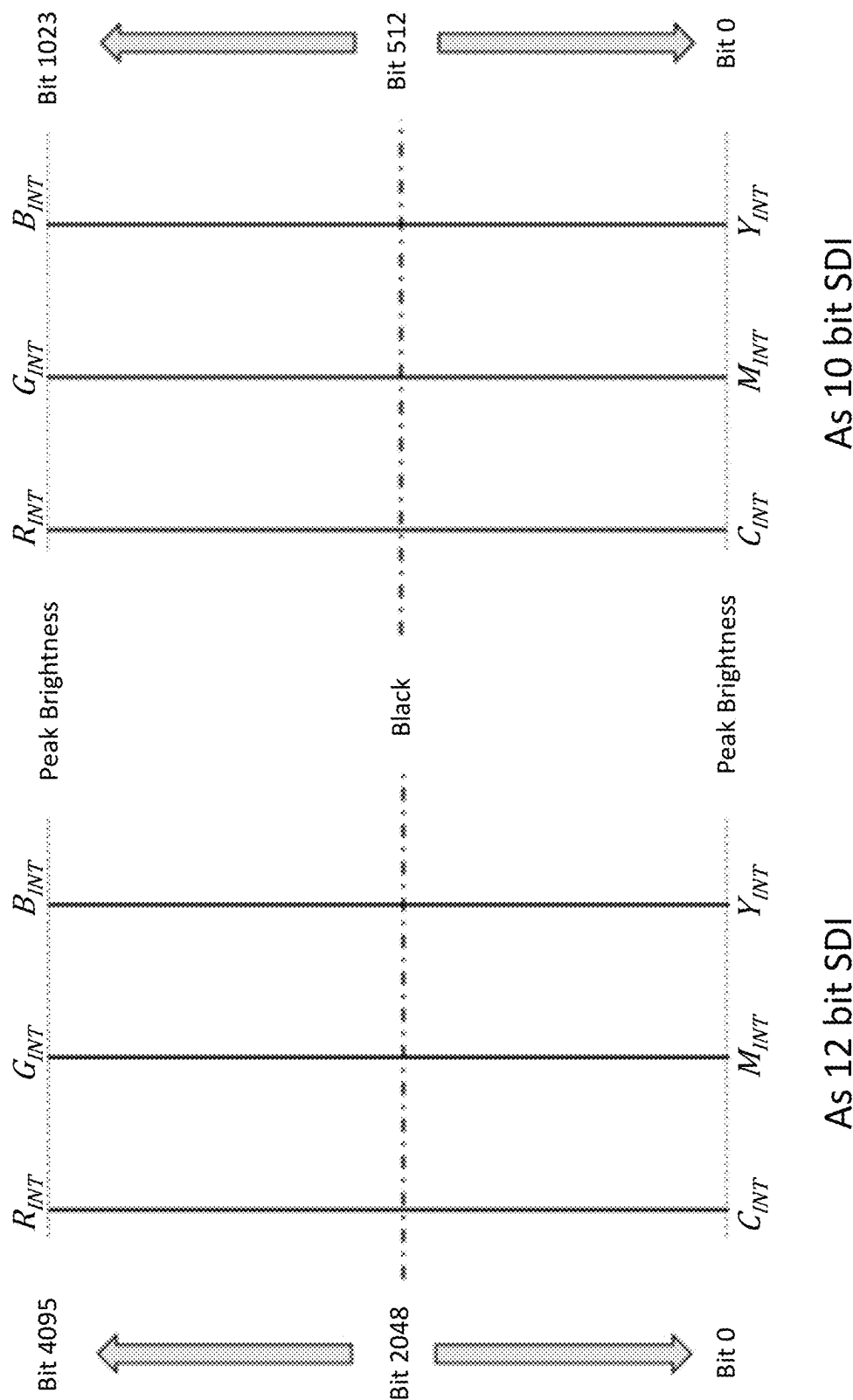


FIG. 12

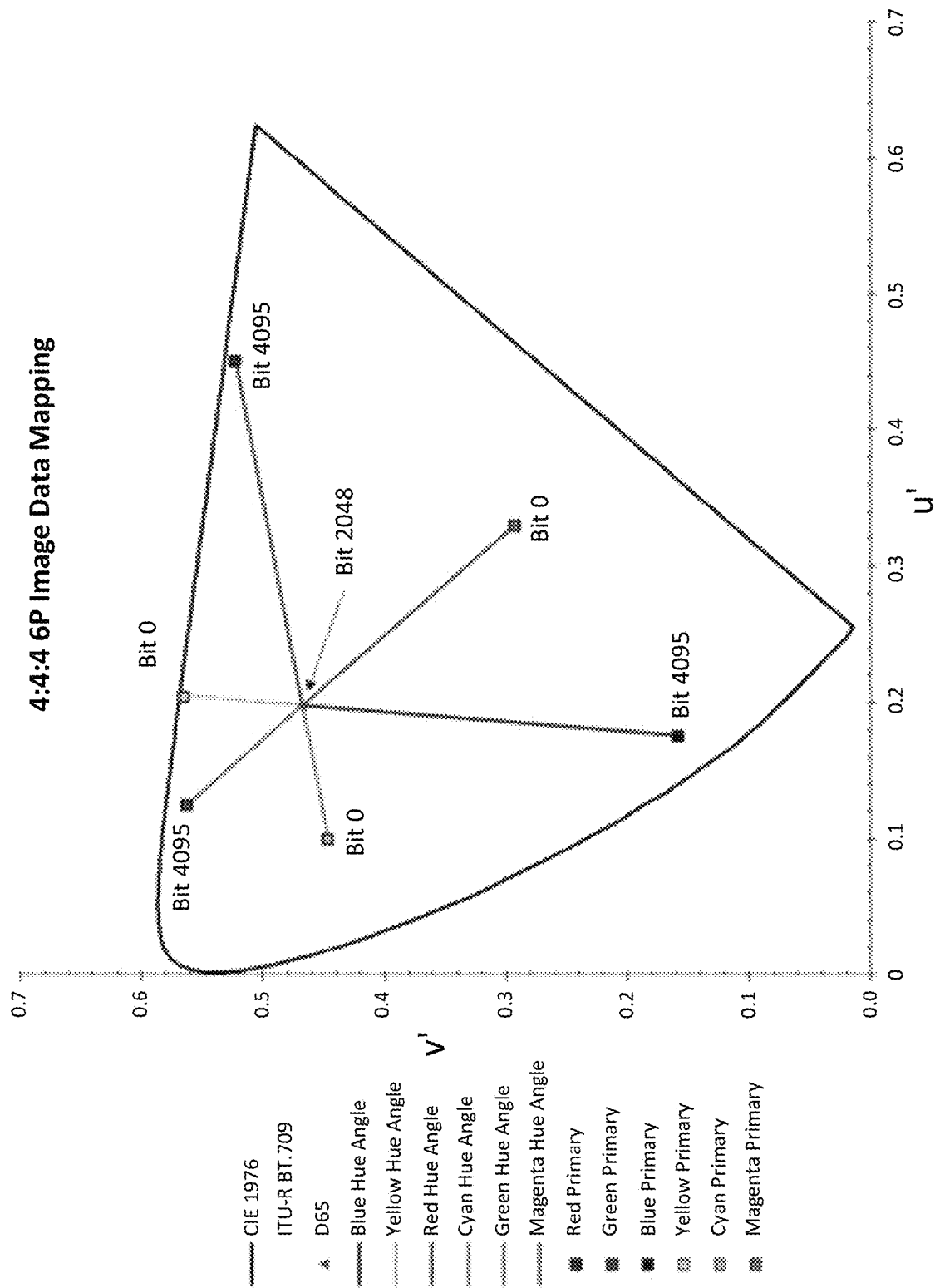


FIG. 13

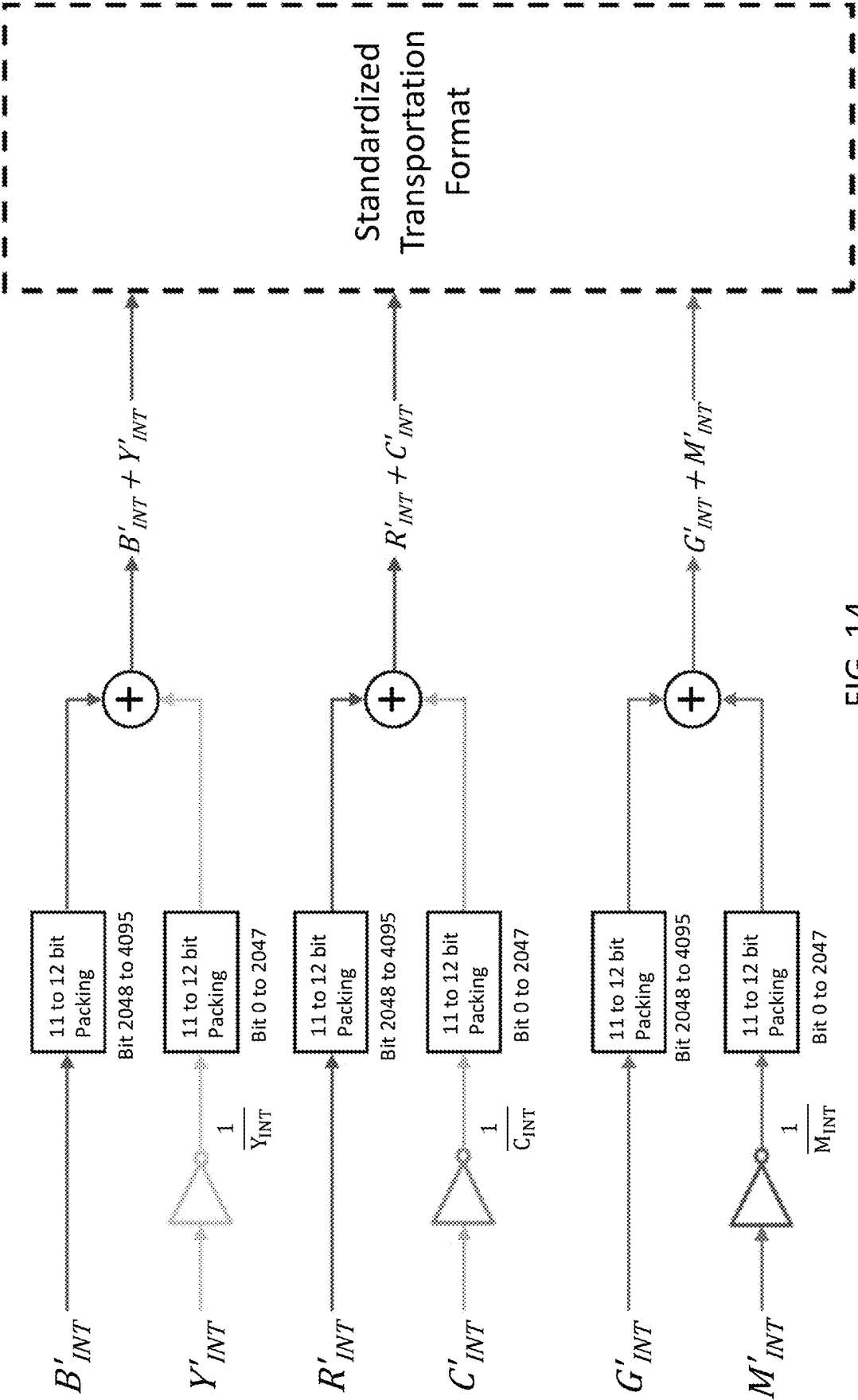


FIG. 14

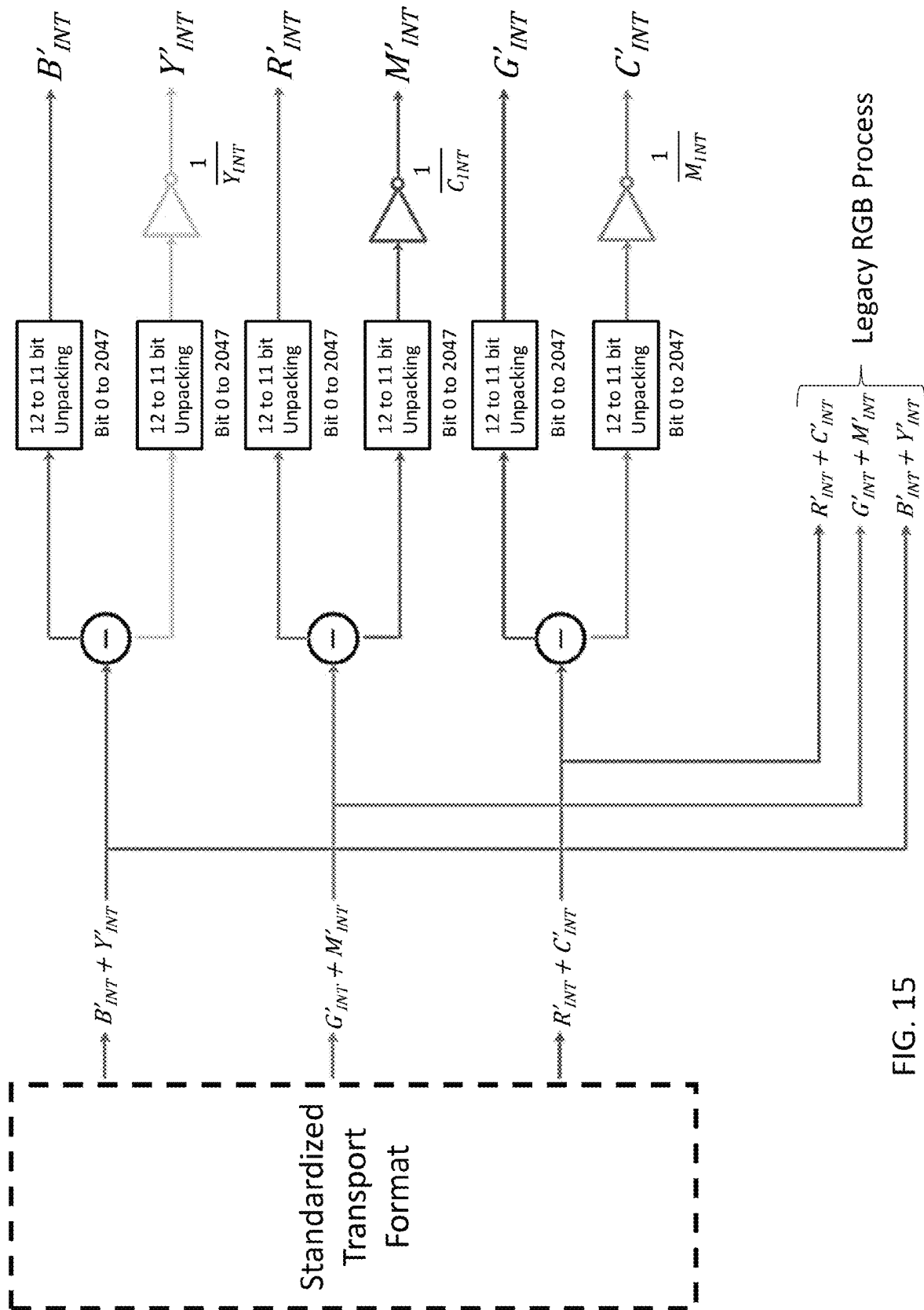


FIG. 15

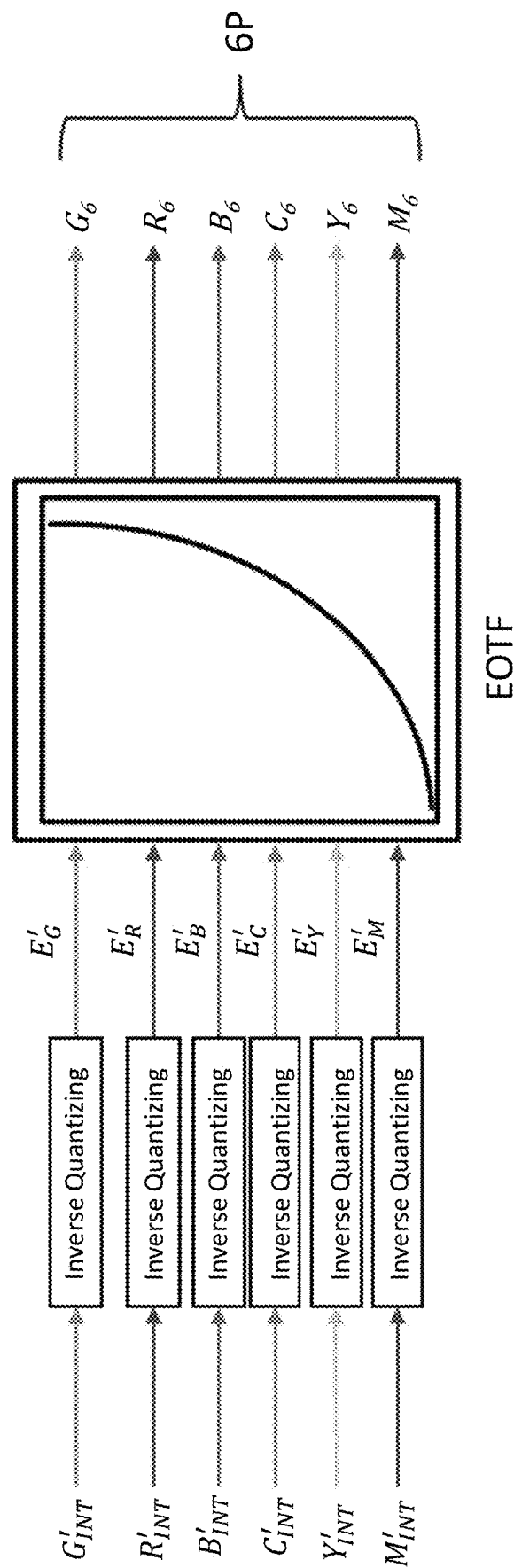
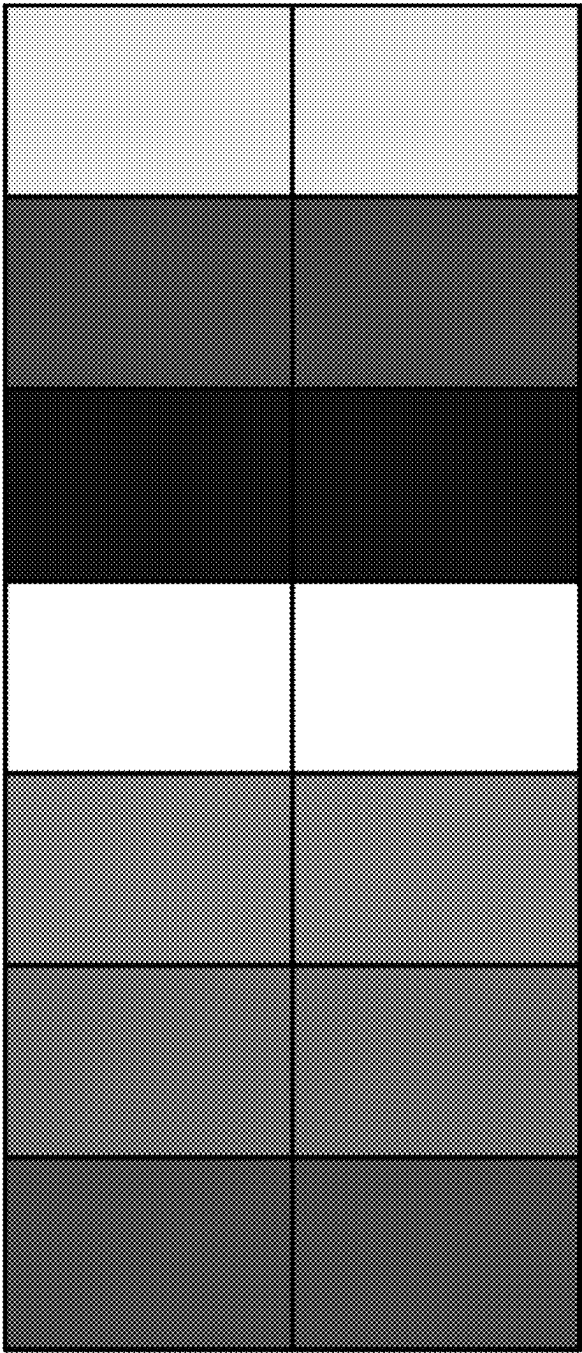


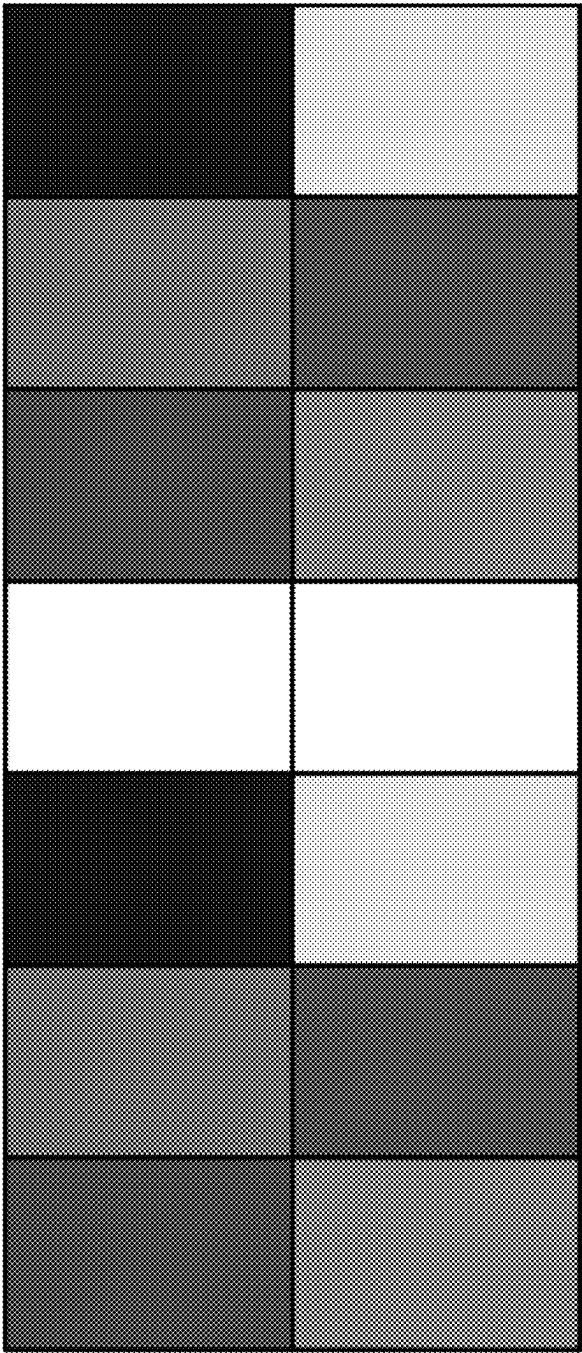
FIG. 16



Line 1

Line 2

FIG. 17



Line 1

Line 2

FIG. 18

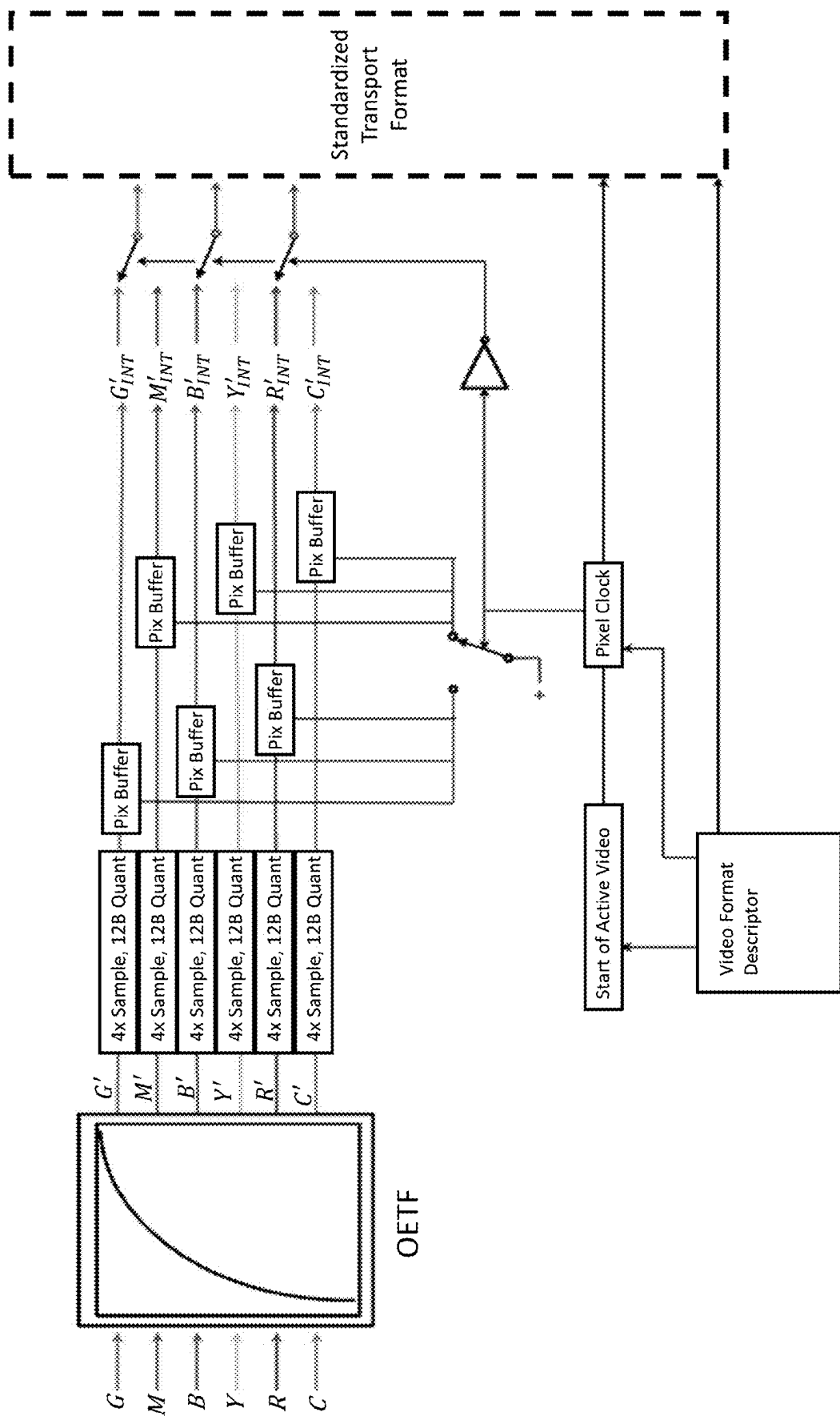


FIG. 19

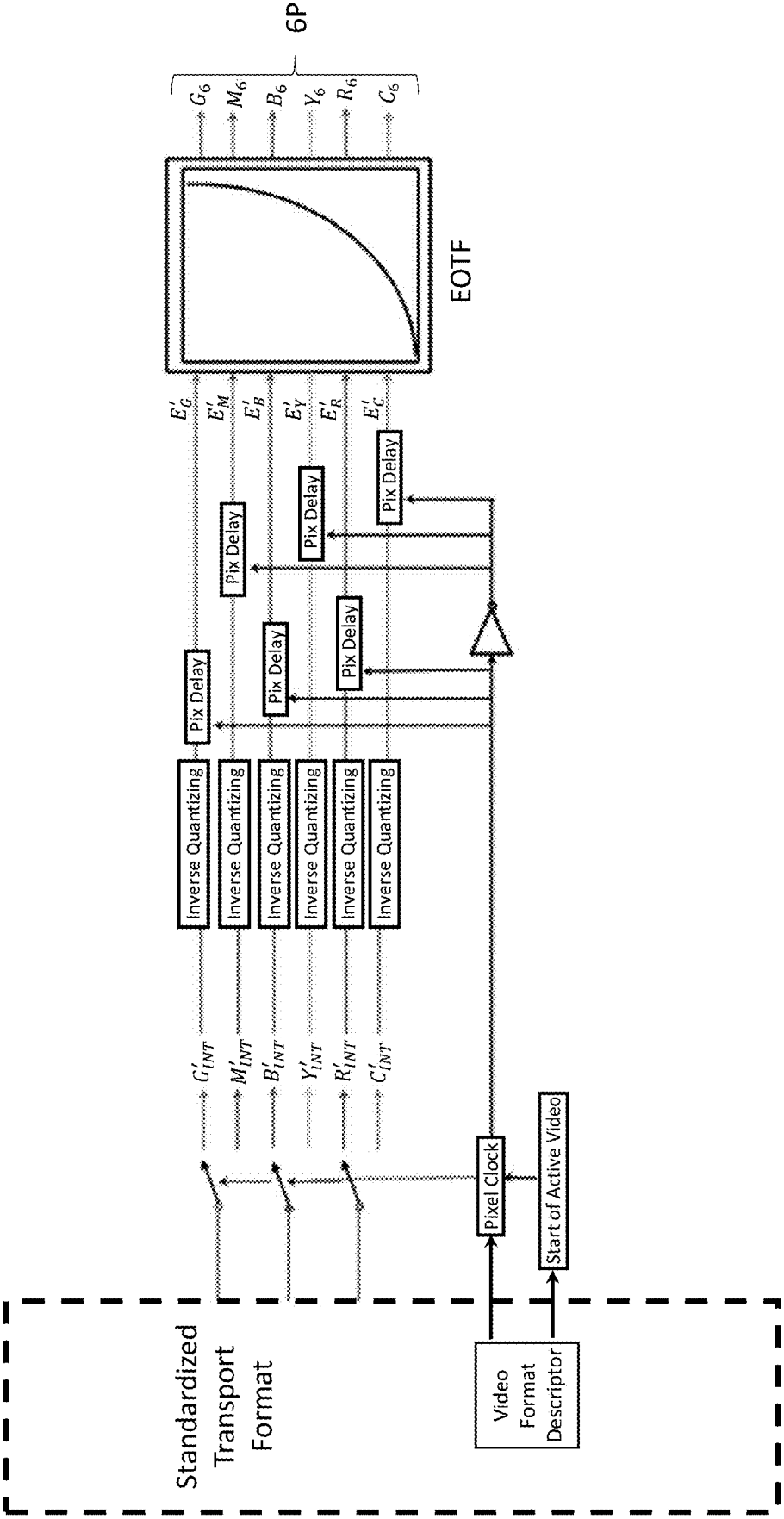


FIG. 20

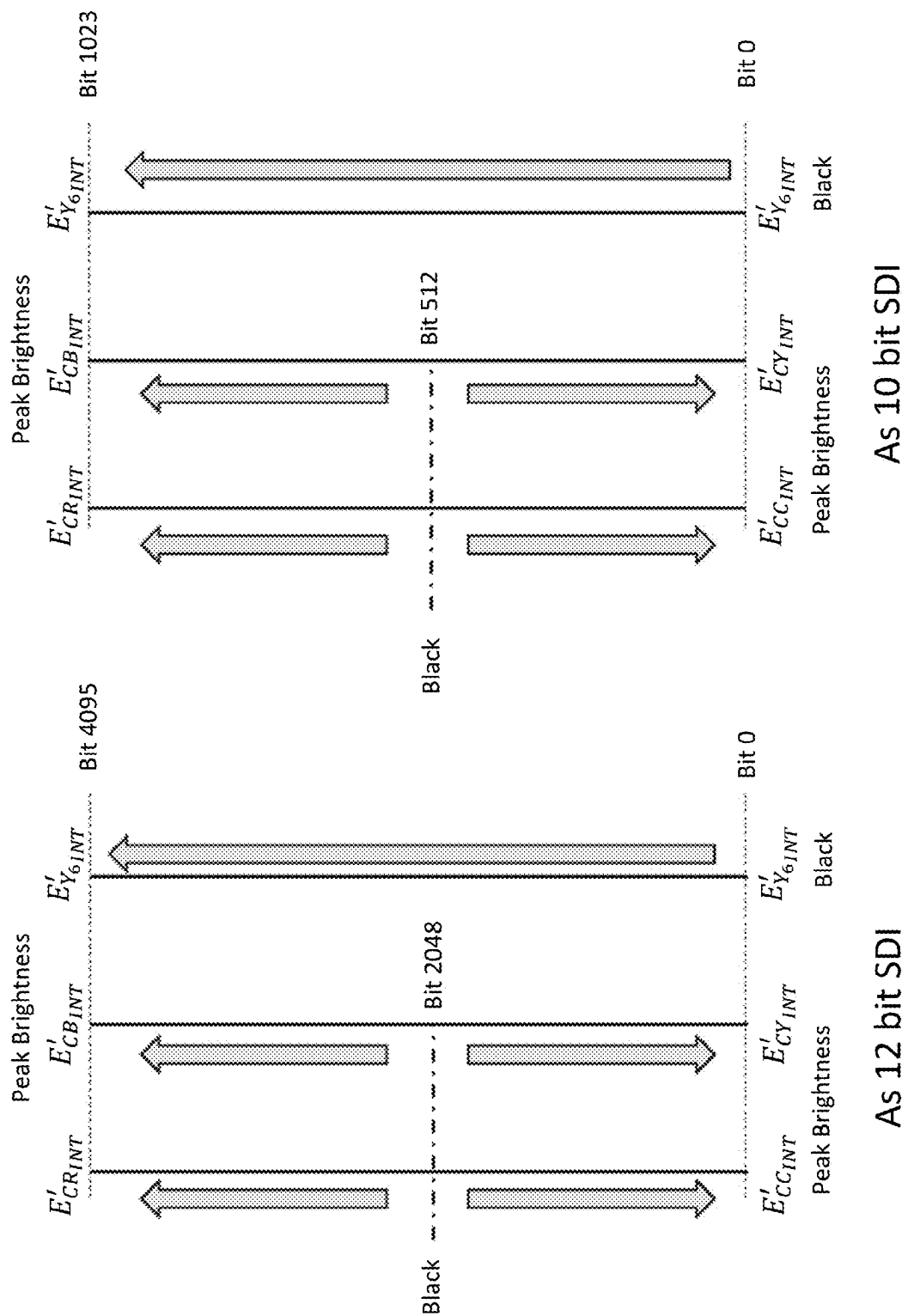


FIG. 21

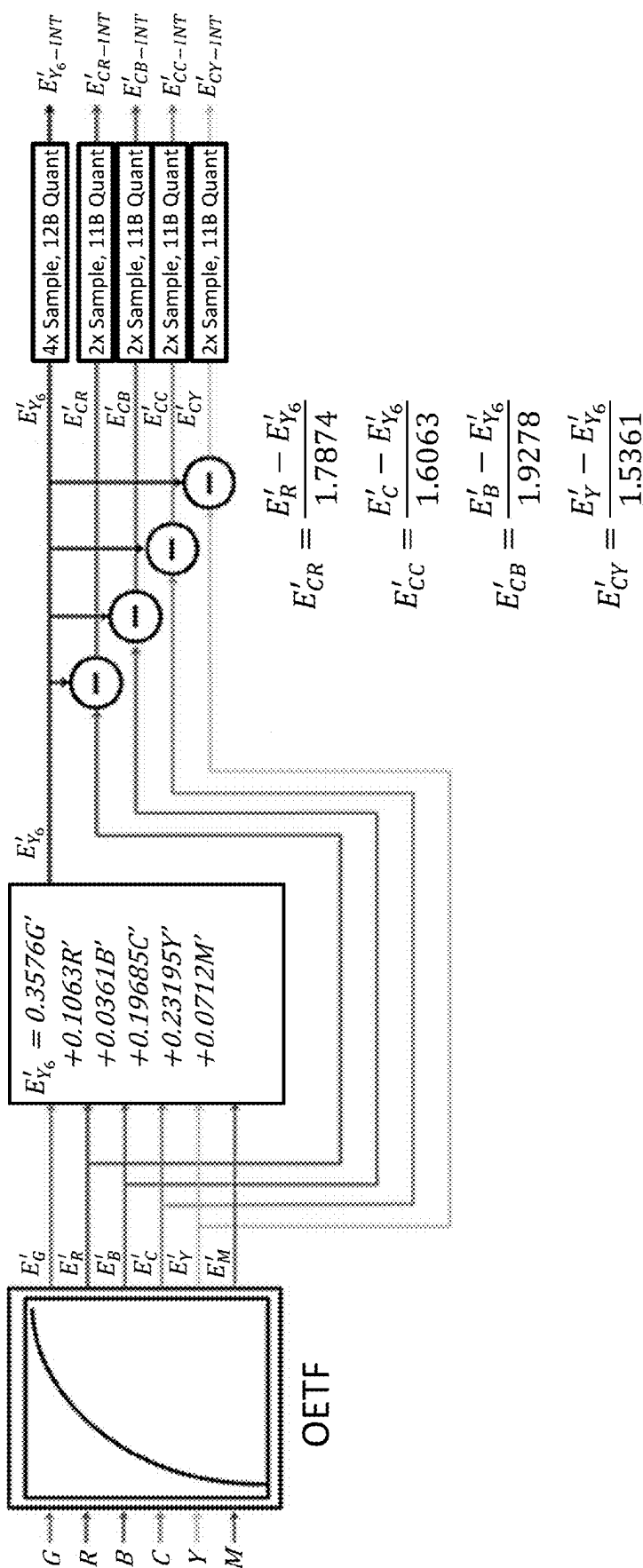


FIG. 22

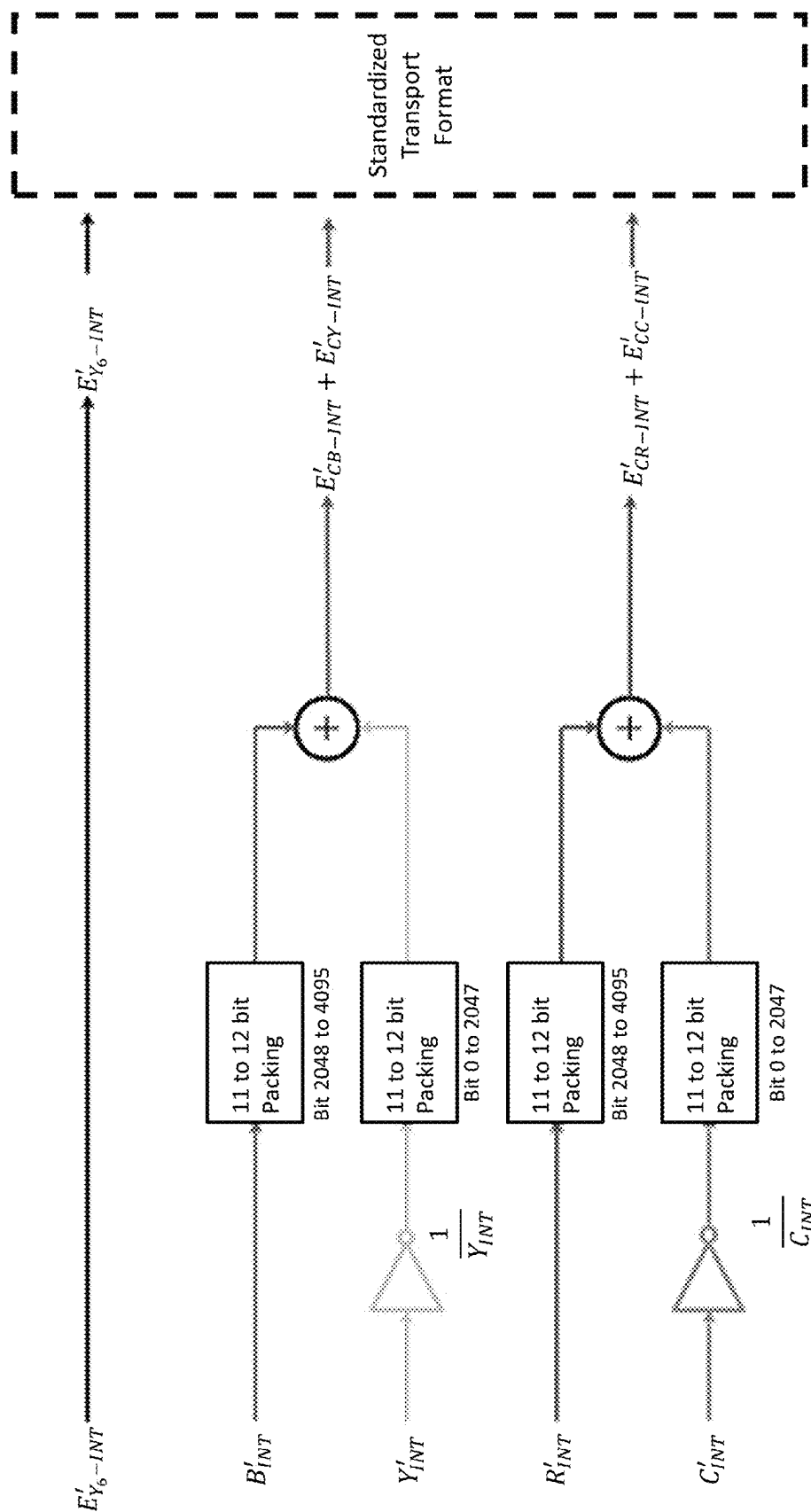


FIG. 23

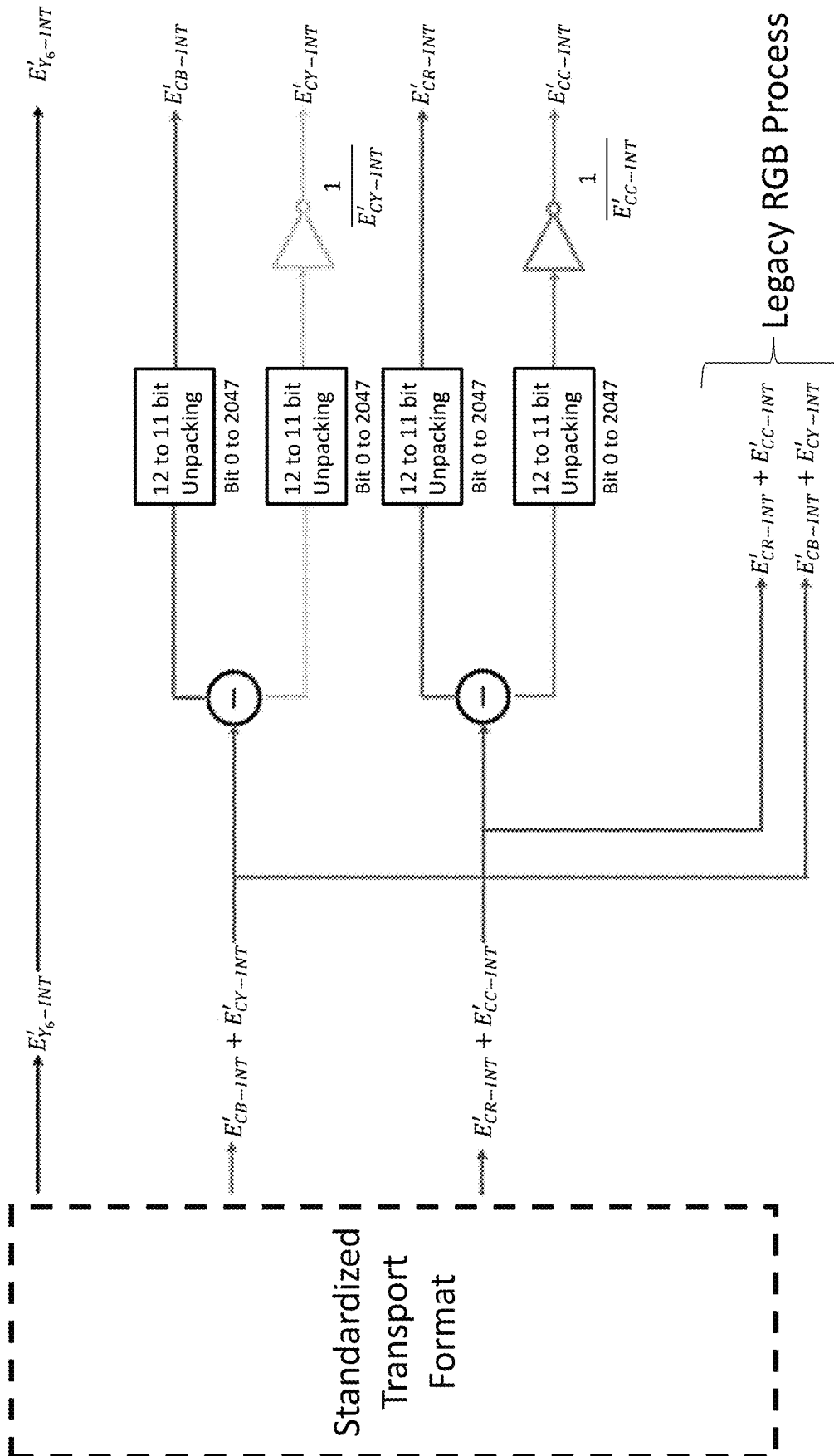


FIG. 24

$$E'_C = \frac{E'_{CC} - E'_{Y_6}}{1.6063}$$
$$E'_M = \frac{\frac{E'_B + \frac{E'_R}{2}}{2} - E'_{CY}}{1.5361}$$

$$E'_B = \frac{E'_{CB} - E'_{Y_6}}{1.9278}$$

$$E'_R = \frac{E'_{CR} - E'_{Y_6}}{1.7874}$$

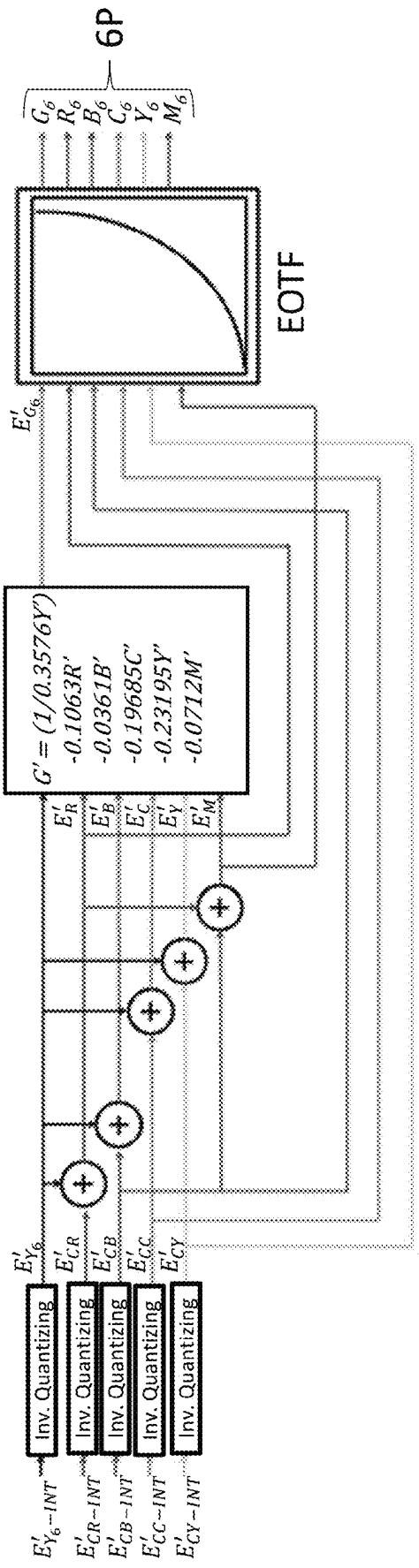


FIG. 25

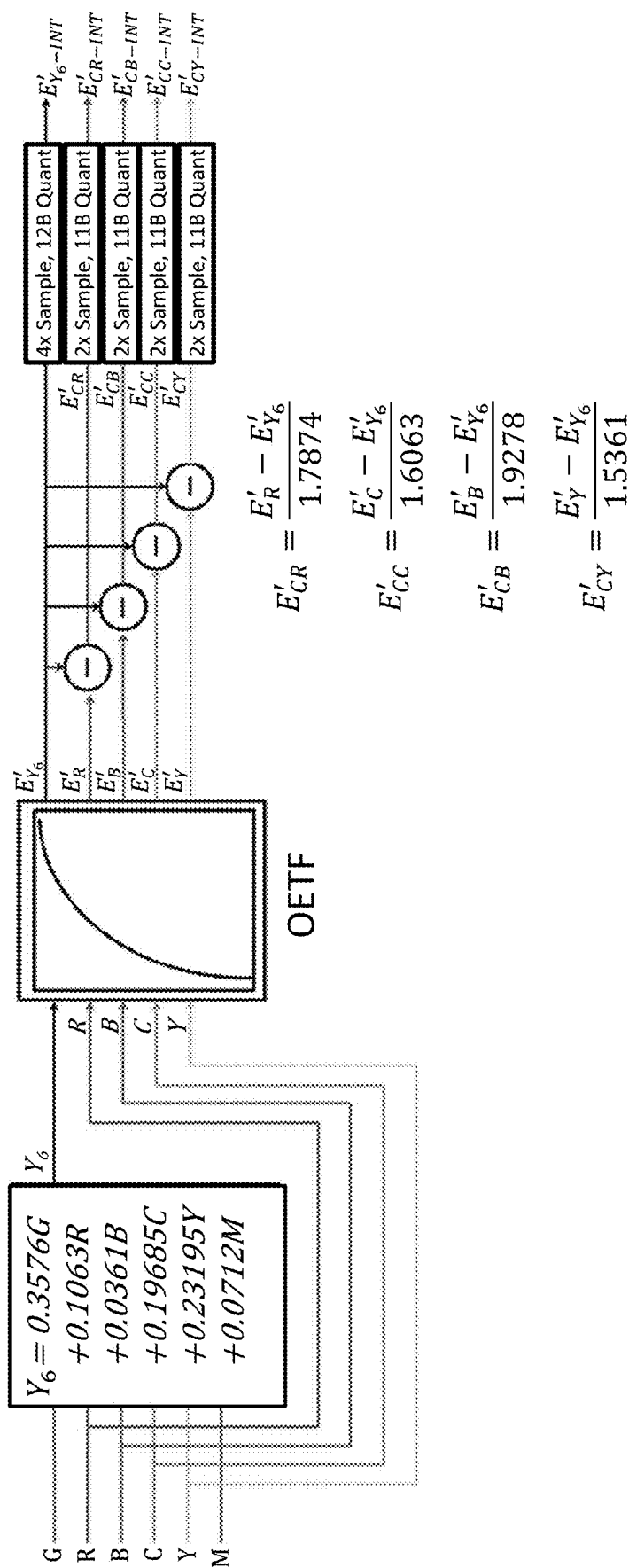


FIG. 26

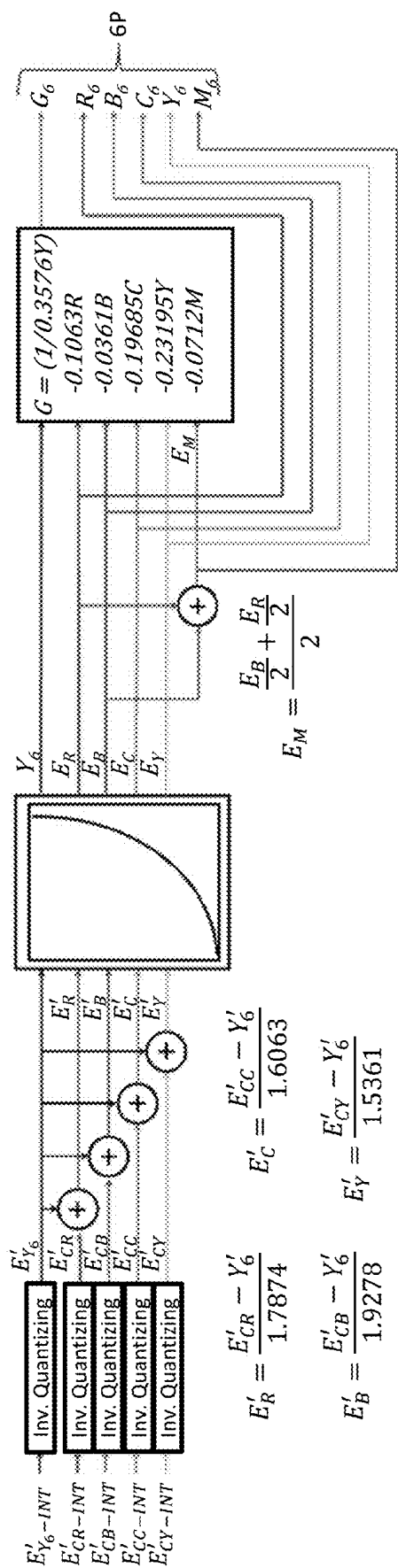


FIG. 27

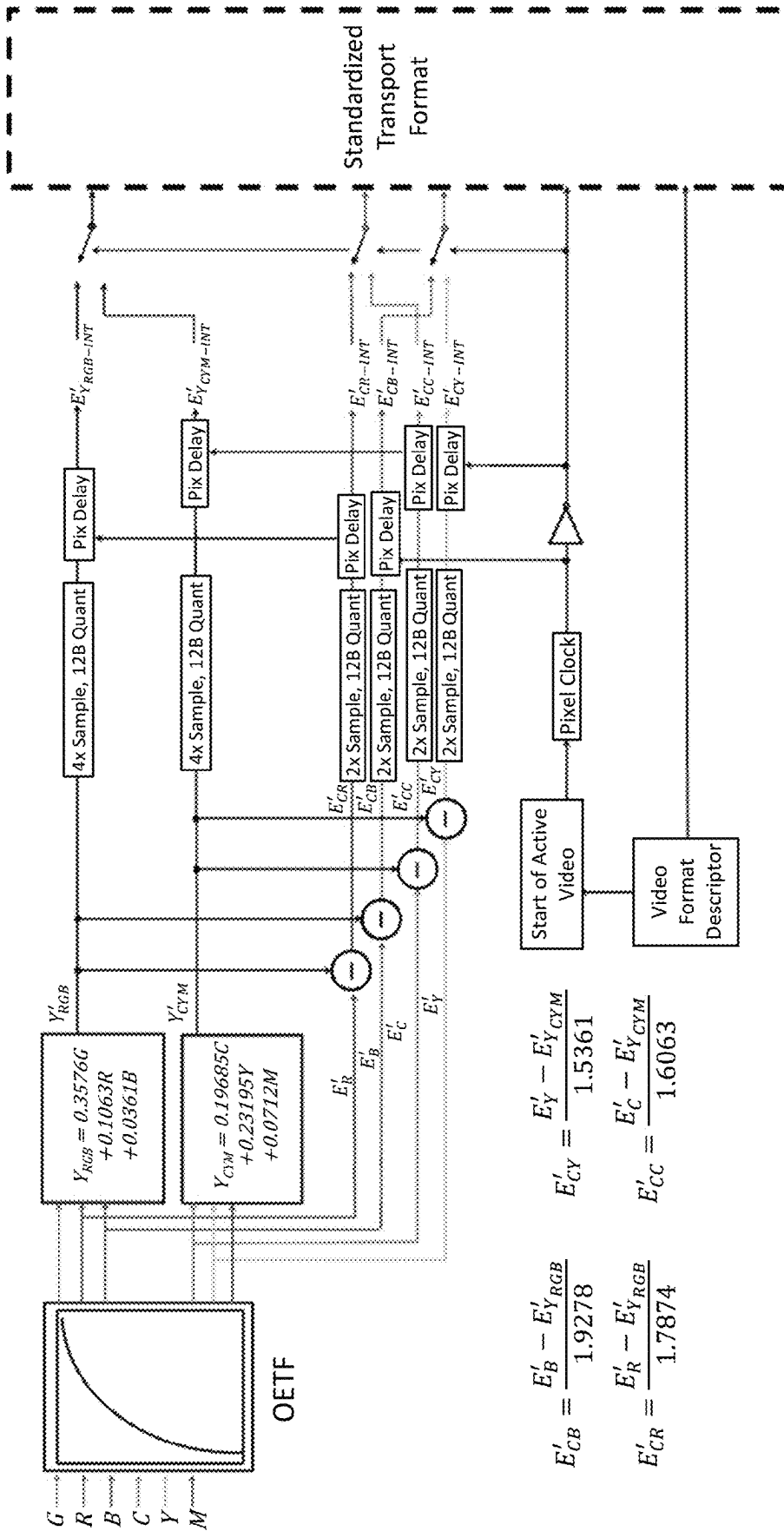


FIG. 28

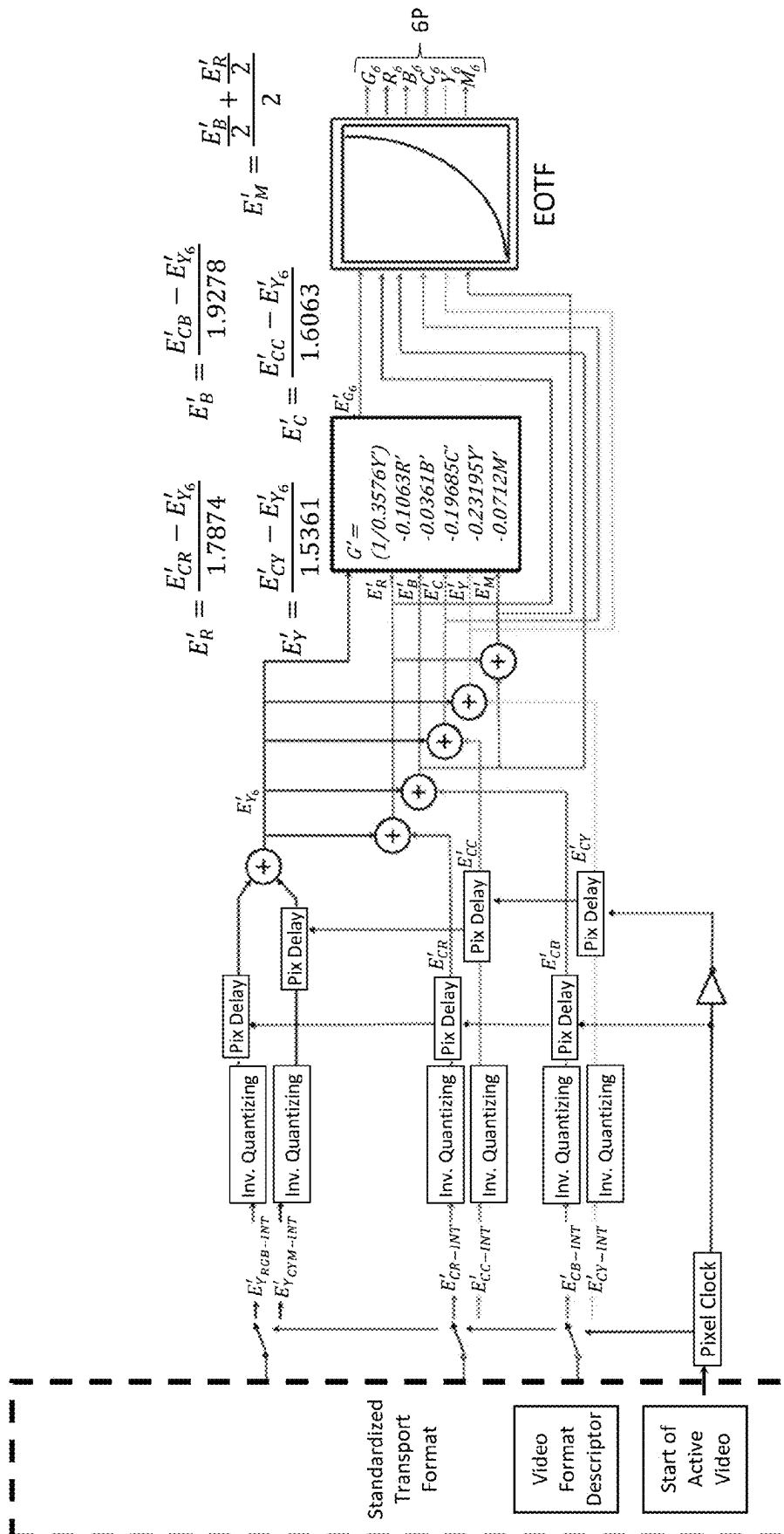


FIG. 29

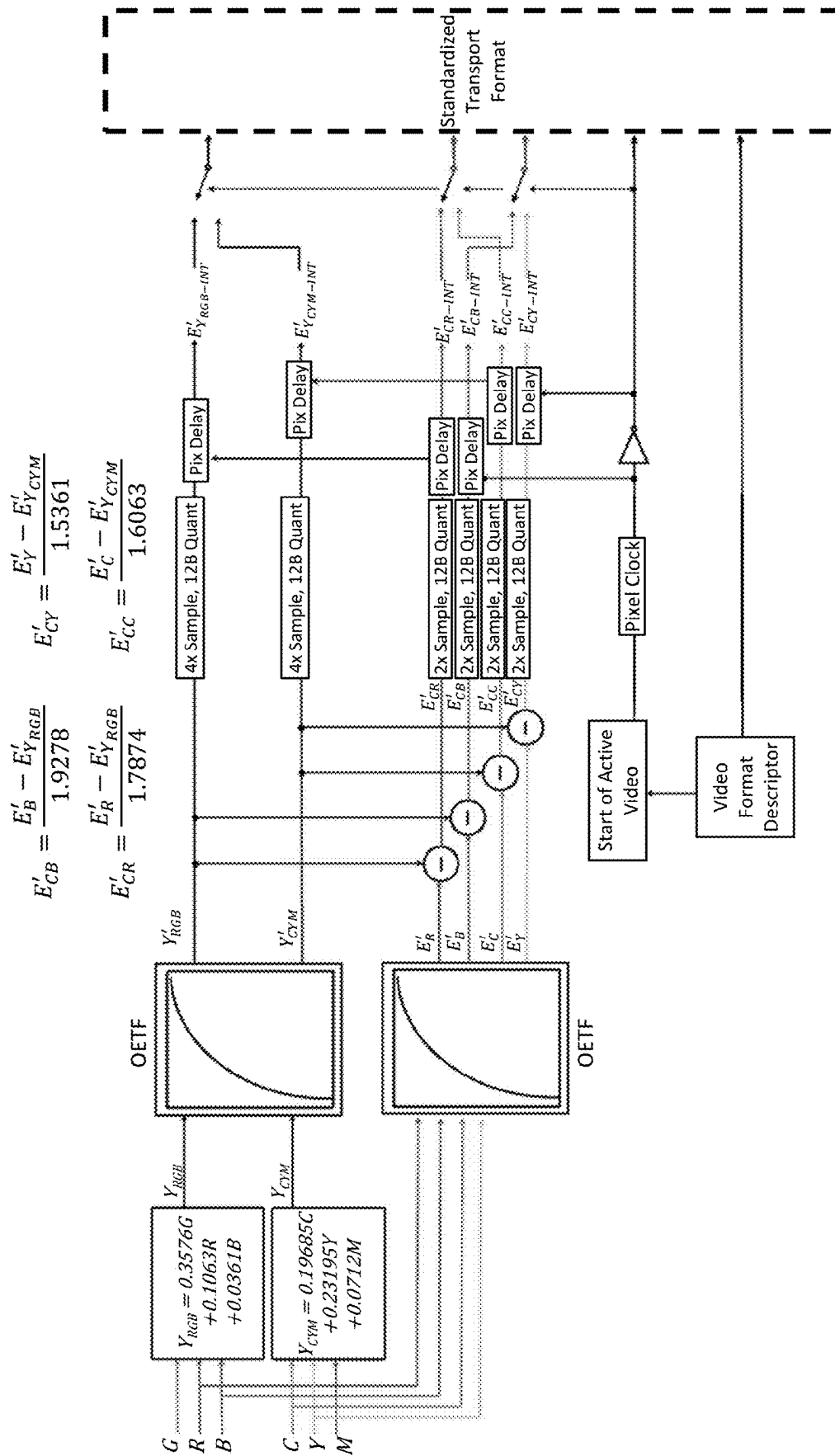


FIG. 30

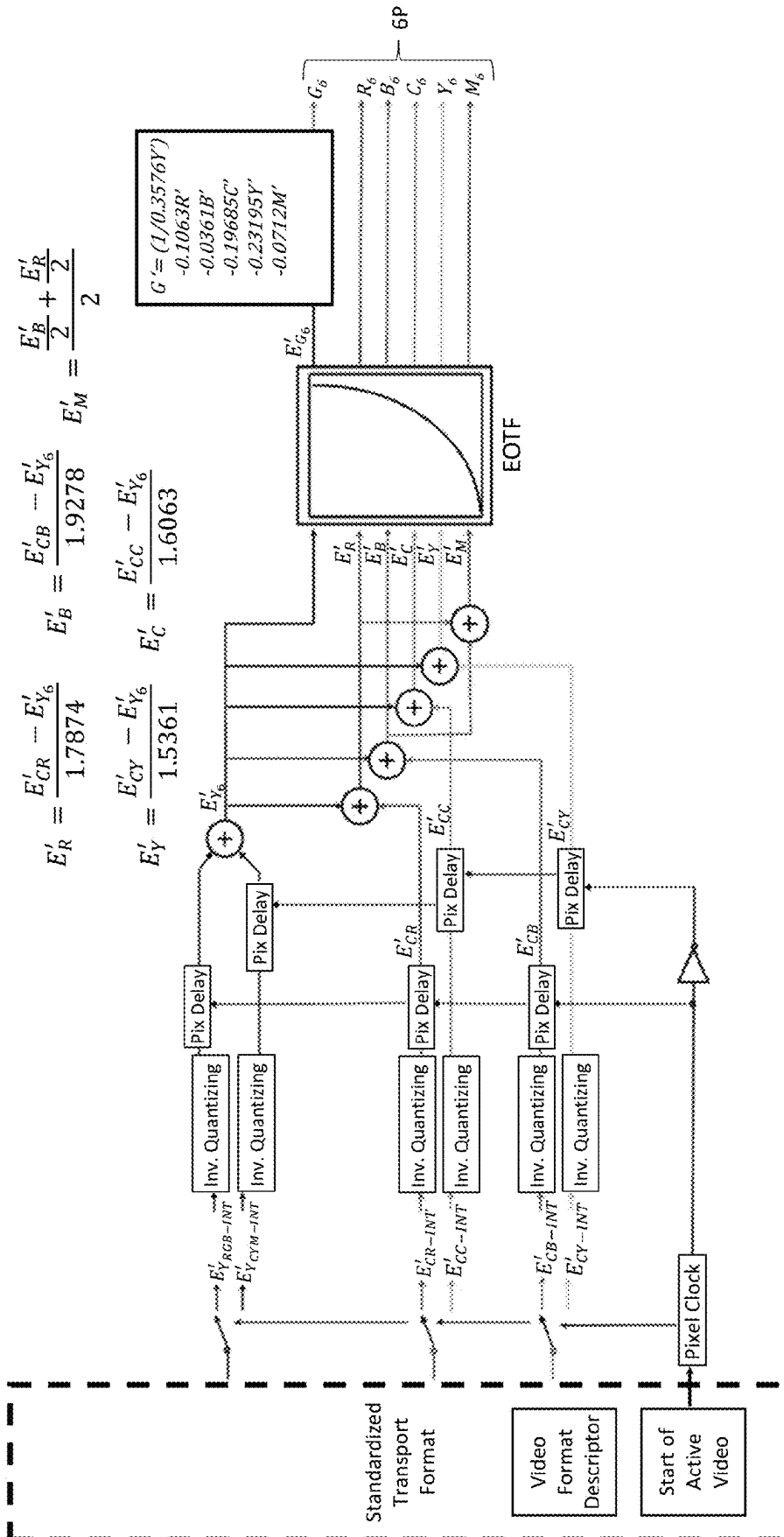


FIG. 31

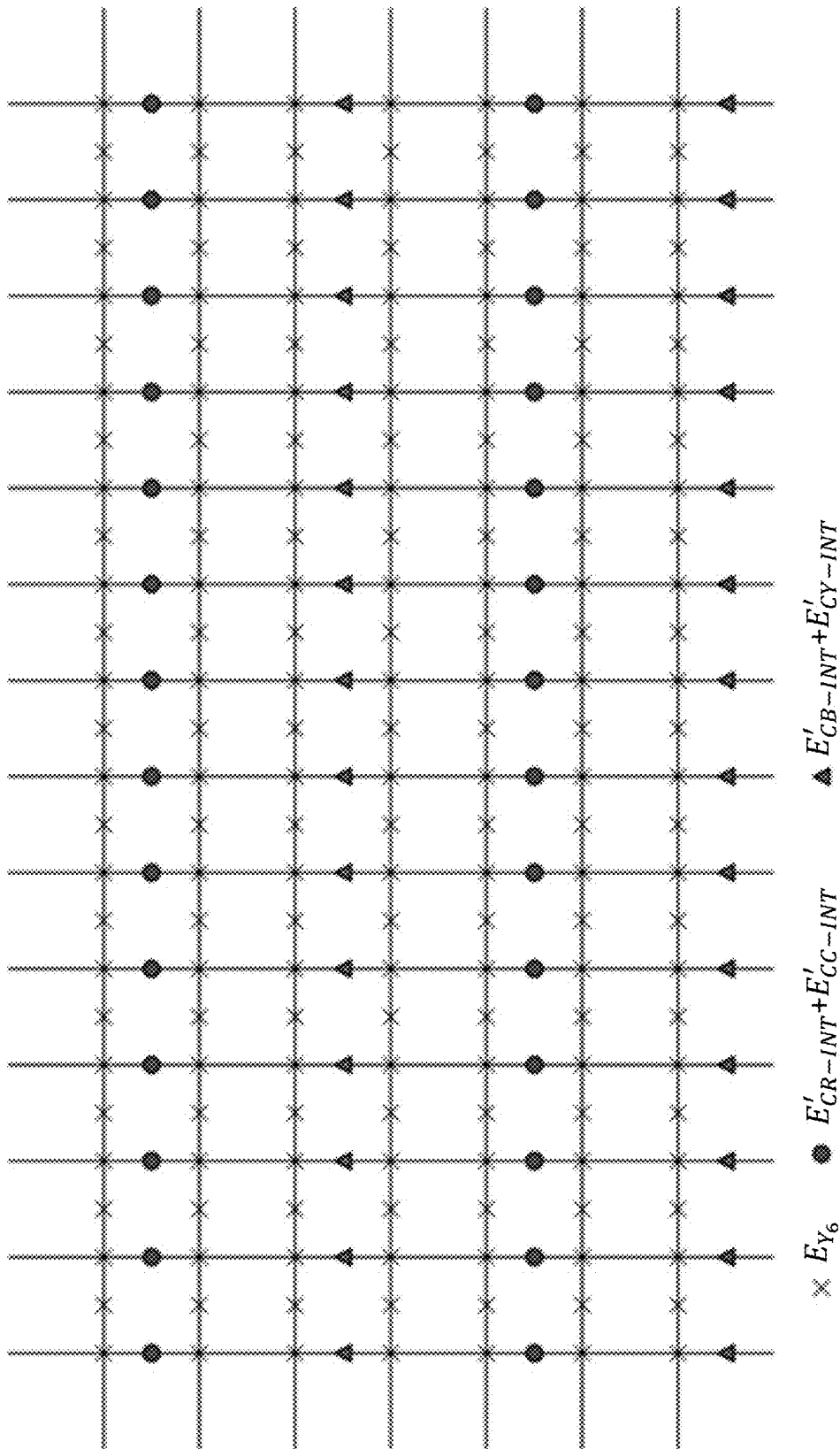


FIG. 32

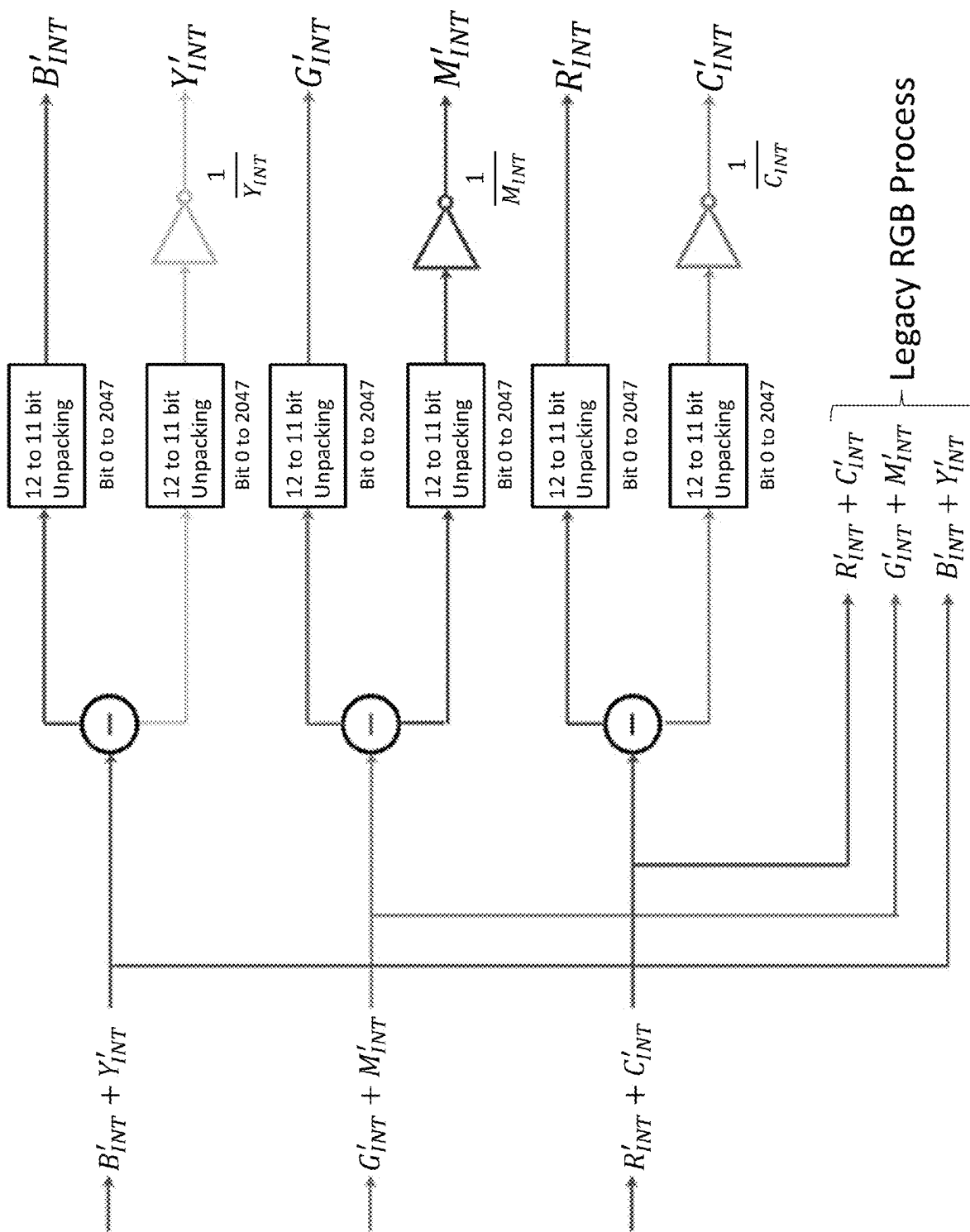


FIG. 33

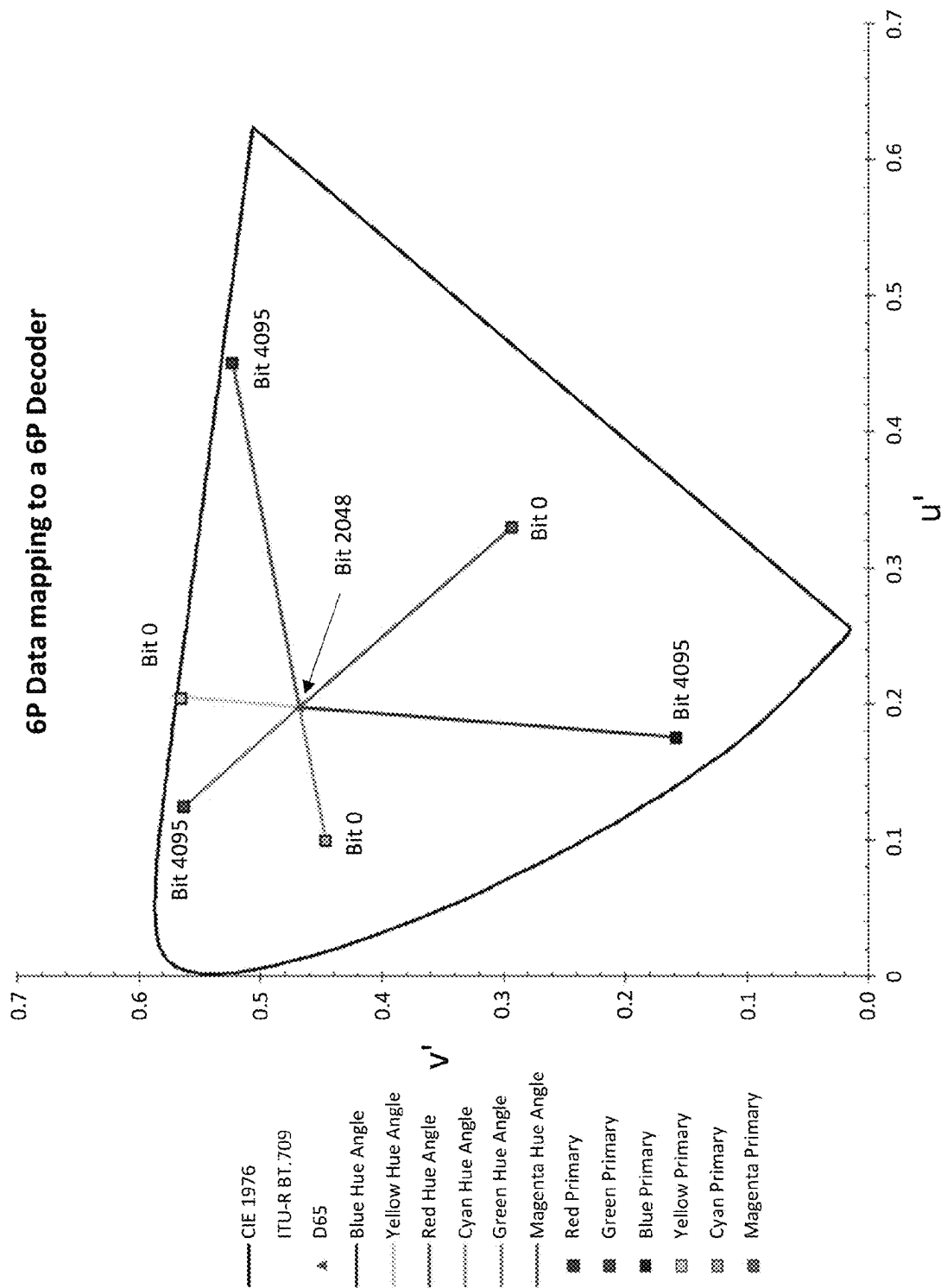


FIG. 34

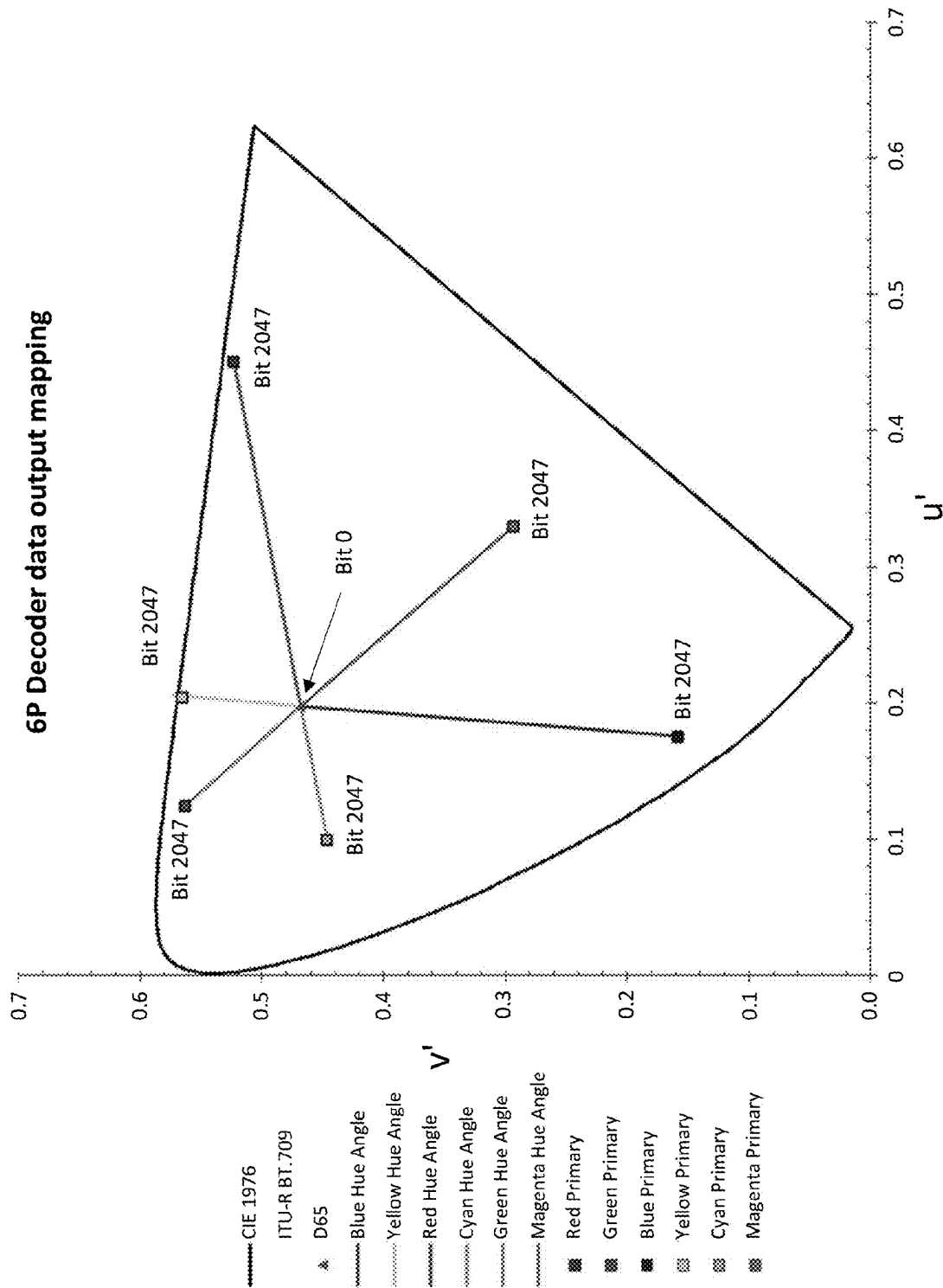


FIG. 35

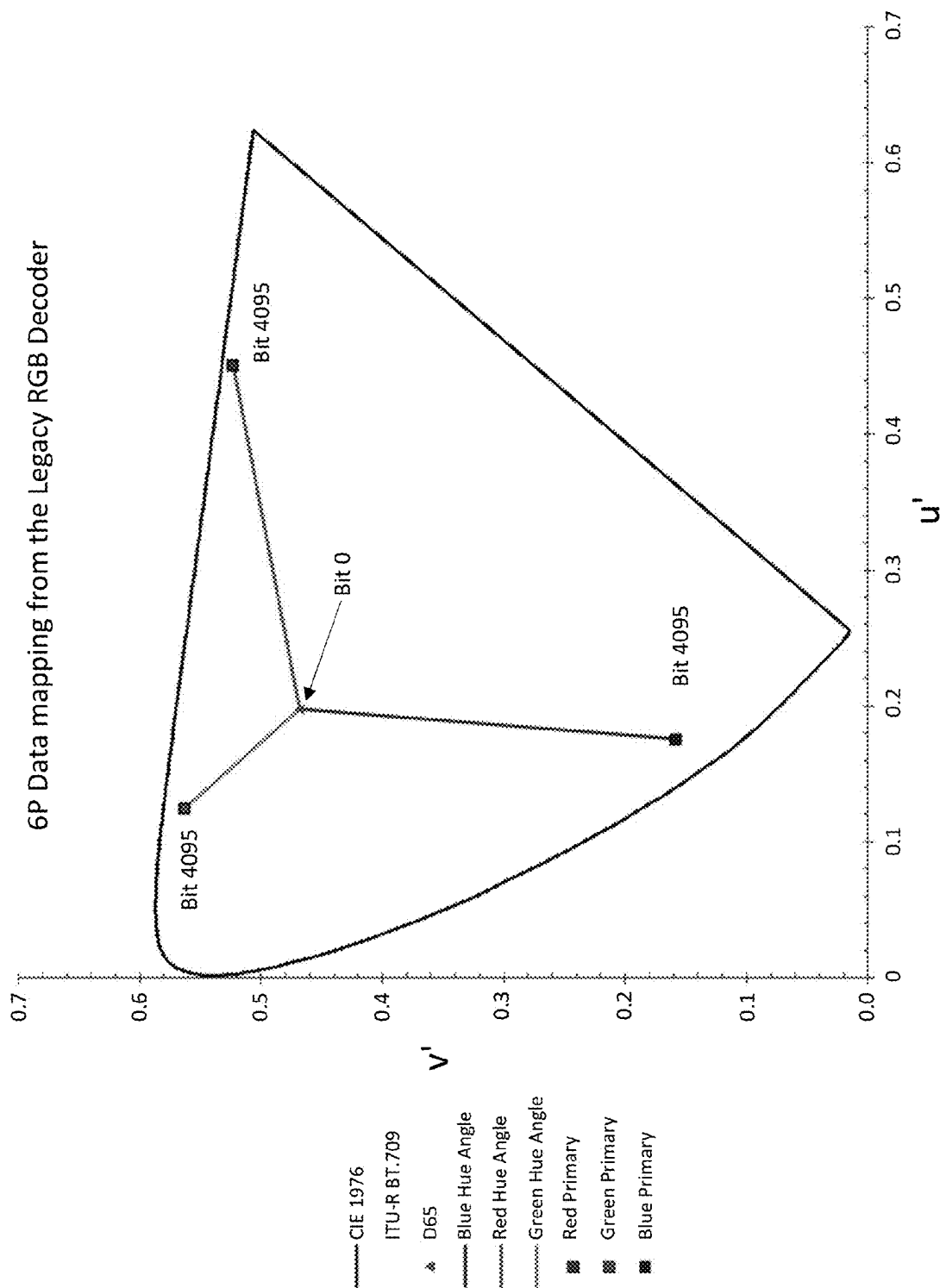


FIG. 36

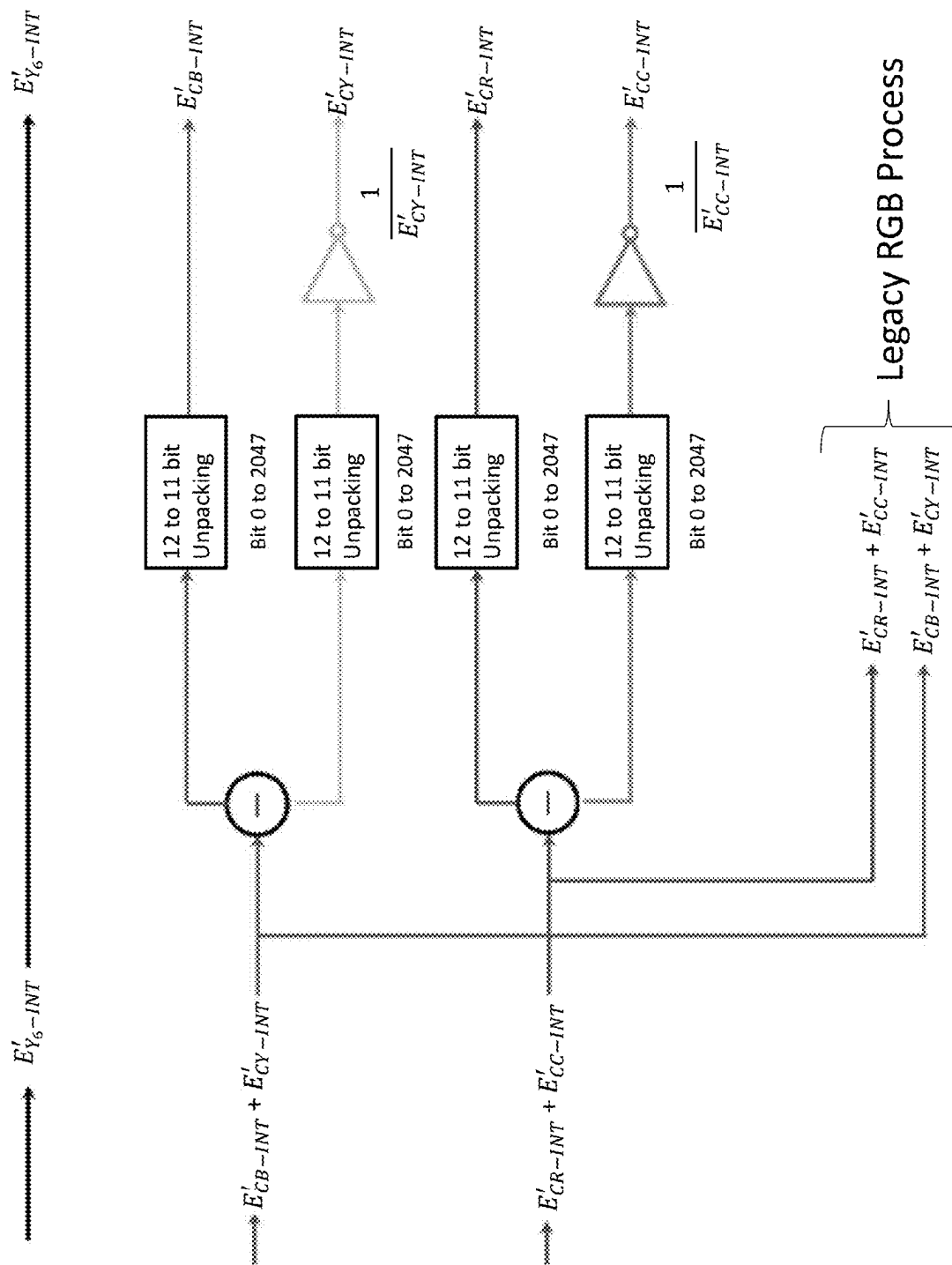


FIG. 37

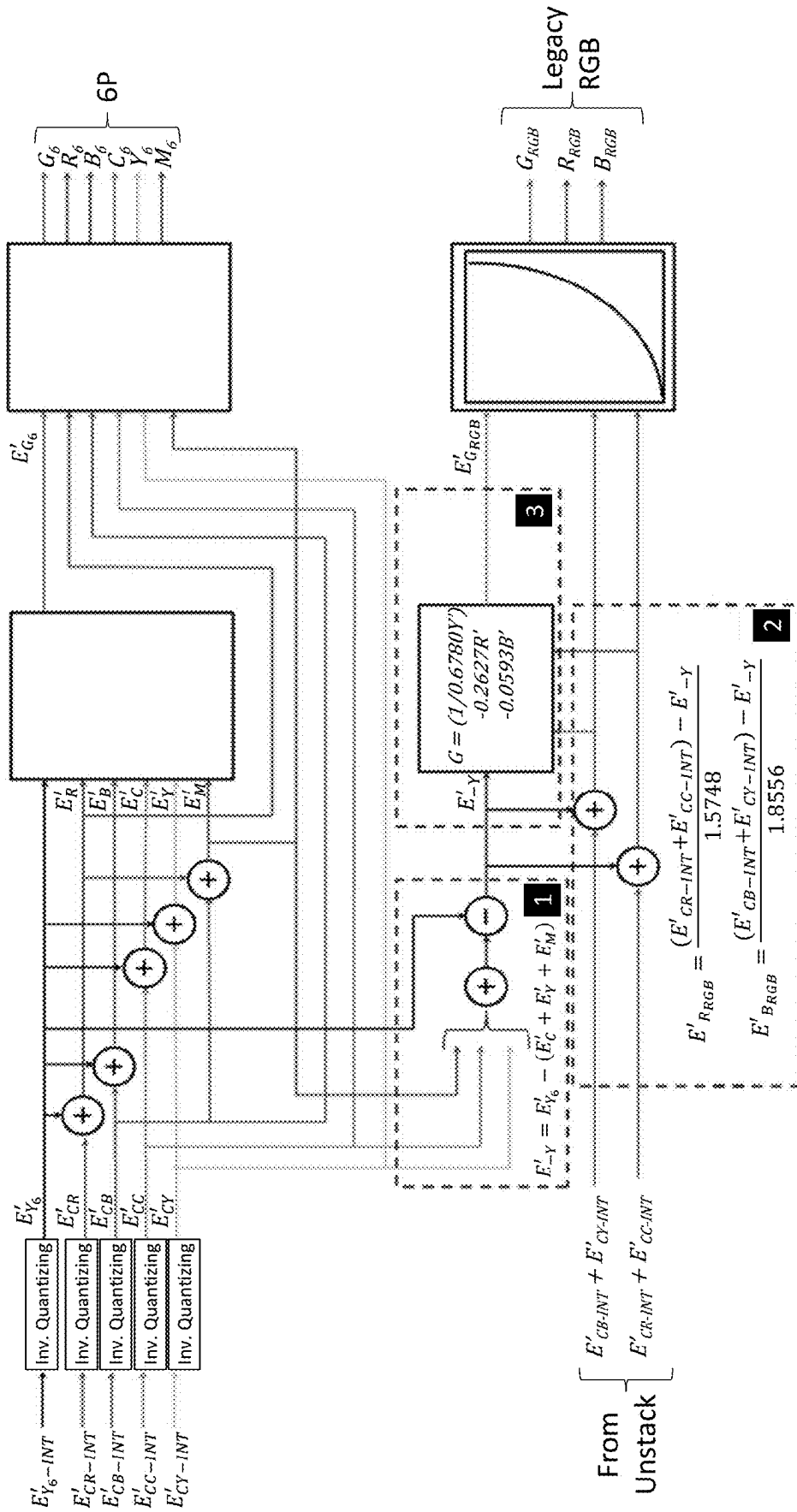


FIG. 38

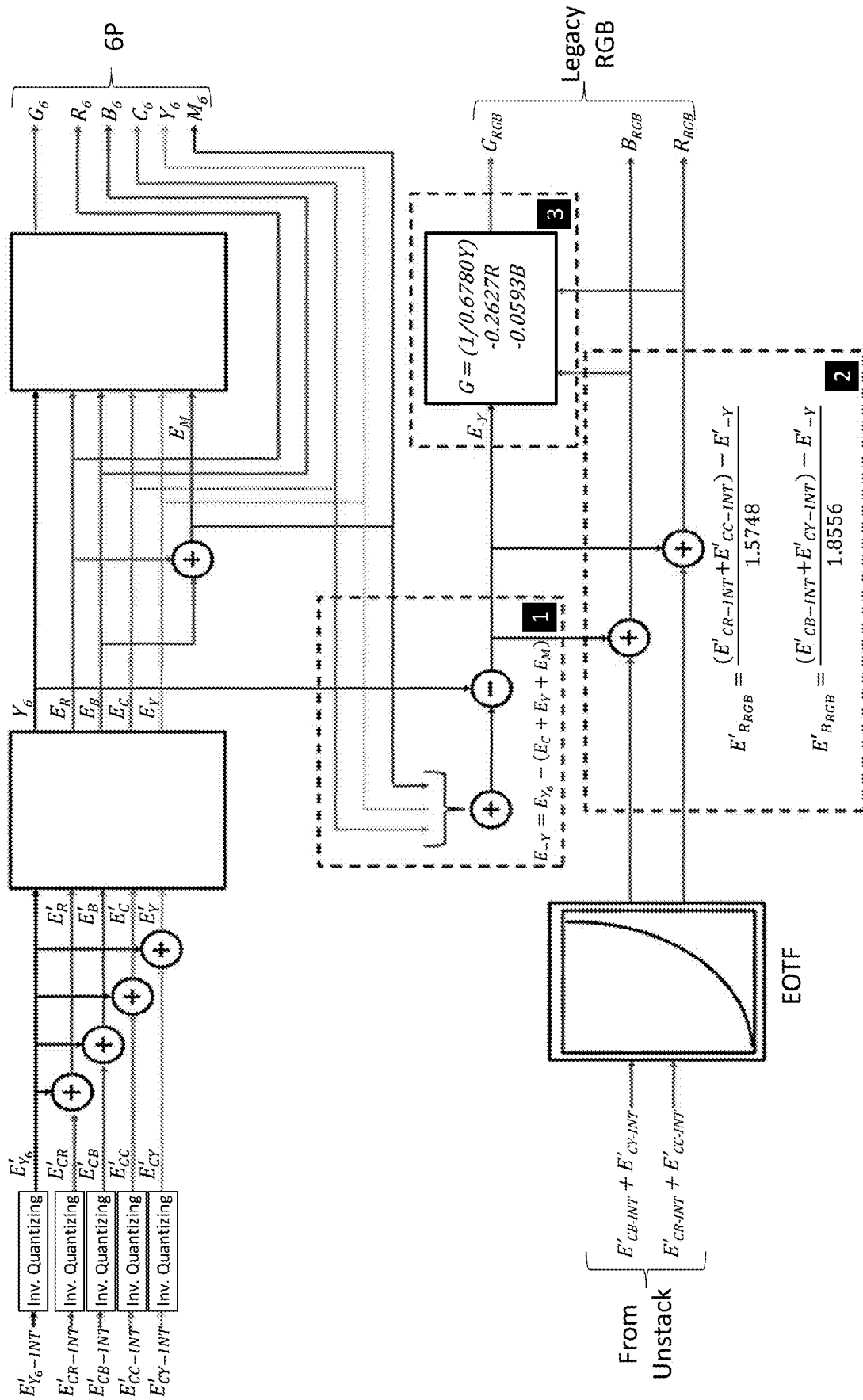


FIG. 39

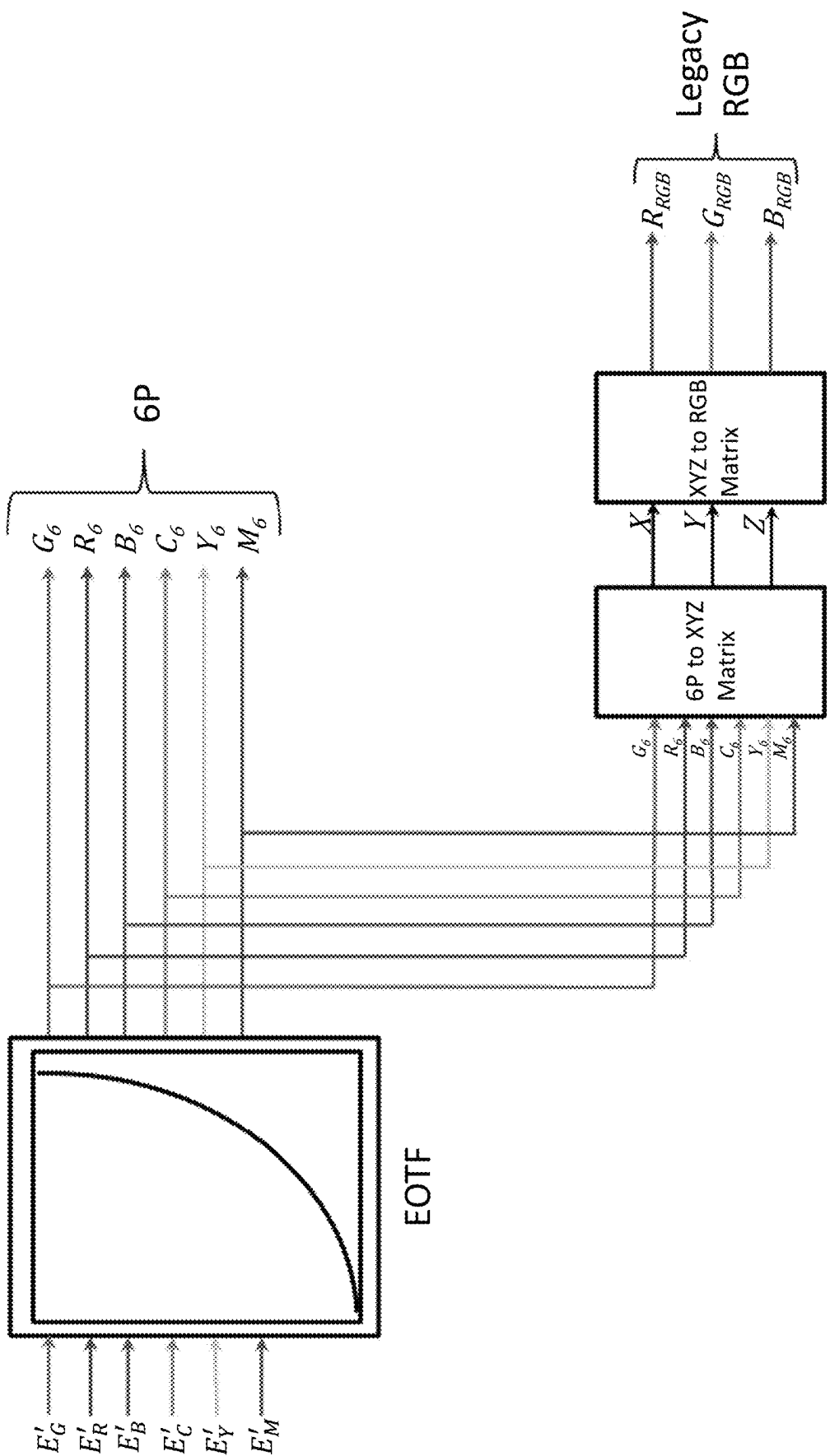


FIG. 40

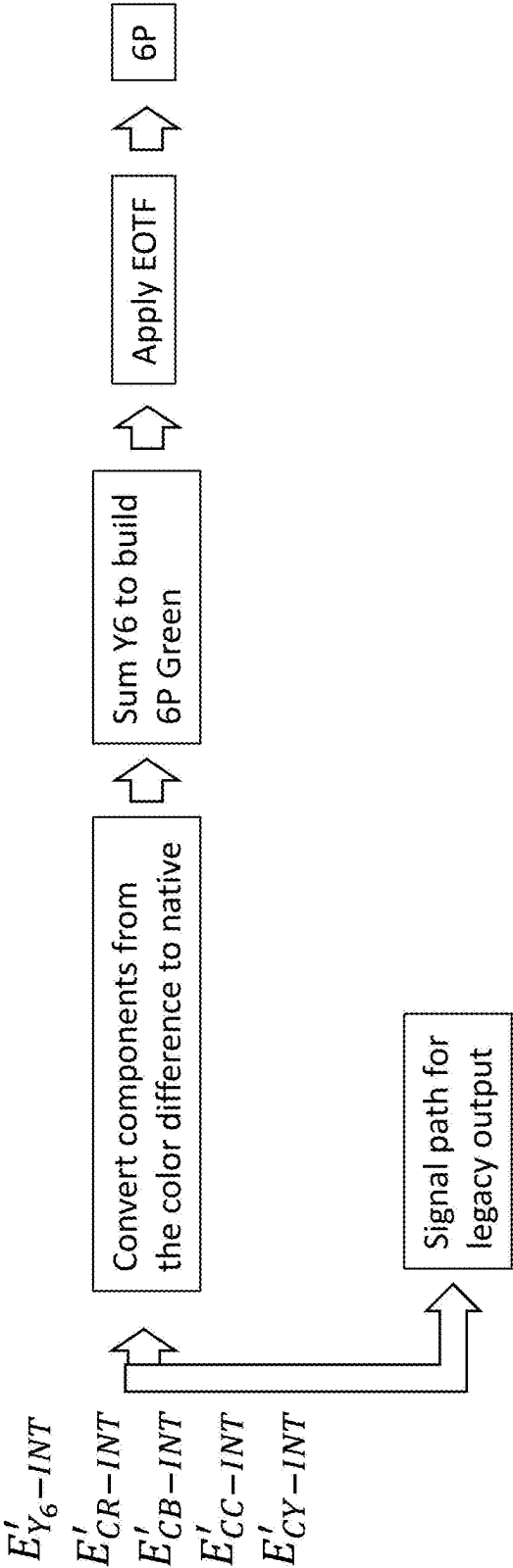


FIG. 41

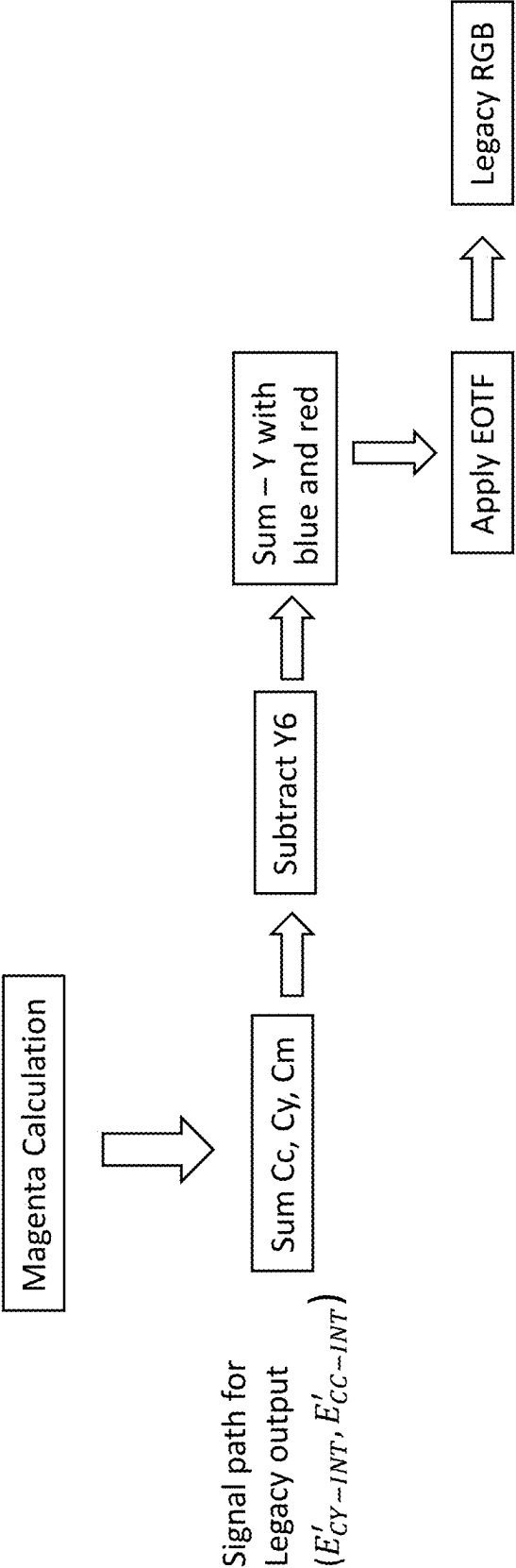


FIG. 42

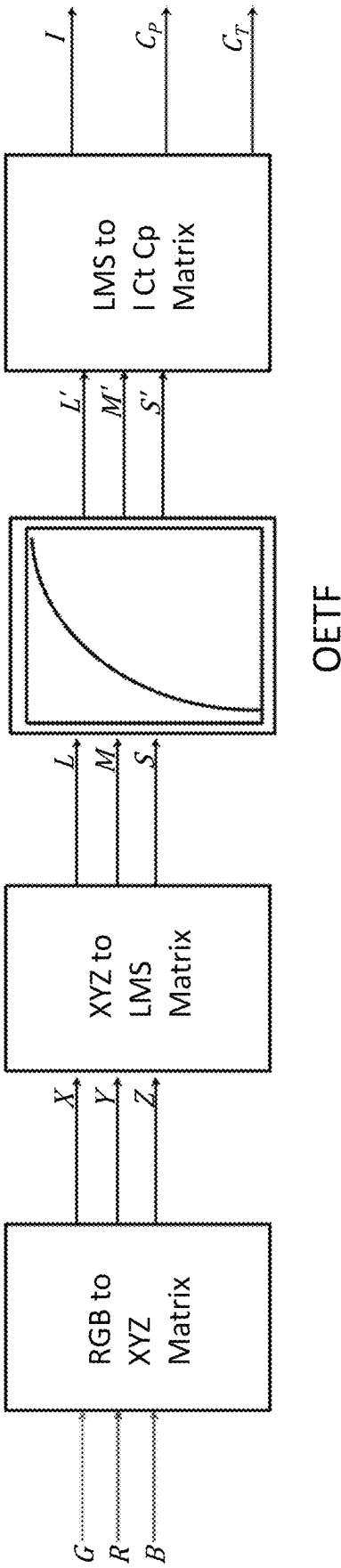


FIG. 43

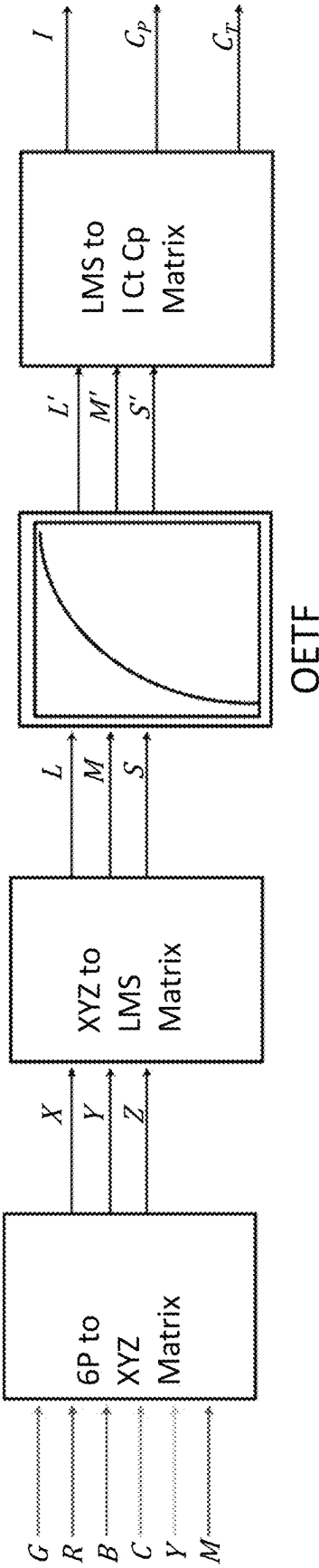


FIG. 44

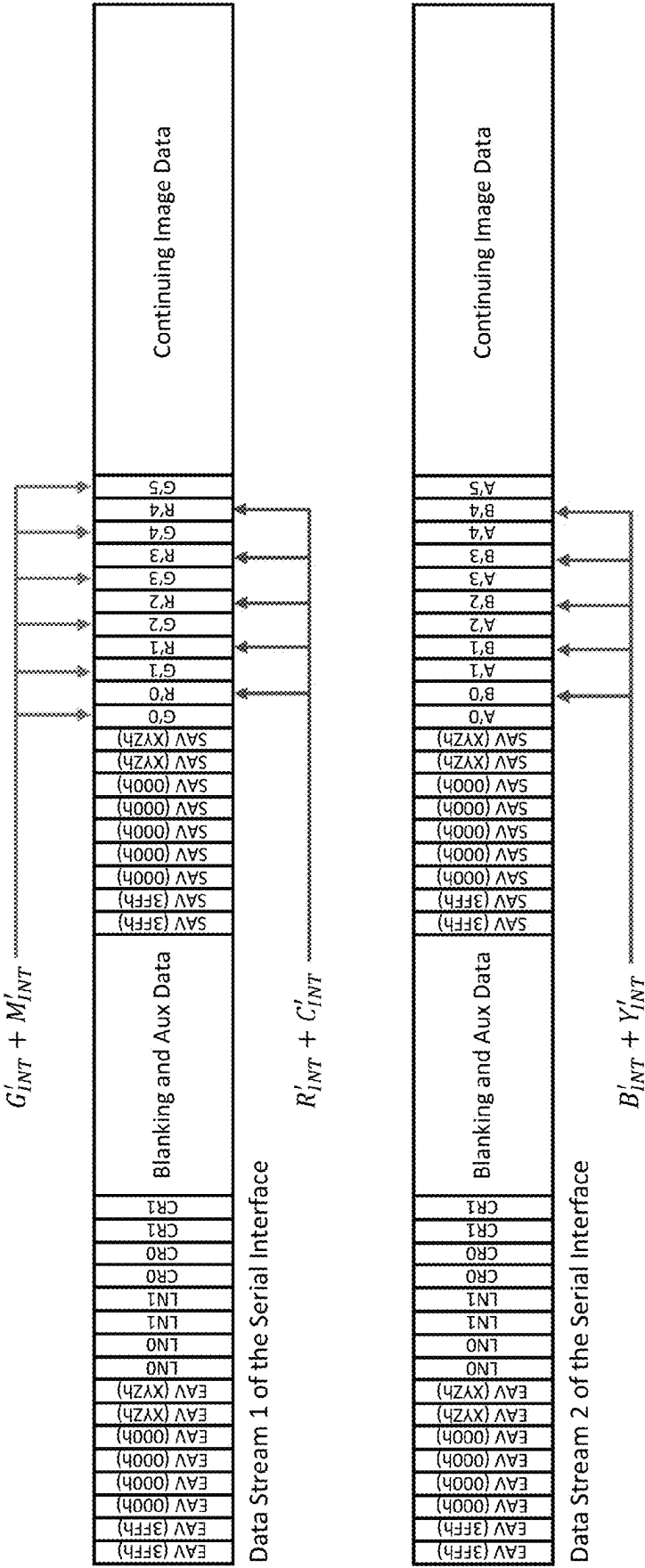


FIG. 45

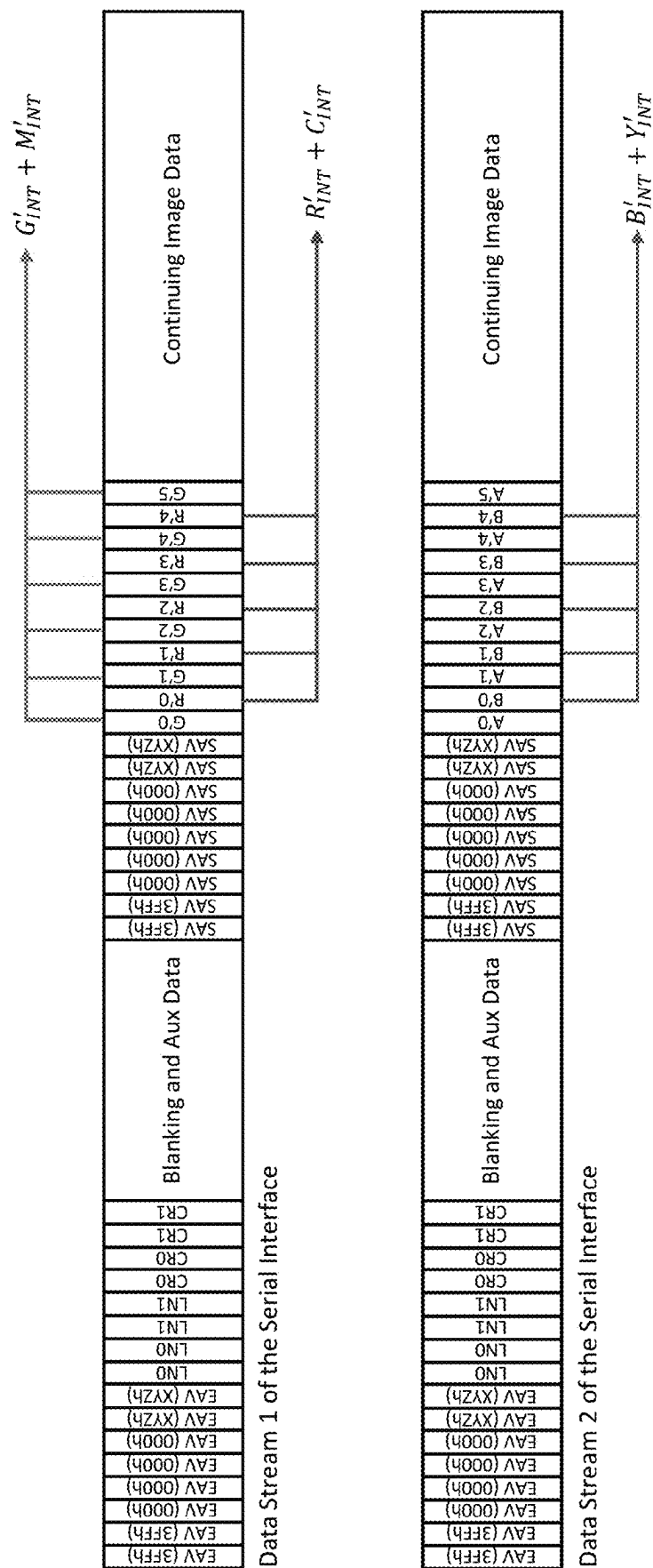


FIG. 46

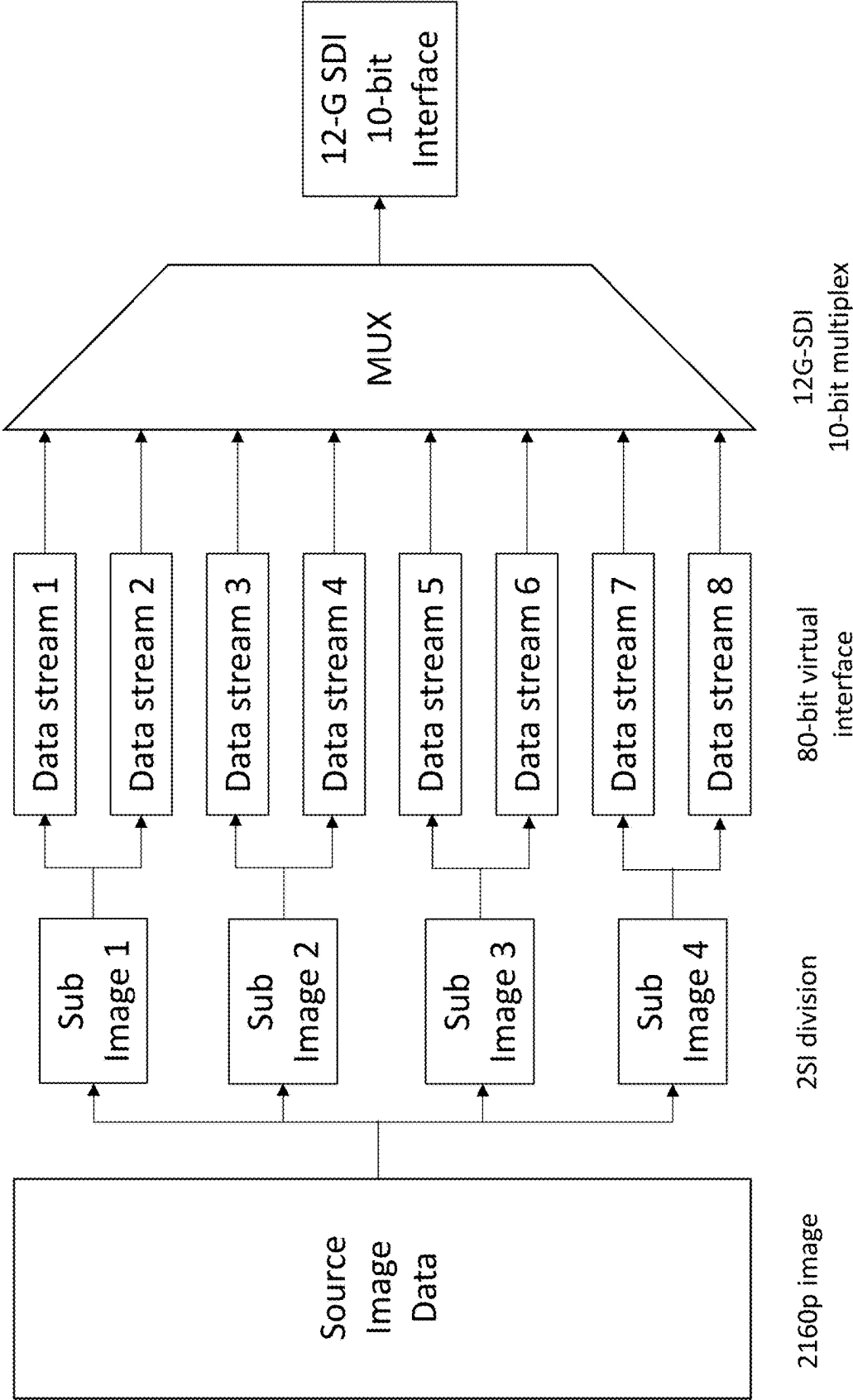


FIG. 47

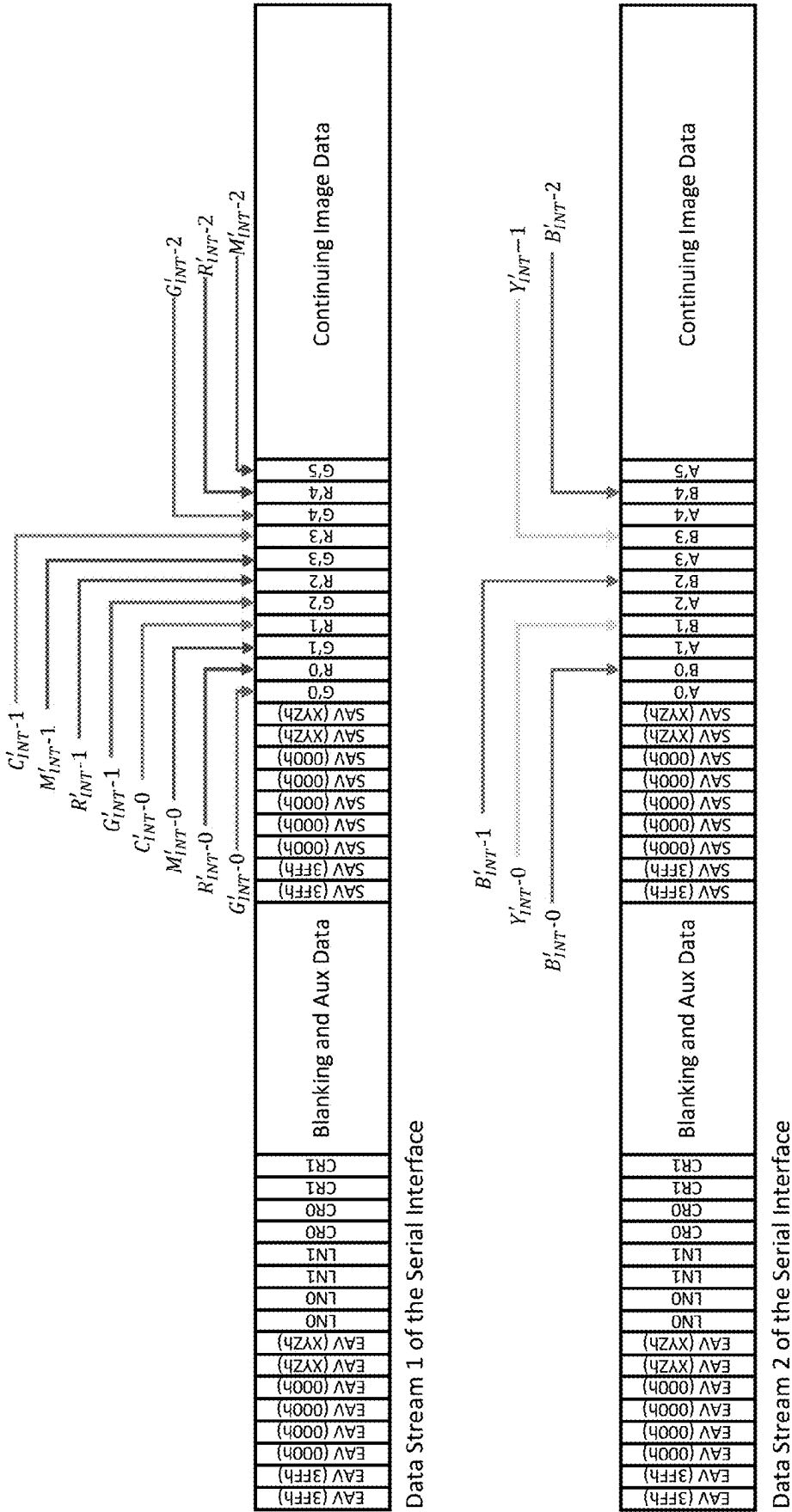


FIG. 48

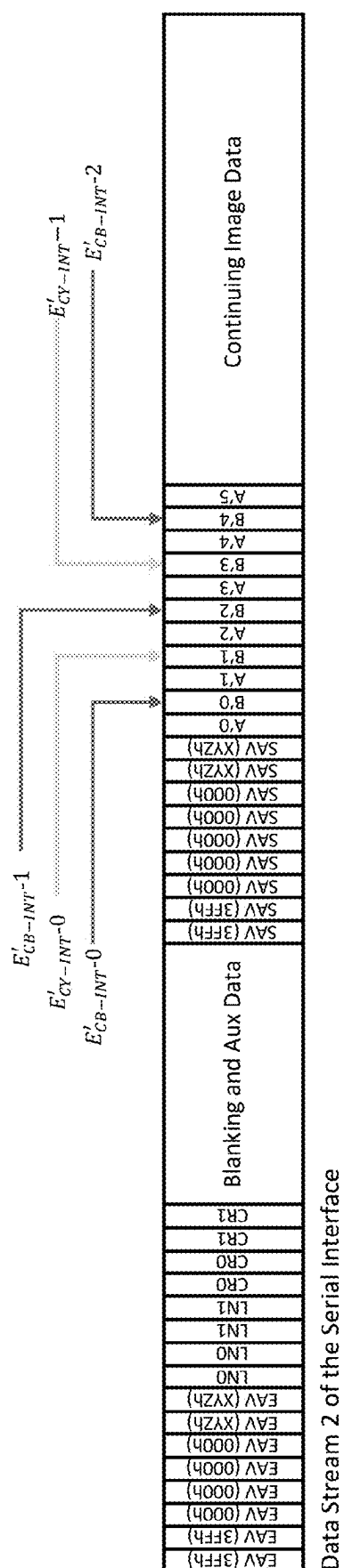
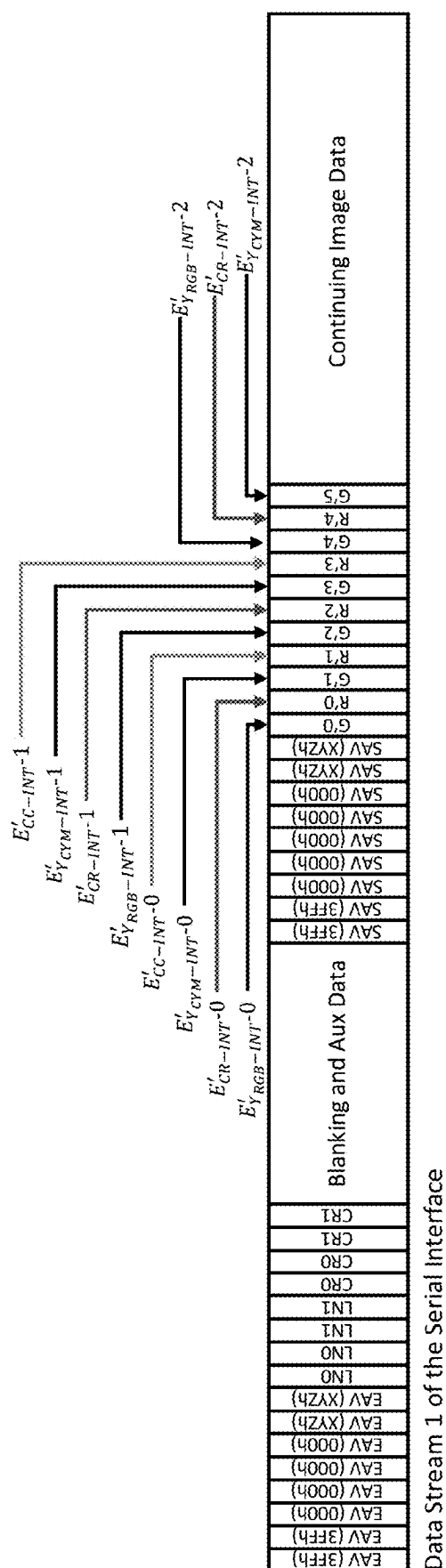


FIG. 49

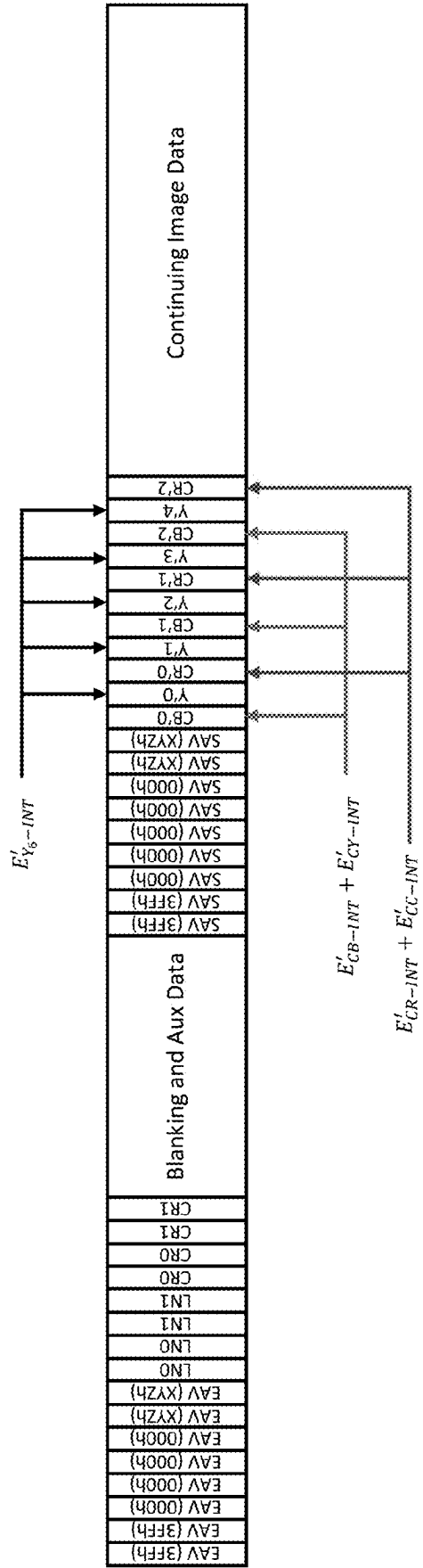


FIG. 50

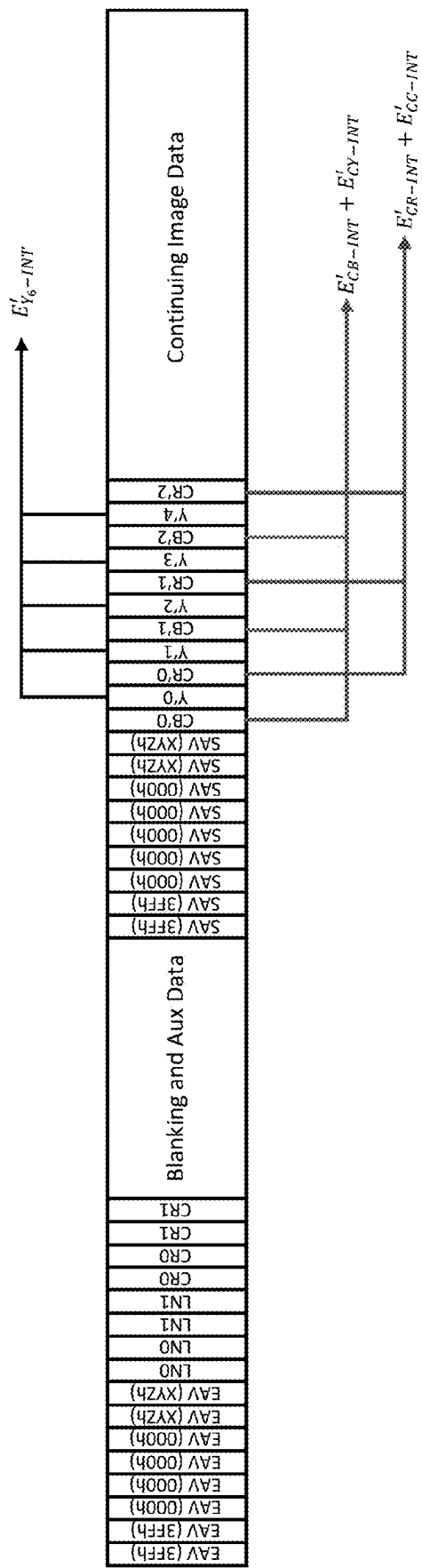


FIG. 51

Value	Sampling	Value	Sampling	Value	Sampling	Value	Sampling
0h	4:2:2 (Y/Cb/Cr)	1h	4:4:4 (Y/Cb/Cr)	2h	4:4:4 (G/B/R)	3h	4:2:0 (Y/Cb/Cr)
4h	4:2:2:4 (Y/Cb/Cr/A)	5h	4:4:4:4 (Y/Cb/Cr/A)	6h	4:4:4:4 (G/B/R/A)	7h	4:2:0 (Y/Cb/Cr/Cc/Cy)
8h	4:2:2:4 (Y/Cb/Cr/D)	9h	4:4:4:4 (Y/Cb/Cr/D)	Ah	4:4:4:4 (G/B/R/D)	Bh	4:2:2 (CL-Y/Cb/Cr/Cc/Cy)
Ch	4:2:2 (Y/Cb/Cr/Cc/Cy)	Dh	4:4:4 (G/B/R/M/Y/C)	Eh	4:4:4 (X'Y'Z')	Fh	Sys 1/Sys2

FIG. 52

Value	Sampling	Value	Sampling	Value	Sampling	Value	Sampling
0h	Unknown/Unspecified	1h	4:2:2 10 bit	2h	4:4:4 10 bit	3h	4:4:4:4 10 bit
4h	Reserved	5h	4:2:2 12 bit	6h	4:4:4 12 bit	7h	4:4:4:4 12 bit
8h	(4:2:2:4) 12 bit	9h	6P 4:2:0 10 bit	Ah	6P 4:2:0 12 bit	Bh	6P 4:2:2 10 bit
Ch	6P 4:2:2 12 bit	Dh	6P 4:4:4 10 bit	Eh	6P 4:4:4 12 bit	Fh	Reserved

FIG. 53

	R0'	G0'	B0'	R1'	G1'	B1'	R2'	G2'	B2'	R3'	G3'	B3'
$G'_{INT} + M'_{INT}$		O			O			O			O	
$R'_{INT} + C'_{INT}$	O			O			O			O		
$B'_{INT} + Y'_{INT}$			O			O			O			O

FIG. 54

	R0'	G0'	B0'	R1'	G1'	B1'	R0'	G0'	B0'	R1'	G1'	B1'
$G'_{INT} + M'_{INT}$		O			O			O			O	
$R'_{INT} + C'_{INT}$	O			O			O			O		
$B'_{INT} + Y'_{INT}$			O			O			O			O

FIG. 55

	Cb0'	Y0'	Cr0'	Y1'	Cb1'	Y2'	Cr1'	Y3'	Cb2'	Y4'	Cr2'	Y5'
E'_{Y_6-INT}		O		O		O		O		O		O
$E'_{Cb-INT} + E'_{Cy-INT}$	O				O				O			
$E'_{Cr-INT} + E'_{Cc-INT}$			O				O				O	

FIG. 56

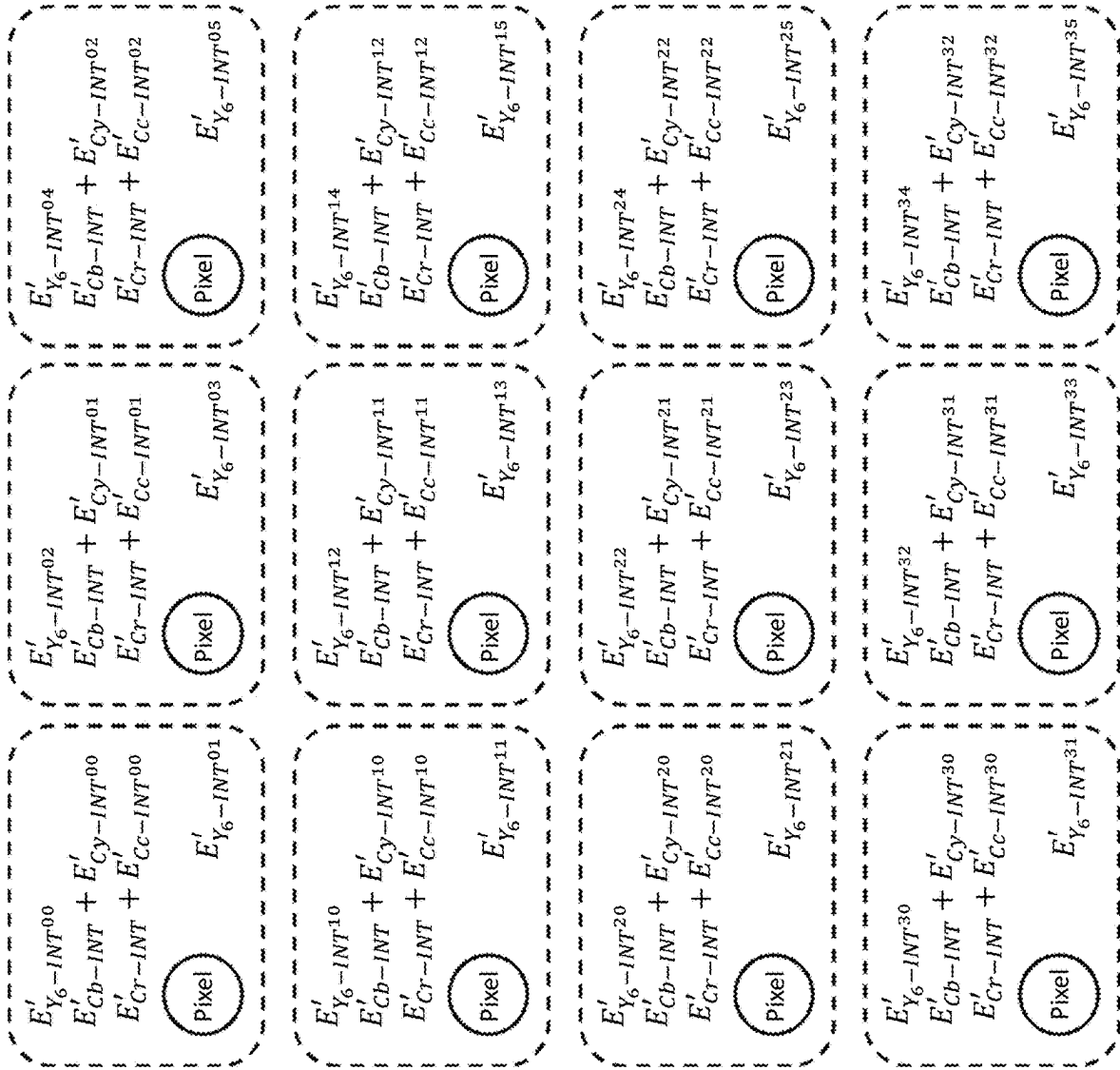


FIG. 57

	Cb0'	Y0'	Cr0'	Y1'	Cb1'	Y2'	Cr1'	Y3'	Cb2'	Y4'	Cr2'	Y5'
E'_{Y_6-INT}		○		○		○		○		○		○
$E'_{Cb-INT} + E'_{Cy-INT}$	○				○				○			
$E'_{Cr-INT} + E'_{Cc-INT}$			○				○				○	

FIG. 58

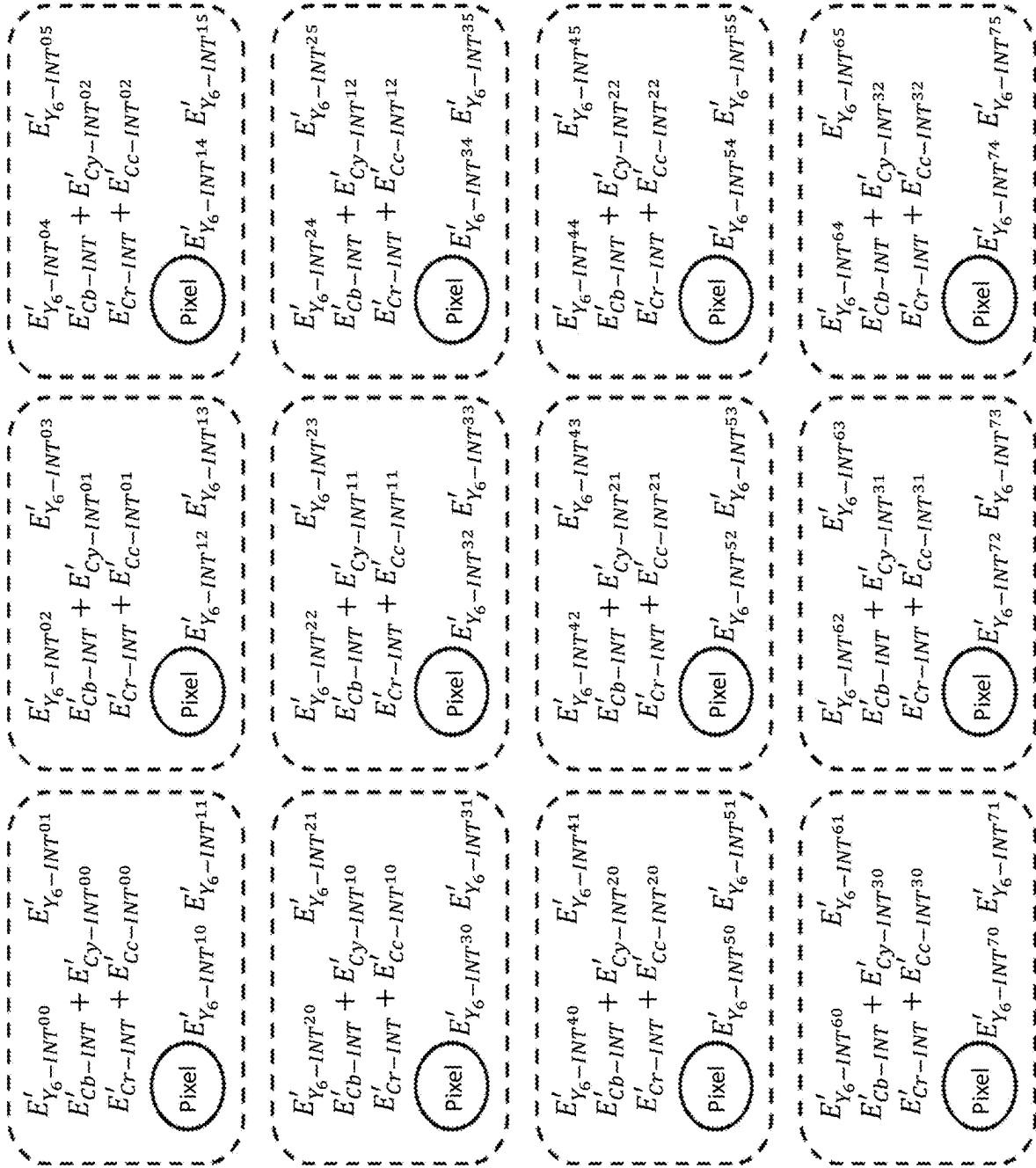


FIG. 59

	R0'	G0'	B0'	R1'	G1'	B1'	R2'	G2'	B2'	R3'	G3'	B3'
G'_{INT}		O						O				
B'_{INT}			O						O			
R'_{INT}	O						O					
M'_{INT}					O						O	
Y'_{INT}						O						O
C'_{INT}				O						O		

FIG. 60

	R0'	G0'	B0'	R1'	G1'	B1'	R0'	G0'	B0'	R1'	G1'	B1'
G'_{INT}		O						O				
R'_{INT}			O						O			
B'_{INT}	O						O					
M'_{INT}					O						O	
Y'_{INT}						O						O
C'_{INT}				O						O		

	R2'	G2'	B2'	R3'	G3'	B3'	R2'	G2'	B2'	R3'	G3'	B3'
G'_{INT}		O						O				
R'_{INT}			O						O			
B'_{INT}	O						O					
M'_{INT}					O						O	
Y'_{INT}						O						O
C'_{INT}				O						O		

FIG. 61

	Cb0'	Y0'	Cr0'	Y1'	Cb1'	Y2'	Cr1'	Y3'	Cb2'	Y4'	Cr2'	Y5'
$E'_{Y_{RGB-INT}}$		O				O				O		
$E'_{Y_{CYM-INT}}$				O				O				O
E'_{Cr-INT}			O						O			
E'_{Cb-INT}	O										O	
E'_{Cc-INT}							O					
E'_{Cy-INT}					O							

FIG. 62

	Cb0'	Y0'	Cr0'	Y1'	Cb1'	Y2'	Cr1'	Y3'	Cb2'	Y4'	Cr2'	Y5'
$E'_{Y_{RGB-INT}}$		○				○				○		
$E'_{Y_{CYM-INT}}$				○				○				○
E'_{Cr-INT}	○								○			
E'_{Cb-INT}			○								○	
E'_{Cc-INT}					○							
E'_{Cy-INT}							○					

FIG. 63

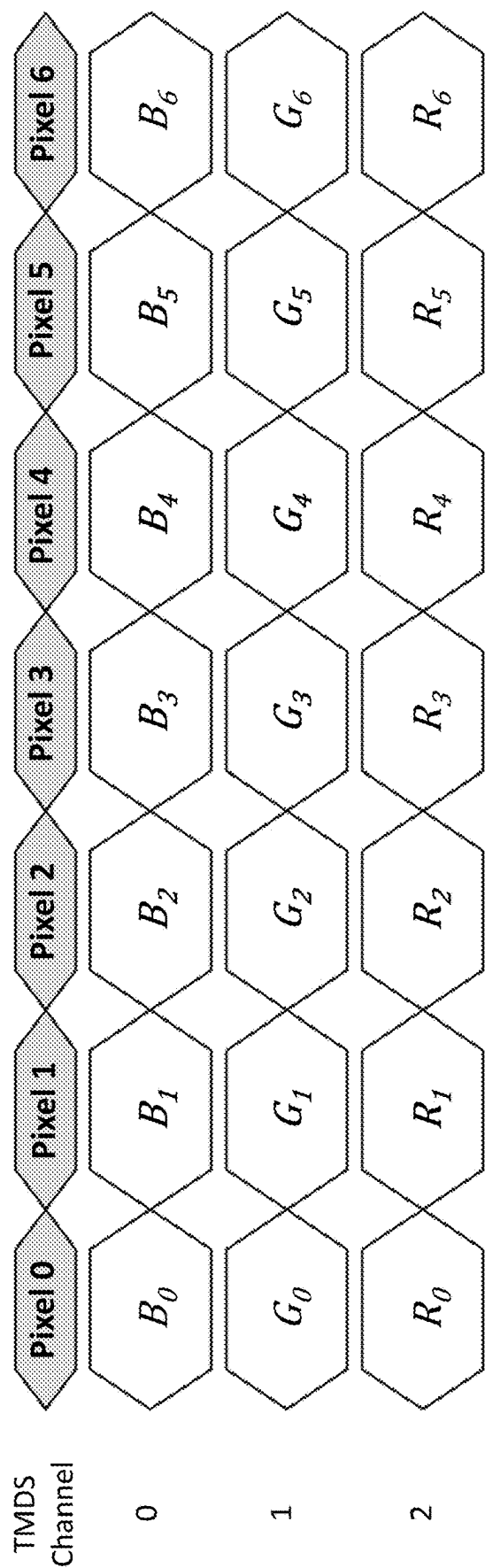


FIG. 64

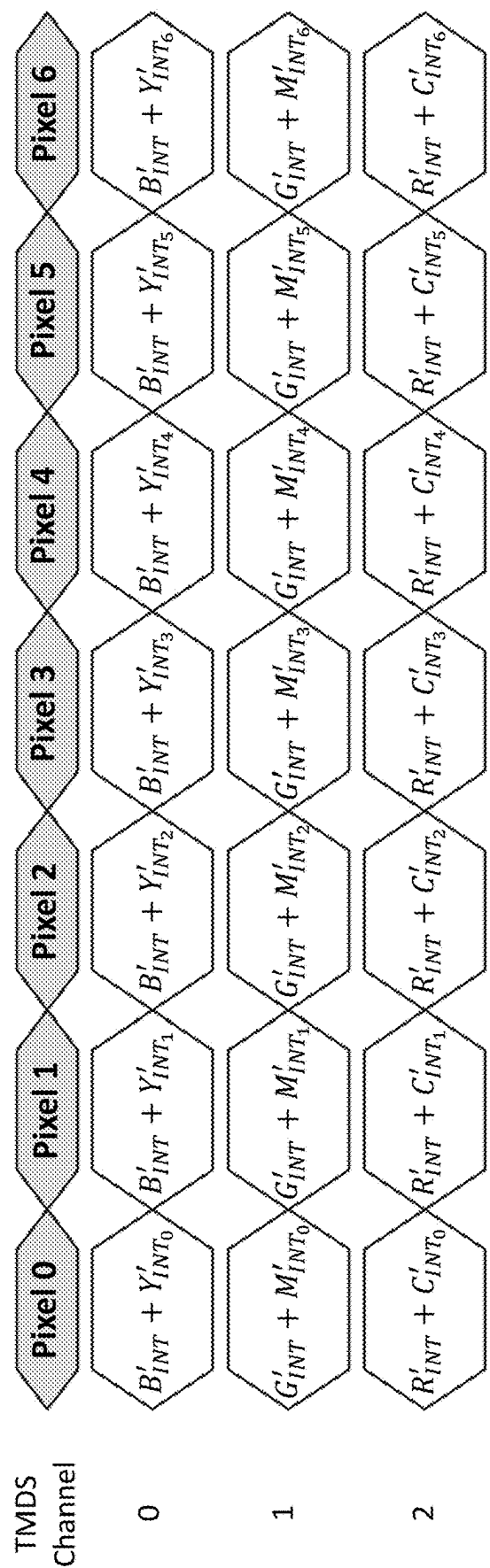


FIG. 65

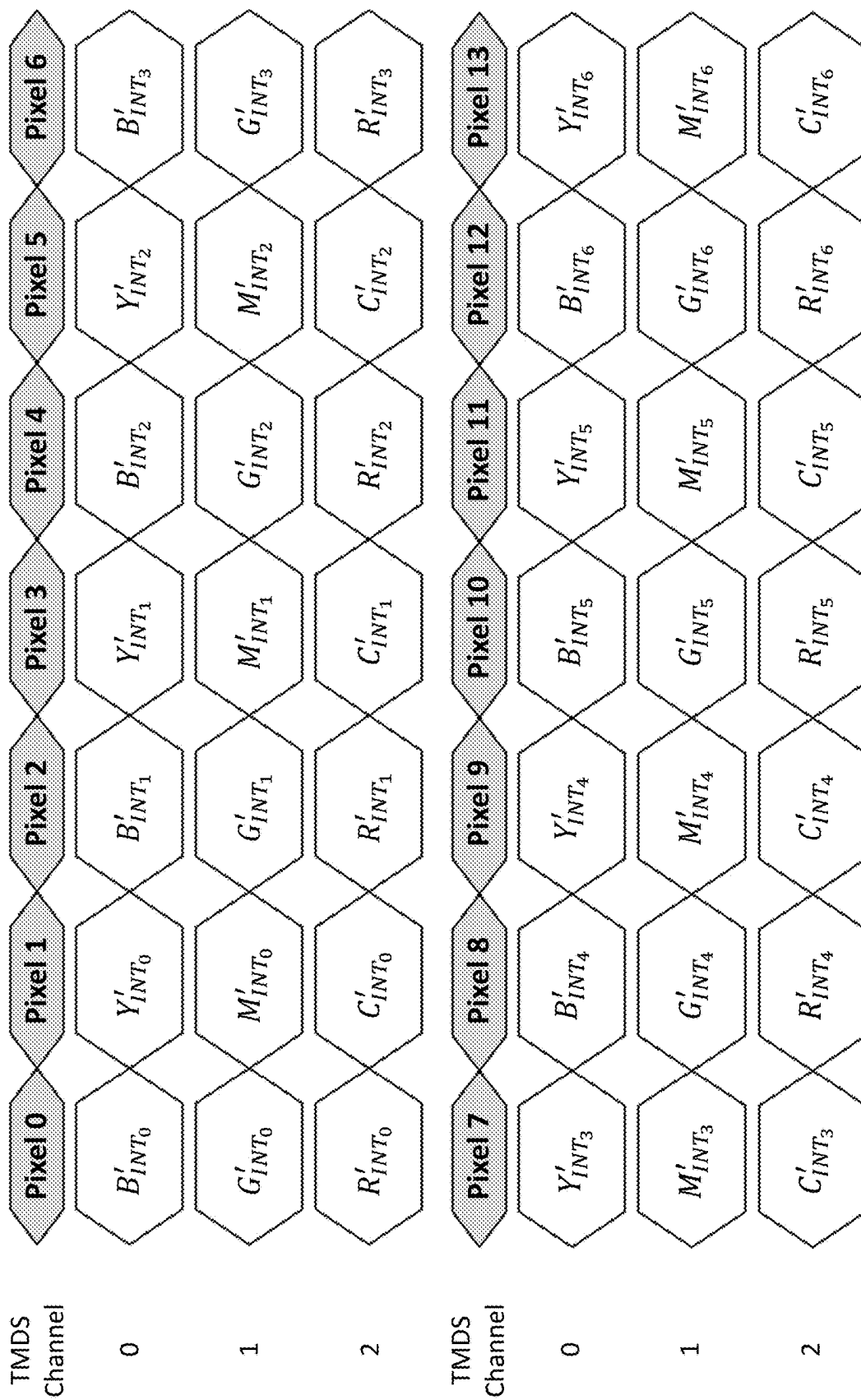


FIG. 66

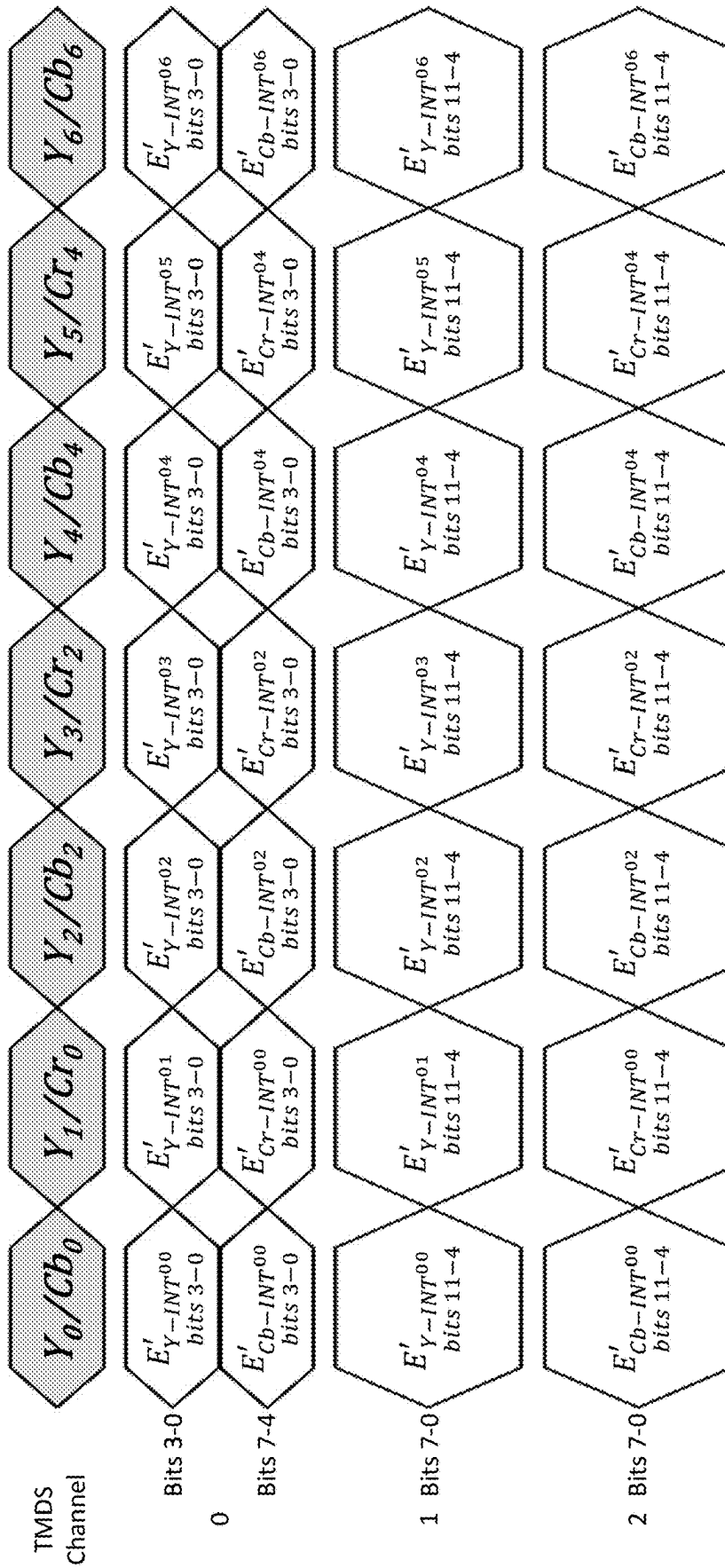
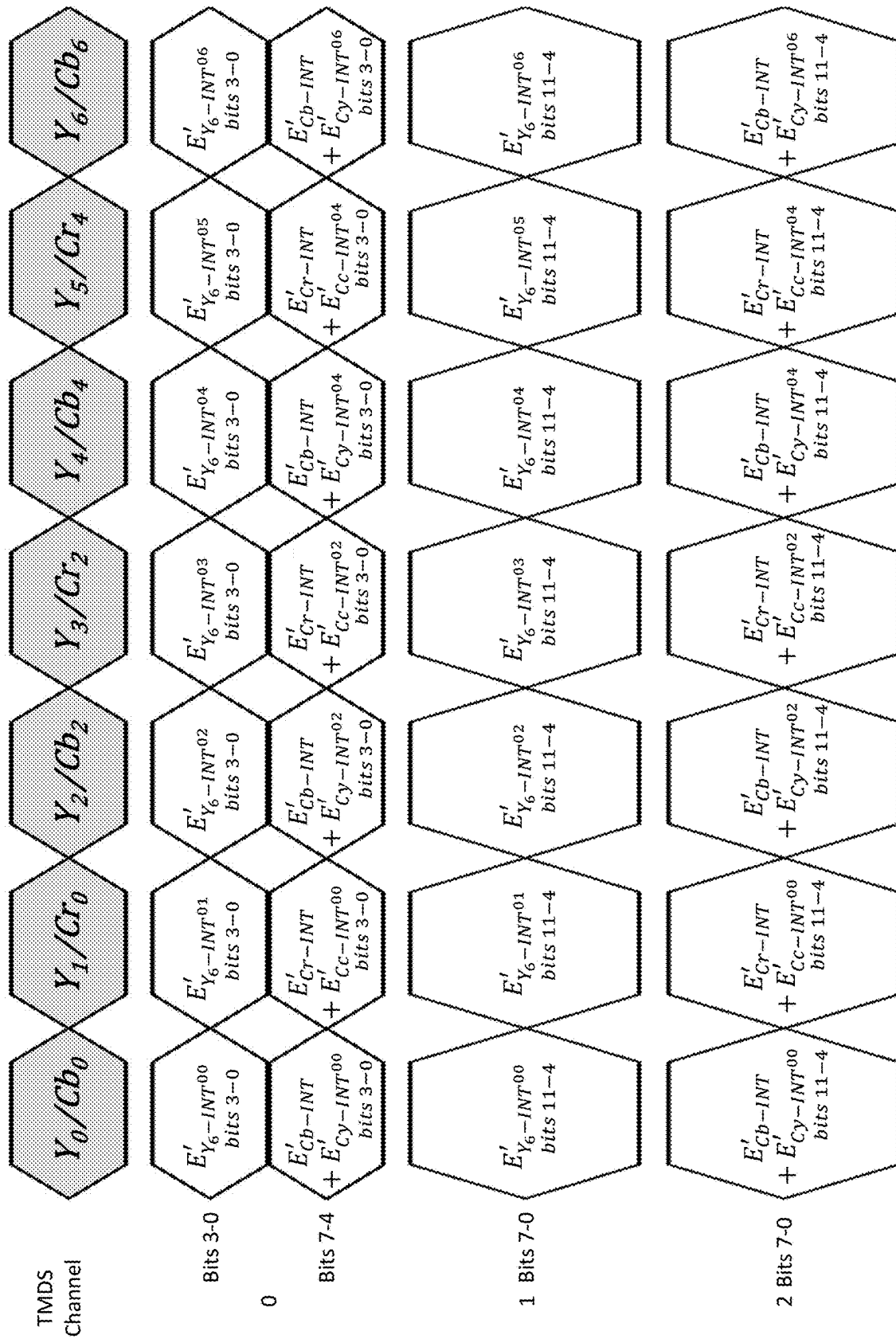


FIG. 67



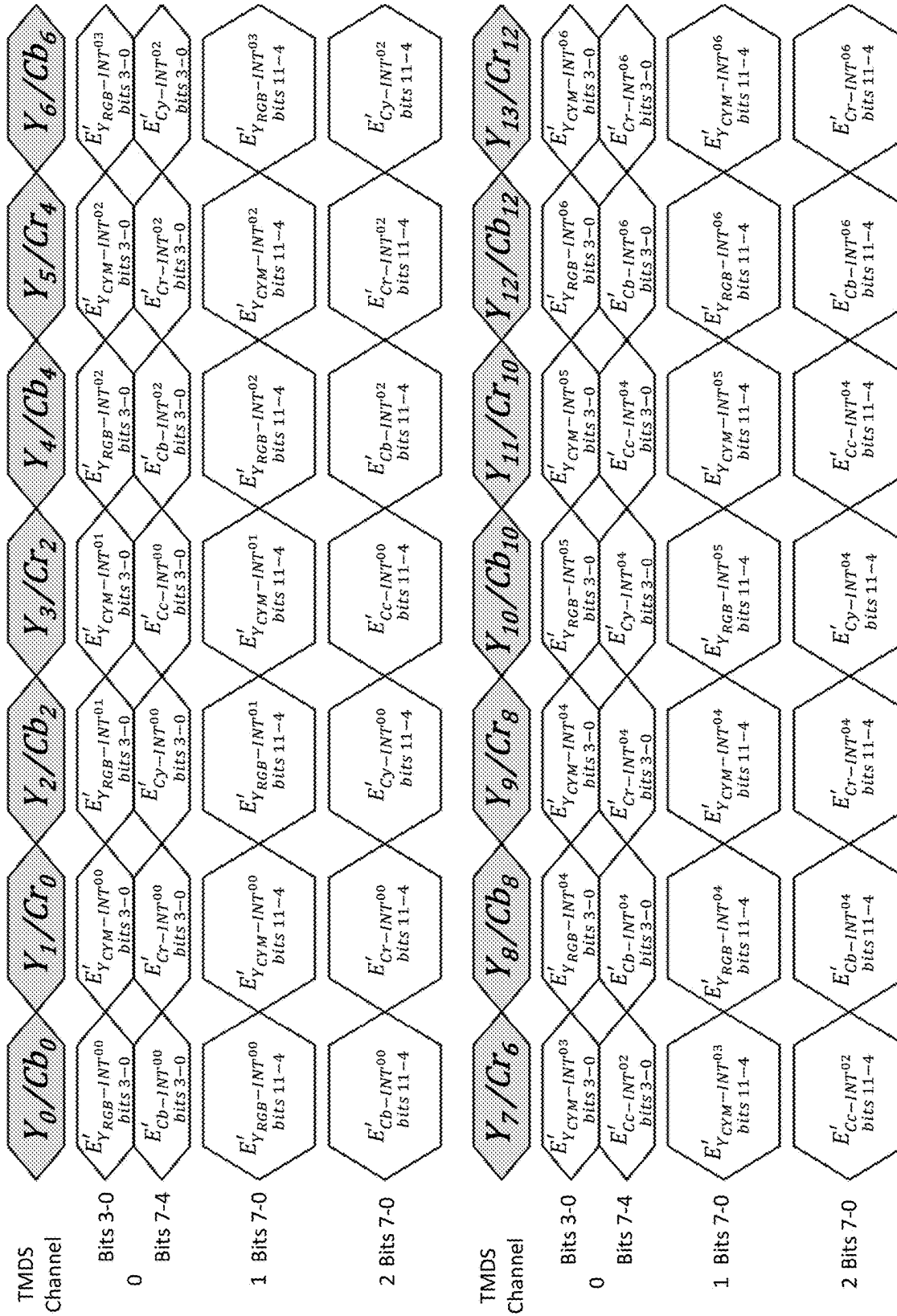


FIG. 69

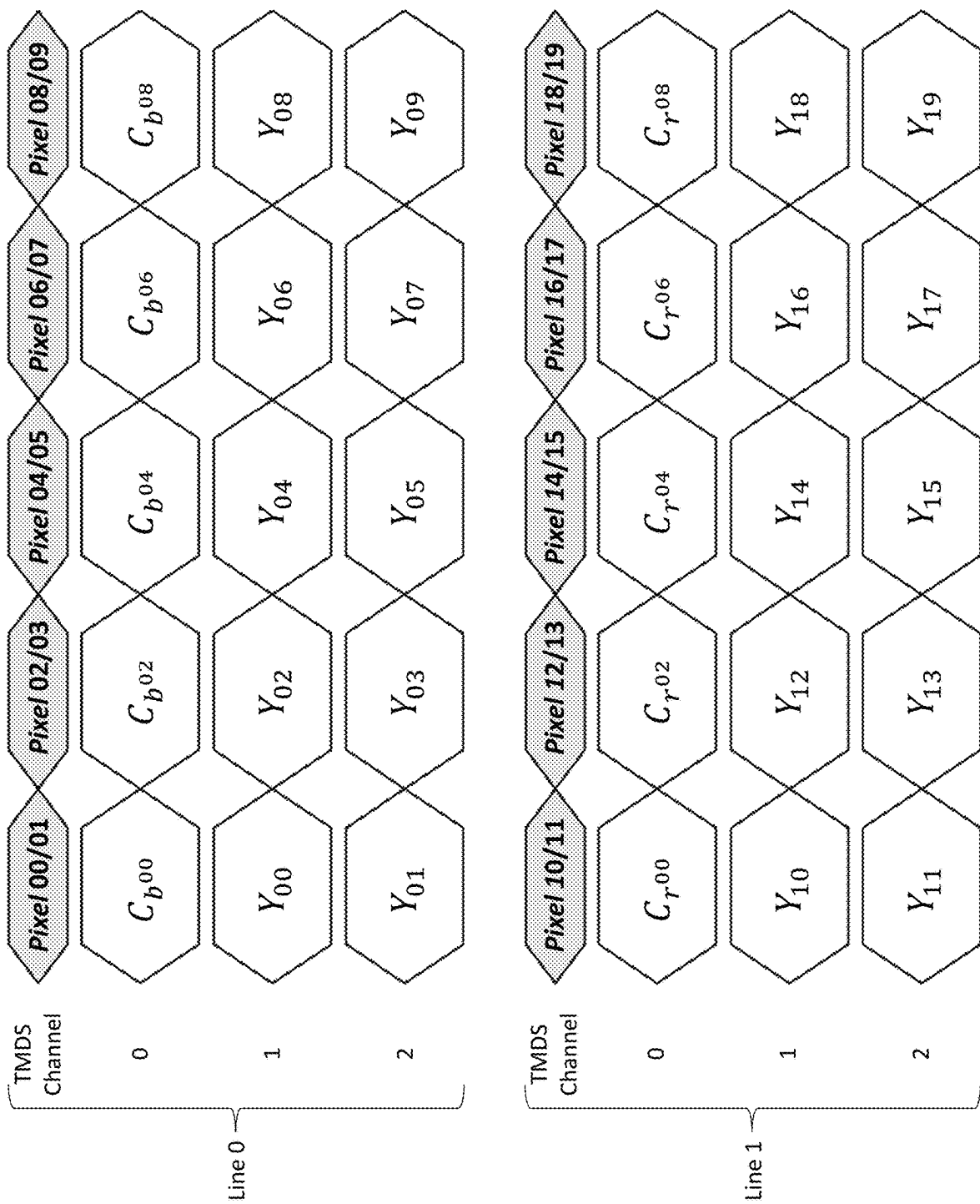


FIG. 70

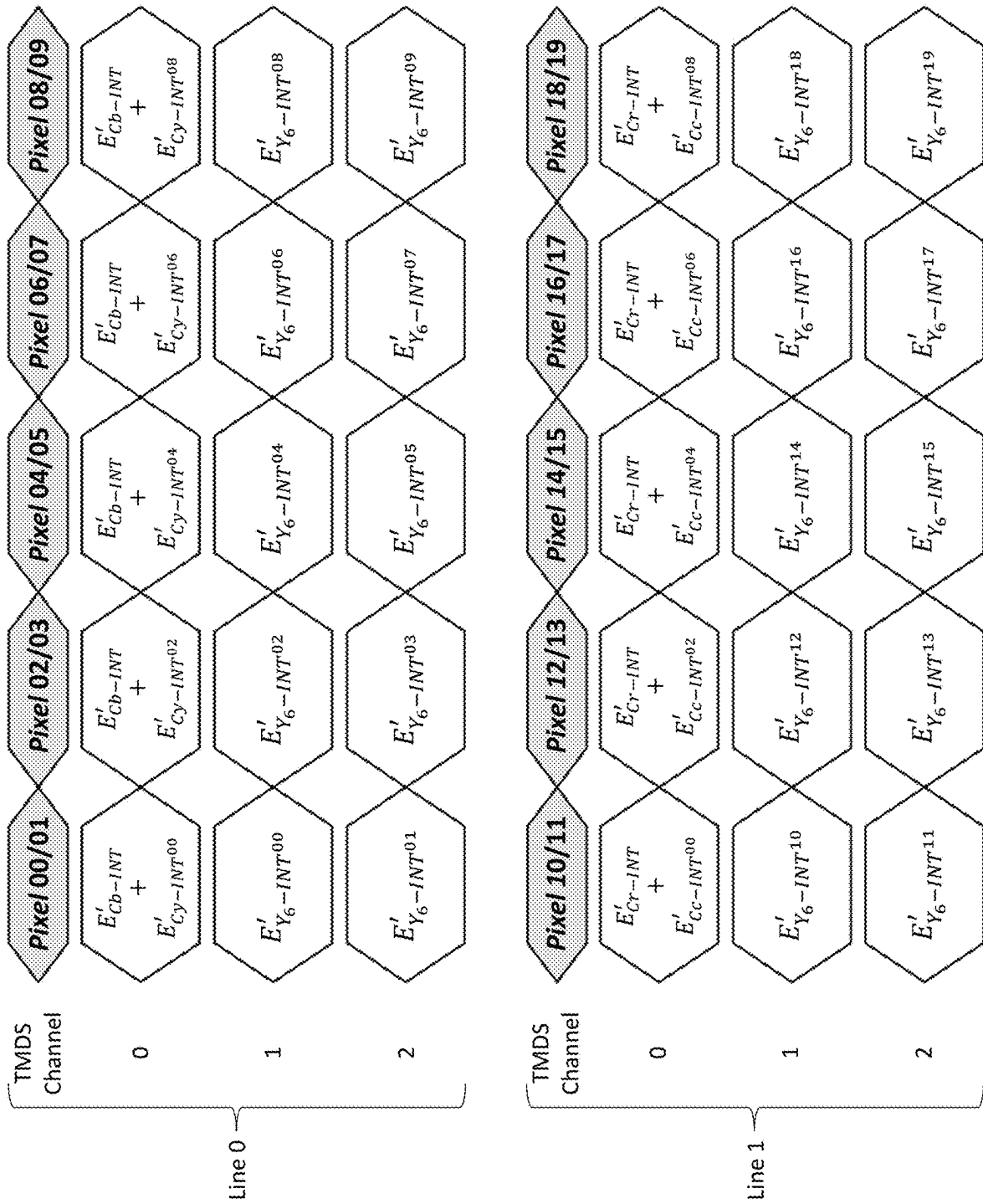


FIG. 71

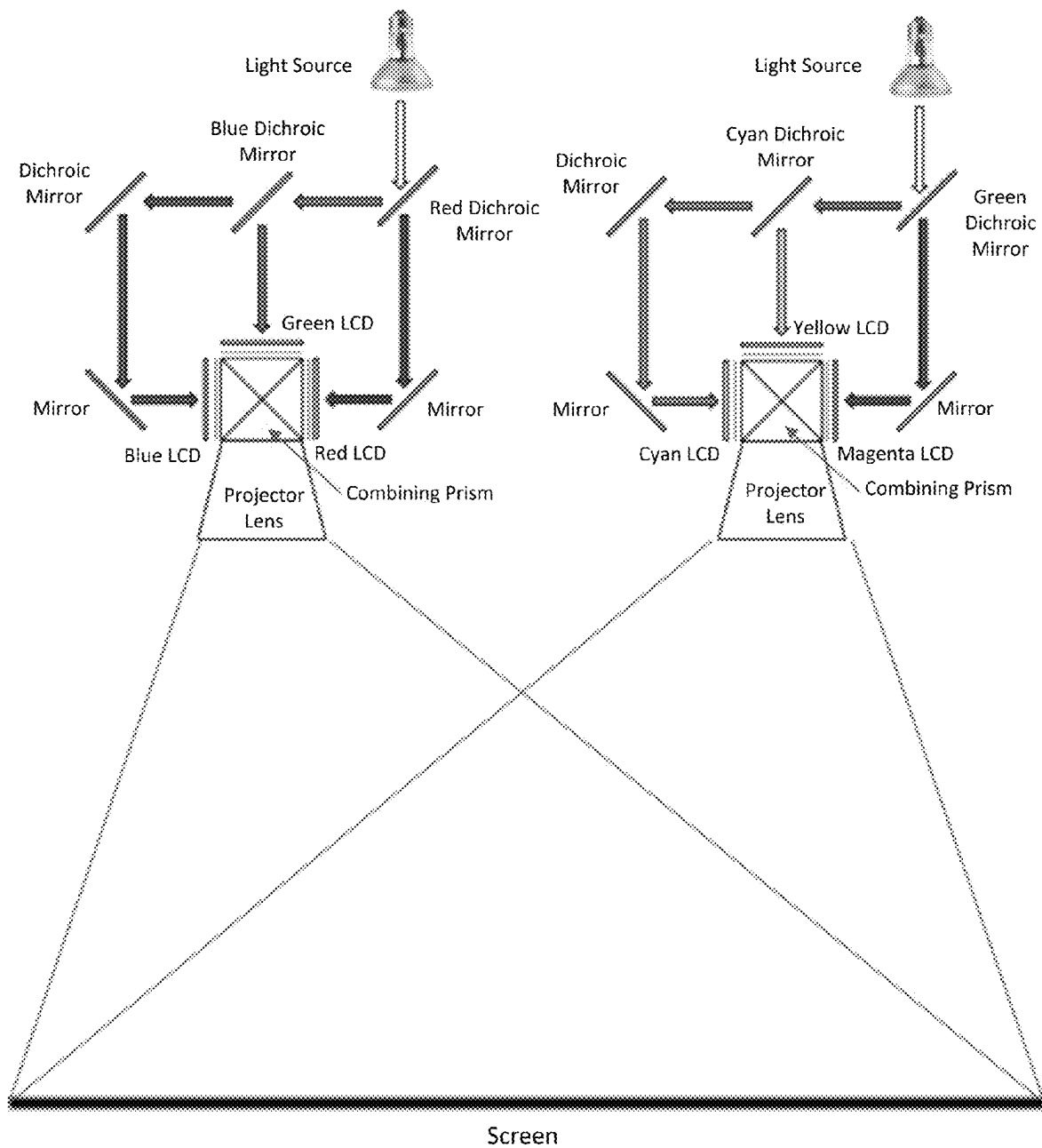


FIG. 72

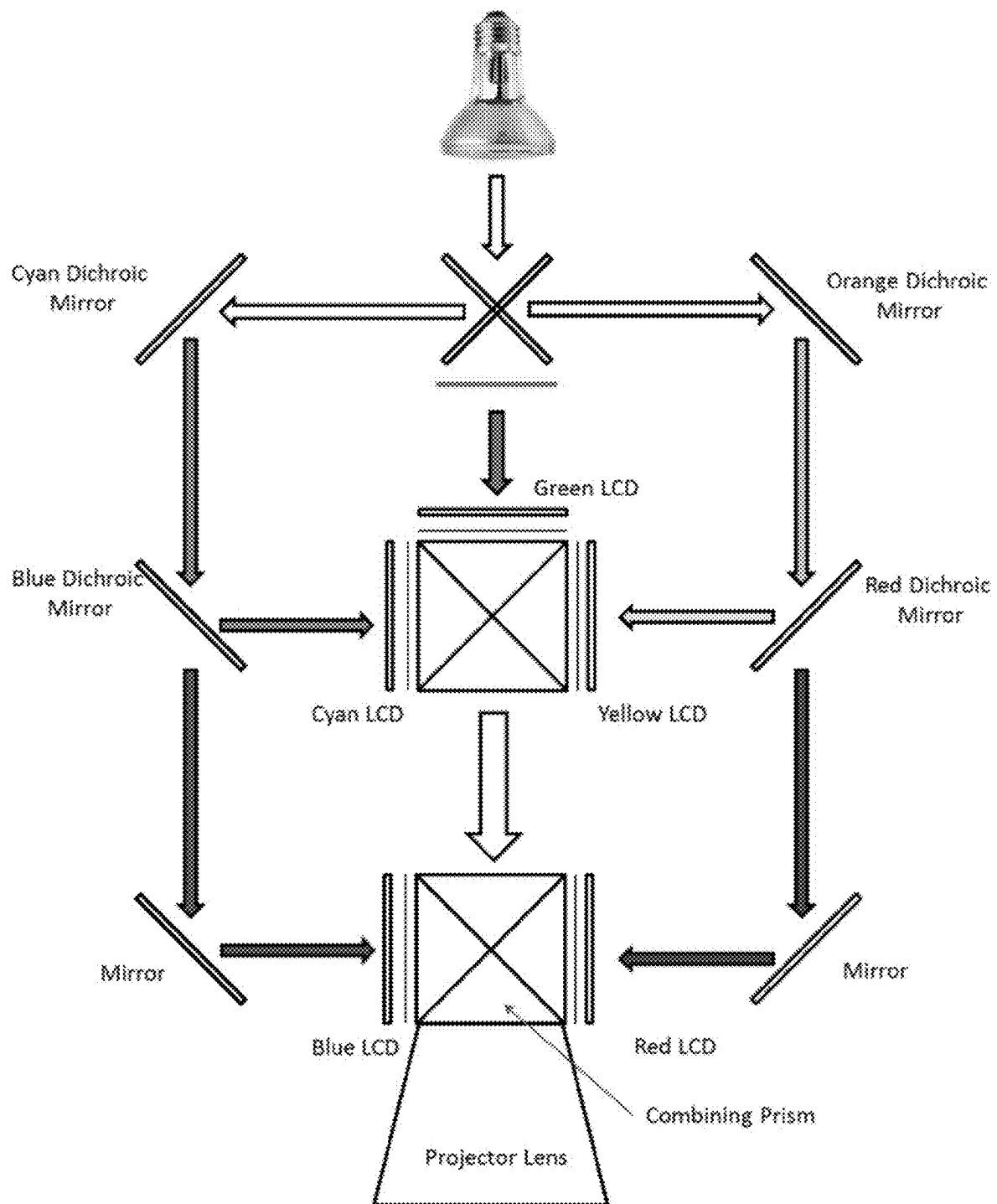


FIG. 73

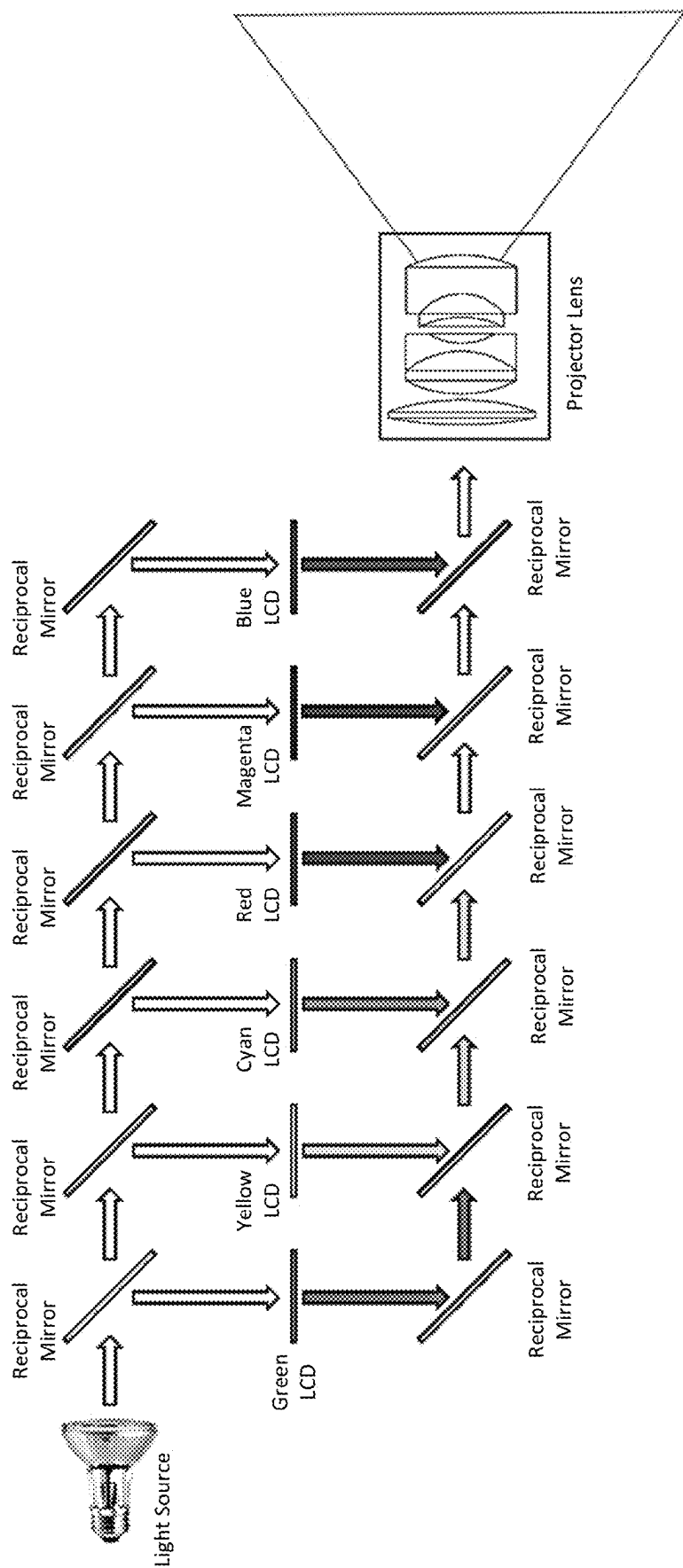


FIG. 74

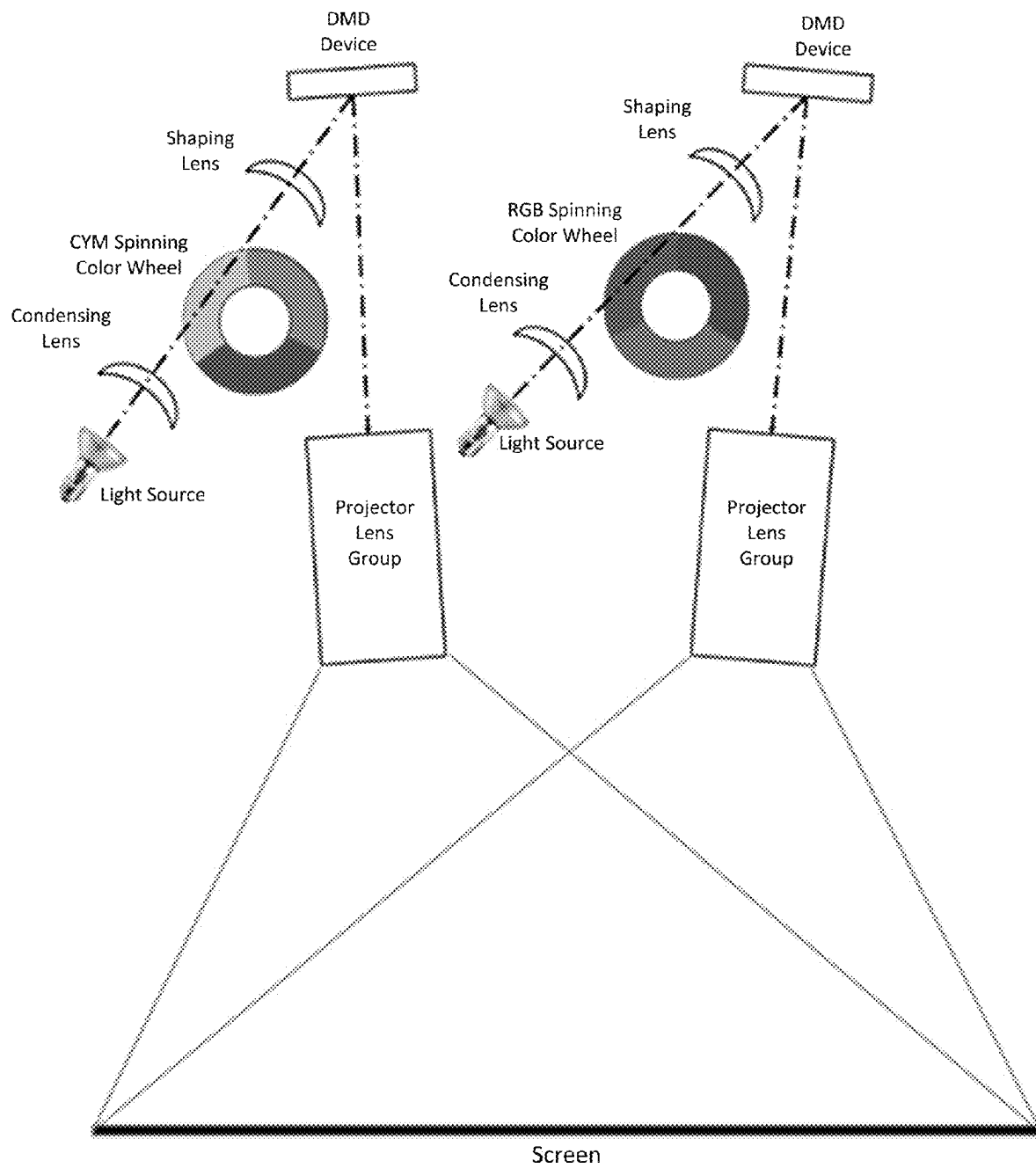


FIG. 75

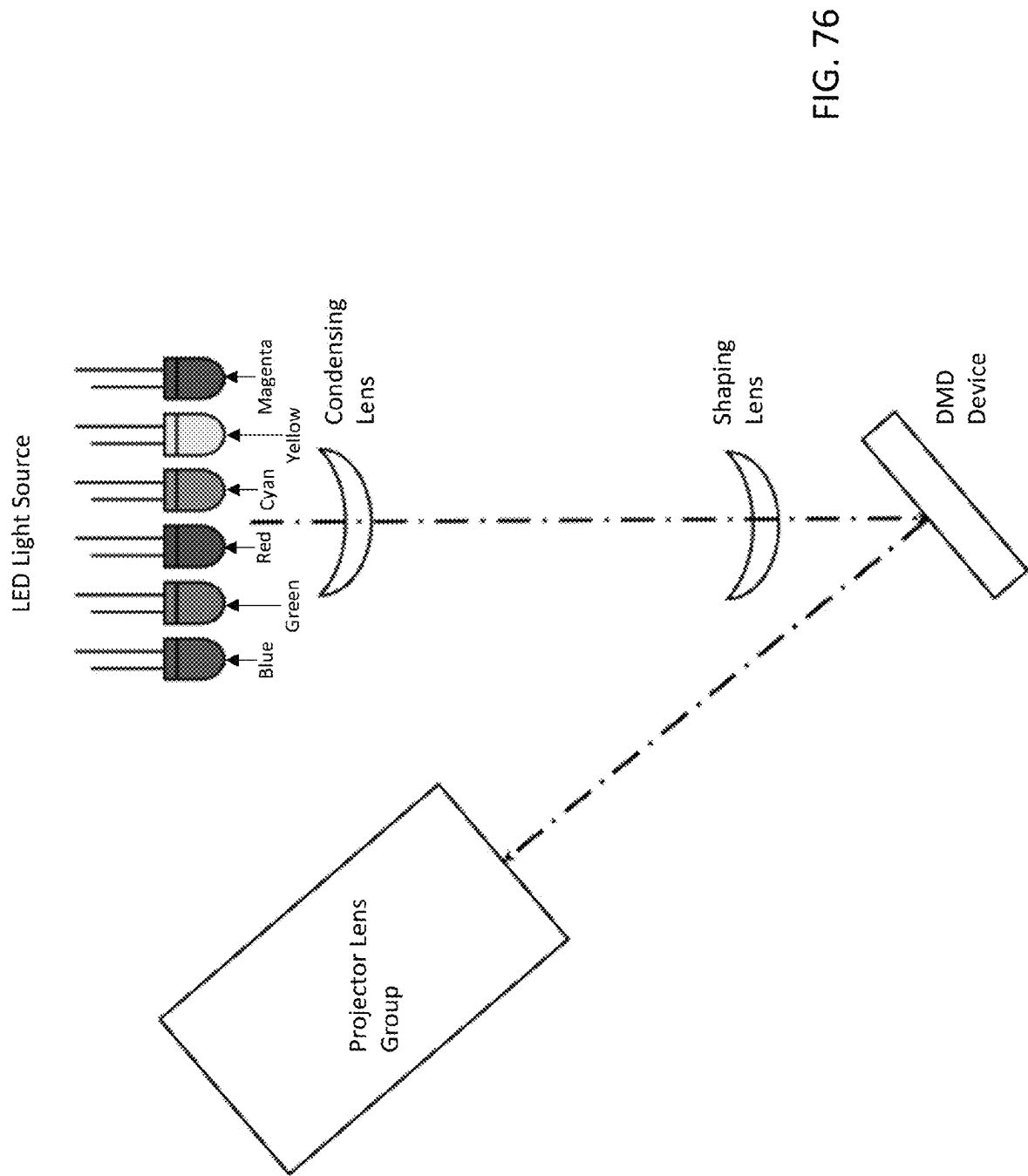


FIG. 76

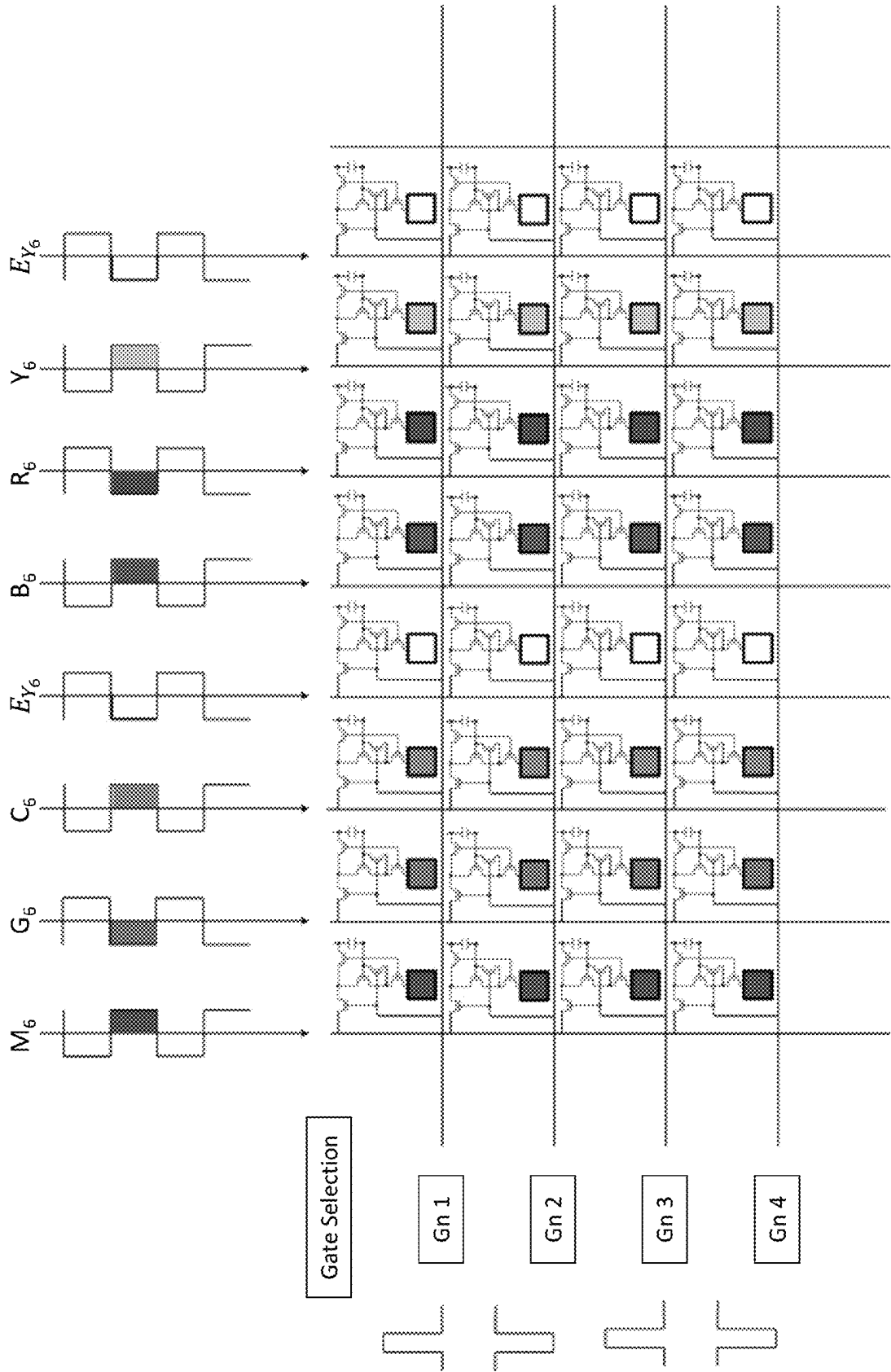


FIG. 77

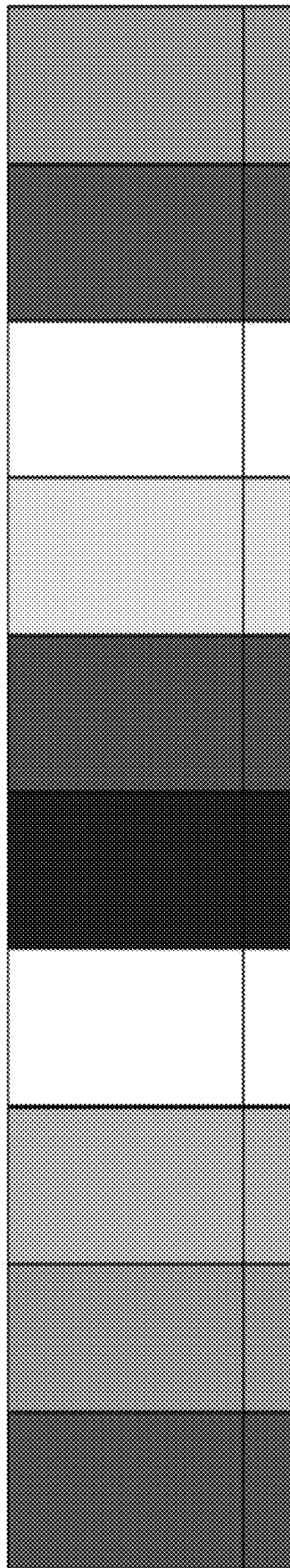


FIG. 78

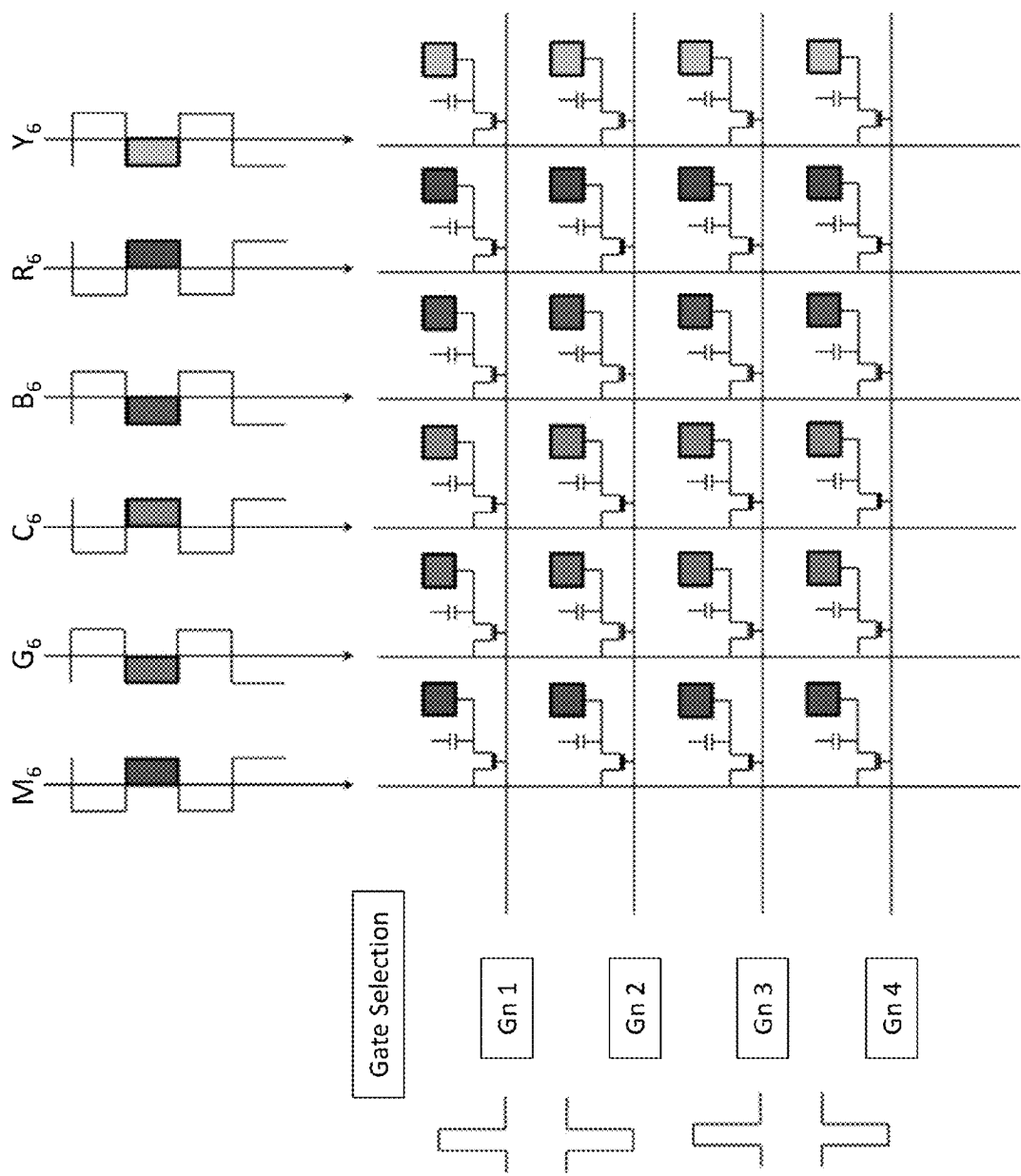


FIG. 79

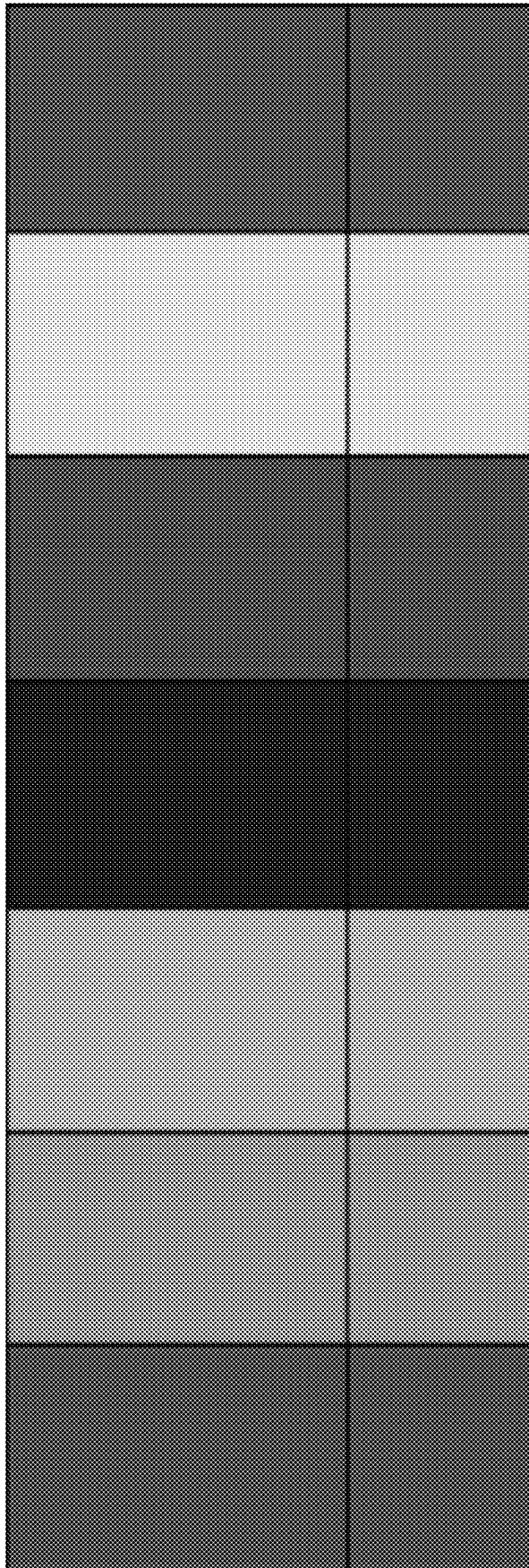


FIG. 80

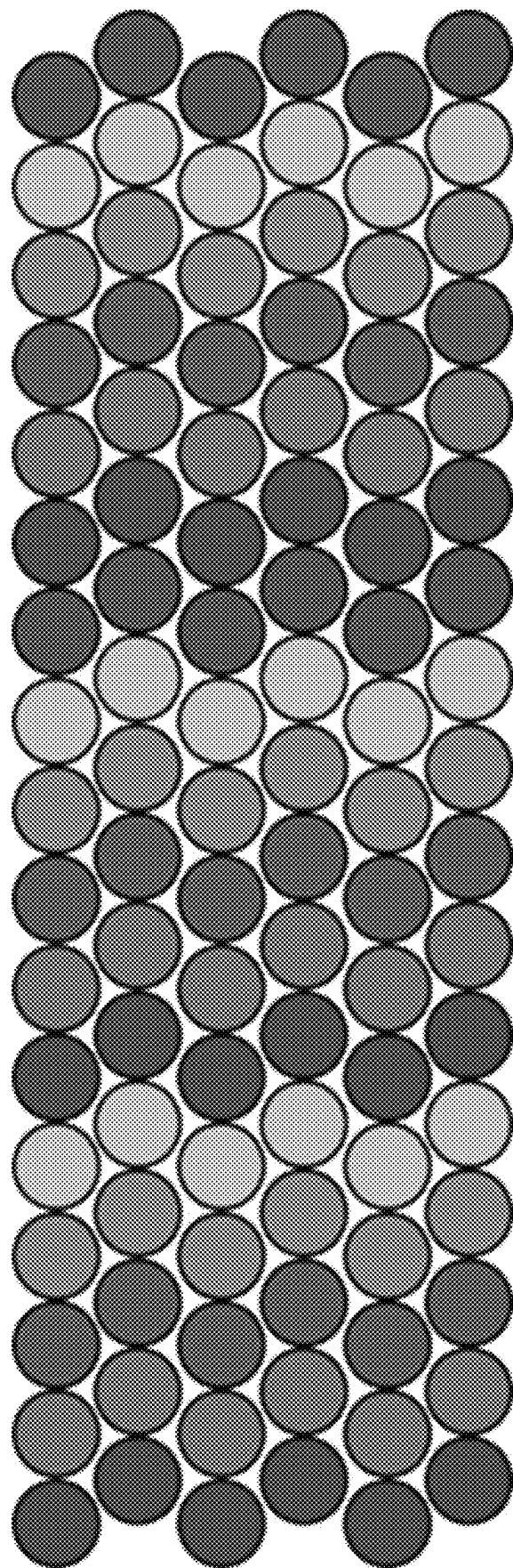


FIG. 81

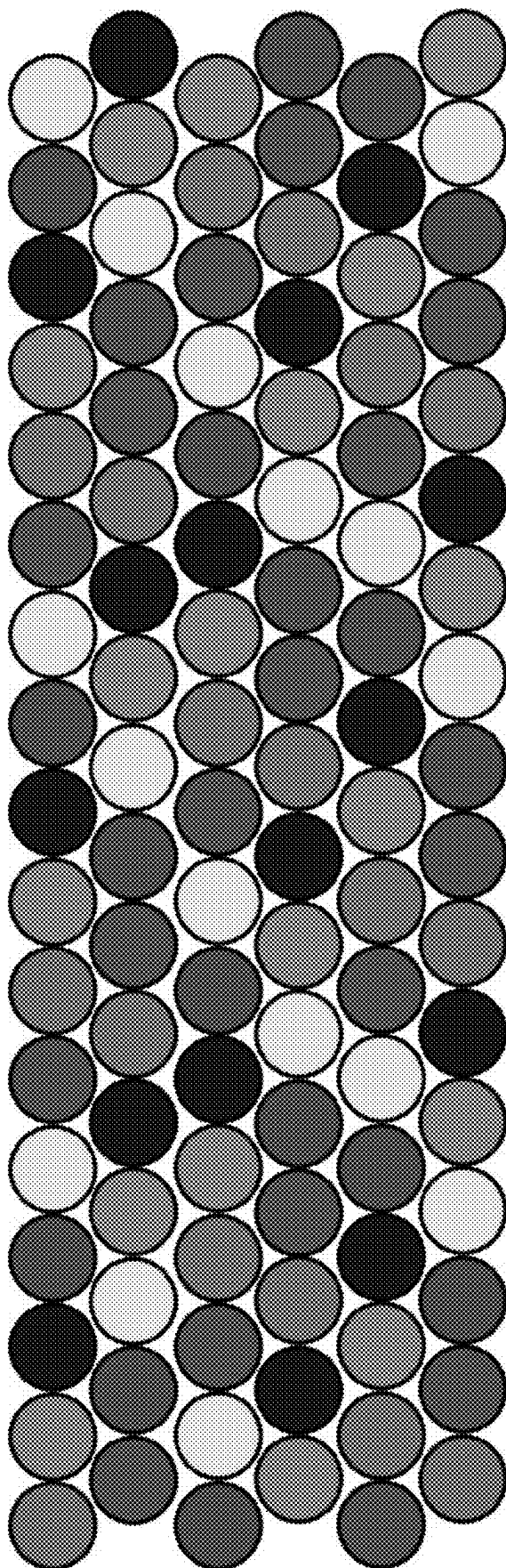


FIG. 82

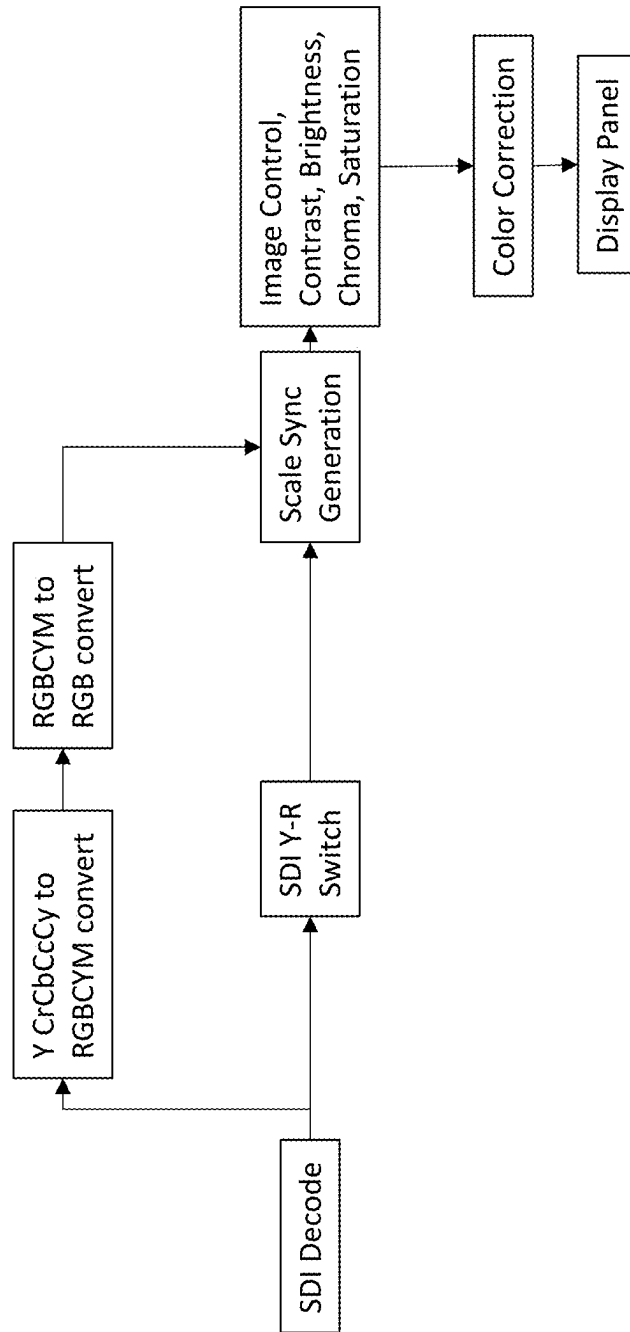


FIG. 83

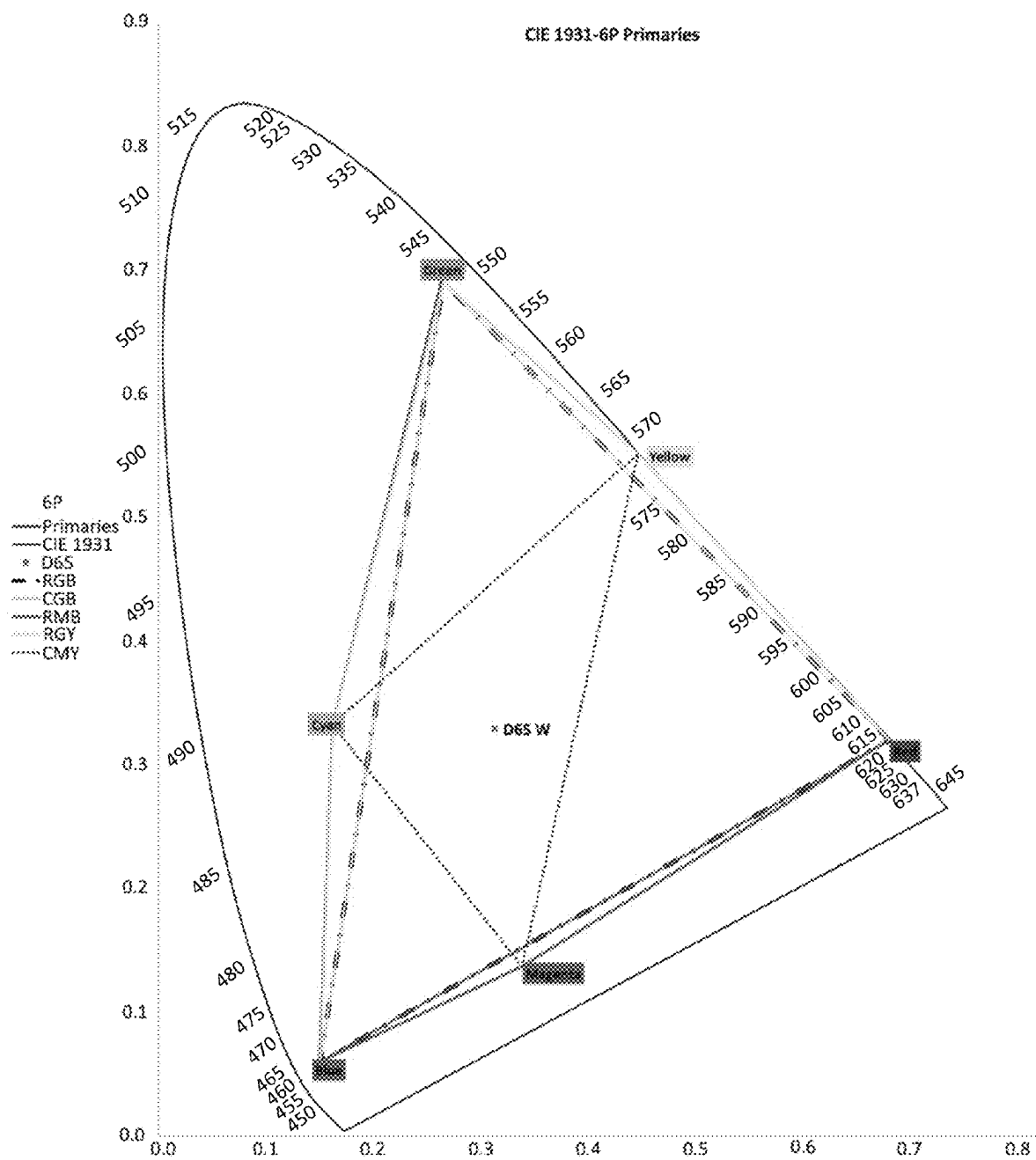


FIG. 84

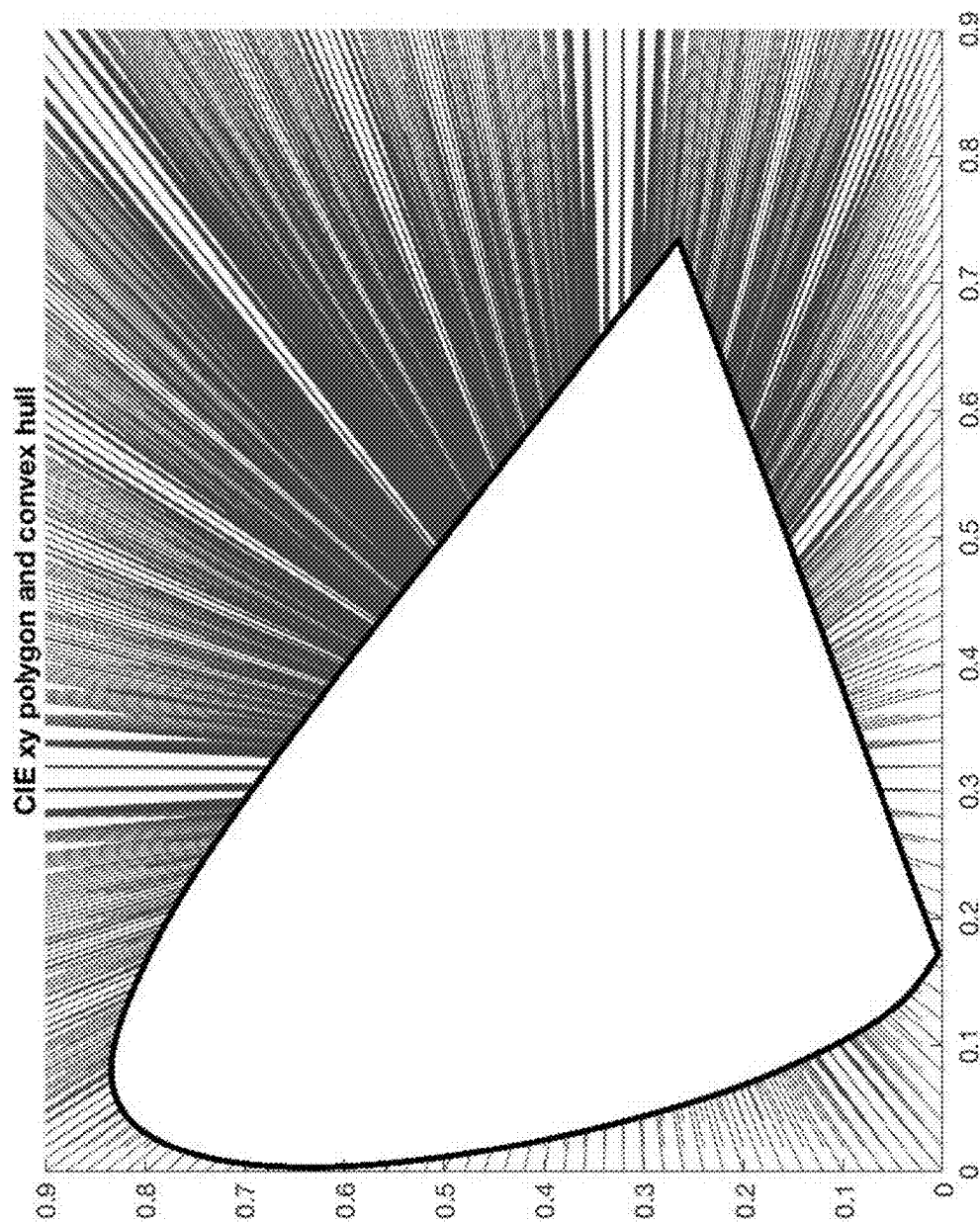


FIG. 85

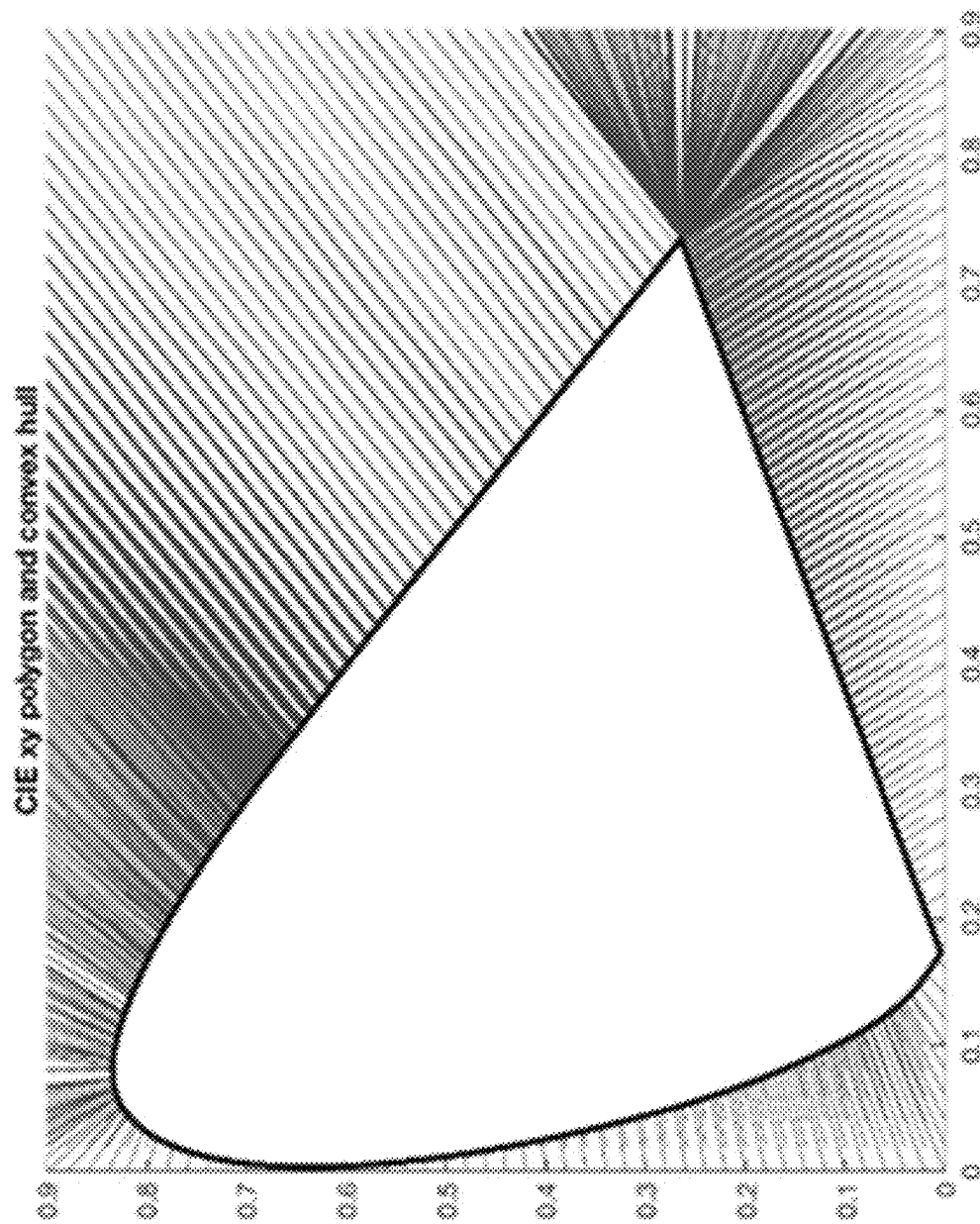


FIG. 86

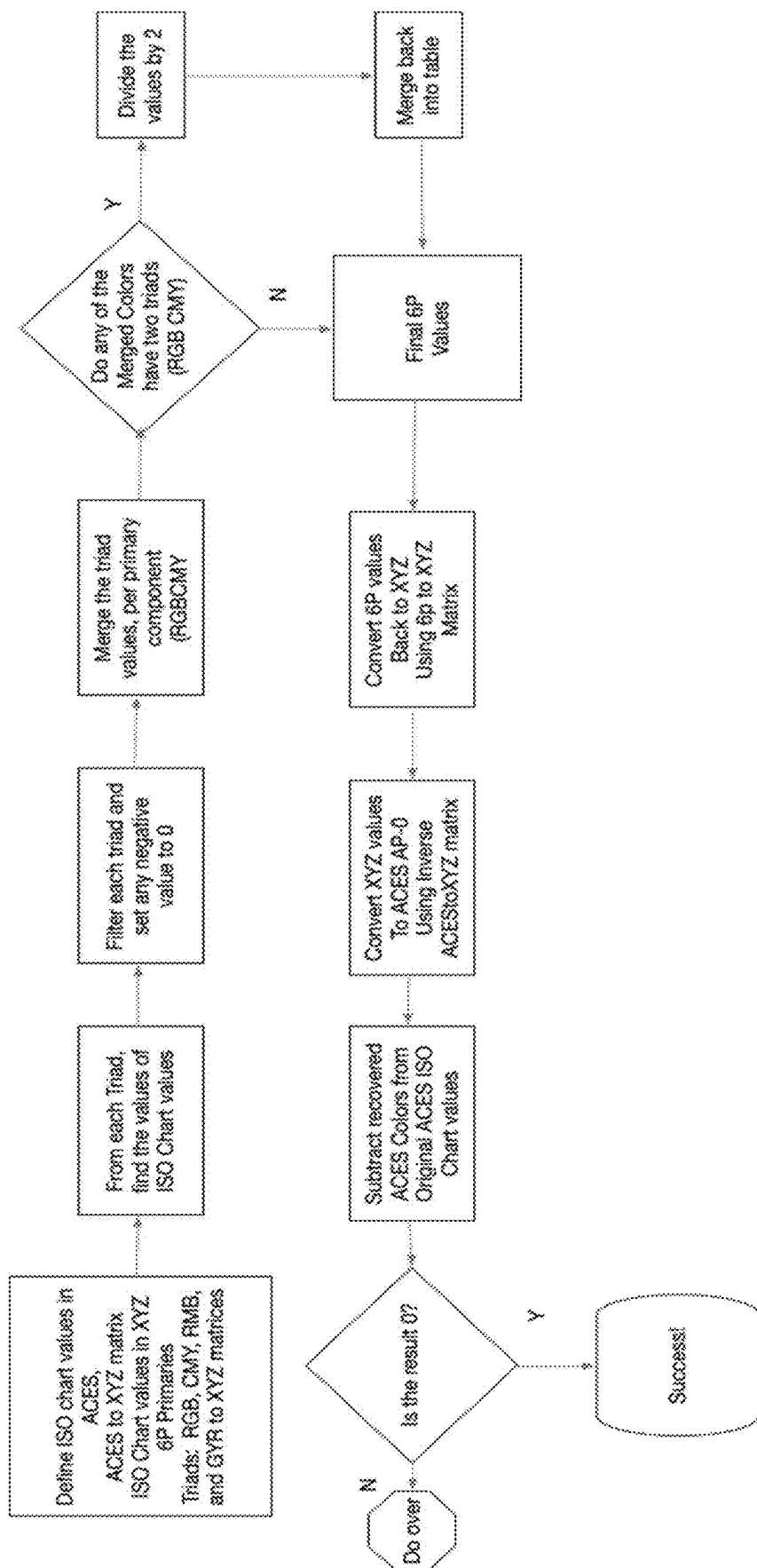
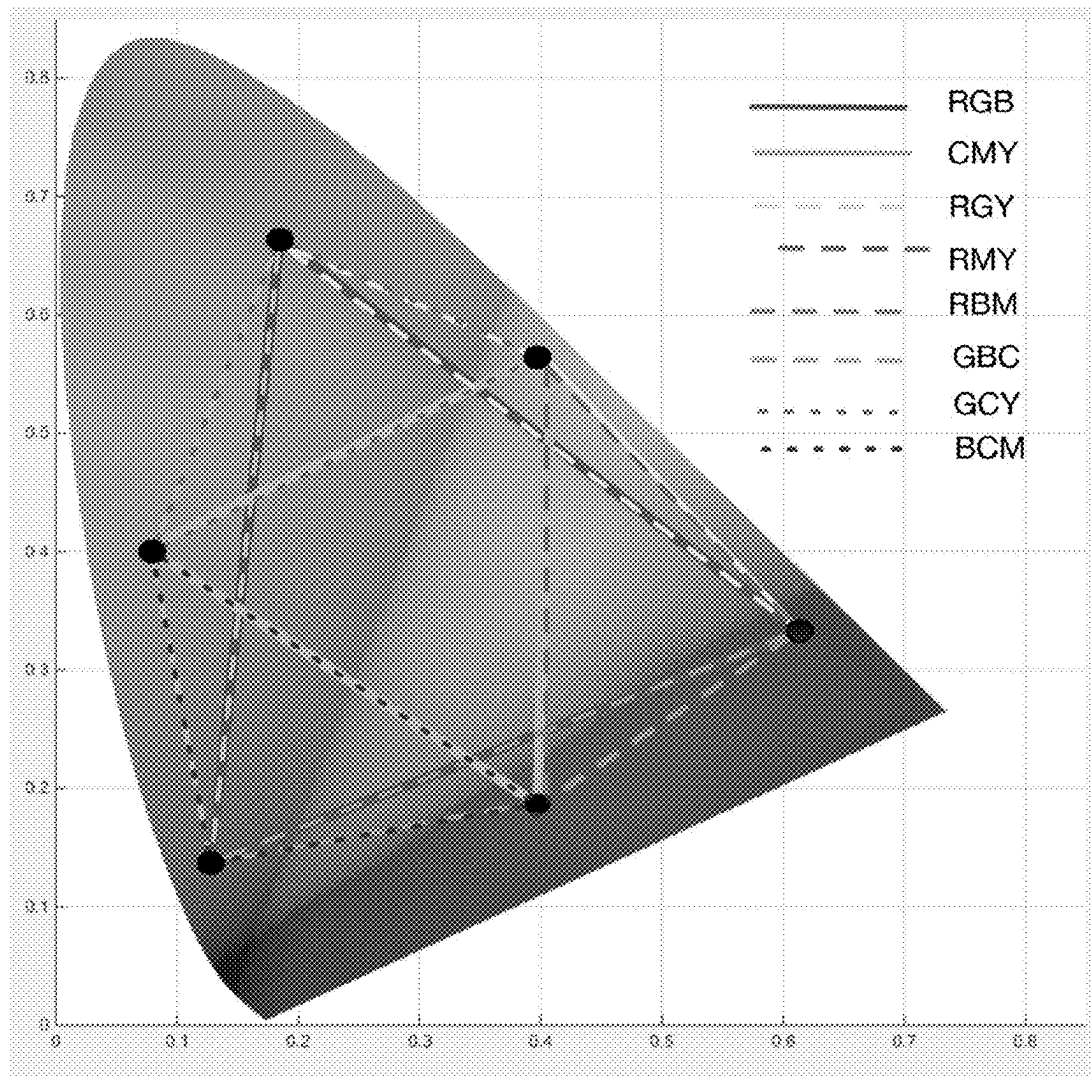


FIG. 87

Schematic Diagram of Triad Color Space



Notes:

1. Assumes all XYZ values are normalized in range of $[0,1]$.
2. Results should be ≥ 0 but may exceed 1.

FIG. 88

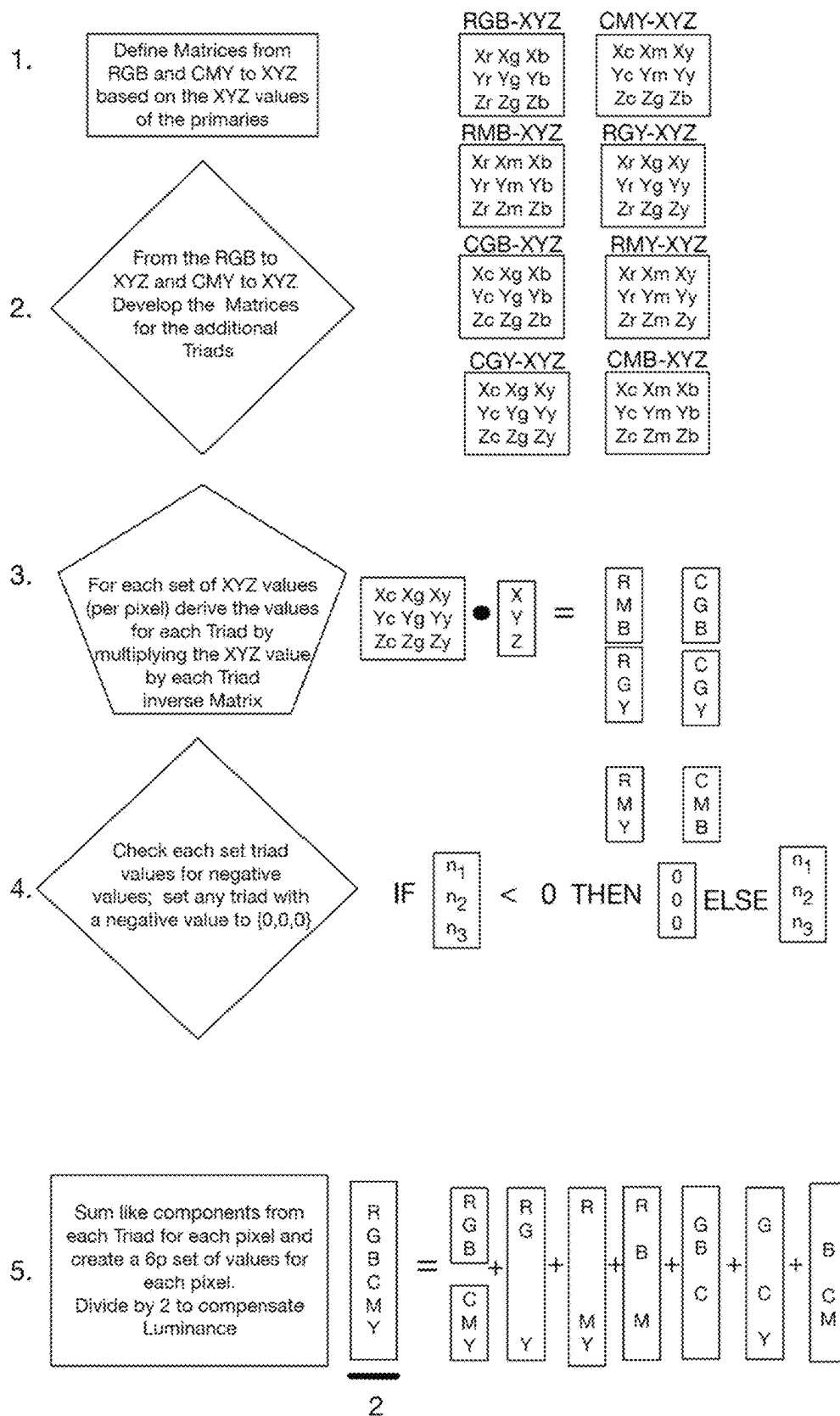


FIG. 89

RGBCMY 6P-N xy polygon and CIE xy convex hull W Triads

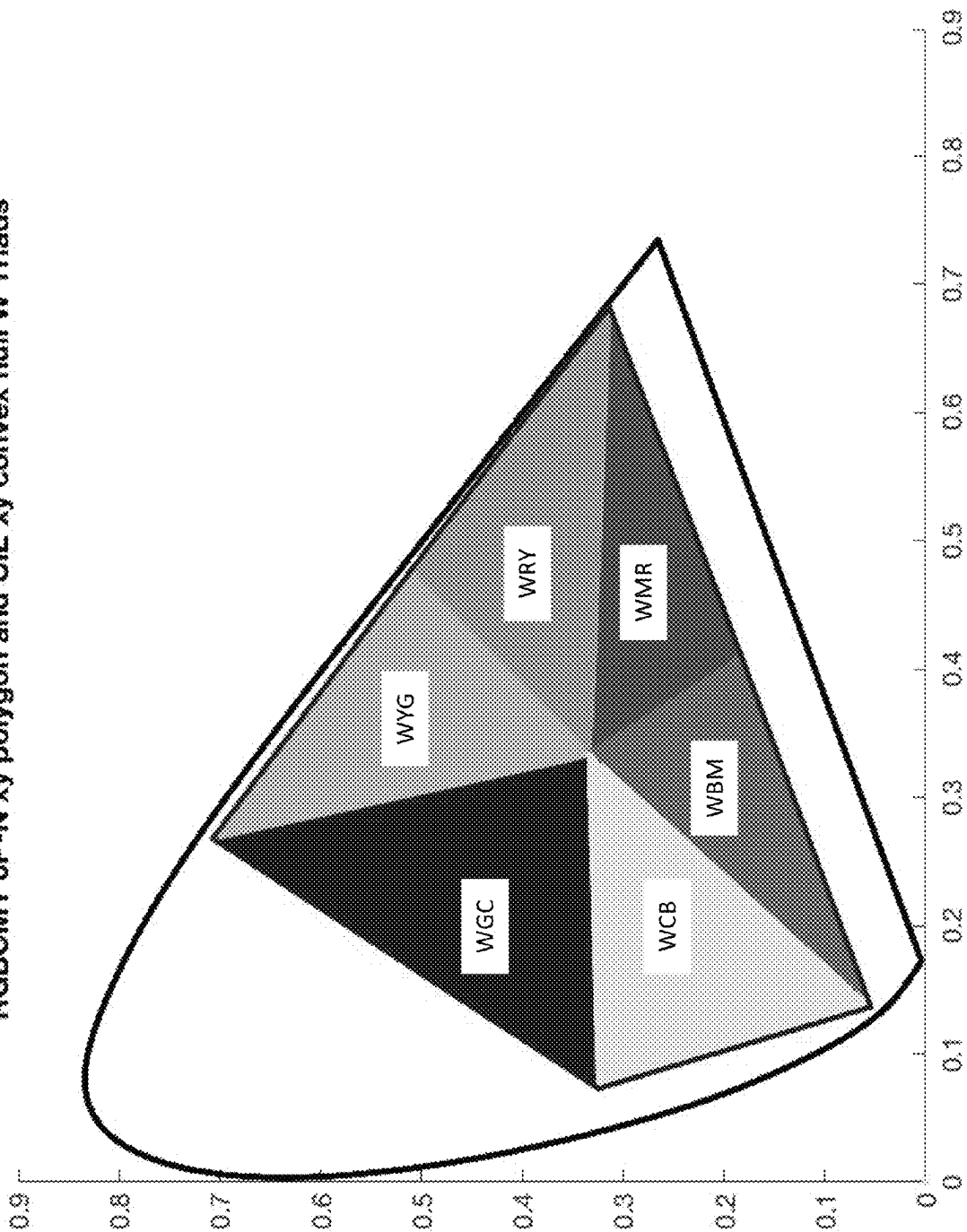
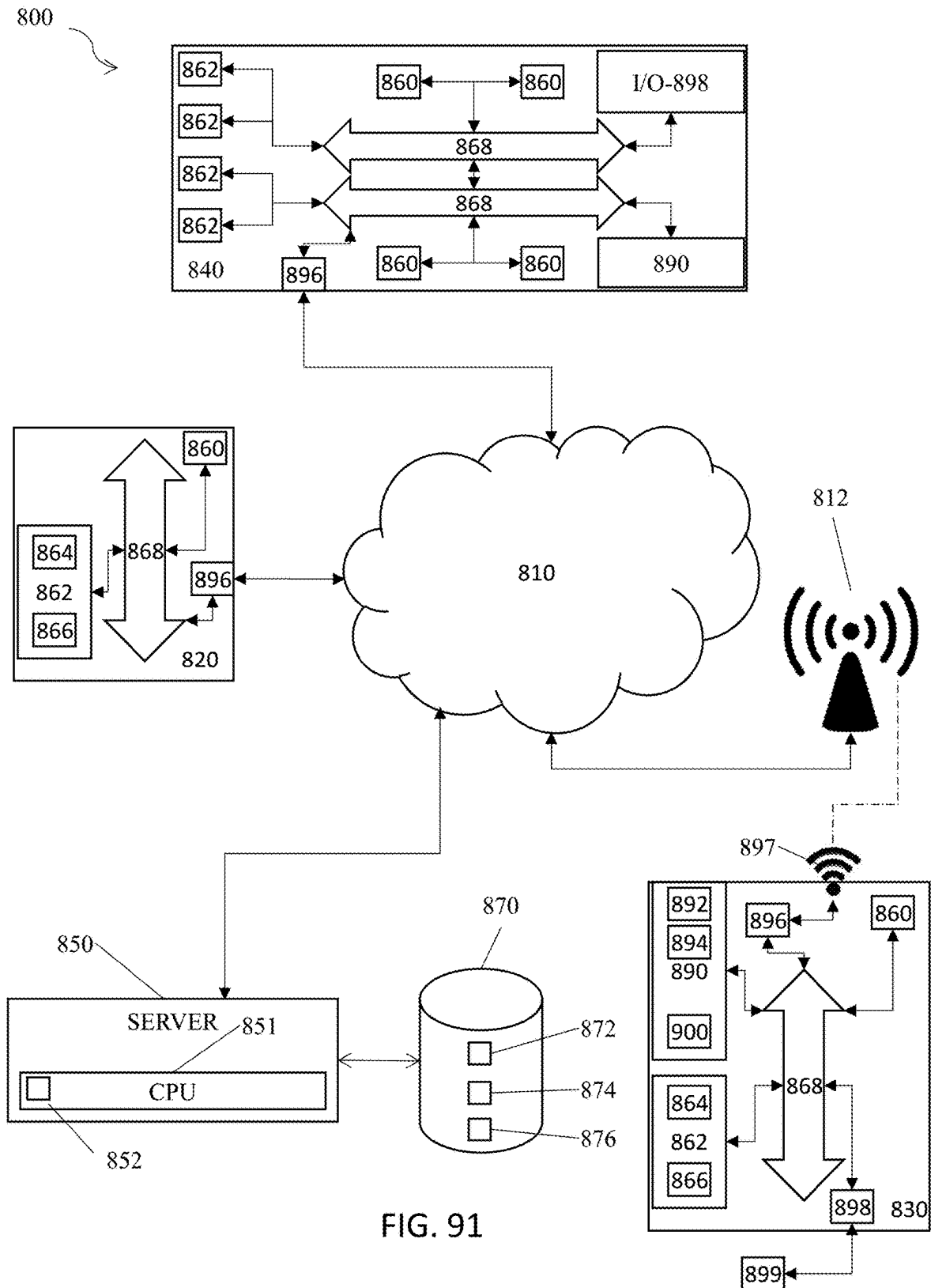


FIG. 90



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SYSTEM AND METHOD FOR A MULTI-PRIMARY WIDE GAMUT COLOR SYSTEM

CROSS REFERENCES TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 17/980,151, filed Nov. 3, 2022, which is a continuation of U.S. application Ser. No. 17/521,380, filed Nov. 8, 2021, which is a continuation of U.S. application Ser. No. 17/345,666, filed Jun. 11, 2021, which is a continuation of U.S. application Ser. No. 17/180,441, filed Feb. 19, 2021, which is a continuation-in-part of U.S. application Ser. No. 17/009,408, filed Sep. 1, 2020, which is a continuation-in-part of U.S. application Ser. No. 16/887,807, filed May 29, 2020, which is a continuation-in-part of U.S. application Ser. No. 16/860,769, filed Apr. 28, 2020, which is a continuation-in-part of U.S. application Ser. No. 16/853,203, filed Apr. 20, 2020, which is a continuation-in-part of U.S. patent application Ser. No. 16/831,157, filed Mar. 26, 2020, which is a continuation of U.S. patent application Ser. No. 16/659,307, filed Oct. 21, 2019, now U.S. Pat. No. 10,607,527, which is related to and claims priority from U.S. Provisional Patent Application No. 62/876,878, filed Jul. 22, 2019, U.S. Provisional Patent Application No. 62/847,630, filed May 14, 2019, U.S. Provisional Patent Application No. 62/805,705, filed Feb. 14, 2019, and U.S. Provisional Patent Application No. 62/750,673, filed Oct. 25, 2018, each of which is incorporated herein by reference in its entirety.

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates to color systems, and more specifically to a wide gamut color system with an increased number of primary colors.

2. Description of the Prior Art

It is generally known in the prior art to provide for an increased color gamut system within a display.

Prior art patent documents include the following:

U.S. Pat. No. 10,222,263 for RGB value calculation device by inventor Yasuyuki Shigezane, filed Feb. 6, 2017 and issued Mar. 5, 2019, is directed to a microcomputer that equally divides the circumference of an RGB circle into $6 \times n$ (n is an integer of 1 or more) parts, and calculates an RGB value of each divided color. (255, 0, 0) is stored as a reference RGB value of a reference color in a ROM in the microcomputer. The microcomputer converts the reference RGB value depending on an angular difference of the RGB circle between a designated color whose RGB value is to be found and the reference color, and assumes the converted RGB value as an RGB value of the designated color.

U.S. Pat. No. 9,373,305 for Semiconductor device, image processing system and program by inventor Hiorfumi Kawaguchi, filed May 29, 2015 and issued Jun. 21, 2016, is directed to an image process device including a display panel operable to provide an input interface for receiving an input of an adjustment value of at least a part of color attributes of each vertex of n axes (n is an integer equal to or greater than 3) serving as adjustment axes in an RGB color space, and an adjustment data generation unit operable to calculate the degree of influence indicative of a following index of each of the n -axis vertices, for each of the n axes,

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on a basis of distance between each of the n -axis vertices and a target point which is an arbitrary lattice point in the RGB color space, and operable to calculate adjusted coordinates of the target point in the RGB color space.

U.S. Publication No. 20130278993 for Color-mixing bi-primary color systems for displays by inventors Heikenfeld, et al., filed Sep. 1, 2011 and published Oct. 24, 2013, is directed to a display pixel. The pixel includes first and second substrates arranged to define a channel. A fluid is located within the channel and includes a first colorant and a second colorant. The first colorant has a first charge and a color. The second colorant has a second charge that is opposite in polarity to the first charge and a color that is complimentary to the color of the first colorant. A first electrode, with a voltage source, is operably coupled to the fluid and configured to moving one or both of the first and second colorants within the fluid and alter at least one spectral property of the pixel.

U.S. Pat. No. 8,599,226 for Device and method of data conversion for wide gamut displays by inventors Ben-Chorin, et al., filed Feb. 13, 2012 and issued Dec. 3, 2013, is directed to a method and system for converting color image data from a, for example, three-dimensional color space format to a format usable by an n -primary display, wherein n is greater than or equal to 3. The system may define a two-dimensional sub-space having a plurality of two-dimensional positions, each position representing a set of n primary color values and a third, scaleable coordinate value for generating an n -primary display input signal. Furthermore, the system may receive a three-dimensional color space input signal including out-of range pixel data not reproducible by a three-primary additive display, and may convert the data to side gamut color image pixel data suitable for driving the wide gamut color display.

U.S. Pat. No. 8,081,835 for Multiprimary color sub-pixel rendering with metameric filtering by inventors Elliot, et al., filed Jul. 13, 2010 and issued Dec. 20, 2011, is directed to systems and methods of rendering image data to multiprimary displays that adjusts image data across metamers. The metamer filtering may be based upon input image content and may optimize sub-pixel values to improve image rendering accuracy or perception. The optimizations may be made according to many possible desired effects. The patent discloses a display system comprising: a display, said display capable of selecting from a set of image data values, said set comprising at least one metamer; an input image data unit; a spatial frequency detection unit, said spatial frequency detection unit extracting a spatial frequency characteristic from said input image data; and a selection unit, said unit selecting image data from said metamer according to said spatial frequency characteristic.

U.S. Pat. No. 7,916,939 for High brightness wide gamut display by inventors Roth, et al., filed Nov. 30, 2009 and issued Mar. 29, 2011, is directed to a device to produce a color image, the device including a color filtering arrangement to produce at least four colors, each color produced by a filter on a color filtering mechanism having a relative segment size, wherein the relative segment sizes of at least two of the primary colors differ.

U.S. Pat. No. 6,769,772 for Six color display apparatus having increased color gamut by inventors Roddy, et al., filed Oct. 11, 2002 and issued Aug. 3, 2004, is directed to a display system for digital color images using six color light sources or two or more multicolor LED arrays or OLEDs to provide an expanded color gamut. Apparatus uses two or more spatial light modulators, which may be cycled between two or more color light sources or LED arrays to provide a

six-color display output. Pairing of modulated colors using relative luminance helps to minimize flicker effects.

SUMMARY OF THE INVENTION

It is an object of this invention to provide an enhancement to the current RGB systems or a replacement for them.

In one embodiment, the present invention is a system for displaying a six primary color system, including a set of image data, wherein the set of image data is comprised of a first set of color channel data and a second set of color channel data, wherein the set of image data further includes a bit level, an image data converter, wherein the image data converter includes a digital interface, wherein the digital interface is operable to encode and decode the set of image data, at least one transfer function (TF) for processing the set of image data, a set of Session Description Protocol (SDP) parameters, wherein the set of SDP parameters is modifiable, at least one display device, wherein the at least one display device and the image data converter are in network communication, wherein the image data converter is operable to convert the bit level of the set of image data, thereby creating an updated bit level, wherein the image data converter is operable to convert the set of image data for display on the at least one display device, wherein once the set of image data has been converted by the image data converter for the at least one display device the set of SDP parameters are modified based on the conversion, and wherein the at least one display device is operable to display a six-primary color system based on the set of image data, such that the SDP parameters indicate that the set of image data being displayed on the at least one display device is using a six-primary color system.

In another embodiment, the present invention is a system for displaying a six-primary color system, including a set of image data, wherein the set of image data includes a first set of color channel data and a second set of color channel data, wherein the set of image data includes a bit level, a magenta primary value, wherein the magenta primary value is derived from the set of image data, an image data converter, wherein the image data converter includes a digital interface, wherein the digital interface is operable to encode and decode the set of image data, at least one transfer function (TF) for processing the set of image data, a set of Session Description Protocol (SDP) parameters, wherein the set of SDP parameters are modifiable, at least one display device, wherein the at least one display device and the image data converter are in network communication, wherein the image data converter is operable to convert the bit level for the set of image data to a new bit level, wherein the at least one data converter is operable to convert the set of image data for display on the at least one display device, wherein once the set of image data has been converted for the at least one display device the set of SDP parameters are modified based on the conversion, and wherein the at least one display device is operable to display a six-primary color system based on the set of image data, such that the SDP parameters indicate the magenta primary value and that the set of image data being displayed on the at least one display device is using a six-primary color system.

In yet another embodiment, the present invention is a system for displaying a set of image data using a six-primary color system, including a set of image data, wherein the set of image data includes a bit level, a magenta primary value, wherein the magenta primary value is derived from the set of image data, an image data converter, wherein the image data converter includes a digital interface, wherein the

digital interface is operable to encode and decode the set of image data, at least one transfer function (TF) for processing the set of image data, a set of Session Description Protocol (SDP) parameters, wherein the set of SDP parameters are modifiable, at least one electronic luminance component, wherein the electronic luminance component is derived from the set of image data, at least one display device, wherein the at least one display device and the image data converter are in network communication, wherein the image data converter is operable to convert the set of image data to a new bit level, wherein the data converter is operable to convert the set of image data for display on the at least one display device, wherein once the set of image data has been converted for the at least one display device the set of SDP parameters are modified based on the conversion, and wherein the at least one display device is operable to display a six-primary color system based on the set of image data, such that the SDP parameters indicate the magenta primary value, the at least one electronic luminance component, and that the set of image data being displayed on the at least one display device is using a six-primary color system.

These and other aspects of the present invention will become apparent to those skilled in the art after a reading of the following description of the preferred embodiment when considered with the drawings, as they support the claimed invention.

BRIEF DESCRIPTION OF THE DRAWINGS

The patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawing(s) will be provided by the Office upon request and payment of the necessary fee.

FIG. 1 illustrates one embodiment of a six primary system including a red primary, a green primary, a blue primary, a cyan primary, a magenta primary, and a yellow primary ("6P-B") compared to ITU-R BT.709-6.

FIG. 2 illustrates another embodiment of a six primary system including a red primary, a green primary, a blue primary, a cyan primary, a magenta primary, and a yellow primary ("6P-C") compared to SMPTE RP431-2 for a D60 white point.

FIG. 3 illustrates yet another embodiment of a six primary system including a red primary, a green primary, a blue primary, a cyan primary, a magenta primary, and a yellow primary ("6P-C") compared to SMPTE RP431-2 for a D65 white point.

FIG. 4 illustrates Super 6 Pa compared to 6P-C.

FIG. 5 illustrates Super 6Pb compared to Super 6 Pa and 6P-C.

FIG. 6 illustrates an embodiment of an encode and decode system for a multi-primary color system.

FIG. 7 illustrates a sequential method where three color primaries are passed to the transport format as full bit level image data and inserted as normal ("System 2").

FIG. 8 illustrates one embodiment of a system encode and decode process using a dual link method ("System 3").

FIG. 9 illustrates one embodiment of an encoding process using a dual link method.

FIG. 10 illustrates one embodiment of a decoding process using a dual link method.

FIG. 11 illustrates one embodiment of a six-primary color system encode using a 4:4:4 sampling method.

FIG. 12 illustrates one embodiment for a method to package six channels of primary information into the three

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standard primary channels used in current serial video standards by modifying bit numbers for a 12-bit SDI and a 10-bit SDI.

FIG. 13 illustrates a simplified diagram estimating perceived viewer sensation as code values define each hue angle.

FIG. 14 illustrates one embodiment for a method of stacking/encoding six-primary color information using a 4:4:4 video system.

FIG. 15 illustrates one embodiment for a method of unstacking/decoding six-primary color information using a 4:4:4 video system.

FIG. 16 illustrates one embodiment of a 4:4:4 decoder for a six-primary color system.

FIG. 17 illustrates one embodiment of an optical filter.

FIG. 18 illustrates another embodiment of an optical filter.

FIG. 19 illustrates an embodiment of the present invention for sending six primary colors to a standardized transport format.

FIG. 20 illustrates one embodiment of a decode process adding a pixel delay to the RGB data for realigning the channels to a common pixel timing.

FIG. 21 illustrates one embodiment of an encode process for 4:2:2 video for packaging five channels of information into the standard three-channel designs.

FIG. 22 illustrates one embodiment for a non-constant luminance encode for a six-primary color system.

FIG. 23 illustrates one embodiment of a packaging process for a six-primary color system.

FIG. 24 illustrates a 4:2:2 unstack process for a six-primary color system.

FIG. 25 illustrates one embodiment of a process to inversely quantize each individual color and pass the data through an electronic optical function transfer (EOTF) in a non-constant luminance system.

FIG. 26 illustrates one embodiment of a constant luminance encode for a six-primary color system.

FIG. 27 illustrates one embodiment of a constant luminance decode for a six-primary color system.

FIG. 28 illustrates one example of 4:2:2 non-constant luminance encoding.

FIG. 29 illustrates one embodiment of a non-constant luminance decoding system.

FIG. 30 illustrates one embodiment of a 4:2:2 constant luminance encoding system.

FIG. 31 illustrates one embodiment of a 4:2:2 constant luminance decoding system.

FIG. 32 illustrates a raster encoding diagram of sample placements for a six-primary color system.

FIG. 33 illustrates one embodiment of the six-primary color unstack process in a 4:2:2 video system.

FIG. 34 illustrates one embodiment of mapping input to the six-primary color system unstack process.

FIG. 35 illustrates one embodiment of mapping the output of a six-primary color system decoder.

FIG. 36 illustrates one embodiment of mapping the RGB decode for a six-primary color system.

FIG. 37 illustrates one embodiment of an unstack system for a six-primary color system.

FIG. 38 illustrates one embodiment of a legacy RGB decoder for a six-primary, non-constant luminance system.

FIG. 39 illustrates one embodiment of a legacy RGB decoder for a six-primary, constant luminance system.

FIG. 40 illustrates one embodiment of a six-primary color system with output to a legacy RGB system.

FIG. 41 illustrates one embodiment of six-primary color output using a non-constant luminance decoder.

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FIG. 42 illustrates one embodiment of a legacy RGB process within a six-primary color system.

FIG. 43 illustrates one embodiment of packing six-primary color system image data into an $IC_T C_P$ (ITP) format.

FIG. 44 illustrates one embodiment of a six-primary color system converting RGBCYM image data into XYZ image data for an ITP format.

FIG. 45 illustrates one embodiment of six-primary color mapping with SMPTE ST424.

FIG. 46 illustrates one embodiment of a six-primary color system readout for a SMPTE ST424 standard.

FIG. 47 illustrates a process of 2160p transport over 12G-SDI.

FIG. 48 illustrates one embodiment for mapping RGBCYM data to the SMPTE ST2082 standard for a six-primary color system.

FIG. 49 illustrates one embodiment for mapping $Y_{RGB} Y_{CYM} C_R C_B C_C C_Y$ data to the SMPTE ST2082 standard for a six-primary color system.

FIG. 50 illustrates one embodiment for mapping six-primary color system data using the SMPTE ST292 standard.

FIG. 51 illustrates one embodiment of the readout for a six-primary color system using the SMPTE ST292 standard.

FIG. 52 illustrates modifications to the SMPTE ST352 standards for a six-primary color system.

FIG. 53 illustrates modifications to the SMPTE ST2022 standard for a six-primary color system.

FIG. 54 illustrates a table of 4:4:4 sampling for a six-primary color system for a 10-bit video system.

FIG. 55 illustrates a table of 4:4:4 sampling for a six-primary color system for a 12-bit video system.

FIG. 56 illustrates sequence substitutions for 10-bit and 12-bit video in 4:2:2 sampling systems in a Y Cb Cr Cc Cy color space.

FIG. 57 illustrates sample placements of six-primary system components for a 4:2:2 sampling system image.

FIG. 58 illustrates sequence substitutions for 10-bit and 12-bit video in 4:2:0 sampling systems using a Y Cb Cr Cc Cy color space.

FIG. 59 illustrates sample placements of six-primary system components for a 4:2:0 sampling system image.

FIG. 60 illustrates modifications to SMPTE ST2110-20 for a 10-bit six-primary color system in 4:4:4 video.

FIG. 61 illustrates modifications to SMPTE ST2110-20 for a 12-bit six-primary color system in 4:4:4 video.

FIG. 62 illustrates modifications to SMPTE ST2110-20 for a 10-bit six primary color system in 4:2:2 video.

FIG. 63 illustrates modifications to SMPTE ST2110-20 for a 12-bit six-primary color system in 4:2:0 video.

FIG. 64 illustrates an RGB sampling transmission for a 4:4:4 sampling system.

FIG. 65 illustrates a RGBCYM sampling transmission for a 4:4:4 sampling system.

FIG. 66 illustrates an example of System 2 to RGBCYM 4:4:4 transmission.

FIG. 67 illustrates a Y Cb Cr sampling transmission using a 4:2:2 sampling system.

FIG. 68 illustrates a Y Cr Cb Cc Cy sampling transmission using a 4:2:2 sampling system.

FIG. 69 illustrates an example of a System 2 to Y Cr Cb Cc Cy 4:2:2 Transmission as non-constant luminance.

FIG. 70 illustrates a Y Cb Cr sampling transmission using a 4:2:0 sampling system.

FIG. 71 illustrates a Y Cr Cb Cc Cy sampling transmission using a 4:2:0 sampling system.

FIG. 72 illustrates a dual stack LCD projection system for a six-primary color system.

FIG. 73 illustrates one embodiment of a single projector.

FIG. 74 illustrates a six-primary color system using a single projector and reciprocal mirrors.

FIG. 75 illustrates a dual stack DMD projection system for a six-primary color system.

FIG. 76 illustrates one embodiment of a single DMD projector solution.

FIG. 77 illustrates one embodiment of a color filter array for a six-primary color system with a white OLED monitor.

FIG. 78 illustrates one embodiment of an optical filter array for a six-primary color system with a white OLED monitor.

FIG. 79 illustrates one embodiment of a matrix of an LCD drive for a six-primary color system with a backlight illuminated LCD monitor.

FIG. 80 illustrates one embodiment of an optical filter array for a six-primary color system with a backlight illuminated LCD monitor.

FIG. 81 illustrates an array for a Quantum Dot (QD) display device.

FIG. 82 illustrates one embodiment of an array for a six-primary color system for use with a direct emissive assembled display.

FIG. 83 illustrates one embodiment of a six-primary color system in an emissive display that does not incorporate color filtered subpixels.

FIG. 84 illustrates one embodiment of a primary triad system for a multi-primary system including red, green, blue, cyan, magenta, and yellow primaries.

FIG. 85 illustrates one embodiment of out-of-gamut color mapping.

FIG. 86 illustrates another embodiment of out-of-gamut color mapping.

FIG. 87 illustrates a process to validate the ACES-to-6P-to-ACES conversion process according to one embodiment of the present invention.

FIG. 88 illustrates one embodiment of a system with at least eight primary triads.

FIG. 89 illustrates a flow chart of an embodiment of a system with eight triads in a six-primary system.

FIG. 90 illustrates one embodiment of a system with primary triads including a virtual primary.

FIG. 91 is a schematic diagram of an embodiment of the invention illustrating a computer system.

DETAILED DESCRIPTION

The present invention is generally directed to a multi-primary color system.

In one embodiment, the present invention is a system for displaying a six primary color system, including a set of image data, wherein the set of image data is comprised of a first set of color channel data and a second set of color channel data, wherein the set of image data further includes a bit level, an image data converter, wherein the image data converter includes a digital interface, wherein the digital interface is operable to encode and decode the set of image data, at least one transfer function (TF) for processing the set of image data, a set of Session Description Protocol (SDP) parameters, wherein the set of SDP parameters is modifiable, at least one display device, wherein the at least one display device and the image data converter are in network communication, wherein the image data converter is operable to convert the bit level of the set of image data, thereby creating an updated bit level, wherein the image data converter is

operable to convert the set of image data for display on the at least one display device, wherein once the set of image data has been converted by the image data converter for the at least one display device the set of SDP parameters are modified based on the conversion, and wherein the at least one display device is operable to display a six-primary color system based on the set of image data, such that the SDP parameters indicate that the set of image data being displayed on the at least one display device is using a six-primary color system.

In another embodiment, the present invention is a system for displaying a six-primary color system, including a set of image data, wherein the set of image data includes a first set of color channel data and a second set of color channel data, wherein the set of image data includes a bit level, a magenta primary value, wherein the magenta primary value is derived from the set of image data, an image data converter, wherein the image data converter includes a digital interface, wherein the digital interface is operable to encode and decode the set of image data, at least one transfer function (TF) for processing the set of image data, a set of Session Description Protocol (SDP) parameters, wherein the set of SDP parameters are modifiable, at least one display device, wherein the at least one display device and the image data converter are in network communication, wherein the image data converter is operable to convert the bit level for the set of image data to a new bit level, wherein the at least one data converter is operable to convert the set of image data for display on the at least one display device, wherein once the set of image data has been converted for the at least one display device the set of SDP parameters are modified based on the conversion, and wherein the at least one display device is operable to display a six-primary color system based on the set of image data, such that the SDP parameters indicate the magenta primary value and that the set of image data being displayed on the at least one display device is using a six-primary color system.

In yet another embodiment, the present invention is a system for displaying a set of image data using a six-primary color system, including a set of image data, wherein the set of image data includes a bit level, a magenta primary value, wherein the magenta primary value is derived from the set of image data, an image data converter, wherein the image data converter includes a digital interface, wherein the digital interface is operable to encode and decode the set of image data, at least one transfer function (TF) for processing the set of image data, a set of Session Description Protocol (SDP) parameters, wherein the set of SDP parameters are modifiable, at least one electronic luminance component, wherein the electronic luminance component is derived from the set of image data, at least one display device, wherein the at least one display device and the image data converter are in network communication, wherein the image data converter is operable to convert the set of image data to a new bit level, wherein the data converter is operable to convert the set of image data for display on the at least one display device, wherein once the set of image data has been converted for the at least one display device the set of SDP parameters are modified based on the conversion, and wherein the at least one display device is operable to display a six-primary color system based on the set of image data, such that the SDP parameters indicate the magenta primary value, the at least one electronic luminance component, and that the set of image data being displayed on the at least one display device is using a six-primary color system.

The present invention relates to color systems. A multitude of color systems are known, but they continue to suffer

numerous issues. As imaging technology is moving forward, there has been a significant interest in expanding the range of colors that are replicated on electronic displays. Enhancements to the television system have expanded from the early CCIR 601 standard to ITU-R BT.709-6, to SMPTE RP431-2, and ITU-R BT.2020. Each one has increased the gamut of visible colors by expanding the distance from the reference white point to the position of the Red (R), Green (G), and Blue (B) color primaries (collectively known as “RGB”) in chromaticity space. While this approach works, it has several disadvantages. When implemented in content presentation, issues arise due to the technical methods used to expand the gamut of colors seen (typically using a more-narrow emissive spectrum) can result in increased viewer metameric errors and require increased power due to lower illumination source. These issues increase both capital and operational costs.

With the current available technologies, displays are limited in respect to their range of color and light output. There are many misconceptions regarding how viewers interpret the display output technically versus real-world sensations viewed with the human eye. The reason we see more than just the three emitting primary colors is because the eye combines the spectral wavelengths incident on it into the three bands. Humans interpret the radiant energy (spectrum and amplitude) from a display and process it so that an individual color is perceived. The display does not emit a color or a specific wavelength that directly relates to the sensation of color. It simply radiates energy at the same spectrum which humans sense as light and color. It is the observer who interprets this energy as color.

When the CIE 2° standard observer was established in 1931, common understanding of color sensation was that the eye used red, blue, and green cone receptors (James Maxwell & James Forbes 1855). Later with the Munsell vision model (Munsell 1915), Munsell described the vision system to include three separate components: luminance, hue, and saturation. Using RGB emitters or filters, these three primary colors are the components used to produce images on today’s modern electronic displays.

There are three primary physical variables that affect sensation of color. These are the spectral distribution of radiant energy as it is absorbed into the retina, the sensitivity of the eye in relation to the intensity of light landing on the retinal pigment epithelium, and the distribution of cones within the retina. The distribution of cones (e.g., L cones, M cones, and S cones) varies considerably from person to person.

Enhancements in brightness have been accomplished through larger backlights or higher efficiency phosphors. Encoding of higher dynamic ranges is addressed using higher range, more perceptually uniform electro-optical transfer functions to support these enhancements to brightness technology, while wider color gamuts are produced by using narrow bandwidth emissions. Narrower bandwidth emitters result in the viewer experiencing higher color saturation. But there can be a disconnect between how saturation is produced and how it is controlled. What is believed to occur when changing saturation is that increasing color values of a color primary represents an increase to saturation. This is not true, as changing saturation requires the variance of a color primary spectral output as parametric. There are no variable spectrum displays available to date as the technology to do so has not been commercially developed, nor has the new infrastructure required to support this been discussed.

Instead, the method that a display changes for viewer color sensation is by changing color luminance. As data values increase, the color primary gets brighter. Changes to color saturation are accomplished by varying the brightness of all three primaries and taking advantage of the dominant color theory.

Expanding color primaries beyond RGB has been discussed before. There have been numerous designs of multi-primary displays. For example, SHARP has attempted this with their four-color QUATTRON TV systems by adding a yellow color primary and developing an algorithm to drive it. Another four primary color display was proposed by Matthew Brennesholtz which included an additional cyan primary, and a six primary display was described by Yan Xiong, Fei Deng, Shan Xu, and Sufang Gao of the School of Physics and Optoelectric Engineering at the Yangtze University Jingzhou China. In addition, AU OPTRONICS has developed a five primary display technology. SONY has also recently disclosed a camera design featuring RGBCMY (red, green, blue, cyan, magenta, and yellow) and RGBCMYW (red, green, blue cyan, magenta, yellow, and white) sensors.

Actual working displays have been shown publicly as far back as the late 1990’s, including samples from Tokyo Polytechnic University, Nagoya City University, and Genoa Technologies. However, all of these systems are exclusive to their displays, and any additional color primary information is limited to the display’s internal processing.

Additionally, the Visual Arts System for Archiving and Retrieval of Images (VASARI) project developed a colorimetric scanner system for direct digital imaging of paintings. The system provides more accurate coloring than conventional film, allowing it to replace film photography. Despite the project beginning in 1989, technical developments have continued. Additional information is available at <https://www.southampton.ac.uk/~km2/orojs/vasari/> (last accessed Mar. 30, 2020), which is incorporated herein by reference in its entirety.

None of the prior art discloses developing additional color primary information outside of the display. Moreover, the system driving the display is often proprietary to the demonstration. In each of these executions, nothing in the workflow is included to acquire or generate additional color primary information. The development of a multi-primary color system is not complete if the only part of the system that supports the added primaries is within the display itself.

Referring now to the drawings in general, the illustrations are for the purpose of describing one or more preferred embodiments of the invention and are not intended to limit the invention thereto.

Additional details about multi-primary systems are available in U.S. Pat. No. 10,607,527, U.S. application Ser. No. 17/082,741, and U.S. Publication Nos. 20200251039, 20210027693, 20210020094, and 20210035487, each of which is incorporated herein by reference in its entirety.

The multi-primary system of the present invention includes at least four primaries. The at least four primaries preferably include at least one red primary, at least one green primary, and/or at least one blue primary. In one embodiment, the at least four primaries include a cyan primary, a magenta primary, and/or a yellow primary.

In one embodiment, the multi-primary system includes six primaries. In one preferred embodiment, the six primaries include a red primary, a green primary, a blue primary, a cyan primary, a magenta primary, and a yellow primary.

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6P-B

6P-B is a color set that uses the same RGB values that are defined in the ITU-R BT.709-6 television standard. The gamut includes these RGB primary colors and then adds three more color primaries orthogonal to these based on the white point. The white point used in 6P-B is D65 (ISO 11664-2).

In one embodiment, the red primary has a dominant wavelength of 609 nm, the yellow primary has a dominant wavelength of 571 nm, the green primary has a dominant wavelength of 552 nm, the cyan primary has a dominant wavelength of 491 nm, and the blue primary has a dominant wavelength of 465 nm as shown in Table 1. In one embodiment, the dominant wavelength is approximately (e.g., within $\pm 10\%$) the value listed in the table below. Alternatively, the dominant wavelength is within $\pm 5\%$ of the value listed in the table below. In yet another embodiment, the dominant wavelength is within $\pm 2\%$ of the value listed in the table below.

TABLE 1

	x	y	u'	v'	λ
W (D65)	0.3127	0.3290	0.1978	0.4683	
R	0.6400	0.3300	0.4507	0.5228	609 nm
G	0.3000	0.6000	0.1250	0.5625	552 nm
B	0.1500	0.0600	0.1754	0.1578	464 nm
C	0.1655	0.3270	0.1041	0.4463	491 nm
M	0.3221	0.1266	0.3325	0.2940	
Y	0.4400	0.5395	0.2047	0.5649	571 nm

FIG. 1 illustrates 6P-B compared to ITU-R BT.709-6.

6P-C

6P-C is based on the same RGB primaries defined in SMPTE RP431-2 projection recommendation. Each gamut includes these RGB primary colors and then adds three more color primaries orthogonal to these based on the white point. The white point used in 6P-B is D65 (ISO 11664-2). Two versions of 6P-C are used. One is optimized for a D60 white point (SMPTE ST2065-1), and the other is optimized for a D65 white point.

In one embodiment, the red primary has a dominant wavelength of 615 nm, the yellow primary has a dominant wavelength of 570 nm, the green primary has a dominant wavelength of 545 nm, the cyan primary has a dominant wavelength of 493 nm, and the blue primary has a dominant wavelength of 465 nm as shown in Table 2. In one embodiment, the dominant wavelength is approximately (e.g., within $\pm 10\%$) the value listed in the table below. Alternatively, the dominant wavelength is within $\pm 5\%$ of the value listed in the table below. In yet another embodiment, the dominant wavelength is within $\pm 2\%$ of the value listed in the table below.

TABLE 2

	x	y	u'	v'	λ
W (D60)	0.3217	0.3377	0.2008	0.4742	
R	0.6800	0.3200	0.4964	0.5256	615 nm
G	0.2650	0.6900	0.0980	0.5777	545 nm
B	0.1500	0.0600	0.1754	0.1579	465 nm
C	0.1627	0.3419	0.0960	0.4540	493 nm
M	0.3523	0.1423	0.3520	0.3200	
Y	0.4502	0.5472	0.2078	0.5683	570 nm

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FIG. 2 illustrates 6P-C compared to SMPTE RP431-2 for a D60 white point.

In one embodiment, the red primary has a dominant wavelength of 615 nm, the yellow primary has a dominant wavelength of 570 nm, the green primary has a dominant wavelength of 545 nm, the cyan primary has a dominant wavelength of 423 nm, and the blue primary has a dominant wavelength of 465 nm as shown in Table 3. In one embodiment, the dominant wavelength is approximately (e.g., within $\pm 10\%$) the value listed in the table below. Alternatively, the dominant wavelength is within $\pm 5\%$ of the value listed in the table below. In yet another embodiment, the dominant wavelength is within $\pm 2\%$ of the value listed in the table below.

TABLE 3

	x	y	u'	v'	λ
W (D65)	0.3127	0.3290	0.1978	0.4683	
R	0.6800	0.3200	0.4964	0.5256	615 nm
G	0.2650	0.6900	0.0980	0.5777	545 nm
B	0.1500	0.0600	0.1754	0.1579	465 nm
C	0.1617	0.3327	0.0970	0.4490	492 nm
M	0.3383	0.1372	0.3410	0.3110	
Y	0.4470	0.5513	0.2050	0.5689	570 nm

FIG. 3 illustrates 6P-C compared to SMPTE RP431-2 for a D65 white point.

Super 6P

One of the advantages of ITU-R BT.2020 is that it can include all of the Pointer colors and that increasing primary saturation in a six-color primary design could also do this. Pointer is described in "The Gamut of Real Surface Colors, M.R. Pointer, Published in Colour Research and Application Volume #5, Issue #3 (1980), which is incorporated herein by reference in its entirety. However, extending the 6P gamut beyond SMPTE RP431-2 ("6P-C") adds two problems. The first problem is the requirement to narrow the spectrum of the extended primaries. The second problem is the complexity of designing a backwards compatible system using color primaries that are not related to current standards. But in some cases, there may be a need to extend the gamut beyond 6P-C and avoid these problems. If the goal is to encompass Pointer's data set, then it is possible to keep most of the 6P-C system and only change the cyan color primary position. In one embodiment, the cyan color primary position is located so that the gamut edge encompasses all of Pointer's data set. In another embodiment, the cyan color primary position is a location that limits maximum saturation. With 6P-C, cyan is positioned as $u'=0.096$, $v'=0.454$. In one embodiment of Super 6P, cyan is moved to $u'=0.075$, $v'=0.430$ ("Super 6 Pa" (S6 Pa)). Advantageously, this creates a new gamut that covers Pointer's data set almost in its entirety. FIG. 4 illustrates Super 6 Pa compared to 6P-C.

Table 4 is a table of values for Super 6 Pa. The definition of x,y are described in ISO 11664-3:2012/CIE S 014 Part 3, which is incorporated herein by reference in its entirety. The definition of u', v' are described in ISO 11664-5:2016/CIE S 014 Part 5, which is incorporated herein by reference in its entirety. λ defines each color primary as dominant color wavelength for RGB and complementary wavelengths CMY.

TABLE 4

	x	y	u'	v'	λ
W (D60)	0.3217	0.3377	0.2008	0.4742	
W (D65)	0.3127	0.3290	0.1978	0.4683	

TABLE 4-continued

	x	y	u'	v'	λ
R	0.6800	0.3200	0.4964	0.5256	615 nm
G	0.2650	0.6900	0.0980	0.5777	545 nm
B	0.1500	0.0600	0.1754	0.1579	465 nm
C	0.1211	0.3088	0.0750	0.4300	490 nm
M	0.3523	0.1423	0.3520	0.3200	
Y	0.4502	0.5472	0.2078	0.5683	570 nm

In an alternative embodiment, the saturation is expanded on the same hue angle as 6P-C as shown in FIG. 5. Advantageously, this makes backward compatibility less complicated. However, this requires much more saturation (i.e., narrower spectra). In another embodiment of Super 6P, cyan is moved to $u'=0.067$, $v'=0.449$ ("Super 6Pb" (S6Pb)). Additionally, FIG. 5 illustrates Super 6Pb compared to Super 6 Pa and 6P-C.

Table 5 is a table of values for Super 6Pb. The definition of x,y are described in ISO 11664-3:2012/CIE S 014 Part 3, which is incorporated herein by reference in its entirety. The definition of u', v' are described in ISO 11664-5:2016/CIE S 014 Part 5, which is incorporated herein by reference in its entirety. λ defines each color primary as dominant color wavelength for RGB and complementary wavelengths CMY.

TABLE 5

	x	y	u'	v'	λ
W (ACES D60)	0.32168	0.33767	0.2008	0.4742	
W (D65)	0.3127	0.3290	0.1978	0.4683	
R	0.6800	0.3200	0.4964	0.5256	615 nm
G	0.2650	0.6900	0.0980	0.5777	545 nm
B	0.1500	0.0600	0.1754	0.1579	465 nm
C	0.1156	0.3442	0.0670	0.4490	493 nm
M	0.3523	0.1423	0.3520	0.3200	
Y	0.4502	0.5472	0.2078	0.5683	570 nm

In a preferred embodiment, a matrix is created from XYZ values of each of the primaries. As the XYZ values of the primaries change, the matrix changes. Additional details about the matrix are described below.

Formatting and Transportation of Multi-Primary Signals

The present invention includes three different methods to format video for transport: System 1, System 2, and System 3. System 1 is comprised of an encode and decode system, which can be divided into base encoder and digitation, image data stacking, mapping into the standard data transport, readout, unstack, and finally image decoding. In one embodiment, the basic method of this system is to combine opposing color primaries within the three standard transport channels and identify them by their code value.

System 2 uses a sequential method where three color primaries are passed to the transport format as full bit level image data and inserted as normal. The three additional channels are delayed by one pixel and then placed into the transport instead of the first colors. This is useful in situations where quantizing artifacts may be critical to image performance. In one embodiment, this system is comprised of the six primaries (e.g., RGB plus a method to delay the CYM colors for injection), image resolution identification to allow for pixel count synchronization, start of video identification, and RGB Delay.

System 3 utilizes a dual link method where two wires are used. In one embodiment, a first set of three channels (e.g., RGB) are sent to link A and a second set of three channels

(e.g., CYM) is sent to link B. Once they arrive at the image destination, they are recombined.

To transport up to six color components (e.g., four, five, or six), System 1, System 2, or System 3 can be used as described. If four color components are used, two of the channels are set to "0". If five color components are used, one of the channels is set to "0". Advantageously, this transportation method works for all primary systems described herein that include up to six color components.

Comparison of Three Systems

Advantageously, System 1 fits within legacy SDI, CTA, and Ethernet transports. Additionally, System 1 has zero latency processing for conversion to an RGB display. However, System 1 is limited to 11-bit words.

System 2 is advantageously operable to transport 6 channels using 16-bit words with no compression. Additionally, System 2 fits within newer SDI, CTA, and Ethernet transport formats. However, System 2 requires double bit rate speed. For example, a 4K image requires a data rate for an 8K RGB image.

In comparison, System 3 is operable to transport up to 6 channels using 16-bit words with compression and at the same data rate required for a specific resolution. For example, a data rate for an RGB image is the same as for a 6P image using System 3. However, System 3 requires a twin cable connection within the video system.

Nomenclature

In one embodiment, a standard video nomenclature is used to better describe each system.

R describes red data as linear light. G describes green data as linear light. B describes blue data as linear light. (describes cyan data as linear light. M describes magenta data as linear light. Y^C and/or Y describe yellow data as linear light.

R' describes red data as non-linear light. G' describes green data as non-linear light. B' describes blue data as non-linear light. C' describes cyan data as non-linear light. M' describes magenta data as non-linear light. $Y^{C'}$ and/or Y' describe yellow data as non-linear light.

Y_6 describes the luminance sum of RGBCMY data. Y_{RGB} describes a System 2 encode that is the linear luminance sum of the RGB data. Y_{CMY} describes a System 2 encode that is the linear luminance sum of the CMY data.

C_R describes the data value of red after subtracting linear image luminance. C_B describes the data value of blue after subtracting linear image luminance. C_C describes the data value of cyan after subtracting linear image luminance. C_Y describes the data value of yellow after subtracting linear image luminance.

Y'_{RGB} describes a System 2 encode that is the nonlinear luminance sum of the RGB data. Y'_{CMY} describes a System 2 encode that is the nonlinear luminance sum of the CMY data. $-Y$ describes the sum of RGB data subtracted from Y_6 .

C'_R describes the data value of red after subtracting nonlinear image luminance. C'_B describes the data value of blue after subtracting nonlinear image luminance. C'_C describes the data value of cyan after subtracting nonlinear image luminance. C'_Y describes the data value of yellow after subtracting nonlinear image luminance.

B+Y describes a System 1 encode that includes either blue or yellow data. G+M describes a System 1 encode that includes either green or magenta data. R+C describes a System 1 encode that includes either red or cyan data.

C_R+C_C describes a System 1 encode that includes either color difference data. C_B+C_Y describes a System 1 encode that includes either color difference data.

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4:4:4 describes full bandwidth sampling of a color in an RGB system. 4:4:4:4:4 describes full sampling of a color in an RGBCMY system. 4:2:2 describes an encode where a full bandwidth luminance channel (Y) is used to carry image detail and the remaining components are half sampled as a Cb Cr encode. 4:2:2:2:2 describes an encode where a full bandwidth luminance channel (Y) is used to carry image detail and the remaining components are half sampled as a Cb Cr Cy Cc encode. 4:2:0 describes a component system similar to 4:2:2, but where Cr and Cb samples alternate per line. 4:2:0:2:0 describes a component system similar to 4:2:2, but where Cr, Cb, Cy, and Cc samples alternate per line.

Constant luminance is the signal process where luminance (Y) are calculated in linear light. Non-constant luminance is the signal process where luminance (Y) are calculated in nonlinear light.

Deriving Color Components

When using a color difference method (4:2:2), several components need specific processing so that they can be used in lower frequency transports. These are derived as:

$$Y'_6 = 0.1063R' + 0.23195Y'^c + 0.3576G' + 0.19685C' + 0.0361B' + 0.0712M'$$

$$G'_6 = \left(\frac{1}{0.3576Y'} \right) - (0.1063R') - (0.0361B') - (0.19685C') - (0.23195Y'^c) - (0.0712M')$$

$$- Y' = Y'_6 - (C' + Y'^c + M')$$

$$C'_R = \frac{R' - Y'_6}{1.7874} \quad C'_B = \frac{B' - Y'_6}{1.9278} \quad C'_C = \frac{C' - Y'_6}{1.6063} \quad C'_Y = \frac{Y'^c - Y'_6}{1.5361}$$

$$R' = \frac{C'_R - Y'_6}{1.7874} \quad B' = \frac{C'_B - Y'_6}{1.9278} \quad C' = \frac{C'_C - Y'_6}{1.6063} \quad Y'^c = \frac{C'_Y - Y'_6}{1.5361}$$

The ratios for Cr, Cb, Cc, and Cy are also valid in linear light calculations.

Magenta can be calculated as follows:

$$M' = \frac{B' + R'}{B' \times R'} \text{ or } M = \frac{B + R}{B \times R}$$

System 1

In one embodiment, the multi-primary color system is compatible with legacy systems. A backwards compatible multi-primary color system is defined by a sampling method. In one embodiment, the sampling method is 4:4:4. In one embodiment, the sampling method is 4:2:2. In another embodiment, the sampling method is 4:2:0. In one embodiment of a backwards compatible multi-primary color system, new encode and decode systems are divided into the steps of performing base encoding and digitization, image data stacking, mapping into the standard data transport, readout, unstacking, and image decoding ("System 1"). In one embodiment, System 1 combines opposing color primaries within three standard transport channels and identifies them by their code value. In one embodiment of a backwards compatible multi-primary color system, the processes are analog processes. In another embodiment of a backwards compatible multi-primary color system, the processes are digital processes.

In one embodiment, the sampling method for a multi-primary color system is a 4:4:4 sampling method. Black and

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white bits are redefined. In one embodiment, putting black at midlevel within each data word allows the addition of CYM color data.

FIG. 6 illustrates an embodiment of an encode and decode system for a multi-primary color system. In one embodiment, the multi-primary color encode and decode system is divided into a base encoder and digitation, image data stacking, mapping into the standard data transport, readout, unstack, and finally image decoding ("System 1"). In one embodiment, the method of this system combines opposing color primaries within the three standard transport channels and identifies them by their code value. In one embodiment, the encode and decode for a multi-primary color system are analog-based. In another embodiment, the encode and decode for a multi-primary color system are digital-based. System 1 is designed to be compatible with lower bandwidth systems and allows a maximum of 11 bits per channel and is limited to sending only three channels of up to six primaries at a time. In one embodiment, it does this by using a stacking system where either the color channel or the complementary channel is decoded depending on the bit level of that one channel.

System 2

FIG. 7 illustrates a sequential method where three color primaries are passed to the transport format as full bit level image data and inserted as normal ("System 2"). The three additional channels are delayed by one pixel and then placed into the transport instead of the first colors. This method is useful in situations where quantizing artifacts is critical to image performance. In one embodiment, this system is comprised of six primaries (RGBCYM), a method to delay the CYM colors for injection, image resolution identification to all for pixel count synchronization, start of video identification, RGB delay, and for YCCCC systems, logic to select the dominant color primary. The advantage of System 2 is that full bit level video can be transported, but at double the normal data rate.

System 3

FIG. 8 illustrates one embodiment of a system encode and decode process using a dual link method ("System 3"). System 3 utilizes a dual link method where two wires are used. In one embodiment, RGB is sent to link A and CYM is sent to link B. After arriving at the image destination, the two links are recombined.

System 3 is simpler and more straight forward than Systems 1 and 2. The advantage with this system is that adoption is simply to format non-RGB primaries (e.g., CYM) on a second link. So, in one example, for an SDI design, RGB is sent on a standard SDI stream just as it is currently done. There is no modification to the transport and this link is operable to be sent to any RGB display requiring only the compensation for the luminance difference because the CYM components are not included. CYM data is transported in the same manner as RGB data. This data is then combined in the display to make up a 6P image. The downside is that the system requires two wires to move one image. This system is operable to work with most any format including SMPTE ST292, 424, 2082, and 2110. It also is operable to work with dual HDMI/CTA connections. In one embodiment, the system includes at least one transfer function (e.g., OETF, EOTF).

FIG. 9 illustrates one embodiment of an encoding process using a dual link method.

FIG. 10 illustrates one embodiment of a decoding process using a dual link method.

Transfer Functions

The system design minimizes limitations to use standard transfer functions for both encode and/or decode processes. Current practices used in standards include, but are not limited to, ITU-R BT.1886, ITU-R BT.2020, SMPTE ST274, SMPTE ST296, SMPTE ST2084, and ITU-R BT.2100. These standards are compatible with this system and require no modification.

Encoding and decoding 6P images is formatted into several different configurations to adapt to image transport frequency limitations. The highest quality transport is obtained by keeping all components as RGBCMY components. This uses the highest sampling frequencies and requires the most signal bandwidth. An alternate method is to sum the image details in a luminance channel at full bandwidth and then send the color difference signals at half or quarter sampling (e.g., Y Cr Cb Cc Cy). This allows a similar image to pass through lower bandwidth transports. Six-Primary Color Encode Using a 4:4:4 Sampling Method

FIG. 11 illustrates one embodiment of a six-primary color system encode using a 4:4:4 sampling method.

Subjective testing during the development and implementation of the current digital cinema system (DCI Version 1.2) showed that perceptible quantizing artifacts were not noticeable with system bit resolutions higher than 11 bits. Current serial digital transport systems support 12 bits. Remapping six color components to a 12-bit stream is accomplished by lowering the bit limit to 11 bits (values 0 to 2047) for 12-bit serial systems or 9 bits (values 0 to 512) for 10-bit serial systems. This process is accomplished by processing RGBCYM video information through a standard Optical Electronic Transfer Function (OETF) (e.g., ITU-R BT.709-6), digitizing the video information as four samples per pixel, and quantizing the video information as 11-bit or 9-bit.

In another embodiment, the RGBCYM video information is processed through a standard Optical Optical Transfer Function (OOTF). In yet another embodiment, the RGBCYM video information is processed through a Transfer Function (TF) other than OETF or OOTF. TFs consist of two components, a Modulation Transfer Function (MTF) and a Phase Transfer Function (PTF). The MTF is a measure of the ability of an optical system to transfer various levels of detail from object to image. In one embodiment, performance is measured in terms of contrast (degrees of gray), or of modulation, produced for a perfect source of that detail level. The PTF is a measure of the relative phase in the image(s) as a function of frequency. A relative phase change of 180°, for example, indicates that black and white in the image are reversed. This phenomenon occurs when the TF becomes negative.

There are several methods for measuring MTF. In one embodiment, MTF is measured using discrete frequency generation. In one embodiment, MTF is measured using continuous frequency generation. In another embodiment, MTF is measured using image scanning. In another embodiment, MTF is measured using waveform analysis.

In one embodiment, the six-primary color system is for a 12-bit serial system. Current practices normally set black at bit value 0 and white at bit value 4095 for 12-bit video. In order to package six colors into the existing three-serial streams, the bit defining black is moved to bit value 2048. Thus, the new encode has RGB values starting at bit value 2048 for black and bit value 4095 for white and CYM values starting at bit value 2047 for black and bit value 0 as white. In another embodiment, the six-primary color system is for a 10-bit serial system.

FIG. 12 illustrates one embodiment for a method to package six channels of primary information into the three standard primary channels used in current serial video standards by modifying bit numbers for a 12-bit SDI and a 10-bit SDI. FIG. 13 illustrates a simplified diagram estimating perceived viewer sensation as code values define each hue angle. TABLE 6 and TABLE 7 list bit assignments for computer, production, and broadcast for a 12-bit system and a 10-bit system, respectively. In one embodiment, "Computer" refers to bit assignments compatible with CTA 861-G, November 2016, which is incorporated herein by reference in its entirety. In one embodiment, "Production" and/or "Broadcast" refer to bit assignments compatible with SMPTE ST 2082-0 (2016), SMPTE ST 2082-1 (2015), SMPTE ST 2082-10 (2015), SMPTE ST 2082-11 (2016), SMPTE ST 2082-12 (2016), SMPTE ST 2110-10 (2017), SMPTE ST 2110-20 (2017), SMPTE ST 2110-21 (2017), SMPTE ST 2110-30 (2017), SMPTE ST 2110-31 (2018), and/or SMPTE ST 2110-40 (2018), each of which is incorporated herein by reference in its entirety.

TABLE 6

	12-Bit Assignments					
	Computer		Production		Broadcast	
	RGB	CYM	RGB	CYM	RGB	CYM
Peak Brightness	4095	0	4076	16	3839	256
Minimum Brightness	2048	2047	2052	2032	2304	1792

TABLE 7

	10-Bit Assignments					
	Computer		Production		Broadcast	
	RGB	CYM	RGB	CYM	RGB	CYM
Peak Brightness	1023	0	1019	4	940	64
Minimum Brightness	512	511	516	508	576	448

In one embodiment, the OETF process is defined in ITU-R BT.709-6, which is incorporated herein by reference in its entirety. In one embodiment, the OETF process is defined in ITU-R BT.709-5, which is incorporated herein by reference in its entirety. In another embodiment, the OETF process is defined in ITU-R BT.709-4, which is incorporated herein by reference in its entirety. In yet another embodiment, the OETF process is defined in ITU-R BT.709-3, which is incorporated herein by reference in its entirety. In yet another embodiment, the OETF process is defined in ITU-R BT.709-2, which is incorporated herein by reference in its entirety. In yet another embodiment, the OETF process is defined in ITU-R BT.709-1, which is incorporated herein by reference in its entirety.

In one embodiment, the encoder is a non-constant luminance encoder. In another embodiment, the encoder is a constant luminance encoder.

Six-Primary Color Packing/Stacking Using a 4:4:4 Sampling Method

FIG. 14 illustrates one embodiment for a method of stacking/encoding six-primary color information using a 4:4:4 video system. Image data must be assembled accord-

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ing the serial system used. This is not a conversion process, but instead is a packing/stacking process. In one embodiment, the packing/stacking process is for a six-primary color system using a 4:4:4 sampling method.

FIG. 15 illustrates one embodiment for a method of unstacking/decoding six-primary color information using a 4:4:4 video system. In one embodiment, the RGB channels and the CYM channels are combined into one 12-bit word and sent to a standardized transport format. In one embodiment, the standardized transport format is SMPTE ST424 SDI. In one embodiment, the decode is for a non-constant luminance, six-primary color system. In another embodiment, the decode is for a constant luminance, six-primary color system. In yet another embodiment, an electronic optical transfer function (EOTF) (e.g., ITU-R BT.1886) converts image data back to linear for display. In one embodiment, the EOTF is defined in ITU-R BT.1886 (2011), which is incorporated herein by reference in its entirety. FIG. 16 illustrates one embodiment of a 4:4:4 decoder.

System 2 uses sequential mapping to the standard transport format, so it includes a delay for the CYM data. The CYM data is recovered in the decoder by delaying the RGB data. Since there is no stacking process, the full bit level video can be transported. For displays that are using optical filtering, this RGB delay could be removed and the process of mapping image data to the correct filter could be eliminated by assuming this delay with placement of the optical filter and the use of sequential filter colors.

Two methods can be used based on the type of optical filter used. Since this system is operating on a horizontal pixel sequence, some vertical compensation is required and pixels are rectangular. This can be either as a line double repeat using the same RGBCYM data to fill the following line as shown in FIG. 17, or could be separated as RGB on line one and CYM on line two as shown in FIG. 18. The format shown in FIG. 18 allows for square pixels, but the CMY components requires a line delay for synchronization. Other patterns eliminating the white subpixel are also compatible with the present invention.

FIG. 19 illustrates an embodiment of the present invention for sending six primary colors to a standardized transport format using a 4:4:4 encoder according to System 2. Encoding is straight forward with a path for RGB sent directly to the transport format. RGB data is mapped to each even numbered data segment in the transport. CYM data is mapped to each odd numbered segment. Because different resolutions are used in all of the standardized transport formats, there must be identification for what they are so that the start of each horizontal line and horizontal pixel count can be identified to time the RGB/CYM mapping to the transport. The identification is the same as currently used in each standardized transport function. TABLE 8, TABLE 9, TABLE 10, and TABLE 11 list 16-bit assignments, 12-bit assignments, 10-bit assignments, and 8-bit assignments, respectively. In one embodiment, "Computer" refers to bit assignments compatible with CTA 861-G, November 2016, which is incorporated herein by reference in its entirety. In one embodiment, "Production" and/or "Broadcast" refer to bit assignments compatible with SMPTE ST 2082-0 (2016), SMPTE ST 2082-1 (2015), SMPTE ST 2082-10 (2015), SMPTE ST 2082-11 (2016), SMPTE ST 2082-12 (2016), SMPTE ST 2110-10 (2017), SMPTE ST 2110-20 (2017), SMPTE ST 2110-21 (2017), SMPTE ST 2110-30 (2017), SMPTE ST 2110-31 (2018), and/or SMPTE ST 2110-40 (2018), each of which is incorporated herein by reference in its entirety.

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TABLE 8

	16-Bit Assignments			
	Computer		Production	
	RGB	CYM	RGB	CYM
Peak Brightness	65536	65536	65216	65216
Minimum Brightness	0	0	256	256

TABLE 9

	12-Bit Assignments					
	Computer		Production		Broadcast	
	RGB	CYM	RGB	CYM	RGB	CYM
Peak Brightness	4095	4095	4076	4076	3839	3839
Minimum Brightness	0	0	16	16	256	256

TABLE 10

	10-Bit Assignments					
	Computer		Production		Broadcast	
	RGB	CYM	RGB	CYM	RGB	CYM
Peak Brightness	1023	1023	1019	1019	940	940
Minimum Brightness	0	0	4	4	64	64

TABLE 11

	8-Bit Assignments					
	Computer		Production		Broadcast	
	RGB	CYM	RGB	CYM	RGB	CYM
Peak Brightness	255	255	254	254	235	235
Minimum Brightness	0	0	1	1	16	16

The decode adds a pixel delay to the RGB data to realign the channels to a common pixel timing. EOTF is applied and the output is sent to the next device in the system. Metadata based on the standardized transport format is used to identify the format and image resolution so that the unpacking from the transport can be synchronized. FIG. 20 shows one embodiment of a decoding with a pixel delay.

In one embodiment, the decoding is 4:4:4 decoding. With this method, the six-primary color decoder is in the signal path, where 11-bit values for RGB are arranged above bit value 2048, while CYM levels are arranged below bit value 2047 as 11-bit. If the same data set is sent to a display and/or process that is not operable for six-primary color processing, the image data is assumed as black at bit value 0 as a full 12-bit word. Decoding begins by tapping image data prior to the unstacking process.

Six-Primary Color Encode Using a 4:2:2 Sampling Method

In one embodiment, the packing/stacking process is for a six-primary color system using a 4:2:2 sampling method. In order to fit the new six-primary color system into a lower

bandwidth serial system, while maintaining backwards compatibility, the standard method of converting from RGBCYM to a luminance and a set of color difference signals requires the addition of at least one new image designator. In one embodiment, the encoding and/or decoding process is compatible with transport through SMPTE ST 292-0 (2011), SMPTE ST 292-1 (2011, 2012, and/or 2018), SMPTE ST 292-2 (2011), SMPTE ST 2022-1 (2007), SMPTE ST 2022-2 (2007), SMPTE ST 2022-3 (2010), SMPTE ST 2022-4 (2011), SMPTE ST 2022-5 (2012 and/or 2013), SMPTE ST 2022-6 (2012), SMPTE ST 2022-7 (2013), and/or and CTA 861-G (2016), each of which is incorporated herein by reference in its entirety.

In order for the system to package all of the image while supporting both six-primary and legacy displays, an electronic luminance component (Y) must be derived. The first component is: E'_{Y_6} . It can be described as:

$$E'_{Y_6} = 0.1063E'_{Red} + 0.23195E'_{Yellow} + 0.3576E'_{Green} + 0.19685E'_{Cyan} + 0.0361E'_{Blue} + 0.0712E'_{Magenta}$$

Critical to getting back to legacy display compatibility, value E'_{-Y} is described as:

$$E'_{-Y} = E'_{Y_6} - (E'_{Cyan} + E'_{Yellow} + E'_{Magenta})$$

In addition, at least two new color components are disclosed. These are designated as Cc and Cy components. The at least two new color components include a method to compensate for luminance and enable the system to function with older Y Cb Cr infrastructures. In one embodiment, adjustments are made to Cb and Cr in a Y Cb Cr infrastructure since the related level of luminance is operable for division over more components. These new components are as follows:

$$E'_{CR} = \frac{(E'_R - E'_{Y_6})}{1.7874}, E'_{CB} = \frac{(E'_B - E'_{Y_6})}{1.9278},$$

$$E'_{CC} = \frac{(E'_C - E'_{Y_6})}{1.6063}, E'_{CY} = \frac{(E'_Y - E'_{Y_6})}{1.5361}$$

Within such a system, it is not possible to define magenta as a wavelength. This is because the green vector in CIE 1976 passes into, and beyond, the CIE designated purple line. Magenta is a sum of blue and red. Thus, in one embodiment, magenta is resolved as a calculation, not as optical data. In one embodiment, both the camera side and

the monitor side of the system use magenta filters. In this case, if magenta were defined as a wavelength, it would not land at the point described. Instead, magenta would appear as a very deep blue which would include a narrow bandwidth primary, resulting in metameric issues from using narrow spectral components. In one embodiment, magenta as an integer value is resolved using the following equation:

$$M_{INT} = \left\lceil \frac{\frac{B_{INT}}{2} + \frac{R_{INT}}{2}}{2} \right\rceil$$

The above equation assists in maintaining the fidelity of a magenta value while minimizing any metameric errors. This is advantageous over prior art, where magenta appears instead as a deep blue instead of the intended primary color value.

Six-Primary Non-Constant Luminance Encode Using a 4:2:2 Sampling Method

In one embodiment, the six-primary color system using a non-constant luminance encode for use with a 4:2:2 sampling method. In one embodiment, the encoding process and/or decoding process is compatible with transport through SMPTE ST 292-0 (2011), SMPTE ST 292-1 (2011, 2012, and/or 2018), SMPTE ST 292-2 (2011), SMPTE ST 2022-1 (2007), SMPTE ST 2022-2 (2007), SMPTE ST 2022-3 (2010), SMPTE ST 2022-4 (2011), SMPTE ST 2022-5 (2012 and/or 2013), SMPTE ST 2022-6 (2012), SMPTE ST 2022-7 (2013), and/or and CTA 861-G (2016), each of which is incorporated herein by reference in its entirety.

Current practices use a non-constant luminance path design, which is found in all the video systems currently deployed. FIG. 21 illustrates one embodiment of an encode process for 4:2:2 video for packaging five channels of information into the standard three-channel designs. For 4:2:2, a similar method to the 4:4:4 system is used to package five channels of information into the standard three-channel designs used in current serial video standards. FIG. 21 illustrates 12-bit SDI and 10-bit SDI encoding for a 4:2:2 system. TABLE 12 and TABLE 13 list bit assignments for a 12-bit and 10-bit system, respectively. In one embodiment, "Computer" refers to bit assignments compatible with CTA 861-G, November 2016, which is incorporated herein by reference in its entirety. In one embodiment, "Production" and/or "Broadcast" refer to bit assignments compatible with SMPTE ST 2082-0 (2016), SMPTE ST 2082-1 (2015), SMPTE ST 2082-10 (2015), SMPTE ST 2082-11 (2016), SMPTE ST 2082-12 (2016), SMPTE ST 2110-10 (2017), SMPTE ST 2110-20 (2017), SMPTE ST 2110-21 (2017), SMPTE ST 2110-30 (2017), SMPTE ST 2110-31 (2018), and/or SMPTE ST 2110-40 (2018), each of which is incorporated herein by reference in its entirety.

TABLE 12

	12-Bit Assignments									
	Computer			Production			Broadcast			
	EY_6	EC_R, EC_B	EC_C, EC_Y	EY_6	EC_R, EC_B	EC_C, EC_Y	EY_6	EC_R, EC_B	EC_C, EC_Y	
Peak Brightness	4095	4095	0	4076	4076	16	3839	3839	256	
Minimum Brightness	0	2048	2047	16	2052	2032	256	2304	1792	

TABLE 13

	10-Bit Assignments								
	Computer			Production			Broadcast		
	EY ₆	EC _R , EC _B	EC _C , EC _Y	EY ₆	EC _R , EC _B	EC _C , EC _Y	EY ₆	EC _R , EC _B	EC _C , EC _Y
Peak Brightness	1023	1023	0	1019	1019	4	940	940	64
Minimum Brightness	0	512	511	4	516	508	64	576	448

FIG. 22 illustrates one embodiment for a non-constant luminance encoding process for a six-primary color system. The design of this process is similar to the designs used in current RGB systems. Input video is sent to the Optical Electronic Transfer Function (OETF) process and then to the EY₆ encoder. The output of this encoder includes all of the image detail information. In one embodiment, all of the image detail information is output as a monochrome image.

The output is then subtracted from E'_R, E'_B, E'_C, and E'_Y to make the following color difference components:

$$E'_{CR}, E'_{CB}, E'_{CC}, E'_{CY}$$

These components are then half sampled (×2) while E'_{Y₆} is fully sampled (×4).

FIG. 23 illustrates one embodiment of a packaging process for a six-primary color system. These components are then sent to the packing/stacking process. Components E'_{CY-INT} and E'_{CC-INT} are inverted so that bit 0 now defines peak luminance for the corresponding component. In one embodiment, this is the same packaging process performed with the 4:4:4 sampling method design, resulting in two 11-bit components combining into one 12-bit component. Six-Primary Non-Constant Luminance Decode Using a 4:2:2 Sampling Method

FIG. 24 illustrates a 4:2:2 unstack process for a six-primary color system. In one embodiment, the image data is extracted from the serial format through the normal processes as defined by the serial data format standard. In another embodiment, the serial data format standard uses a 4:2:2 sampling structure. In yet another embodiment, the serial data format standard is SMPTE ST292. The color difference components are separated and formatted back to valid 11-bit data. Components E'_{CY-INT} and E'_{CC-INT} are inverted so that bit value 2047 defines peak color luminance.

FIG. 25 illustrates one embodiment of a process to inversely quantize each individual color and pass the data through an electronic optical function transfer (EOTF) in a non-constant luminance system. The individual color components, as well as E'_{Y₆-INT}, are inversely quantized and summed to breakout each individual color. Magenta is then calculated and E'_{Y₆-INT} is combined with these colors to resolve green. These calculations then go back through an Electronic Optical Transfer Function (EOTF) process to output the six-primary color system.

In one embodiment, the decoding is 4:2:2 decoding. This decode follows the same principles as the 4:4:4 decoder. However, in 4:2:2 decoding, a luminance channel is used instead of discrete color channels. Here, image data is still taken prior to unstack from the E'_{CB-INT}+E'_{CY-INT} and E'_{CR-INT}+E'_{CC-INT} channels. With a 4:2:2 decoder, a new component, called E'_{-Y}, is used to subtract the luminance

levels that are present from the CYM channels from the E'_{CB-INT}+E'_{CY-INT} and E'_{CR-INT}+E'_{CC-INT} components. The resulting output is now the R and B image components of the EOTF process. E'_{-Y} is also sent to the G matrix to convert the luminance and color difference components to a green output. Thus, R'G'B' is input to the EOTF process and output as G_{RGB}, R_{RGB}, and B_{RGB}. In another embodiment, the decoder is a legacy RGB decoder for non-constant luminance systems.

In one embodiment, the standard is SMPTE ST292. In one embodiment, the standard is SMPTE RP431-2. In one embodiment, the standard is ITU-R BT.2020. In another embodiment, the standard is SMPTE RP431-1. In another embodiment, the standard is ITU-R BT.1886. In another embodiment, the standard is SMPTE ST274. In another embodiment, the standard is SMPTE ST296. In another embodiment, the standard is SMPTE ST2084. In yet another embodiment, the standard is ITU-R BT.2100. In yet another embodiment, the standard is SMPTE ST424. In yet another embodiment, the standard is SMPTE ST425. In yet another embodiment, the standard is SMPTE ST2110.

Six-Primary Constant Luminance Decode Using a 4:2:2 Sampling Method

FIG. 26 illustrates one embodiment of a constant luminance encode for a six-primary color system. FIG. 27 illustrates one embodiment of a constant luminance decode for a six-primary color system. The process for constant luminance encode and decode are very similar. The main difference being that the management of E'_{Y₆} is linear. The encode and decode processes stack into the standard serial data streams in the same way as is present in a non-constant luminance, six-primary color system. In one embodiment, the stacker design is the same as with the non-constant luminance system.

System 2 operation is using a sequential method of mapping to the standard transport instead of the method in System 1 where pixel data is combined to two color primaries in one data set as an 11-bit word. The advantage of System 1 is that there is no change to the standard transport. The advantage of System 2 is that full bit level video can be transported, but at double the normal data rate.

The difference between the systems is the use of two Y channels in System 2. Y_{RGB} and Y_{CYM} are used to define the luminance value for RGB as one group and CYM for the other.

FIG. 28 illustrates one example of 4:2:2 non-constant luminance encoding. Because the RGB and CYM components are mapped at different time intervals, there is no requirement for a stacking process and data is fed directly to the transport format. The development of the separate color difference components is identical to System 1.

The encoder for System 2 takes the formatted color components in the same way as System 1. Two matrices are used to build two luminance channels. Y_{RGB} contains the

luminance value for the RGB color primaries. Y_{YCM} contains the luminance value for the CYM color primaries. A set of delays are used to sequence the proper channel for Y_{RGB} , Y_{CMY} , and the RBCY channels. Because the RGB and CYM components are mapped at different time intervals, there is no requirement for a stacking process, and data is fed directly to the transport format. The development of the separate color difference components is identical to System 1. The Encoder for System 2 takes the formatted color components in the same way as System 1. Two matrices are used to build two luminance channels: Y_{RGB} contains the luminance value for the RGB color primaries and Y_{CMY} contains the luminance value for the CMY color primaries. This sequences Y_{RGB} , CR, and CC channels into the even segments of the standardized transport and Y_{CMY} , CB, and CY into the odd numbered segments. Since there is no combining color primary channels, full bit levels can be used limited only by the design of the standardized transport method. In addition, for use in matrix driven displays, there is no change to the input processing and only the method of outputting the correct color is required if the filtering or emissive subpixel is also placed sequentially.

Timing for the sequence is calculated by the source format descriptor which then flags the start of video and sets the pixel timing.

FIG. 29 illustrates one embodiment of a non-constant luminance decoding system. Decoding uses timing synchronization from the format descriptor and start of video flags that are included in the payload ID, SDP, or EDID tables. This starts the pixel clock for each horizontal line or identify which set of components are routed to the proper part of the decoder. A pixel delay is used to realign the color primary data of each subpixel. Y_{RGB} and Y_{CMY} are combined to assemble a new Y_6 component which is used to decode the CR, CB, CC, CY, and CM components into RGBCYM.

The constant luminance system is not different from the non-constant luminance system in regard to operation. The difference is that the luminance calculation is done as a linear function instead of including the OOTF. FIG. 30 illustrates one embodiment of a 4:2:2 constant luminance encoding system. FIG. 31 illustrates one embodiment of a 4:2:2 constant luminance decoding system.

Six-Primary Color System Using a 4:2:0 Sampling System

In one embodiment, the six-primary color system uses a 4:2:0 sampling system. The 4:2:0 format is widely used in H.262/MPEG-2, H.264/MPEG-4 Part 10 and VC-1 compression. The process defined in SMPTE RP2050-1 provides a direct method to convert from a 4:2:2 sample structure to a 4:2:0 structure. When a 4:2:0 video decoder and encoder are connected via a 4:2:2 serial interface, the 4:2:0 data is decoded and converted to 4:2:2 by up-sampling the color difference component. In the 4:2:0 video encoder, the 4:2:2 video data is converted to 4:2:0 video data by down-sampling the color difference component.

There typically exists a color difference mismatch between the 4:2:0 video data from the 4:2:0 video data to be encoded. Several stages of codec concatenation are common through the processing chain. As a result, color difference signal mismatch between 4:2:0 video data input to 4:2:0 video encoder and 4:2:0 video output from 4:2:0 video decoder is accumulated and the degradation becomes visible.

Filtering within a Six-Primary Color System Using a 4:2:0 Sampling Method

When a 4:2:0 video decoder and encoder are connected via a serial interface, 4:2:0 data is decoded and the data is converted to 4:2:2 by up-sampling the color difference

component, and then the 4:2:2 video data is mapped onto a serial interface. In the 4:2:0 video encoder, the 4:2:2 video data from the serial interface is converted to 4:2:0 video data by down-sampling the color difference component. At least one set of filter coefficients exists for 4:2:0/4:2:2 up-sampling and 4:2:2/4:2:0 down-sampling. The at least one set of filter coefficients provide minimally degraded 4:2:0 color difference signals in concatenated operations.

Filter Coefficients in a Six-Primary Color System Using a 4:2:0 Sampling Method

FIG. 32 illustrates one embodiment of a raster encoding diagram of sample placements for a six-primary color 4:2:0 progressive scan system. Within this compression process, horizontal lines show the raster on a display matrix. Vertical lines depict drive columns. The intersection of these is a pixel calculation. Data around a particular pixel is used to calculate color and brightness of the subpixels. Each "X" shows placement timing of the E_{Y_6-INT} sample. Red dots depict placement of the $E'_{CR-INT}+E'_{CC-INT}$ sample. Blue triangles show placement of the $E'_{CB-INT}+E'_{CY-INT}$ sample.

In one embodiment, the raster is an RGB raster. In another embodiment, the raster is a RGBCYM raster.

Six-Primary Color System Backwards Compatibility

By designing the color gamut within the saturation levels of standard formats and using inverse color primary positions, it is easy to resolve an RGB image with minimal processing. In one embodiment for six-primary encoding, image data is split across three color channels in a transport system. In one embodiment, the image data is read as six-primary data. In another embodiment, the image data is read as RGB data. By maintaining a standard white point, the axis of modulation for each channel is considered as values describing two colors (e.g., blue and yellow) for a six-primary system or as a single color (e.g., blue) for an RGB system. This is based on where black is referenced. In one embodiment of a six-primary color system, black is decoded at a mid-level value. In an RGB system, the same data stream is used, but black is referenced at bit zero, not a mid-level.

In one embodiment, the RGB values encoded in the 6P stream are based on ITU-R BT.709. In another embodiment, the RGB values encoded are based on SMPTE RP431. Advantageously, these two embodiments require almost no processing to recover values for legacy display.

Two decoding methods are proposed. The first is a preferred method that uses very limited processing, negating any issues with latency. The second is a more straightforward method using a set of matrices at the end of the signal path to conform the 6P image to RGB.

In one embodiment, the decoding is for a 4:4:4 system. In one embodiment, the assumption of black places the correct data with each channel. If the 6P decoder is in the signal path, 11-bit values for RGB are arranged above bit value 2048, while CYM level are arranged below bit value 2047 as 11-bit. However, if this same data set is sent to a display or process that does not understand 6P processing, then that image data is assumed as black at bit value 0 as a full 12-bit word.

FIG. 33 illustrates one embodiment of the six-primary color unstack process in a 4:2:2 video system. Decoding starts by tapping image data prior to the unstacking process. The input to the 6P unstack will map as shown in FIG. 34. The output of the 6P decoder will map as shown in FIG. 35. This same data is sent uncorrected as the legacy RGB image data. The interpretation of the RGB decode will map as shown in FIG. 36.

Alternatively, the decoding is for a 4:2:2 system. This decode uses the same principles as the 4:4:4 decoder, but because a luminance channel is used instead of discrete color channels, the processing is modified. Legacy image data is still taken prior to unstack from the $E'_{CB-INT}+E'_{CY-INT}$ and $E'_{CR-INT}+E'_{CC-INT}$ channels as shown in FIG. 37.

FIG. 38 illustrates one embodiment of a non-constant luminance decoder with a legacy process. The dotted box marked (1) shows the process where a new component called E'_Y is used to subtract the luminance levels that are present from the CYM channels from the $E'_{CB-INT}+E'_{CY-INT}$ and $E'_{CR-INT}+E'_{CC-INT}$ components as shown in box (2). The resulting output is now the R and B image components of the EOTF process. E'_Y is also sent to the G matrix to convert the luminance and color difference components to a green output as shown in box (3). Thus, R'G'B' is input to the EOTF process and output as G_{RGB} , R_{RGB} , and B_{RGB} . In another embodiment, the decoder is a legacy RGB decoder for non-constant luminance systems.

For a constant luminance system, the process is very similar with the exception that green is calculated as linear as shown in FIG. 39.

Six-Primary Color System Using a Matrix Output

In one embodiment, the six-primary color system outputs a legacy RGB image. This requires a matrix output to be built at the very end of the signal path. FIG. 40 illustrates one embodiment of a legacy RGB image output at the end of the signal path. The design logic of the C, M, and Y primaries is in that they are substantially equal in saturation and placed at substantially inverted hue angles compared to R, G, and B primaries, respectively. In one embodiment, substantially equal in saturation refers to a $\pm 10\%$ difference in saturation values for the C, M, and Y primaries in comparison to saturation values for the R, G, and B primaries, respectively. In addition, substantially equal in saturation covers additional percentage differences in saturation values falling within the $\pm 10\%$ difference range. For example, substantially equal in saturation further covers a $\pm 7.5\%$ difference in saturation values for the C, M, and Y primaries in comparison to the saturation values for the R, G, and B primaries, respectively; a $\pm 5\%$ difference in saturation values for the C, M, and Y primaries in comparison to the saturation values for the R, G, and B primaries, respectively; a $\pm 2\%$ difference in saturation values for the C, M, and Y primaries in comparison to the saturation values for the R, G, and B primaries, respectively; a $\pm 1\%$ difference in saturation values for the C, M, and Y primaries in comparison to the saturation values for the R, G, and B primaries, respectively; and/or a $\pm 0.5\%$ difference in saturation values for the C, M, and Y primaries in comparison to the saturation values for the R, G, and B primaries, respectively. In a preferred embodiment, the C, M, and Y primaries are equal in saturation to the R, G, and B primaries, respectively. For example, the cyan primary is equal in saturation to the red primary, the magenta primary is equal in saturation to the green primary, and the yellow primary is equal in saturation to the blue primary.

In an alternative embodiment, the saturation values of the C, M, and Y primaries are not required to be substantially equal to their corollary primary saturation value among the R, G, and B primaries, but are substantially equal in saturation to a primary other than their corollary R, G, or B primary value. For example, the C primary saturation value is not required to be substantially equal in saturation to the R primary saturation value, but rather is substantially equal in saturation to the G primary saturation value and/or the B primary saturation value. In one embodiment, two different

color saturations are used, wherein the two different color saturations are based on standardized gamuts already in use.

In one embodiment, substantially inverted hue angles refers to a $\pm 10\%$ angle range from an inverted hue angle (e.g., 180 degrees). In addition, substantially inverted hue angles cover additional percentage differences within the $\pm 10\%$ angle range from an inverted hue angle. For example, substantially inverted hue angles further covers a $\pm 7.5\%$ angle range from an inverted hue angle, a $\pm 5\%$ angle range from an inverted hue angle, a $\pm 2\%$ angle range from an inverted hue angle, a $\pm 1\%$ angle range from an inverted hue angle, and/or a $\pm 0.5\%$ angle range from an inverted hue angle. In a preferred embodiment, the C, M, and Y primaries are placed at inverted hue angles (e.g., 180 degrees) compared to the R, G, and B primaries, respectively.

In one embodiment, the gamut is the ITU-R BT.709-6 gamut. In another embodiment, the gamut is the SMPTE RP431-2 gamut.

The unstack process includes output as six, 11-bit color channels that are separated and delivered to a decoder. To convert an image from a six-primary color system to an RGB image, at least two matrices are used. One matrix is a 3×3 matrix converting a six-primary color system image to XYZ values. A second matrix is a 3×3 matrix for converting from XYZ to the proper RGB color space. In one embodiment, XYZ values represent additive color space values, where XYZ matrices represent additive color space matrices. Additive color space refers to the concept of describing a color by stating the amounts of primaries that, when combined, create light of that color.

When a six-primary display is connected to the six-primary output, each channel will drive each color. When this same output is sent to an RGB display, the CYM channels are ignored and only the RGB channels are displayed. An element of operation is that both systems drive from the black area. At this point in the decoder, all are coded as bit value 0 being black and bit value 2047 being peak color luminance. This process can also be reversed in a situation where an RGB source can feed a six-primary display. The six-primary display would then have no information for the CYM channels and would display the input in a standard RGB gamut. FIG. 41 illustrates one embodiment of six-primary color output using a non-constant luminance decoder. FIG. 42 illustrates one embodiment of a legacy RGB process within a six-primary color system.

The design of this matrix is a modification of the CIE process to convert RGB to XYZ. First, u'y' values are converted back to CIE 1931 xyz values using the following formulas:

$$x = \frac{9u'}{(6u' - 16v' + 12)} \quad y = \frac{4v'}{(6u' - 16v' + 12)} \quad z = 1 - x - y$$

Next, RGBCYM values are mapped to a matrix. The mapping is dependent upon the gamut standard being used. In one embodiment, the gamut is ITU-R BT.709-6. The mapping for RGBCYM values for an ITU-R BT.709-6 (6P-B) gamut are:

$$\begin{pmatrix} x & y & z \\ R & 0.640 & 0.330 & 0.030 \\ G & 0.300 & 0.600 & 0.100 \\ B & 0.150 & 0.060 & 0.790 \\ C & 0.439 & 0.540 & 0.021 \\ Y & 0.165 & 0.327 & 0.509 \\ M & 0.320 & 0.126 & 0.554 \end{pmatrix}$$

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-continued

$$\begin{bmatrix} R & G & B & C & Y & M \\ x & 0.640 & 0.300 & 0.150 & 0.439 & 0.165 & 0.319 \\ y & 0.330 & 0.600 & 0.060 & 0.540 & 0.327 & 0.126 \\ z & 0.030 & 0.100 & 0.790 & 0.021 & 0.509 & 0.554 \end{bmatrix} = \begin{bmatrix} 0.519 & 0.393 & 0.140 \\ 0.393 & 0.460 & 0.160 \\ 0.140 & 0.160 & 0.650 \end{bmatrix} \quad 5$$

In one embodiment, the gamut is SMPTE RP431-2. The mapping for RGBCYM values for a SMPTE RP431-2 (6P-C) gamut are:

$$\begin{bmatrix} x & y & z \\ R & 0.680 & 0.320 & 0.000 \\ G & 0.264 & 0.691 & 0.045 \\ B & 0.150 & 0.060 & 0.790 \\ C & 0.450 & 0.547 & 0.026 \\ Y & 0.163 & 0.342 & 0.496 \\ M & 0.352 & 0.142 & 0.505 \end{bmatrix}$$

$$\begin{bmatrix} R & G & B & C & Y & M \\ x & 0.680 & 0.264 & 0.150 & 0.450 & 0.163 & 0.352 \\ y & 0.320 & 0.690 & 0.060 & 0.547 & 0.342 & 0.142 \\ z & 0.000 & 0.045 & 0.790 & 0.026 & 0.496 & 0.505 \end{bmatrix} = \begin{bmatrix} 0.565 & 0.400 & 0.121 \\ 0.400 & 0.549 & 0.117 \\ 0.121 & 0.117 & 0.650 \end{bmatrix} \quad 25$$

Following mapping the RGBCYM values to a matrix, a white point conversion occurs:

$$X = \frac{x}{y} \quad Y = 1 \quad Z = 1 - x - y \quad 35$$

For a six-primary color system using an ITU-R BT.709-6 (6P-B) color gamut, the white point is D65:

$$0.9504 = \frac{0.3127}{0.3290} \quad 0.3584 = 1 - 0.3127 - 0.3290 \quad 40$$

For a six-primary color system using a SMPTE RP431-2 (6P-C) color gamut, the white point is D60:

$$0.9541 = \frac{0.3218}{0.3372} \quad 0.3410 = 1 - 0.3218 - 0.3372 \quad 50$$

Following the white point conversion, a calculation is required for RGB saturation values, SR, SG, and SB. The results from the second operation are inverted and multiplied with the white point XYZ values. In one embodiment, the color gamut used is an ITU-R BT.709-6 color gamut. The values calculate as:

$$\begin{bmatrix} S_R \\ S_G \\ S_B \end{bmatrix}^{ITU-R \text{ BT.709-6}} = \begin{bmatrix} 5.445 & -4.644 & -0.0253 \\ -4.644 & 6.337 & -0.563 \\ -0.0253 & -0.563 & 1.682 \end{bmatrix} \begin{bmatrix} 0.950 \\ 1 \\ 0.358 \end{bmatrix} \quad 65$$

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-continued

Where

$$\begin{bmatrix} S_R \\ S_G \\ S_B \end{bmatrix}^{ITU-R \text{ BT.709-6}} = \begin{bmatrix} 0.522 \\ 1.722 \\ 0.015 \end{bmatrix}$$

In one embodiment, the color gamut is a SMPTE RP431-2 color gamut. The values calculate as:

$$\begin{bmatrix} S_R \\ S_G \\ S_B \end{bmatrix}^{SMPTE \text{ RP431-2}} = \begin{bmatrix} 3.692 & -2.649 & -0.211 \\ -2.649 & 3.795 & -0.189 \\ -0.211 & -0.189 & 1.611 \end{bmatrix} \begin{bmatrix} 0.954 \\ 1 \\ 0.341 \end{bmatrix} \quad 15$$

Where

$$\begin{bmatrix} S_R \\ S_G \\ S_B \end{bmatrix}^{SMPTE \text{ RP431-2}} = \begin{bmatrix} 0.802 \\ 1.203 \\ 0.159 \end{bmatrix} \quad 20$$

Next, a six-primary color-to-XYZ matrix must be calculated. For an embodiment where the color gamut is an ITU-R BT.709-6 color gamut, the calculation is as follows:

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} 0.519 & 0.393 & 0.140 \\ 0.393 & 0.460 & 0.160 \\ 0.140 & 0.160 & 0.650 \end{bmatrix}^{ITU-R \text{ BT.709-6}} \begin{bmatrix} 0.522 & 1.722 & 0.153 \\ 0.522 & 1.722 & 0.153 \\ 0.522 & 1.722 & 0.153 \end{bmatrix}^{D65} \quad 30$$

Wherein the resulting matrix is multiplied by the $S_R S_G S_B$ matrix:

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} 0.271 & 0.677 & 0.002 \\ 0.205 & 0.792 & 0.003 \\ 0.073 & 0.276 & 0.010 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \\ C \\ Y \\ M \end{bmatrix}^{ITU-R \text{ BT.709-6}} \quad 40$$

For an embodiment where the color gamut is a SMPTE RP431-2 color gamut, the calculation is as follows:

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} 0.565 & 0.401 & 0.121 \\ 0.401 & 0.549 & 0.117 \\ 0.121 & 0.117 & 0.650 \end{bmatrix}^{SMPTE \text{ RP431-2}} \begin{bmatrix} 0.802 & 1.203 & 0.159 \\ 0.802 & 1.203 & 0.159 \\ 0.802 & 1.203 & 0.159 \end{bmatrix}^{D60} \quad 50$$

Wherein the resulting matrix is multiplied by the $S_R S_G S_B$ matrix:

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} 0.453 & 0.482 & 0.019 \\ 0.321 & 0.660 & 0.019 \\ 0.097 & 0.141 & 0.103 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \\ C \\ Y \\ M \end{bmatrix}^{SMPTE \text{ RP431-2}} \quad 60$$

Finally, the XYZ matrix must converted to the correct standard color space. In an embodiment where the color gamut used is an ITU-R BT709.6 color gamut, the matrices are as follows:

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$$\begin{bmatrix} R \\ G \\ B \end{bmatrix}^{ITU-R BT709-6} = \begin{bmatrix} 3.241 & -1.537 & -0.499 \\ -0.969 & 1.876 & 0.042 \\ 0.056 & -0.204 & 1.057 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

In an embodiment where the color gamut used is a SMPTE RP431-2 color gamut, the matrices are as follows:

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix}^{SMPTE RP431-2} = \begin{bmatrix} 2.73 & -1.018 & -0.440 \\ -0.795 & 1.690 & 0.023 \\ 0.041 & -0.088 & 1.101 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

Packing a Six-Primary Color System into IC_7C_P

IC_7C_P (ITP) is a color representation format specified in the Rec. ITU-R BT.2100 standard that is used as a part of the color image pipeline in video and digital photography systems for high dynamic range (HDR) and wide color gamut (WCG) imagery. The I (intensity) component is a luma component that represents the brightness of the video. C_T and C_P are blue-yellow (“tritanopia”) and red-green (“protanopia”) chroma components. The format is derived from an associated RGB color space by a coordination transformation that includes two matrix transformations and an intermediate non-linear transfer function, known as a gamma pre-correction. The transformation produces three signals: I, C_T , and C_P . The ITP transformation can be used with RGB signals derived from either the perceptual quantizer (PQ) or hybrid log-gamma (HLG) nonlinearity functions. The PQ curve is described in ITU-R BT2100-2:2018, Table 4, which is incorporated herein by reference in its entirety.

FIG. 43 illustrates one embodiment of packing six-primary color system image data into an IC_7C_P (ITP) format. In one embodiment, RGB image data is converted to an XYZ matrix. The XYZ matrix is then converted to an LMS matrix. The LMS matrix is then sent to an optical electronic transfer function (OETF). The conversion process is represented below:

$$\begin{bmatrix} L \\ M \\ S \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} 0.359 & 0.696 & -0.036 \\ -0.192 & 1.100 & 0.075 \\ 0.007 & 0.075 & 0.843 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

Output from the OETF is converted to ITP format. The resulting matrix is:

$$\begin{pmatrix} 0.5 & 0.5 & 0 \\ 1.614 & -3.323 & 1.710 \\ 4.378 & -4.246 & -0.135 \end{pmatrix}$$

FIG. 44 illustrates one embodiment of a six-primary color system converting RGBCYM image data into XYZ image data for an ITP format (e.g., 6P-B, 6P-C). For a six-primary color system, this is modified by replacing the RGB to XYZ matrix with a process to convert RGBCYM to XYZ. This is the same method as described in the legacy RGB process. The new matrix is as follows for an ITU-R BT.709-6 (6P-B) color gamut:

$$\begin{bmatrix} L \\ M \\ S \end{bmatrix} =$$

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-continued

$$\begin{pmatrix} 0.271 & 0.677 & 0.002 \\ 0.205 & 0.792 & 0.003 \\ 0.073 & 0.277 & 0.100 \end{pmatrix} \begin{pmatrix} 0.359 & 0.696 & -0.036 \\ -0.192 & 1.100 & 0.075 \\ 0.007 & 0.075 & 0.843 \end{pmatrix} \begin{bmatrix} R \\ G \\ B \\ C \\ Y \\ M \end{bmatrix}^{ITU-R BT.709-6}$$

RGBCYM data, based on an ITU-R BT.709-6 color gamut, is converted to an XYZ matrix. The resulting XYZ matrix is converted to an LMS matrix, which is sent to an OETF. Once processed by the OETF, the LMS matrix is converted to an ITP matrix. The resulting ITP matrix is as follows:

$$\begin{pmatrix} 0.5 & 0.5 & 0 \\ 1.614 & -3.323 & 1.710 \\ 4.378 & -4.246 & -0.135 \end{pmatrix}$$

In another embodiment, the LMS matrix is sent to an Optical Optical Transfer Function (OOTF). In yet another embodiment, the LMS matrix is sent to a Transfer Function other than OOTF or OETF.

In another embodiment, the RGBCYM data is based on the SMPTE ST431-2 (6P-C) color gamut. The matrices for an embodiment using the SMPTE ST431-2 color gamut are as follows:

$$\begin{bmatrix} L \\ M \\ S \end{bmatrix} =$$

$$\begin{pmatrix} 0.453 & 0.481 & 0.019 \\ 0.321 & 0.660 & 0.019 \\ 0.097 & 0.141 & 0.103 \end{pmatrix} \begin{pmatrix} 0.359 & 0.696 & -0.036 \\ -0.192 & 1.100 & 0.075 \\ 0.007 & 0.075 & 0.843 \end{pmatrix} \begin{bmatrix} R \\ G \\ B \\ C \\ Y \\ M \end{bmatrix}^{SMPTE ST431-2}$$

The resulting ITP matrix is:

$$\begin{pmatrix} 0.5 & 0.5 & 0 \\ 1.614 & -3.323 & 1.710 \\ 4.378 & -4.246 & -0.135 \end{pmatrix}$$

The decode process uses the standard ITP decode process, as the $S_R S_G S_B$ cannot be easily inverted. This makes it difficult to recover the six RGBCYM components from the ITP encode. Therefore, the display is operable to use the standard ICtCp decode process as described in the standards and is limited to just RGB output.

Converting to a Five-Color Multi-Primary Display

In one embodiment, the system is operable to convert image data incorporating five primary colors. In one embodiment, the five primary colors include Red (R), Green (G), Blue (B), Cyan (C), and Yellow (Y), collectively referred to as RGBCY. In another embodiment, the five primary colors include Red (R), Green (G), Blue (B), Cyan (C), and Magenta (M), collectively referred to as RGBCM. In one embodiment, the five primary colors do not include Magenta (M).

In one embodiment, the five primary colors include Red (R), Green (G), Blue (B), Cyan (C), and Orange (O),

collectively referred to as RGBCO. RGBCO primaries provide optimal spectral characteristics, transmittance characteristics, and makes use of a D65 white point. See, e.g., Moon-Cheol Kim et al., *Wide Color Gamut Five Channel Multi-Primary for HDTV Application*, Journal of Imaging Sci. & Tech. Vol. 49, No. 6, November/December 2005, at 594-604, which is hereby incorporated by reference in its entirety.

In one embodiment, a five-primary color model is expressed as $F=M \cdot C$, where F is equal to a tristimulus color vector, $F=(X, Y, Z)^T$, and C is equal to a linear display control vector, $C=(C1, C2, C3, C4, C5)^T$. Thus, a conversion matrix for the five-primary color model is represented as

$$M = \begin{pmatrix} X_1 & X_2 & X_3 & X_4 & X_5 \\ Y_1 & Y_2 & Y_3 & Y_4 & Y_5 \\ Z_1 & Z_2 & Z_3 & Z_4 & Z_5 \end{pmatrix}$$

Using the above equation and matrix, a gamut volume is calculated for a set of given control vectors on the gamut boundary. The control vectors are converted into CIELAB uniform color space. However, because matrix M is non-square, the matrix inversion requires splitting the color gamut into a specified number of pyramids, with the base of each pyramid representing an outer surface and where the control vectors are calculated using linear equation for each given XYZ triplet present within each pyramid. By separating regions into pyramids, the conversion process is normalized. In one embodiment, a decision tree is created in order to determine which set of primaries are best to define a specified color. In one embodiment, a specified color is defined by multiple sets of primaries. In order to locate each pyramid, 2D chromaticity look-up tables are used, with corresponding pyramid numbers for input chromaticity values in xy or $u'y'$. Typical methods using pyramids require 1000×1000 address ranges in order to properly search the boundaries of adjacent pyramids with look-up table memory. The system of the present invention uses a combination of parallel processing for adjacent pyramids and at least one algorithm for verifying solutions by checking constraint conditions. In one embodiment, the system uses a parallel computing algorithm. In one embodiment, the system uses a sequential algorithm. In another embodiment, the system uses a brightening image transformation algorithm. In another embodiment, the system uses a darkening image transformation algorithm. In another embodiment, the system uses an inverse sinusoidal contrast transformation algorithm. In another embodiment, the system uses a hyperbolic tangent contrast transformation algorithm. In yet another embodiment, the system uses a sine contrast transformation execution times algorithm. In yet another embodiment, the system uses a linear feature extraction algorithm. In yet another embodiment, the system uses a JPEG2000 encoding algorithm. In yet another embodiment, the system uses a parallelized arithmetic algorithm. In yet another embodiment, the system uses an algorithm other than those previously mentioned. In yet another embodiment, the system uses any combination of the aforementioned algorithms. Mapping a Six-Primary Color System into Standardized Transport Formats

Each encode and/or decode system fits into existing video serial data streams that have already been established and standardized. This is key to industry acceptance. Encoder and/or decoder designs require little or no modification for a six-primary color system to map to these standard serial formats.

FIG. 45 illustrates one embodiment of a six-primary color system mapping to a SMPTE ST424 standard serial format. The SMPTE ST424/ST425 set of standards allow very high sampling systems to be passed through a single cable. This is done by using alternating data streams, each containing different components of the image. For use with a six-primary color system transport, image formats are limited to RGB due to the absence of a method to send a full bandwidth Y signal.

The process for mapping a six-primary color system to a SMPTE ST425 format is the same as mapping to a SMPTE ST424 format. To fit a six-primary color system into a SMPTE ST425/424 stream involves the following substitutions: $G'_{INT}+M'_{INT}$ is placed in the Green data segments, $R'_{INT}+C'_{INT}$ is placed in the Red data segments, and $B'_{INT}+Y'_{INT}$ is placed into the Blue data segments. FIG. 46 illustrates one embodiment of an SMPTE 424 6P readout.

System 2 requires twice the data rate as System 1, so it is not compatible with SMPTE 424. However, it maps easily into SMPTE ST2082 using a similar mapping sequence. In one example, System 2 is used to have the same data speed defined for 8K imaging to show a 4K image.

In one embodiment, sub-image and data stream mapping occur as shown in SMPTE ST2082. An image is broken into four sub-images, and each sub-image is broken up into two data streams (e.g., sub-image 1 is broken up into data stream 1 and data stream 2). The data streams are put through a multiplexer and then sent to the interface as shown in FIG. 47.

FIG. 48 and FIG. 49 illustrate serial digital interfaces for a six-primary color system using the SMPTE ST2082 standard. In one embodiment, the six-primary color system data is RGBCYM data, which is mapped to the SMPTE ST2082 standard (FIG. 48). Data streams 1, 3, 5, and 7 follow the pattern shown for data stream 1. Data streams 2, 4, 6, and 8 follow the pattern shown for data stream 2. In one embodiment, the six-primary color system data is $Y_{RGB} Y_{CYM} C_R C_B C_C C_Y$ data, which is mapped to the SMPTE ST2082 standard (FIG. 49). Data streams 1, 3, 5, and 7 follow the pattern shown for data stream 1. Data streams 2, 4, 6, and 8 follow the pattern shown for data stream 2.

In one embodiment, the standard serial format is SMPTE ST292. SMPTE ST292 is an older standard than ST424 and is a single wire format for 1.5 GB video, whereas ST424 is designed for up to 3 GB video. However, while ST292 can identify the payload ID of SMPTE ST352, it is constrained to only accepting an image identified by a hex value, 0h. All other values are ignored. Due to the bandwidth and identifications limitations in ST292, a component video six-primary color system incorporates a full bit level luminance component. To fit a six-primary color system into a SMPTE ST292 stream involves the following substitutions: $E'_{Y_{6-INT}}$ is placed in the Y data segments, $E'_{Cb-INT}+E'_{Cy-INT}$ is placed in the Cb data segments, and $E'_{Cr-INT}+E'_{Cc-INT}$ is placed in the Cr data segments. In another embodiment, the standard serial format is SMPTE ST352.

SMPTE ST292 and ST424 Serial Digital Interface (SDI) formats include payload identification (ID) metadata to help the receiving device identify the proper image parameters. The tables for this need modification by adding at least one flag identifying that the image source is a six-primary color RGB image. Therefore, six-primary color system format additions need to be added. In one embodiment, the standard is the SMPTE ST352 standard.

FIG. 50 illustrates one embodiment of an SMPTE ST292 6P mapping. FIG. 51 illustrates one embodiment of an SMPTE ST292 6P readout.

FIG. 52 illustrates modifications to the SMPTE ST352 standards for a six-primary color system. Hex code “Bh” identifies a constant luminance source and flag “Fh” indicates the presence of a six-primary color system. In one embodiment, Fh is used in combination with at least one other identifier located in byte 3. In another embodiment, the Fh flag is set to 0 if the image data is formatted as System 1 and the Fh flag is set to 1 if the image data is formatted as System 2.

In another embodiment, the standard serial format is SMPTE ST2082. Where a six-primary color system requires more data, it may not always be compatible with SMPTE ST424. However, it maps easily into SMPTE ST2082 using the same mapping sequence. This usage would have the same data speed defined for 8K imaging in order to display a 4K image.

In another embodiment, the standard serial format is SMPTE ST2022. Mapping to ST2022 is similar to mapping to ST292 and ST242, but as an ETHERNET format. The output of the stacker is mapped to the media payload based on Real-time Transport Protocol (RTP) 3550, established by the Internet Engineering Task Force (IETF). RTP provides end-to-end network transport functions suitable for applications transmitting real-time data, including, but not limited to, audio, video, and/or simulation data, over multicast or unicast network services. The data transport is augmented by a control protocol (RTCP) to allow monitoring of the data delivery in a manner scalable to large multicast networks, and to provide control and identification functionality. There are no changes needed in the formatting or mapping of the bit packing described in SMPTE ST 2022-6: 2012 (HBRMT).

FIG. 53 illustrates one embodiment of a modification for a six-primary color system using the SMPTE ST2202 standard. For SMPTE ST2202-6:2012 (HBRMT), there are no changes needed in formatting or mapping of the bit packing. ST2022 relies on header information to correctly configure the media payload. Parameters for this are established within the payload header using the video source format fields including, but not limited to, MAP, FRAME, FRATE, and/or SAMPLE. MAP, FRAME, and FRATE remain as described in the standard. MAP is used to identify if the input is ST292 or ST425 (RGB or Y Cb Cr). SAMPLE is operable for modification to identify that the image is formatted as a six-primary color system image. In one embodiment, the image data is sent using flag “0h” (unknown/unspecified).

In another embodiment, the standard is SMPTE ST2110. SMPTE ST2110 is a relatively new standard and defines moving video through an Internet system. The standard is based on development from the IETF and is described under RFC3550. Image data is described through “pgroup” construction. Each pgroup consists of an integer number of octets. In one embodiment, a sample definition is RGB or YCbCr and is described in metadata. In one embodiment, the metadata format uses a Session Description Protocol (SDP) format. Thus, pgroup construction is defined for 4:4:4, 4:2:2, and 4:2:0 sampling as 8-bit, 10-bit, 12-bit, and in some cases 16-bit and 16-bit floating point wording. In one embodiment, six-primary color image data is limited to a 10-bit depth. In another embodiment, six-primary color image data is limited to a 12-bit depth. Where more than one sample is used, it is described as a set. For example, 4:4:4 sampling for blue, as a non-linear RGB set, is described as C0'B, C1'B, C2'B, C3'B, and C4'B. The lowest number index being left most within the image. In another embodiment, the method of substitution is the same method used to map six-primary color content into the ST2110 standard.

In another embodiment, the standard is SMPTE ST2110. SMPTE ST2110-20 describes the construction for each pgroup. In one embodiment, six-primary color system content arrives for mapping as non-linear data for the SMPTE ST2110 standard. In another embodiment, six-primary color system content arrives for mapping as linear data for the SMPTE ST2110 standard.

FIG. 54 illustrates a table of 4:4:4 sampling for a six-primary color system for a 10-bit video system. For 4:4:4 10-bit video, 15 octets are used and cover 4 pixels.

FIG. 55 illustrates a table of 4:4:4 sampling for a six-primary color system for a 12-bit video system. For 4:4:4 12-bit video, 9 octets are used and cover 2 pixels before restarting the sequence.

Non-linear GRBMYC image data would arrive as: $G'_{INT} + M'_{INT}$, $R'_{INT} + C'_{INT}$, and $B'_{INT} + Y'_{INT}$. Component substitution would follow what has been described for SMPTE ST424, where $G'_{INT} + M'_{INT}$ is placed in the Green data segments, $R'_{INT} + C'_{INT}$ is placed in the Red data segments, and $B'_{INT} + Y'_{INT}$ is placed in the Blue data segments. The sequence described in the standard is shown as R0', G0', B0', R1', G1', B1', etc.

FIG. 56 illustrates sequence substitutions for 10-bit and 12-bit video in 4:2:2 sampling systems in a Y Cb Cr Cc Cy color space. Components are delivered to a 4:2:2 pgroup including, but not limited to, E'_{Y6-INT} , $E'_{Cb-INT} + E'_{Cy-INT}$, and $E'_{Cr-INT} + E'_{Cc-INT}$. For 4:2:2 10-bit video, 5 octets are used and cover 2 pixels before restarting the sequence. For 4:2:2 12-bit video, 6 octets are used and cover 2 pixels before restarting the sequence. Component substitution follows what has been described for SMPTE ST292, where E'_{Y6-INT} is placed in the Y data segments, $E'_{Cb-INT} + E'_{Cy-INT}$ is placed in the Cb data segments, and $E'_{Cr-INT} + E'_{Cc-INT}$ is placed in the Cr data segments. The sequence described in the standard is shown as Cb0', Y0', Cr0', Y1', Cr1', Y3', Cb2', Y4', Cr2', Y5', etc. In another embodiment, the video data is represented at a bit level other than 10-bit or 12-bit. In another embodiment, the sampling system is a sampling system other than 4:2:2. In another embodiment, the standard is SMPTE ST2110.

FIG. 57 illustrates sample placements of six-primary system components for a 4:2:2 sampling system image. This follows the substitutions illustrated in FIG. 56, using a 4:2:2 sampling system.

FIG. 58 illustrates sequence substitutions for 10-bit and 12-bit video in 4:2:0 sampling systems using a Y Cb Cr Cc Cy color space. Components are delivered to a pgroup including, but not limited to, E'_{Y6-INT} , $E'_{Cb-INT} + E'_{Cy-INT}$, and $E'_{Cr-INT} + E'_{Cc-INT}$. For 4:2:0 10-bit video data, 15 octets are used and cover 8 pixels before restarting the sequence. For 4:2:0 12-bit video data, 9 octets are used and cover 4 pixels before restarting the sequence. Component substitution follows what is described in SMPTE ST292 where E'_{Y6-INT} is placed in the Y data segments, $E'_{Cb-INT} + E'_{Cy-INT}$ is placed in the Cb data segments, and $E'_{Cr-INT} + E'_{Cc-INT}$ is placed in the Cr data segments. The sequence described in the standard is shown as Y'00, Y'01, Y', etc.

FIG. 59 illustrates sample placements of six-primary system components for a 4:2:0 sampling system image. This follows the substitutions illustrated in FIG. 58, using a 4:2:0 sampling system.

FIG. 60 illustrates modifications to SMPTE ST2110-20 for a 10-bit six-primary color system in 4:4:4 video. SMPTE ST2110-20 describes the construction of each “pgroup”. Normally, six-primary color system data and/or content would arrive for mapping as non-linear. However, with the present system there is no restriction on mapping data and/or

content. For 4:4:4, 10-bit video, 15 octets are used and cover 4 pixels before restarting the sequence. Non-linear, six-primary color system image data would arrive as G'_{INT} , B'_{INT} , R'_{INT} , M'_{INT} , Y'_{INT} , and C'_{INT} . The sequence described in the standard is shown as $R0'$, $G0'$, $B0'$, $R1'$, $G1'$, $B1'$, etc.

FIG. 61 illustrates modifications to SMPTE ST2110-20 for a 12-bit six-primary color system in 4:4:4 video. For 4:4:4, 12-bit video, 9 octets are used and cover 2 pixels before restarting the sequence. Non-linear, six-primary color system image data would arrive as G'_{INT} , B'_{INT} , R'_{INT} , M'_{INT} , Y'_{INT} , and C'_{INT} . The sequence described in the standard is shown as $R0'$, $G0'$, $B0'$, $R1'$, $G1'$, $B1'$, etc.

FIG. 62 illustrates modifications to SMPTE ST2110-20 for a 10-bit six primary color system in 4:2:2 video. Components that are delivered to a SMPTE ST2110 pgroup include, but are not limited to, $E'_{Yrgb-INT}$, $E'_{Ycym-INT}$, E'_{Cb-INT} , E'_{Cr-INT} , E'_{Cy-INT} , and E'_{Cc-INT} . For 4:2:2, 10-bit video, 5 octets are used and cover 2 pixels before restarting the sequence. For 4:2:2:2, 12-bit video, 6 octets are used and cover 2 pixels before restarting the sequence. Component substitution follows what is described for SMPTE ST292, where $E'_{Yrgb-INT}$ or $E'_{Ycym-INT}$ are placed in the Y data

segments, E'_{Cr-INT} or E'_{Cc-INT} are placed in the Cr data segments, and E'_{Cb-INT} or E'_{Cy-INT} are placed in the Cb data segments. The sequence described in the standard is shown as $Cb'0$, $Y'0$, $Cr'0$, $Y'1$, $Cb'1$, $Y'2$, $Cr'1$, $Y'3$, $Cb'2$, $Y'4$, $Cr'2$, etc.

FIG. 63 illustrates modifications to SMPTE ST2110-20 for a 12-bit six-primary color system in 4:2:0 video. Components that are delivered to a SMPTE ST2110 pgroup are the same as with the 4:2:2 method. For 4:2:0, 10-bit video, 15 octets are used and cover 8 pixels before restarting the sequence. For 4:2:0, 12-bit video, 9 octets are used and cover 4 pixels before restarting the sequence. Component substitution follows what is described for SMPTE ST292, where $E'_{Yrgb-INT}$ or $E'_{Ycym-INT}$ are placed in the Y data segments, E'_{Cr-INT} or E'_{Cc-INT} are placed in the Cr data segments, and E'_{Cb-INT} or E'_{Cy-INT} are placed in the Cb data segments. The sequence described in the standard is shown as $Y'00$, $Y'01$, Y' , etc.

Table 14 summarizes mapping to SMPTE ST2110 for 4:2:2:2 and 4:2:0:2:0 sampling for System 1 and Table 15 summaries mapping to SMPTE ST2110 for 4:4:4:4:4:4 sampling (linear and non-linear) for System 1.

TABLE 14

Sampling	Depth	Pgroup		Octets	Pixels	Y PbPr Sample Order	6P Sample Order
		Bit					
4:2:2:2:2	8	4	2	$C_B'Y0', C_R'Y1'$			
	10	5	2	$C_B'Y0', C_R'Y1'$			$C_B'+Cy', Y0', C_R'+Cc', Y1'$
	12	6	2	$C_B'Y0', C_R'Y1'$			$C_B'+Cy', Y0', C_R'+Cc', Y1'$
	16, 16f	8	2	$C_B'Y0', C_R'Y1'$			$C_B'+Cy', Y0', C_R'+Cc', Y1'$
4:2:0:2:0	8	6	4	$Y'00, Y'01, Y'10, Y'11, C_B'00, C_R'00$			
	10	15	8	$Y'00, Y'01, Y'10, Y'11, C_B'00, C_R'00$			$Y'00, Y'01, Y'10, Y'11, C_B'00+Cy'00, C_R'00+Cc'00$
				$Y'02, Y'03, Y'12, Y'13, C_B'01, C_R'01$			$Y'02, Y'03, Y'12, Y'13, C_B'01+Cy'01, C_R'01+Cc'01$
	12	9	4	$Y'00, Y'01, Y'10, Y'11, C_B'00, C_R'00$			$Y'00, Y'01, Y'10, Y'11, C_B'00+Cy'00, C_R'00+Cc'00$

TABLE 15

Sampling	Depth	pgroup		Octets	pixels	RGB Sample Order	6P Sample Order
		Bit					
4:4:4:4:4:4	8	3	1	R, G, B			
	Linear	10	4	$R0, G0, B0, R1, G1, B1, R2, G2, B2$			$R + C0, G + M0, B + Y0, R + C1, R2, G2, B2$
							$G + M1, B + Y1, R + C2, G + M2, B + Y2$
	12	9	2	$R0, G0, B0, R1, G1, B1$			$R + C0, G + M0, B + Y0, R + C1, G + M1, B + Y1$
4:4:4:4:4:4	16, 16f	6	1	R, G, B			$R + C, G + M, B + Y$
	8	3	1	R', G', B'			
	Non-Linear	10	4	$R0', G0', B0', R1', G1', B1', R2', G2', B2'$			$R' + C'0, G' + M'0, B' + Y'0, R' + C'1, G' + M'1, B' + Y'1, R' + C'2, G' + M'2, B' + Y'2$
							$R' + C'0, G' + M'0, B' + Y'0, R' + C'1, G' + M'1, B' + Y'1$
4:4:4:4:4:4	12	9	2	$R0', G0', B0', R1', G1', B1'$			$R' + C'0, G' + M'0, B' + Y'0, R' + C'1, G' + M'1, B' + Y'1$
	16, 16f	6	1	R', G', B'			$R' + C', G' + M', B' + Y'$

Table 16 summarizes mapping to SMPTE ST2110 for 4:2:2:2:2 sampling for System 2 and Table 17 summaries mapping to SMPTE ST2110 for 4:4:4:4:4:4 sampling (linear and non-linear) for System 2.

TABLE 16

		pgroup			
Sampling	Bit Depth	octets	pixels	Y PbPr Sample Order	6P Sample Order
4:2:2:2:2	8	8	2	C_B', Y_0', C_R', Y_1'	$C_B', C_Y', Y_{RGB}^{0'}, C_R', C_c, Y_{CM}^{0'} C_B', C_Y', Y_{RGB}^{1'}$
	10	10	2	C_B', Y_0', C_R', Y_1'	$C_B', C_Y', Y_{RGB}^{0'}, C_R', C_c, Y_{CM}^{0'} C_B', C_Y', Y_{RGB}^{1'}$
	12	12	2	C_B', Y_0', C_R', Y_1'	$C_B', C_Y', Y_{RGB}^{0'}, C_R', C_c, Y_{CM}^{0'} C_B', C_Y', Y_{RGB}^{1'}$
	16, 16f	16	2	C_B', Y_0', C_R', Y_1'	$C_B', C_Y', Y_{RGB}^{0'}, C_R', C_c, Y_{CM}^{0'} C_B', C_Y', Y_{RGB}^{1'}$

TABLE 17

	Bit	pgroup				
Sampling	Depth	octets	pixels	RGB Sample Order		6P Sample Order
4:4:4:4:4:4 Linear	8	3	1	R,G,B		R,C,G,M,B,Y
	10	15	4	R0, G0, B0, R1, G1, B1, R2, G2,B2		R0, C0, G0, M0, B0, Y0, R1, C1, G1, M1, B1, Y1, R2, C2, G2, M2,B2+Y2
	12	9	2	R0, G0, B0, R1, G1, B1		R0,C0,G0,M0,B0,Y0, R1,C1,G1,M1,B1,Y1
	16, 16f	6	1	R, G, B		R,C,G,M,B,Y
4:4:4:4:4:4 Non-Linear	8	3	1	R', G', B'		R',C',G',M',B',Y'
	10	15	4	R0', G0', B0', R1', G1', B1', R2', G2',B2'		R0', C0', G0', M0', B0', Y0', R1', C1', G1', M1', B1', Y1', R2', C2', G2',M2',B2'+Y2'
	12	9	2	R0', G0', B0', R1', G1', B1'		R0', C0', G0', M0',B0', Y0', R1',C1',G1',M1',B1',Y1'
	16, 16f	6	1	R', G', B'		R',C',G',M',B',Y'

Session Description Protocol (SDP) Modification for a Six-Primary Color System

SDP is derived from IETF RFC 4566 which sets parameters including, but not limited to, bit depth and sampling parameters. In one embodiment, SDP parameters are contained within the RTP payload. In another embodiment, SDP parameters are contained within the media format and transport protocol. This payload information is transmitted as text. Therefore, modifications for the additional sampling identifiers requires the addition of new parameters for the sampling statement. SDP parameters include, but are not limited to, color channel data, image data, framerate data, a sampling standard, a flag indicator, an active picture size code, a timestamp, a clock frequency, a frame count, a scrambling indicator, and/or a video format indicator. For non-constant luminance imaging, the additional parameters include, but are not limited to, RGBCYM-4:4:4, YBRCY-4:2:2, and YBRCY-4:2:0. For constant luminance signals, the additional parameters include, but are not limited to, CLYBRCY-4:2:2 and CLYBRCY-4:2:0.

Additionally, differentiation is included with the colorimetry identifier in one embodiment. For example, 6PB1 defines 6P with a color gamut limited to ITU-R BT.709 formatted as System 1, 6PB2 defines 6P with a color gamut limited to ITU-R BT.709 formatted as System 2, 6PB3 defines 6P with a color gamut limited to ITU-R BT.709 formatted as System 3, 6PC1 defines 6P with a color gamut limited to SMPTE RP 431-2 formatted as System 1, 6PC2

defines 6P with a color gamut limited to SMPTE RP 431-2 formatted as System 2, 6PC3 defines 6P with a color gamut limited to SMPTE RP 431-2 formatted as System 3, 6PS1 defines 6P with a color gamut as Super 6P formatted as

System 1, 6PS2 defines 6P with a color gamut as Super 6P formatted as System 2, and 6PS3 defines 6P with a color gamut as Super 6P formatted as System 3.

Colorimetry can also be defined between a six-primary color system using the ITU-R BT.709-6 standard and the SMPTE ST431-2 standard, or colorimetry can be left defined as is standard for the desired standard. For example, the SDP parameters for a 1920x1080 six-primary color system using the ITU-R BT.709-6 standard with a 10-bit signal as System 1 are as follows: m=video 30000 RTP/AVP 112, a=rtpmap: 112 raw/90000, a=fmtp: 112, sampling=YBRCY-4:2:2, width=1920, height=1080, exactframerate=30000/1001, depth=10, TCS=SDR, colorimetry=6PB1, PM=2110GPM, SSN=ST2110-20:2017.

In one embodiment, the six-primary color system is integrated with a Consumer Technology Association (CTA) 861-based system. CTA-861 establishes protocols, requirements, and recommendations for the utilization of uncompressed digital interfaces by consumer electronics devices including, but not limited to, digital televisions (DTVs), digital cable, satellite or terrestrial set-top boxes (STBs), and related peripheral devices including, but not limited to, DVD players and/or recorders, and other related Sources or Sinks.

These systems are provided as parallel systems so that video content is parsed across several line pairs. This enables each video component to have its own transition-minimized differential signaling (TMDS) path. TMDS is a technology for transmitting high-speed serial data and is

used by the Digital Visual Interface (DVI) and High-Definition Multimedia Interface (HDMI) video interfaces, as well as other digital communication interfaces. TMDS is similar to low-voltage differential signaling (LVDS) in that it uses differential signaling to reduce electromagnetic interference (EMI), enabling faster signal transfers with increased accuracy. In addition, TMDS uses a twisted pair for noise reduction, rather than a coaxial cable that is conventional for carrying video signals. Similar to LVDS, data is transmitted serially over the data link. When transmitting video data, and using HDMI, three TMDS twisted pairs are used to transfer video data.

In such a system, each pixel packet is limited to 8 bits only. For bit depths higher than 8 bits, fragmented packs are used. This arrangement is no different than is already described in the current CTA-861 standard.

Based on CTA extension Version 3, identification of a six-primary color transmission would be performed by the sink device (e.g., the monitor). Adding recognition of the additional formats would be flagged in the CTA Data Block Extended Tag Codes (byte 3). Since codes 33 and above are reserved, any two bits could be used to identify that the format is RGB, RGBCYM, Y Cb Cr, or Y Cb Cr Cc Cy and/or identify System 1 or System 2. Should byte 3 define a six-primary sampling format, and where the block 5 extension identifies byte 1 as ITU-R BT.709, then logic assigns as 6P-B. However, should byte 4 bit 7 identify colorimetry as DCI-P3, the color gamut would be assigned as 6P-C.

In one embodiment, the system alters the AVI Infoframe Data to identify content. AVI Infoframe Data is shown in Table 10 of CTA 861-G. In one embodiment, Y2=1, Y1=0, and Y0=0 identifies content as 6P 4:2:0:2:0. In another embodiment, Y2=1, Y1=0, and Y0=1 identifies content as Y Cr Cb Cc Cy. In yet another embodiment, Y2=1, Y1=1, and Y0=0 identifies content as RGBCMY.

Byte 2 C1=1, C0=1 identifies extended colorimetry in Table 11 of CTA 861-G. Byte 3 EC2, EC1, EC0 identifies additional colorimetry extension valid in Table 13 of CTA 861-G. Table 14 of CTA 861-G reserves additional extensions. In one embodiment, ACE3=1, ACE2=0, ACE1=0, and ACE0=X identifies 6P-B. In one embodiment, ACE3=0, ACE2=1, ACE1=0, and ACE0=X identifies 6P-C. In one embodiment, ACE3=0, ACE2=0, ACE1=1, and ACE0=X identifies System 1. In one embodiment, ACE3=1, ACE2=1, ACE1=0, and ACE0=X identifies System 2.

FIG. 64 illustrates the current RGB sampling structure for 4:4:4 sampling video data transmission. For HDMI 4:4:4 sampling, video data is sent through three TMDS line pairs. FIG. 65 illustrates a six-primary color sampling structure, RGBCYM, using System 1 for 4:4:4 sampling video data transmission. In one embodiment, the six-primary color sampling structure complies with CTA 861-G, November 2016, Consumer Technology Association, which is incorporated herein by reference in its entirety. FIG. 66 illustrates an example of System 2 to RGBCYM 4:4:4 transmission. FIG. 67 illustrates current Y Cb Cr 4:2:2 sampling transmission as non-constant luminance. FIG. 68 illustrates a six-primary color system (System 1) using Y Cr Cb Cc Cy 4:2:2 sampling transmission as non-constant luminance. FIG. 69 illustrates an example of a System 2 to Y Cr Cb Cc Cy 4:2:2 transmission as non-constant luminance. In one embodiment, the Y Cr Cb Cc Cy 4:2:2 sampling transmission complies with CTA 861-G, November 2016, Consumer Technology Association. FIG. 70 illustrates current Y Cb Cr

4:2:0 sampling transmission. FIG. 71 illustrates a six-primary color system (System 1) using Y Cr Cb Cc Cy 4:2:0 sampling transmission.

HDMI sampling systems include Extended Display Identification Data (EDID) metadata. EDID metadata describes the capabilities of a display device to a video source. The data format is defined by a standard published by the Video Electronics Standards Association (VESA). The EDID data structure includes, but is not limited to, manufacturer name and serial number, product type, phosphor or filter type, timings supported by the display, display size, luminance data, and/or pixel mapping data. The EDID data structure is modifiable and modification requires no additional hardware and/or tools.

EDID information is transmitted between the source device and the display through a display data channel (DDC), which is a collection of digital communication protocols created by VESA. With EDID providing the display information and DDC providing the link between the display and the source, the two accompanying standards enable an information exchange between the display and source.

In addition, VESA has assigned extensions for EDID. Such extensions include, but are not limited to, timing extensions (00), additional time data black (CEA EDID Timing Extension (02)), video timing block extensions (VTB-EXT (10)), EDID 2.0 extension (20), display information extension (DI-EXT (40)), localized string extension (LS-EXT (50)), microdisplay interface extension (MI-EXT (60)), display ID extension (70), display transfer characteristics data block (DTCDB (A7, AF, BF)), block map (F0), display device data block (DDDB (FF)), and/or extension defined by monitor manufacturer (FF).

In one embodiment, SDP parameters include data corresponding to a payload identification (ID) and/or EDID information.

Multi-Primary Color System Display

FIG. 72 illustrates a dual stack LCD projection system for a six-primary color system. In one embodiment, the display is comprised of a dual stack of projectors. This display uses two projectors stacked on top of one another or placed side by side. Each projector is similar, with the only difference being the color filters in each unit. Refresh and pixel timings are synchronized, enabling a mechanical alignment between the two units so that each pixel overlays the same position between projector units. In one embodiment, the two projectors are Liquid-Crystal Display (LCD) projectors. In another embodiment, the two projectors are Digital Light Processing (DLP) projectors. In yet another embodiment, the two projectors are Liquid-Crystal on Silicon (LCOS) projectors. In yet another embodiment, the two projectors are Light-Emitting Diode (LED) projectors.

In one embodiment, the display is comprised of a single projector. A single projector six-primary color system requires the addition of a second cross block assembly for the additional colors. One embodiment of a single projector (e.g., single LCD projector) is shown in FIG. 73. A single projector six-primary color system includes a cyan dichroic mirror, an orange dichroic mirror, a blue dichroic mirror, a red dichroic mirror, and two additional standard mirrors. In one embodiment, the single projector six-primary color system includes at least six mirrors. In another embodiment, the single projector six-primary color system includes at least two cross block assembly units.

FIG. 74 illustrates a six-primary color system using a single projector and reciprocal mirrors. In one embodiment, the display is comprised of a single projector unit working

in combination with at first set of at least six reciprocal mirrors, a second set of at least six reciprocal mirrors, and at least six LCD units. Light from at least one light source emits towards the first set of at least six reciprocal mirrors. The first set of at least six reciprocal mirrors reflects light towards at least one of the at least six LCD units. The at least six LCD units include, but are not limited to, a Green LCD, a Yellow LCD, a Cyan LCD, a Red LCD, a Magenta LCD, and/or a Blue LCD. Output from each of the at least six LCDs is received by the second set of at least six reciprocal mirrors. Output from the second set of at least six reciprocal mirrors is sent to the single projector unit. Image data output by the single projector unit is output as a six-primary color system. In another embodiment, there are more than two sets of reciprocal mirrors. In another embodiment, more than one projector is used.

In another embodiment, the display is comprised of a dual stack Digital Micromirror Device (DMD) projector system. FIG. 75 illustrates one embodiment of a dual stack DMD projector system. In this system, two projectors are stacked on top of one another. In one embodiment, the dual stack DMD projector system uses a spinning wheel filter. In another embodiment, the dual stack DMD projector system uses phosphor technology. In one embodiment, the filter systems are illuminated by a xenon lamp. In another embodiment, the filter system uses a blue laser illuminator system. Filter systems in one projector are RGB, while the second projector uses a CYM filter set. The wheels for each projector unit are synchronized using at least one of an input video sync or a projector to projector sync, and timed so that the inverted colors are output of each projector at the same time.

In one embodiment, the projectors are phosphor wheel systems. A yellow phosphor wheel spins in time with a DMD imager to output sequential RG. The second projector is designed the same, but uses a cyan phosphor wheel. The output from this projector becomes sequential BG. Combined, the output of both projectors is YRGGCB. Magenta is developed by synchronizing the yellow and cyan wheels to overlap the flashing DMD.

In another embodiment, the display is a single DMD projector solution. A single DMD device is coupled with an RGB diode light source system. In one embodiment, the DMD projector uses LED diodes. In one embodiment, the DMD projector includes CYM diodes. In another embodiment, the DMD projector creates CYM primaries using a double flashing technique. FIG. 76 illustrates one embodiment of a single DMD projector solution.

FIG. 77 illustrates one embodiment of a six-primary color system using a white OLED display. In yet another embodiment, the display is a white OLED monitor. Current emissive monitor and/or television designs use a white emissive OLED array covered by a color filter. Changes to this type of display only require a change to pixel indexing and new six color primary filters. Different color filter arrays are used, placing each subpixel in a position that provides the least light restrictions, color accuracy, and off axis display.

FIG. 78 illustrates one embodiment of an optical filter array for a white OLED display.

FIG. 79 illustrates one embodiment of a matrix of an LCD drive for a six-primary color system with a backlight illuminated LCD monitor. In yet another embodiment, the display is a backlight illuminated LCD display. The design of an LCD display involves adding the CYM subpixels. Drives for these subpixels are similar to the RGB matrix drives. With the advent of 8K LCD televisions, it is tech-

nically feasible to change the matrix drive and optical filter and have a 4K six-primary color TV.

FIG. 80 illustrates one embodiment of an optical filter array for a six-primary color system with a backlight illuminated LCD monitor. The optical filter array includes the additional CYM subpixels.

In yet another embodiment, the display is a direct emissive assembled display. The design for a direct emissive assembled display includes a matrix of color emitters grouped as a six-color system. Individual channel inputs drive each Quantum Dot (QD) element illuminator and/or micro LED element.

FIG. 81 illustrates an array for a Quantum Dot (QD) display device.

FIG. 82 illustrates one embodiment of an array for a six-primary color system for use with a direct emissive assembled display.

FIG. 83 illustrates one embodiment of a six-primary color system in an emissive display that does not incorporate color filtered subpixels. For LCD and WOLED displays, this can be modified for a six-primary color system by expanding the RGB or WRGB filter arrangement to an RGBCYM matrix. For WRGB systems, the white subpixel could be removed as the luminance of the three additional primaries will replace it. SDI video is input through an SDI decoder. In one embodiment, the SDI decoder outputs to a Y CrCbCcCy-RGBCYM converter. The converter outputs RGBCYM data, with the luminance component (Y) subtracted. RGBCYM data is then converted to RGB data. This RGB data is sent to a scale sync generation component, receives adjustments to image controls, contrast, brightness, chroma, and saturation, is sent to a color correction component, and output to the display panel as LVDS data. In another embodiment the SDI decoder outputs to an SDI Y-R switch component. The SDI Y-R switch component outputs RGBCYM data. The RGBCYM data is sent to a scale sync generation component, receives adjustments to image controls, contrast, brightness, chroma, and saturation, is sent to a color correction component, and output to a display panel as LVDS data.

Primary Triads

A conversion between XYZ and any three primary system (e.g., RGB) yields an exact solution. However, in systems with more than three primaries, there is not an exact solution. For example, a six primary system with an RGBCMY to XYZ is overdetermined. This requires a mathematical pseudo-inverse (e.g., Moore-Penrose pseudo-inverse), which provides one of an infinite number of solutions. An algorithm is required to go from XYZ to RGBCMY.

In one embodiment, the system uses at least one triad in the algorithm. Each of the at least one triad is formed using three points (e.g., primaries). Although a larger number of permutations is possible for the three points, the order of the three points is not important (e.g., ABC functions the same as BCA). Therefore, for each set of primaries, a number of possible triads is calculated using the following equation:

$$C(n, r) = {}^nC_r = \frac{n!}{(n-r)!r!}$$

where n=the number of primaries and r=3. Thus, for three primaries, there is one possible triad; for four primaries, there are 4 possible triads; for five primaries, there are 10 possible triads; for six primaries, there are 20 possible triads; for seven primaries, there are 35 possible triads; for eight

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primaries, there are 56 possible triads; for nine primaries, there are 84 possible triads; for ten primaries, there are 120 possible triads; for eleven primaries, there are 165 possible triads; and for twelve primaries, there are 220 possible triads.

In one embodiment, the system uses at least five primary triads. In one embodiment, the at least five primary triads are formed using at least six primaries (e.g., P_1 - P_6). In one embodiment, the first triad is formed using P_1 , P_2 , and P_3 ; the second triad is formed using P_4 , P_5 , and P_6 ; the third triad is formed using P_1 , P_3 , and P_5 ; the fourth triad is formed using P_2 , P_3 , and P_4 ; and the fifth triad is formed using P_1 , P_2 , and P_6 . For example, in a six primary system using RGBCMY, the first triad is formed using RGB, the second triad is formed using CMY, the third triad is formed using RMB, the fourth triad is formed using CGB, and the fifth triad is formed using RGY. Alternate numbers of primaries, numbers of triads, and alternate triads are compatible with the present invention.

In a preferred embodiment, a color value (e.g., ACES APO) is converted to a colorimetric position (e.g., XYZ), and corresponding values for the at least five primary triads are calculated. In one embodiment, the conversion process first converts an RGB value in ACES APO to XYZ with a D65 white point. Alternative white points are compatible with the present invention. To convert the RGB value in ACES APO to XYZ for a 6P-C system using a D65 white point, the following equation is used:

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix}_{D65} = \begin{bmatrix} 0.938653 & -0.005740 & 0.017517 \\ 0.338102 & 0.727227 & -0.065328 \\ 0.000721 & 0.000816 & 1.087264 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}_{ACES_{APO}}$$

In one embodiment, a color value in ACES APO is converted to XYZ with a D65 white point. In one embodiment, the conversion from ACES APO to XYZ with a D65 white point is performed using the following equation:

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix}_{D65} = \begin{bmatrix} 0.950348 & 0.000000 & 0.000101 \\ 0.343170 & 0.734689 & -0.077859 \\ 0.000000 & 0.000000 & 1.088917 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}_{ACES_{APO}}$$

In one embodiment, a color value in ACES APO is converted to XYZ with a D60 white point. In one embodiment, the conversion from ACES APO to XYZ with a D60 white point is performed using the following equation:

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix}_{D60} = \begin{bmatrix} 0.952552 & 0.000000 & 0.000094 \\ 0.343966 & 0.728166 & -0.072133 \\ 0.000000 & 0.000000 & 1.008825 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}_{ACES_{APO}}$$

In one embodiment, a color value in ACES APO with a D60 white point is converted to ACES APO with a D65 white point. In one embodiment, the conversion from ACES APO with a D60 white point to ACES APO with a D65 white point is performed using a Bradford chromatic adaptation. In one embodiment, the conversion from ACES APO with a D60 white point to ACES APO with a D65 white point is performed using the following equation:

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$$\begin{bmatrix} R \\ G \\ B \end{bmatrix}_{ACES_{APO-D65}} = \begin{bmatrix} 0.986779 & -0.005817 & 0.003823 \\ 0.001868 & 0.996064 & -0.000986 \\ 0.015613 & 0.003559 & 1.001637 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}_{ACES_{APO-D60}}$$

In one embodiment, a color value in XYZ with a D60 white point is converted to XYZ with a D65 white point. In one embodiment, the conversion from XYZ with a D60 white point to XYZ with a D65 white point is performed using a Bradford chromatic adaptation. In one embodiment, the conversion from XYZ with a D60 white point to XYZ with a D65 white point is performed using the following equation:

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix}_{D65} = \begin{bmatrix} 0.987239 & -0.007592 & 0.003067 \\ -0.006107 & 1.001864 & -0.005086 \\ 0.015927 & 0.005321 & 1.081537 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}_{D60}$$

In one embodiment, a color value in ACES APO is converted to XYZ with a D65 white point using a Bradford chromatic adaptation. In one embodiment, the conversion from ACES APO to XYZ with a D65 white point is performed using the following equation:

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix}_{D65} = \begin{bmatrix} 0.937785 & -0.005528 & 0.003734 \\ 0.338790 & 0.729523 & -0.077399 \\ 0.017001 & 0.003875 & 1.090699 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}_{ACES_{APO}}$$

In one embodiment, the system uses equations found in RP 177 to generate a normalized set of values for color conversion matrices. "RP 177:1993—SMPTE Recommended Practice—Derivation of Basic Television Color Equations," in RP 177:1993, vol., no., pp. 1-4, 1 Nov. 1993, doi: 10.5594/SMPTE.RP177.1993, is incorporated herein by reference in its entirety. In one embodiment, the calculation of the normalized set of values for the color conversion matrices uses the following equations:

$$X = \frac{xY}{y} \quad Y = 1 \quad Z = \frac{(1-x-y)Y}{y}$$

An RGB-to-XYZ matrix and a CMY-to-XYZ matrix are created using a white point (e.g., D65) of the system. In each of these triads, the white point is within the triad. The RGB-to-XYZ matrix determines what ratio of each normalized primary (R, G, and B) is required to achieve the D65 white with an RGB value of [1 1 1], and then scales the matrix such that the input of [1 1 1] yields the XYZ of D65. Similarly, the CMY-to-XYZ matrix determines what ratio of each normalized primary (C, M, and Y) is required to achieve the D65 white with a CMY value of [1 1 1], and then scales the matrix such that the input of [1 1 1] yields the XYZ of D65. With these two matrices, additional triad-to-XYZ matrices are created by using the columns in the RGB-to-XYZ matrix and the CMY-to-XYZ matrix. For example, an RMB-to-XYZ matrix is created by taking the RGB-to-XYZ matrix and replacing the second column with the second column from the CMY-to-XYZ matrix. In this case, as well as all other triads, the D65 white point is not inside the triad, so in that case no combination of RMB can achieve the D65 color point.

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In one embodiment, a hardware calibration is required to create the conversion matrices. For example, if a projector has 7 signals (e.g., RGBCMYW) for a six primary system, the output is measured as real XYZ for the individual primaries at maximum and for white when all six primaries are at maximum. A projector calibration is required to adjust the intensities of the primaries to achieve a white point (e.g., D65) from a [1 1 1 1 1 1] position. The matrices are then created as described above.

A normalized matrix is used when a multi-primary display device is calibrated such that full power of all primaries yields the intended white point (e.g., D65). If the multi-primary display device has primaries such that the result from all primaries being at full power gives a white point that is not the desired white point, then the non-normalized method is used and the calibration is done (e.g., via software, a look-up table (LUT)) to scale the primaries such that the full power values result in the desired white point. In one embodiment, the LUT is a three-dimensional (3D) LUT.

Alternatively, the system uses a non-normalized method to generate a set of values for the color conversion matrices. For example, a set of six primaries P_1 - P_6 has a set of xyz primary values as shown in Table 18. Although this example shows a set of six primaries, this process is operable to be used with at least four primaries.

TABLE 18

P_{1xyz}	P_{1x}	P_{1y}	P_{1z}
P_{2xyz}	P_{2x}	P_{2y}	P_{2z}
P_{3xyz}	P_{3x}	P_{3y}	P_{3z}
P_{4xyz}	P_{4x}	P_{4y}	P_{4z}
P_{5xyz}	P_{5x}	P_{5y}	P_{5z}
P_{6xyz}	P_{6x}	P_{6y}	P_{6z}

In one embodiment, an XYZ-to- $P_1P_2P_3$ matrix is created using the inverse of the result from transposing a P_1XYZ matrix created from the primary values in Table 18, a P_2XYZ matrix created from the primary values in Table 18, and a P_3XYZ matrix created from the primary values in Table 18. The conversion from XYZ color space to $P_1P_2P_3$ is shown below in the following equation:

$$\begin{bmatrix} P_1 \\ P_2 \\ P_3 \end{bmatrix} = \begin{bmatrix} P_{1x} & P_{1y} & P_{1z} \\ P_{2x} & P_{2y} & P_{2z} \\ P_{3x} & P_{3y} & P_{3z} \end{bmatrix}^{-1} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} P_{1x} & P_{2x} & P_{3x} \\ P_{1y} & P_{2y} & P_{3y} \\ P_{1z} & P_{2z} & P_{3z} \end{bmatrix}^{-1} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

Similarly, in one embodiment, an XYZ-to- $P_4P_5P_6$ matrix is created using the inverse of the result from transposing a P_4xyz matrix created from the primary values in Table 18, a P_5xyz matrix created from the primary values in Table 18, and a P_6xyz matrix created from the primary values in Table 18. The conversion from XYZ color space to $P_4P_5P_6$ is shown below in the following equation:

$$\begin{bmatrix} P_4 \\ P_5 \\ P_6 \end{bmatrix} = \begin{bmatrix} P_{4x} & P_{4y} & P_{4z} \\ P_{5x} & P_{5y} & P_{5z} \\ P_{6x} & P_{6y} & P_{6z} \end{bmatrix}^{-1} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} P_{4x} & P_{5x} & P_{6x} \\ P_{4y} & P_{5y} & P_{6y} \\ P_{4z} & P_{5z} & P_{6z} \end{bmatrix}^{-1} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

This process is used to create additional matrices for alternative triads. As stated above, although this process is shown using six primaries, alternate numbers of primaries, numbers of triads, and alternate triads are compatible with the present invention.

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FIG. 84 illustrates one embodiment of a primary triad system for a multi-primary system including red, green, blue, cyan, magenta, and yellow primaries. The primary triad system includes an RGB triad, a CMY triad, an RMB triad, a CGB triad, and an RGY triad.

Primary values for each color value (RGBCMY) are listed in Table 19 below.

TABLE 19

Rxyz	0.68	0.32	0
Gxyz	0.265	0.69	0.045
Bxyz	0.15	0.06	0.79
Cxyz	0.1617	0.3327	0.5056
Mxyz	0.3383	0.1372	0.5245
Yxyz	0.4470	0.5513	0.0017

In one embodiment, an XYZ-to-RGB matrix is created using the inverse of the result from transposing an Rxyz matrix created from the primary values in Table 19, a Gxyz matrix created from the primary values in Table 19, and a Bxyz matrix created from the primary values in Table 19. For example, to obtain the RGB triad, an inverse is taken of the transpose of a matrix created using the first three rows of Table 19 for 6P-C. The conversion from D65 XYZ color space to RGB in a 6P-C system using a D65 white point is shown below in the following equation:

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix}_{6P-C} = \begin{bmatrix} 1.78421 & -0.666447 & -0.288158 \\ -0.831579 & 1.76711 & 0.0236842 \\ 0.0473684 & -0.100658 & 1.26447 \end{bmatrix}^{-1} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}_{D65}$$

In one embodiment, an XYZ-to-CMY matrix is created using the inverse of the result from transposing a Cxyz matrix created from the primary values in Table 19, an Mxyz matrix created from the primary values in Table 19, and a Yxyz matrix created from the primary values in Table 19. The conversion from D65 XYZ color space to CMY in a 6P-C system using a D65 white point is shown below in the following equation:

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix}_{6P-C} = \begin{bmatrix} -3.06125 & 2.47801 & 1.32629 \\ 2.94733 & -2.39167 & 0.631184 \\ 1.11392 & 0.913668 & -0.957473 \end{bmatrix}^{-1} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}_{D65}$$

In one embodiment, an XYZ-to-RMB matrix is created using the inverse of the result from transposing an Rxyz matrix created from the primary values in Table 19, an Mxyz matrix created from the primary values in Table 19, and a Bxyz matrix created from the primary values in Table 19. The conversion from D65 XYZ color space to RMB in a 6P-C system using a D65 white point is shown below in the following equation:

$$\begin{bmatrix} R \\ M \\ B \end{bmatrix}_{6P-C} = \begin{bmatrix} -9.56454 & 23.4496 & 0.0350659 \\ 31.435 & -66.7993 & -0.8953 \\ -20.8704 & 44.3497 & 1.86023 \end{bmatrix}^{-1} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}_{D65}$$

In one embodiment, an XYZ-to-CBG matrix is created using the inverse of the result from transposing a Cxyz matrix created from the primary values in Table 19, a Bxyz matrix created from the primary values in Table 19, and a Gxyz matrix created from the primary values in Table 19.

The conversion from D65 XYZ color space to CBG in a 6P-C system using a D65 white point is shown below in the following equation:

$$\begin{bmatrix} C \\ B \\ G \end{bmatrix}_{6P-C} = \begin{bmatrix} -22.6099 & 8.44536 & 3.6516 \\ 13.9183 & -5.28179 & -0.975739 \\ 9.69162 & -2.16357 & -1.67586 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}_{D65}$$

In one embodiment, an XYZ-to-RGY matrix is created using the inverse of the result from transposing an Rxyz matrix created from the primary values in Table 19, a Gxyz matrix created from the primary values in Table 19, and a Yxyz matrix created from the primary values in Table 19. The conversion from D65 XYZ color space to RGY in a 6P-C system using a D65 white point is shown below in the following equation:

$$\begin{bmatrix} R \\ G \\ Y \end{bmatrix}_{6P-C} = \begin{bmatrix} 2.41684 & -2.01079 & 16.5996 \\ 0.0556266 & -0.118206 & 23.7071 \\ -1.47247 & 3.12899 & -39.3067 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}_{D65}$$

The XYZ value is multiplied by each of the XYZ-to-triad matrices shown above (i.e., the XYZ-to-RGB matrix, the XYZ-to-CMY matrix, the XYZ-to-RMB matrix, the XYZ-to-CBG matrix, the XYZ-to-RGY matrix). The result of each multiplication is filtered for negative values. If a resulting matrix includes one or more negative values, all three values in the resulting matrix are set to zero. This results in a set of triad vectors. The primary components of the at least five primary triads are added together on a per-component basis (e.g., {SUM (R), SUM (B), SUM (G), SUM (C), SUM (M), SUM (Y)}). In one embodiment, if any of the merged colors are present in at least two triads, the primary components are divided by a number of the at least two primary triads to create a set of final values. For example, if any of the merged colors are present in two triads (e.g., RGB and RMB), the values are divided by two (2) and

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix}_{D65} =$$

$$\begin{bmatrix} 0.48657095 & 0.26566769 & 0.19821729 & 0.32295962 & -0.54969800 & 1.177199435 \\ 0.22897456 & 0.69173852 & 0.07928691 & 0.67867175 & -0.22203240 & 0.543360700 \\ 0.00000000 & 0.04511338 & 1.04394437 & 0.98336936 & -0.78858190 & 0.894270250 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \\ C \\ M \\ Y \end{bmatrix}_{6P-C}$$

merged back into the matrices, resulting in a set of final 6P values. If the merged colors are only present in one triad, this represents the set of final 6P values.

If the signal values for two triads are non-negative, the sums for each component are divided by two. In the embodiment described above (i.e., RGB, CMY, RBM, CBG, and RGY), it is not possible for more than two triads to be completely non-negative. In alternative embodiments, with a different set of at least five triads, it is possible for more than two triads to be completely non-negative. In the six primary system described above, the result is output as an RGBCMY value.

If all of the values for the at least five primary triads resulting from each multiplication are negative, the color is

out-of-gamut for the at least six primaries and must be mapped to an in-gamut color. In one embodiment, signals from the triad with the least negative minimum value are used, with the negative signal values clipped to zero. These signal values are then used on a per-component basis for each out-of-gamut signal. After including all of the out-of-gamut signals, the signal values for each of the at least six primaries (e.g., RGBCMY) are clipped to one.

In one embodiment, the system includes a three-dimensional look up table that converts an out-of-gamut color value to an in-gamut color value for the at least six primary system. In one embodiment, XYZ values for the out-of-gamut color value are converted to xy Y. The out-of-gamut color value is substituted to an in-gamut color value with a new xy and the original Y, thereby creating a new xyY. The new xyY is then transformed to XYZ. If any channel is greater than 1.0, the complete triad is divided by the maximum, scaling the channel to a maximum value of 1.0.

In a preferred embodiment, the system maps the out-of-gamut color to an in-gamut color by defining a vector using a white point and the out-of-gamut color and mapping the out-of-gamut color to a locus edge of the multi-primary system along the vector as shown in FIG. 85. Points inside the shape are white and represent points that do not need to be remapped. Advantageously, mapping along the straight line to the white point provides a more radially consistent and more operationally reasonable position. Alternatively, the system maps the out-of-gamut color value to the nearest in-gamut color regardless of a location of the white point (e.g., perpendicular reference) as shown in FIG. 86. In one embodiment, the mapping is done in xy Y color space. The colorimetric xy values are mapped from the out-of-gamut color to the in-gamut color. In a preferred embodiment, the Y is carried along unchanged and merged with the new remapped x,y in-gamut color.

The set of final 6P values is converted back to XYZ space with a D65 white point using a 6P-to-XYZ matrix. The conversion from 6P-to-XYZ for 6P-C is shown below in the following equation:

$$\begin{bmatrix} R \\ G \\ B \end{bmatrix}_{ACES_{D65}} = \begin{bmatrix} 1.062343 & 0.008404 & -0.016611 \\ -0.493934 & 1.371087 & 0.090340 \\ -0.000334 & -0.001034 & 0.919683 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}_{D65}$$

In one embodiment, the ISO 17321 matrix/chart values are defined using a perfect reflecting diffuser matrix

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(0.97784 0.97784 0.97784), an 18% grey card matrix (0.18 0.18 0.18), and the 24 patches of the ISO 17321-1 chart, as illuminated using CIE D60:

(0.11877 0.08709 0.05895), (0.40002 0.31916 0.23737),
 (0.18476 0.20398 0.31311), (0.10901 0.13511 0.06493),
 (0.26684 0.24604 0.40932), (0.32283 0.46208 0.40606),
 (0.38606 0.22744 0.05777), (0.13822 0.13037 0.33703),
 (0.30203 0.13752 0.12758), (0.0931 0.06347 0.13525),
 (0.34876 0.43654 0.10614), (0.48656 0.36686 0.08061),
 (0.08732 0.07443 0.27274), (0.15366 0.25691 0.09071),
 (0.21742 0.0707 0.0513), (0.5892 0.53943 0.09157),
 (0.30904 0.14818 0.27426), (0.14901 0.23378 0.35939),
 (0.86653 0.86792 0.85818), (0.57356 0.57256 0.57169),
 (0.35346 0.35337 0.35391), (0.20253 0.20243 0.20287),
 (0.09467 0.0952 0.09637), (0.03745 0.03766 0.03895).

These are the ACES RICD values for a perfect reflecting diffuser, an 18% gray card, and the 24 patches of the ISO 17321-1 chart, as illuminated using CIE D60.

While the process shown in FIG. 87 is using the ACES RICD values for a perfect reflecting diffuser, an 18% gray card, and the 24 patches of the ISO 17321-1 chart, as illuminated using CIE D60, it can be used to validate any conversion from ACES-to-6P-to-ACES.

It is important to note that 6P {1,1, 1,1, 1,1} converts to ACES-0 {2,2,2}. In one embodiment using ITU-R BT.2100 color space, it is necessary to perform a scaling operation of the linear display-referred RGB values, followed by applying an inverse Perceptual Quantizer (PQ) EOTF. The scaling is such that 6P data {1,1,1,1,1,1} maps to 10-bit PQ {668, 668,668}. In one embodiment, the scaling maps at a rate of 403 candelas per square meter (cd/m²).

In a more preferred embodiment, the system uses at least eight primary triads. FIG. 88 illustrates one embodiment of a system with at least eight primary triads. In one embodiment, the at least eight primary triads are formed using at least six primaries. In one embodiment, a multi-primary system includes six primaries (e.g., P₁-P₆) and eight primary triads. In one embodiment, the first triad is formed using P₁, P₂, and P₃; the second triad is formed using P₄, P₅, and P₆; the third triad is formed using P₁, P₃, and P₅; the fourth triad is formed using P₂, P₃, and P₄; the fifth triad is formed using P₁, P₂, and P₆; the sixth triad is formed using P₃, P₄, and P₅; the seventh triad is formed using P₂, P₄, and P₆; and the eighth triad is formed using P₁, P₅, and P₆. Any point is included in a first triad of a first set of triads (e.g., P₁P₂P₃, P₁P₂P₅, P₁P₃P₆, or P₂P₃P₄) and a second triad of a second set of triads (e.g., P₄P₅P₆, P₂P₄P₅, P₁P₅P₆, or P₃P₄P₆). For example, a multi-primary system includes a red primary (R), a green primary (G), a blue primary (B), a cyan primary (C), a yellow primary (Y), and a magenta primary (M). Any point is present in one of RGB, RGY, RMB, or CGB and in one of CMY, CGY, RMY, or CMB. In one embodiment, the resulting values are added together and divided by two as shown in the equation below (e.g., for an XYZ input), wherein a point is present in only one triad in each set shown in brackets:

$$RGBCMY = \left(\begin{bmatrix} RGB \\ RGY \\ RMB \\ CGB \end{bmatrix} + \begin{bmatrix} CMY \\ CGY \\ RMY \\ CMB \end{bmatrix} \right) / 2$$

In a preferred embodiment, the third triad, the fourth triad, and the fifth triad are formed by doing one replacement in the first triad, and the sixth triad, the seventh triad, and the

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eighth triad are formed by doing one replacement in the second triad. In one embodiment, red replaces cyan, blue replaces yellow, green replaces magenta, cyan replaces red, yellow replaces blue, and/or magenta replaces green. For example, RGB is modified to RMB, RGY, and CGB, and CMY is modified to RMY, CGY, and CMB. In a preferred embodiment, the primary triads are selected such that none of the primary triads include both a primary and its complement (e.g., no triad contains R and C, no triad contains B and Y, no triad contains G and M). Primary triads including both a primary and its complement are not needed for a full color description of the system. In a preferred embodiment, the primary triads are selected such that the first triad includes the white point, the second triad includes the white point, and no other triads include the white point. For example, RGB includes the white point, CMY includes the white point, and no other triads include the white point.

Alternate numbers of primaries, alternate numbers of triads, and alternate triads are compatible with the present invention. Advantageously, this embodiment provides an easier method of calculating luminance because every color is included in two triads and a simple divide by two provides a final result while still describing the full gamut using the primary triads. Out-of-gamut colors are mapped to in-gamut colors as previously described.

The XYZ-to-RGB matrix, the XYZ-to-CMY matrix, the XYZ-to-RMB matrix, the XYZ-to-CBG matrix, and the XYZ-to-RGY matrix are calculated as previously described. Additionally, in one embodiment, an XYZ-to-CMB matrix, an XYZ-to-RMY matrix, and an XYZ-to-CGY matrix are calculated as described below.

In one embodiment, an XYZ-to-CMB matrix is created using the inverse of the result from transposing a Cxyz matrix created from the primary values in Table 19, an Mxyz matrix created from the primary values in Table 19, and a Bxyz matrix created from the primary values in Table 19. The conversion from D65 XYZ color space to CMB in a 6P-C system using a D65 white point is shown below in the following equation:

$$\begin{bmatrix} C \\ M \\ B \end{bmatrix}_{6P-C} = \begin{bmatrix} -1.52475 & 3.73828 & 0.00559012 \\ 4.60881 & -1.02888 & -0.796949 \\ -2.08406 & -1.7094 & 1.79136 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}_{D65}$$

In one embodiment, an XYZ-to-RMY matrix is created using the inverse of the result from transposing an Rxyz matrix created from the primary values in Table 19, an Mxyz matrix created from the primary values in Table 19, and a Yxyz matrix created from the primary values in Table 19. The conversion from D65 XYZ color space to RMY in a 6P-C system using a D65 white point is shown below in the following equation:

$$\begin{bmatrix} R \\ M \\ Y \end{bmatrix}_{6P-C} = \begin{bmatrix} 2.37548 & -1.92289 & -1.02918 \\ 0.00447267 & -0.00950442 & 1.90618 \\ -1.37995 & 2.93239 & 0.122998 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}_{D65}$$

In one embodiment, an XYZ-to-CGY matrix is created using the inverse of the result from transposing a Cxyz matrix created from the primary values in Table 19, a Gxyz matrix created from the primary values in Table 19, and a Yxyz matrix created from the primary values in Table 19. The conversion from D65 XYZ color space to CGY in a 6P-C system using a D65 white point is shown below in the following equation:

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$$\begin{bmatrix} C \\ G \\ Y \end{bmatrix}_{6P-C} = \begin{bmatrix} 0.29784 & -0.2478 & 2.04565 \\ -3.50535 & 2.84449 & -0.750686 \\ 4.20751 & -1.59669 & -0.294967 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}_{D65}$$

A set of matrices is defined for the at least eight triads based on the XYZ values of the at least six primaries in a multi-primary system. For each set of XYZ values (per pixel), the values for each of the at least eight triads are derived by multiplying the XYZ value by each triad inverse matrix. Each set of triad values is checked for negative values, and any triad with a negative value is set to (0, 0, 0). Like components from each triad are summed for each pixel and a set of values in the multi-primary system are created for each pixel. The sum is divided by a number of triads with non-negative values to compensate for luminance. For example, if a point is found in one triad, the sum is the final value; if a point is found in two triads, the sum is divided by two to provide the final value; if a point is found in three triads, the sum is divided by three to provide the final value; if a point is found in four triads, the sum is divided by four to provide the final value; etc.

As previously described, in a six primary system, there are a total of 20 possible triads. Thus, points may be included in more than two triads of the total of 20 possible triads. However, in the embodiment of the present invention described above using eight triads (i.e., RGB, CMY, RMB, CBG, RGY, CMB, RMY, and CGY), no point is included in more than two triads. This is advantageous over a system that uses all possible triads (e.g., 20 for RGBCMY) because it requires significantly less processing for calculations. Further, a simple divide by two is always required for calculations, which simplifies logic and reduces processing required to complete the calculations.

FIG. 89 illustrates a flow chart of an embodiment of a system with eight triads in a six-primary system (i.e., RGBCMY).

Triads in Four Primary and Five Primary Systems

In one embodiment, a multi-primary system includes four primaries (e.g., P_1 - P_4). Any point is included in a first triad of a first set of triads including P_1P_4 (e.g., $P_1P_4P_2$ or $P_1P_4P_3$) and a second triad of a second set of triads including P_2P_3 (e.g., $P_2P_3P_1$ or $P_2P_3P_4$). For example, a multi-primary system includes a red primary (R), a green primary (G), a blue primary (B), and a cyan primary (C). Any point is present in one of RGB or CGB and in one of RCB or RCG. In one embodiment, the resulting values are added together and divided by two as shown in the equation below (e.g., for an XYZ input), wherein a point is present in only one triad in each set shown in brackets:

$$RGBC = \left(\begin{bmatrix} RGB \\ CGB \end{bmatrix} + \begin{bmatrix} RCB \\ RCG \end{bmatrix} \right) / 2$$

In one embodiment, a multi-primary system includes five primaries (e.g., P_1 - P_5). Any point is included in a first triad of a first set of triads (e.g., $P_1P_4P_3$, $P_1P_4P_5$, or $P_4P_2P_5$) and a second triad of a second set of triads (e.g., $P_1P_2P_5$, $P_1P_2P_3$, or $P_2P_3P_4$). For example, a multi-primary system includes a red primary (R), a green primary (G), a blue primary (B), a cyan primary (C), and a yellow primary (Y). Any point is present in one of RCY, RCB, or CGY and in one of RGY, RGB, or CGB. In one embodiment, the resulting values are added together and divided by two as shown in the equation

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below (e.g., for an XYZ input), wherein a point is present in only one triad in each set shown in brackets:

$$RGBC = \left(\begin{bmatrix} RCY \\ RCB \\ CGY \end{bmatrix} + \begin{bmatrix} RGY \\ RGB \\ CGB \end{bmatrix} \right) / 2$$

Advantageously, each color point resides within only two triads. As previously described, in a five primary system, there are a total of 10 possible triads. Thus, points may be included in more than two triads of the total of 10 possible triads. However, in the embodiment of the present invention described above using six triads (i.e., RGB, RGY, CGB, RCB, CGY, and CRY), no point is included in more than two triads. This is advantageous over a system that uses all possible triads (e.g., 10 for RGBCY) because it requires significantly less processing for calculations. Further, a simple divide by two is always required for calculations, which simplifies logic and reduces processing required to complete the calculations.

In one embodiment, the system uses at least one virtual primary in at least one triad. In one example, an RGBC system uses a virtual magenta primary and a virtual yellow primary. In one embodiment, the virtual magenta primary is an average of a red primary point and a blue primary point. In one embodiment, the virtual yellow primary is an average of a red primary point and a green primary point. In another example, an RGBY system uses a virtual magenta primary and a virtual cyan primary. In one embodiment, the virtual cyan primary is an average of a green primary point and a blue primary point. In yet another example, an RGBCY system uses a virtual magenta primary. In one embodiment, the at least one virtual primary is located on a line connecting two non-virtual primaries. Advantageously, using the virtual primaries allows for the eight triads described above (e.g., RGB, CMY, RBM, CBG, RGY, CMB, RMY, and CGY) to be used within a four primary system and/or a five primary system.

Triads with Two Primaries and a Virtual Primary

In another embodiment, the three points forming a triad include two primaries and a point within the color gamut used as a virtual primary. In a preferred embodiment, the virtual primary is a white point (e.g., D65). Each triad is formed using two adjacent primaries and the virtual primary (e.g., the white point). For example, in an RGBCMY system with a white point ("W"), the triads include WRY, WYG, WGC, WCB, WBM, and WMR as shown in FIG. 90. Advantageously, using the white point as a virtual primary covers the complete color area, and points are only within one triangle. Thus, a four primary system with an additional virtual primary results in four triads, a five primary system with an additional virtual primary results in five triads, a six primary system with an additional virtual primary results in six triads, a seven primary system with an additional virtual primary results in seven triads, an eight primary system with an additional virtual primary results in eight triads, a nine primary system with an additional virtual primary results in nine triads, a ten primary system with an additional virtual primary results in ten triads, an eleven primary system with an additional virtual primary results in eleven triads, a twelve primary system with an additional virtual primary results in twelve triads, etc.

The white point W is a virtual primary defined as $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ of RGBCMY. All color points are in only one of the six triads for a six primary system. If a color point is on the

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line between two triads, the system determines in which triad the color point resides (e.g., based on significant figures and precision of a processor).

For example, when reconstructing the RGBCMY from a RYW triad, the XYZ-to-RYW gives the following matrix:

$$[R \ Y \ W] = [0.244 \ 0.572 \ 0.345]$$

All RGBCMY values are set to the virtual primary value, which is 0.345 in the example above. Thus, RGBCMY(R) = RGBCMY(R)+R (=0.244 in the example above) and RGBCMY(Y)=RGBCMY(Y)+Y (=0.572 in the example above), which yields the following matrix:

$$[R \ G \ B \ C \ M \ Y] = [0.589 \ 0.345 \ 0.345 \ 0.345 \ 0.345 \ 0.917]$$

Further away from the white point, the W value decreases substantially. The W gives positive RGBCMY, and if the magnitude of W is greater than the magnitude of one of the other elements, the result might still be all positive in RGBCMY. From the above example, if R=-0.244, then the final R value would still be positive (i.e., R=0.345-0.244=0.101).

Advantageously, the addition of the virtual primary (e.g., white point) constrains the system to a finite number of values when converting from XYZ to a multi-primary color gamut (e.g., RGBCMY). The virtual primary is not necessary when converting from the multi-primary color gamut (e.g., RGBCMY) to XYZ because this operation is well-defined by a 3x6 matrix (e.g., RGBCMY-to-XYZ matrix) and an absolute inverse of the 3x6 matrix is not possible.

In one embodiment, the multi-primary system includes at least one internal primary (e.g., at least one white point) and at least three peripheral primaries. For example, the multi-primary system includes P peripheral primaries and I internal primaries. There will be I sets of P triads formed from the P peripheral primaries and each of the I internal primaries. Any point will only be within one of the P triads including each of the I internal primaries. These resulting values are then averaged together.

In one embodiment, a multi-primary system includes three peripheral primaries (e.g., P₁-P₃) and an internal primary (I). Any point is included in one triad of a set of triads including I (e.g., P₁P₂I, P₁P₃I, or P₂P₃I) and a triad formed with the three peripheral primaries (e.g., P₁P₂P₃). These resulting values from the one triad of the set of triads including I and the triad formed with the three peripheral primaries are added together and divided by two. For example, a multi-primary system includes a red primary (R), a green primary (G), a blue primary (B), and a white primary (W). Any point is present in one of RGW, RBW, or GBW and in RGB. In one embodiment, the resulting values are added together and divided by two as shown in the equation below (e.g., for an XYZ input), wherein a point is present in only one triad in each set shown in brackets:

$$RGBW = \left(\left[\begin{array}{c} RGW \\ RBW \\ GBW \end{array} \right] + RGB \right) / 2$$

In another embodiment, the value of RGW, RBW, or GBW is used without dividing by two to produce RGBW.

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In one embodiment, a multi-primary system includes three peripheral primaries (e.g., P₁-P₃) and two internal primaries (e.g., I₁-I₂). Any point is included in a first triad of a first set of triads including I₁ (e.g., P₁P₂I₁, P₁P₃I₁, or P₂P₃I₁) and a second triad of a second set of triads including I₂ (e.g., P₁P₂I₂, P₁P₃I₂, or P₂P₃I₂). These resulting values from the first triad of the first set of triads and the second triad of the second set of triads are added together and divided by two. For example, a multi-primary system includes a red primary (R), a green primary (G), a blue primary (B), a first white primary (W₁), and a second white primary (W₂). Any point is present in one of RGW₁, RBW₁, or GBW₁ and in one of RGW₂, RBW₂, or GBW₂. In one embodiment, the resulting values are added together and divided by two as shown in the equation below (e.g., for an XYZ input), wherein a point is present in only one triad in each set shown in brackets:

$$RGBW_1W_2 = \left(\left[\begin{array}{c} RGW_1 \\ RBW_1 \\ GBW_1 \end{array} \right] + \left[\begin{array}{c} RGW_2 \\ RBW_2 \\ GBW_2 \end{array} \right] \right) / 2$$

In one embodiment, a multi-primary system includes four peripheral primaries (e.g., P₁-P₄) and two internal primaries (e.g., I₁-I₂). Any point is included in a first triad of a first set of triads including I₁ (e.g., P₁P₂I₁, P₂P₄I₁, P₃P₄I₁, or P₁P₃I₁) and a second triad of a second set of triads including I₂ (e.g., P₁P₂I₂, P₂P₄I₂, P₃P₄I₂, or P₁P₃I₂). These resulting values from the first triad of the first set of triads and the second triad of the second set of triads are added together and divided by two. For example, a multi-primary system includes a red primary (R), a green primary (G), a blue primary (B), a cyan primary (C), a first white primary (W₁), and a second white primary (W₂). Any point is present in one of RGW₁, GCW₁, BCW₁, or RBW₁ and in one of RGW₂, GCW₂, BCW₂, or RBW₂. In one embodiment, the resulting values are added together and divided by two as shown in the equation below (e.g., for an XYZ input), wherein a point is present in only one triad in each set shown in brackets:

$$RGBCW_1W_2 = \left(\left[\begin{array}{c} RGW_1 \\ GCW_1 \\ BCW_1 \\ RBW_1 \end{array} \right] + \left[\begin{array}{c} RGW_2 \\ GCW_2 \\ BCW_2 \\ RBW_2 \end{array} \right] \right) / 2$$

In one embodiment, a multi-primary system includes three peripheral primaries (e.g., P₁-P₃) and three internal primaries (e.g., I₁-I₃). Any point is included in a first triad of a first set of triads including I₁ (e.g., P₁P₂I₁, P₁P₃I₁, or P₂P₃I₁), a second triad of a second set of triads including I₂ (e.g., P₁P₂I₂, P₁P₃I₂, or P₂P₃I₂), and a third triad of a third set of triads including I₃ (e.g., P₁P₂I₃, P₁P₃I₃, or P₂P₃I₃). These resulting values from the first triad of the first set of triads, the second triad of the second set of triads, and the third triad of the third set of triads are added together and divided by three. For example, a multi-primary system includes a first red primary (R₁), a second red primary (R₂), a first green primary (G₁), a second green primary (G₂), a first blue primary (B₁), and a second blue primary (B₂). The second red primary, the second green primary, and the second blue primary are contained within a triad formed by the first red primary, the first green primary, and the first blue primary. Any point is present in one of R₁G₁R₂, R₁B₁R₂, or G₁B₁R₂, in one of R₁G₁G₂, R₁B₁G₂, or G₁B₁G₂, and in one

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of $R_1G_1B_2$, $R_1B_1B_2$, or $G_1B_1B_2$. In one embodiment, the resulting values are added together and divided by three as shown in the equation below (e.g., for an XYZ input), wherein a point is present in only one triad in each set shown in brackets:

$$R_1R_2G_1G_2B_1B_2 = \left(\begin{bmatrix} R_1G_1R_2 \\ R_1B_1R_2 \\ G_1B_1R_2 \end{bmatrix} + \begin{bmatrix} R_1G_1G_2 \\ R_1B_1G_2 \\ G_1B_1G_2 \end{bmatrix} + \begin{bmatrix} R_1G_1B_2 \\ R_1B_1B_2 \\ G_1B_1B_2 \end{bmatrix} \right) / 3$$

In another embodiment, the multi-primary system includes the three peripheral primaries (e.g., P_1 - P_3) and the three internal primaries (e.g., I_1 - I_3). Any point is included in a first triad of a first set of triads (e.g., $P_1P_2I_1$, $P_1I_1I_3$, $P_2P_3I_2$, $P_2I_1I_2$, $P_1P_3I_3$, $P_3I_2I_3$, or $I_1I_2I_3$) and a second triad (e.g., $P_1P_2P_3$). These resulting values from the first triad of the first set of triads and the second triad are added together and divided by two. For example, a multi-primary system includes a first red primary (R_1), a second red primary (R_2), a first green primary (G_1), a second green primary (G_2), a first blue primary (B_1), and a second blue primary (B_2). The second red primary, the second green primary, and the second blue primary are contained within a triad formed by the first red primary, the first green primary, and the first blue primary. Any point is present in one of $R_1G_1R_2$, $R_1R_2B_2$, $G_1B_1G_2$, $G_1R_2G_2$, $R_1B_1B_2$, $B_1G_1B_2$, or $R_2G_2B_2$ and in $R_1G_1B_1$. In one embodiment, the resulting values are added together and divided by three as shown in the equation below (e.g., for an XYZ input), wherein a point is present in only one triad in each set shown in brackets:

$$R_1R_2G_1G_2B_1B_2 = \left(\begin{bmatrix} R_1G_1R_2 \\ R_1R_2B_2 \\ G_1B_1G_2 \\ G_1R_2G_2 \\ R_1B_1B_2 \\ B_1G_1B_2 \\ R_2G_2B_2 \end{bmatrix} + [R_1G_1B_1] \right) / 2$$

FIG. 91 is a schematic diagram of an embodiment of the invention illustrating a computer system, generally described as **800**, having a network **810**, a plurality of computing devices **820**, **830**, **840**, a server **850**, and a database **870**.

The server **850** is constructed, configured, and coupled to enable communication over a network **810** with a plurality of computing devices **820**, **830**, **840**. The server **850** includes a processing unit **851** with an operating system **852**. The operating system **852** enables the server **850** to communicate through network **810** with the remote, distributed user devices. Database **870** may house an operating system **872**, memory **874**, and programs **876**.

In one embodiment of the invention, the system **800** includes a network **810** for distributed communication via a wireless communication antenna **812** and processing by at least one mobile communication computing device **830**. Alternatively, wireless and wired communication and connectivity between devices and components described herein include wireless network communication such as WI-FI, WORLDWIDE INTEROPERABILITY FOR MICRO-WAVE ACCESS (WIMAX), Radio Frequency (RF) communication including RF identification (RFID), NEAR FIELD COMMUNICATION (NFC), BLUETOOTH including BLUETOOTH LOW ENERGY (BLE), ZIGBEE, Infrared (IR) communication, cellular communication, satellite

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communication, Universal Serial Bus (USB), Ethernet communications, communication via fiber-optic cables, coaxial cables, twisted pair cables, and/or any other type of wireless or wired communication. In another embodiment of the invention, the system **800** is a virtualized computing system capable of executing any or all aspects of software and/or application components presented herein on the computing devices **820**, **830**, **840**. In certain aspects, the computer system **800** may be implemented using hardware or a combination of software and hardware, either in a dedicated computing device, or integrated into another entity, or distributed across multiple entities or computing devices.

By way of example, and not limitation, the computing devices **820**, **830**, **840** are intended to represent various forms of electronic devices including at least a processor and a memory, such as a server, blade server, mainframe, mobile phone, personal digital assistant (PDA), smartphone, desktop computer, notebook computer, tablet computer, workstation, laptop, and other similar computing devices. The components shown here, their connections and relationships, and their functions, are meant to be exemplary only, and are not meant to limit implementations of the invention described and/or claimed in the present application.

In one embodiment, the computing device **820** includes components such as a processor **860**, a system memory **862** having a random access memory (RAM) **864** and a read-only memory (ROM) **866**, and a system bus **868** that couples the memory **862** to the processor **860**. In another embodiment, the computing device **830** may additionally include components such as a storage device **890** for storing the operating system **892** and one or more application programs **894**, a network interface unit **896**, and/or an input/output controller **898**. Each of the components may be coupled to each other through at least one bus **868**. The input/output controller **898** may receive and process input from, or provide output to, a number of other devices **899**, including, but not limited to, alphanumeric input devices, mice, electronic styluses, display units, touch screens, signal generation devices (e.g., speakers), or printers.

By way of example, and not limitation, the processor **860** may be a general-purpose microprocessor (e.g., a central processing unit (CPU)), a graphics processing unit (GPU), a microcontroller, a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA), a Programmable Logic Device (PLD), a controller, a state machine, gated or transistor logic, discrete hardware components, or any other suitable entity or combinations thereof that can perform calculations, process instructions for execution, and/or other manipulations of information.

In another implementation, shown as **840** in FIG. 91 multiple processors **860** and/or multiple buses **868** may be used, as appropriate, along with multiple memories **862** of multiple types (e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core).

Also, multiple computing devices may be connected, with each device providing portions of the necessary operations (e.g., a server bank, a group of blade servers, or a multiprocessor system). Alternatively, some steps or methods may be performed by circuitry that is specific to a given function.

According to various embodiments, the computer system **800** may operate in a networked environment using logical connections to local and/or remote computing devices **820**, **830**, **840** through a network **810**. A computing device **830** may connect to a network **810** through a network interface

unit **896** connected to a bus **868**. Computing devices may communicate communication media through wired networks, direct-wired connections or wirelessly, such as acoustic, RF, or infrared, through an antenna **897** in communication with the network antenna **812** and the network interface unit **896**, which may include digital signal processing circuitry when necessary. The network interface unit **896** may provide for communications under various modes or protocols.

In one or more exemplary aspects, the instructions may be implemented in hardware, software, firmware, or any combinations thereof. A computer readable medium may provide volatile or non-volatile storage for one or more sets of instructions, such as operating systems, data structures, program modules, applications, or other data embodying any one or more of the methodologies or functions described herein. The computer readable medium may include the memory **862**, the processor **860**, and/or the storage media **890** and may be a single medium or multiple media (e.g., a centralized or distributed computer system) that store the one or more sets of instructions **900**. Non-transitory computer readable media includes all computer readable media, with the sole exception being a transitory, propagating signal per se. The instructions **900** may further be transmitted or received over the network **810** via the network interface unit **896** as communication media, which may include a modulated data signal such as a carrier wave or other transport mechanism and includes any deliver media. The term “modulated data signal” means a signal that has one or more of its characteristics changed or set in a manner as to encode information in the signal.

Storage devices **890** and memory **862** include, but are not limited to, volatile and non-volatile media such as cache, RAM, ROM, EPROM, EEPROM, FLASH memory, or other solid state memory technology, discs (e.g., digital versatile discs (DVD), HD-DVD, BLU-RAY, compact disc (CD), or CD-ROM) or other optical storage; magnetic cassettes, magnetic tape, magnetic disk storage, floppy disks, or other magnetic storage devices; or any other medium that can be used to store the computer readable instructions and which can be accessed by the computer system **800**.

In one embodiment, the computer system **800** is within a cloud-based network. In one embodiment, the server **850** is a designated physical server for distributed computing devices **820**, **830**, and **840**. In one embodiment, the server **850** is a cloud-based server platform. In one embodiment, the cloud-based server platform hosts serverless functions for distributed computing devices **820**, **830**, and **840**.

In another embodiment, the computer system **800** is within an edge computing network. The server **850** is an edge server, and the database **870** is an edge database. The edge server **850** and the edge database **870** are part of an edge computing platform. In one embodiment, the edge server **850** and the edge database **870** are designated to distributed computing devices **820**, **830**, and **840**. In one embodiment, the edge server **850** and the edge database **870** are not designated for computing devices **820**, **830**, and **840**. The distributed computing devices **820**, **830**, and **840** are connected to an edge server in the edge computing network based on proximity, availability, latency, bandwidth, and/or other factors.

It is also contemplated that the computer system **800** may not include all of the components shown in FIG. **91** may include other components that are not explicitly shown in FIG. **91** or may utilize an architecture completely different than that shown in FIG. **91**. The various illustrative logical

blocks, modules, elements, circuits, and algorithms described in connection with the embodiments discussed herein may be implemented as electronic hardware, computer software, or combinations of both. To clearly illustrate the interchangeability of hardware and software, various illustrative components, blocks, modules, circuits, and steps have been described above generally in terms of their functionality. Whether such functionality is implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system. Skilled artisans may implement the described functionality in varying ways for each particular application (e.g., arranged in a different order or positioned in a different way), but such implementation decisions should not be interpreted as causing a departure from the scope of the present invention.

The above-mentioned examples are provided to serve the purpose of clarifying the aspects of the invention, and it will be apparent to one skilled in the art that they do not serve to limit the scope of the invention. By nature, this invention is highly adjustable, customizable and adaptable. The above-mentioned examples are just some of the many configurations that the mentioned components can take on. All modifications and improvements have been deleted herein for the sake of conciseness and readability but are properly within the scope of the present invention.

The invention claimed is:

1. A method for color conversion in a multi-primary color system, comprising:

providing a color signal corresponding to XYZ values of a color;

providing at least four triads and at least four XYZ-to-triad matrices, wherein at least one of the at least four triads includes at least three primaries, wherein each of the at least four XYZ-to-triad matrices corresponds to one of the at least four triads;

multiplying the XYZ values by the at least four XYZ-to-triad matrices to determine one or more of the at least four triads in which the XYZ values are located;

determining a sum of primary components of the one or more of the at least four triads;

dividing the sum by a number of the one or more of the at least four triads, thereby creating an updated color signal; and

displaying the updated color signal on at least one display device.

2. The method of claim 1, wherein the at least three primaries include at least one white point, and wherein the at least four triads each include two adjacent color and the at least one white point.

3. The method of claim 1, wherein each of the at least four triads does not contain a primary and its complement.

4. The method of claim 1, wherein the at least four triads are eight triads.

5. The method of claim 1, wherein the at least three primaries include red (R), green (G), blue (B), cyan (C), magenta (M), and yellow (Y), wherein the at least four triads are five triads, and wherein the five triads include RGB, CMY, RBM, CGB, and RGY.

6. The method of claim 1, wherein the at least three primaries include red (R), green (G), blue (B), cyan (C), and yellow (Y), wherein the at least four triads are six triads, and wherein the six triads include RGB, RGY, CGB, RCB, CGY, and CRY.

7. The method of claim 1, wherein the one or more of the at least four triads is not more than two of the at least four triads.

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8. The method of claim 1, wherein the color is an out-of-gamut color, further including remapping the out-of-gamut color to an in-gamut color along a straight line between the out-of-gamut color and a white point.

9. The method of claim 1, further including setting all values in a resulting matrix from the multiplying the XYZ values by the at least four XYZ-to-triad matrices to zero if any value in the resulting matrix is negative.

10. The method of claim 1, further including selecting the at least four triads such that a first triad includes a white point, a second triad includes the white point, and no other triads include the white point.

11. The method of claim 1, wherein the at least three primaries include at least one virtual primary.

12. The method of claim 11, wherein the at least one virtual primary is a virtual magenta primary, a virtual yellow primary, a virtual cyan primary, and/or a white point.

13. A method for color conversion in a multi-primary color system, comprising:

providing a color signal corresponding to XYZ values of a color;

providing eight triads and eight XYZ-to-triad matrices, wherein each of at least eight triads includes at least three primaries, wherein each of the eight XYZ-to-triad matrices corresponds to one of the eight triads;

multiplying the XYZ values by the eight XYZ-to-triad matrices to determine two or more of the eight triads in which the XYZ values are located;

determining a sum of primary components of the two or more triads;

dividing the sum by a number of the two or more of the at least eight triads, thereby creating an updated color signal; and

displaying the updated color signal on at least one display device.

14. The method of claim 13, wherein each of the at least eight triads does not contain a primary and its complement.

15. The method of claim 13, wherein the at least three primaries include red (R), green (G), blue (B), cyan (C), magenta (M), and yellow (Y).

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16. The method of claim 15, wherein the at least eight triads include RGB, CMY, RBM, CGB, RGY, CMB, RMY, and CGY.

17. A method for color conversion in a multi-primary color system, comprising:

providing a color signal corresponding to XYZ values of a color;

providing at least four triads and at least four XYZ-to-triad matrices, wherein each of the at least four triads includes three primaries, wherein the at least four triads are selected such that at least one of the at least four triads includes a white point, wherein each of the at least four XYZ-to-triad matrices corresponds to one of the at least four triads;

determining a sum of primary components of one or more of the at least four triads;

dividing the sum by a number of the one or more of the at least four triads, thereby creating an updated color signal; and

displaying the updated color signal on at least one display device.

18. The method of claim 17, wherein the at least three primaries include red (R), green (G), blue (B), cyan (C), magenta (M), and yellow (Y), wherein the at least four triads are eight triads, and wherein the eight triads include RGB, CMY, RBM, CGB, RGY, CMB, RMY, and CGY.

19. The method of claim 17, wherein the at least three primaries include red (R), green (G), blue (B), cyan (C), magenta (M), and yellow (Y), wherein the at least four triads are five triads, and wherein the five triads include RGB, CMY, RBM, CGB, and RGY.

20. The method of claim 17, wherein the at least three primaries include red (R), green (G), blue (B), cyan (C), and yellow (Y), wherein the at least four triads are six triads, and wherein the six triads include RGB, RGY, CGB, RCB, CGY, and CRY.

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